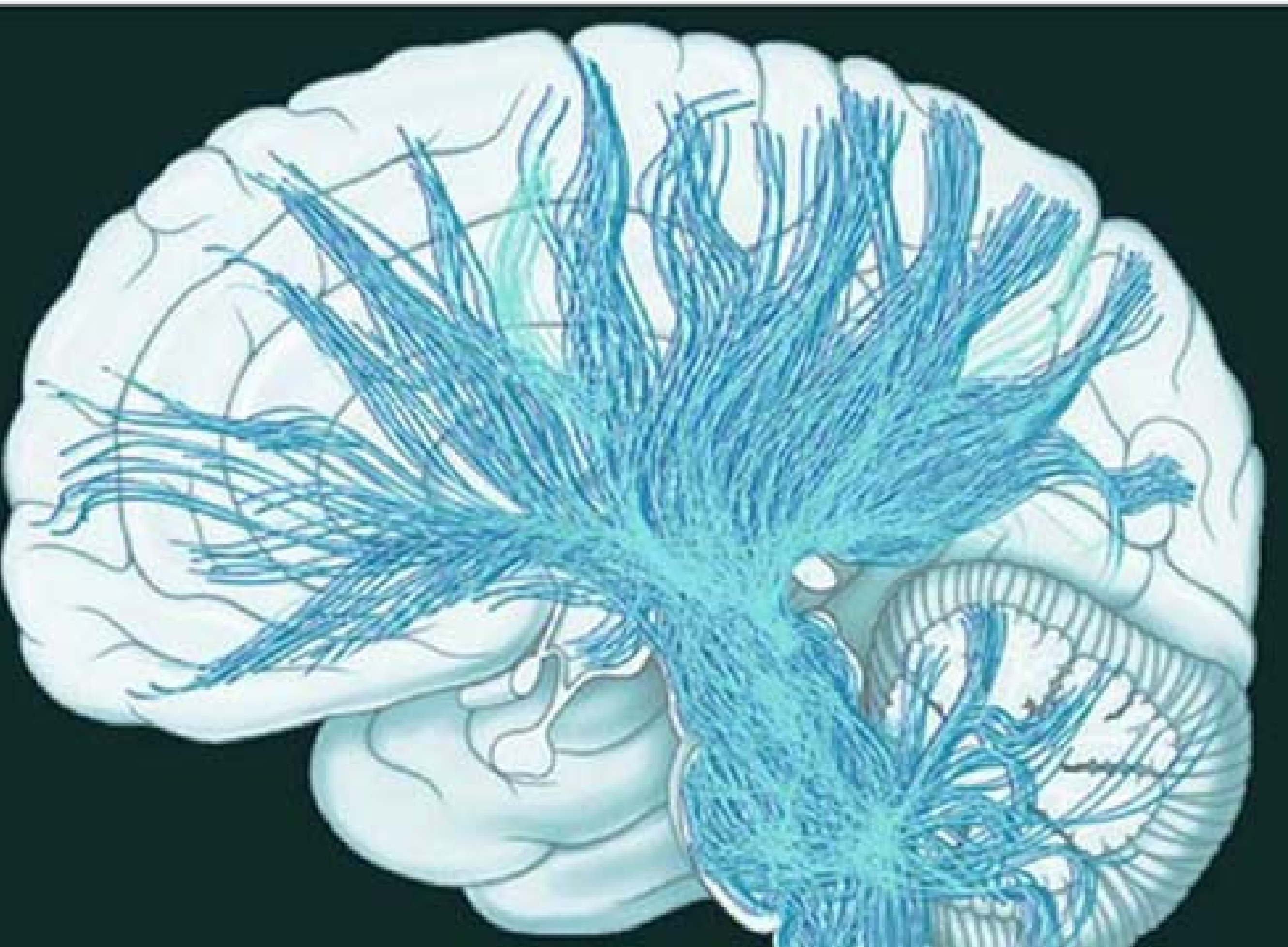
FIFTH EDITION

NEUROLOGY AND NEUROSURGERY ILLUSTRATED

KENNETH W LINDSAY • IAN BONE • GERAINT FULLER



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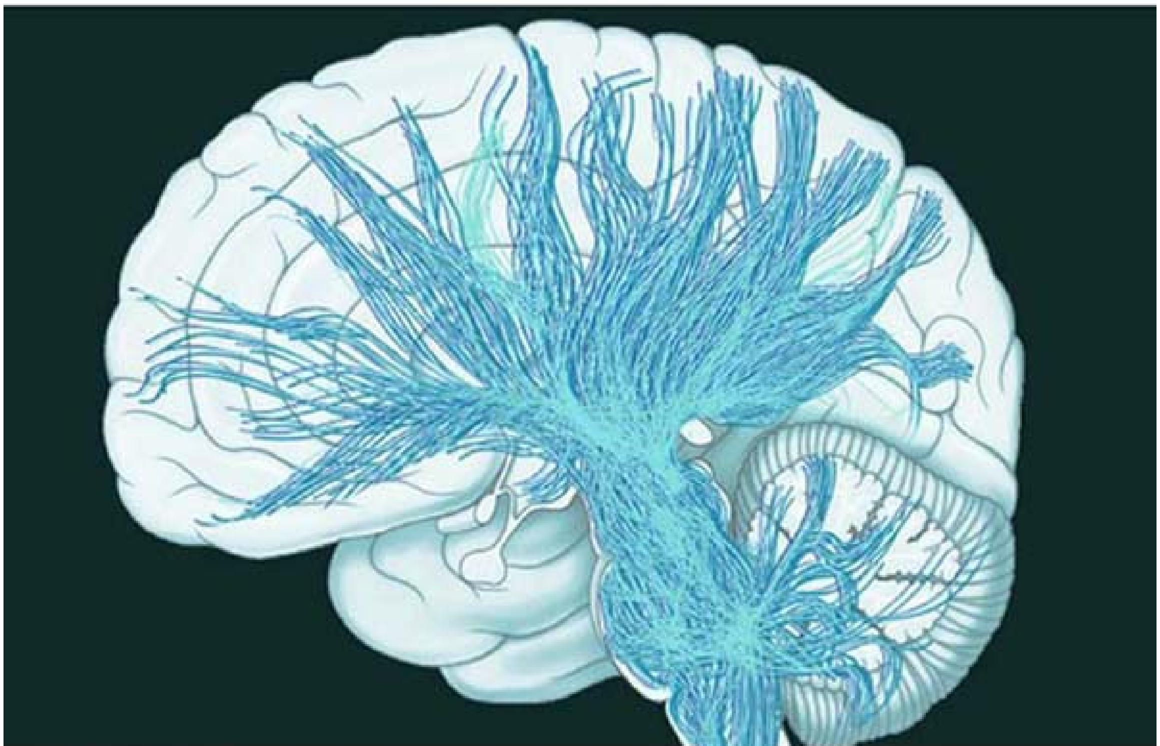

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Front Matter

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Emeritus Professor of Neurology, Utrecht, The Netherlands **FIFTH EDITION**

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FOREWORD

Students often tend to regard diseases of the nervous system as a difficult subject. This book has surely dispelled that traditional belief, as testified by the success of the four previous editions, spanning almost a quarter of a century. The authors have managed to make the nervous system and its disorders accessible in several ways. First and foremost, they have used every possible opportunity to include illustrations, especially simple line drawings, whenever the subject allowed it. In this way the structure and functions of the nervous system, baffling at first sight, are lucidly explained, part by part. Thanks to their didactic guidance, the student will eventually find the matter less complicated than the street map of inner London. Secondly, the text has been restricted to bare essentials. Students do not have to wade through a wilderness of words in order to grasp the key elements they need to know. Finally, between the traditional signposts of physical examination, technical investigations and traditional disease categories, the authors have made ample room for a didactic discussion of the variety of symptoms that bring patients to the neurologist or neurosurgeon - from loss of smell to problems of memory. After all, the patient is the point of departure in medicine. Like a convenient travel guide that leads the tourist to memorable sights, the book will teach the student – and remind the physician - how to understand, recognize and treat disorders of the brain, spinal cord, nerves and muscle. In this fifth edition the authors have taken account of new developments, while preserving the admirable clarity and simplicity that make it stand out from other textbooks.

J. van Gijn, MD FRCP FRCP(Edin) , Emeritus Professor of Neurology, Utrecht, The Netherlands

PREFACE

It has been 24 years since the first edition of *Neurology and Neurosurgery Illustrated* was published. On writing each new edition, we are always surprised at the number of changes required. For this edition there is an additional change. Ian Bone has retired from clinical practice and Geraint Fuller has joined to edit and update this edition. As in all previous editions there have been updates in many areas.

With the increasing trend to sub-specialise within clinical neuroscience, we have become increasingly dependent on colleagues for advice. The following have provided many valuable suggestions – Laurence Dunn, Patricia Littlechild and Jerome St George (neurosurgery), Colin Smith (neuropathology), Alison Wagstaff (neuroanaesthetics), Donald Hadley (neuroradiology) and Roy Rampling (oncology). We would like to offer sincere thanks to all. Finally we are indebted to Ailsa Laing of Elsevier for her patience and gentle encouragement.

2010

K.W. Lindsay, I. Bone, G. Fuller

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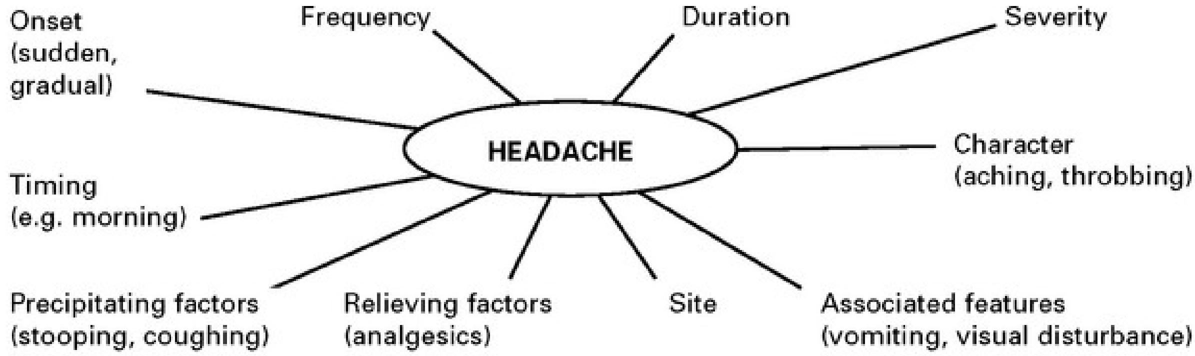
SECTION I

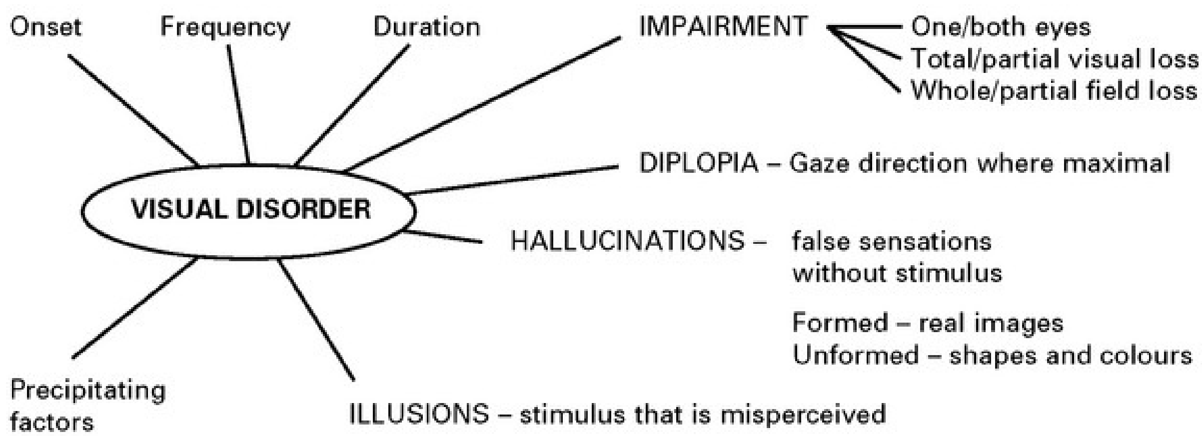
GENERAL APPROACH TO HISTORY AND EXAMINATION

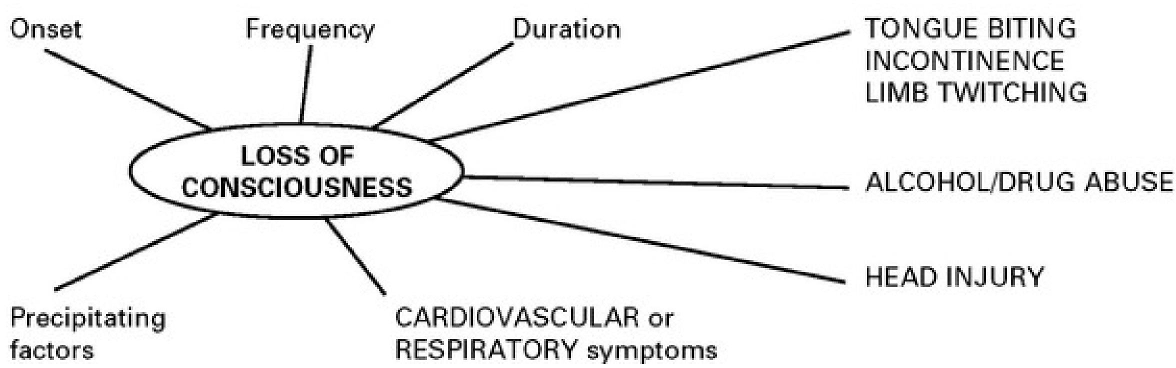
NERVOUS SYSTEM – HISTORY

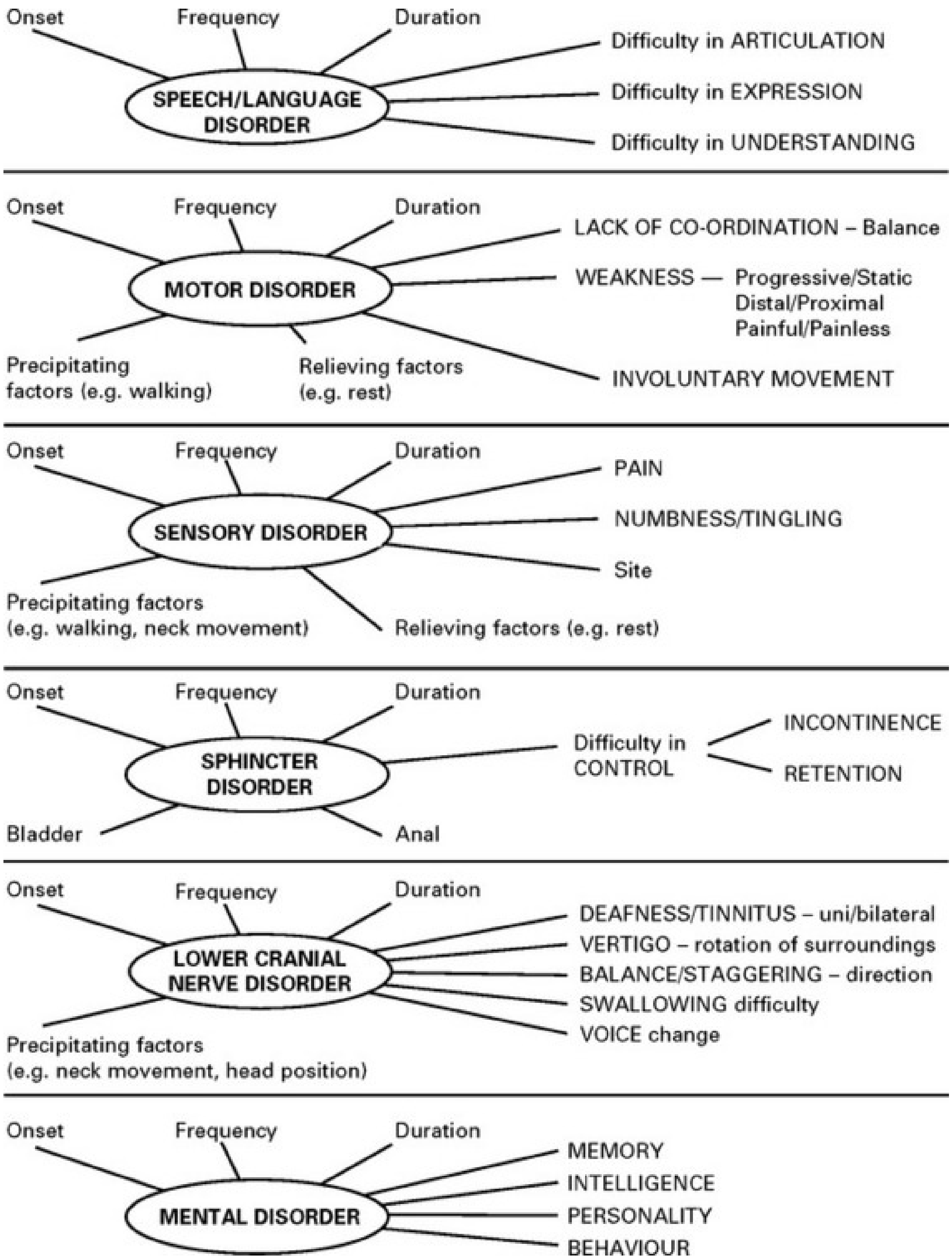
An accurate description of the patient's neurological symptoms is an important aid in establishing the diagnosis; but this must be taken in conjunction with information from other systems, previous medical history, family and social history and current medication. Often the patient's history requires confirmation from a relative or friend.

The following outline indicates the relevant information to obtain for each symptom, although some may require further clarification.


HEADACHE
Onset (sudden, gradual)
Frequency
Duration
Severity
Character (aching, throbbing)
Timing (e.g. morning)
Precipitating factors (stooping, coughing)
Relieving factors (analgesics)
Site
Associated features (vomiting, visual disturbance)


VISUAL DISORDER
Onset
Frequency
Duration
IMPAIRMENT
One/both eyes
Total/partial visual loss
Whole/partial field loss
DIPLOPIA – Gaze direction where maximal
HALLUCINATIONS – false sensations without stimulus
Formed – real images
Unformed – shapes and colours
Precipitating factors
ILLUSIONS – stimulus that is misperceived


LOSS OF CONSCIOUSNESS
Onset
Frequency
Duration
TONGUE BITING
INCONTINENCE
LIMB TWITCHING
ALCOHOL/DRUG ABUSE
HEAD INJURY
Precipitating factors
CARDIOVASCULAR or RESPIRATORY symptoms

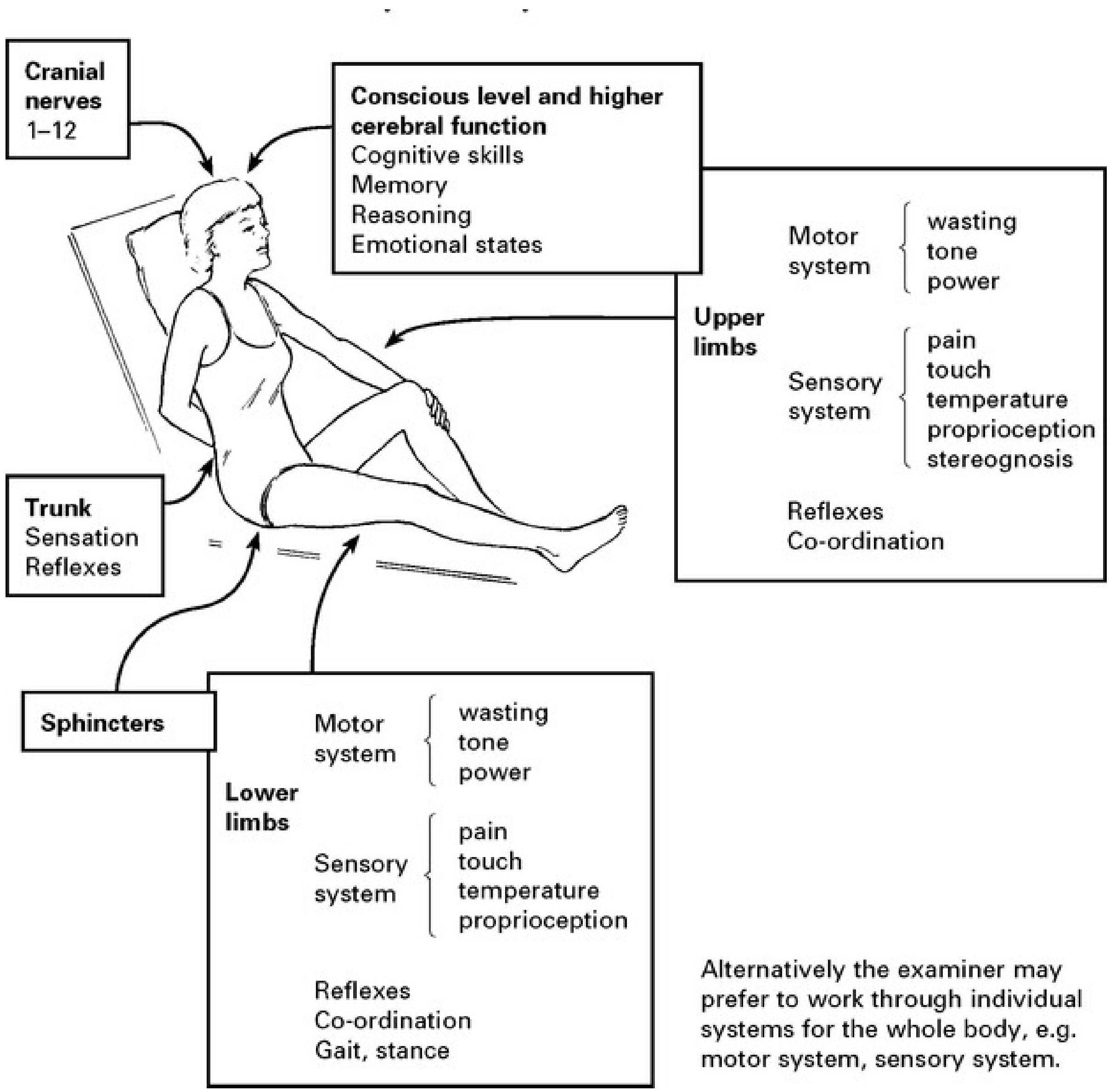


NERVOUS SYSTEM – EXAMINATION

Neurological disease may produce systemic signs and systemic disease may affect the nervous system. A complete general examination must therefore accompany that of the central nervous system. In particular, note the following

Temperature	Evidence of weight loss	Septic source, e.g. teeth, ears,	
Blood pressure	Breast lumps	Skin marks, e.g. rashes	
Neck stiffness	Lymphadenopathy	café-au-lait spots	
Pulse irregularity	Hepatic and splenic enlargement	angiomata	
Carotid bruit		Anterior fontanelle	in baby
Cardiac murmurs	Prostatic irregularity	Head circumference	
Cyanosis/respiratory insufficiency			

CNS examination is described systematically from the head downwards and includes:

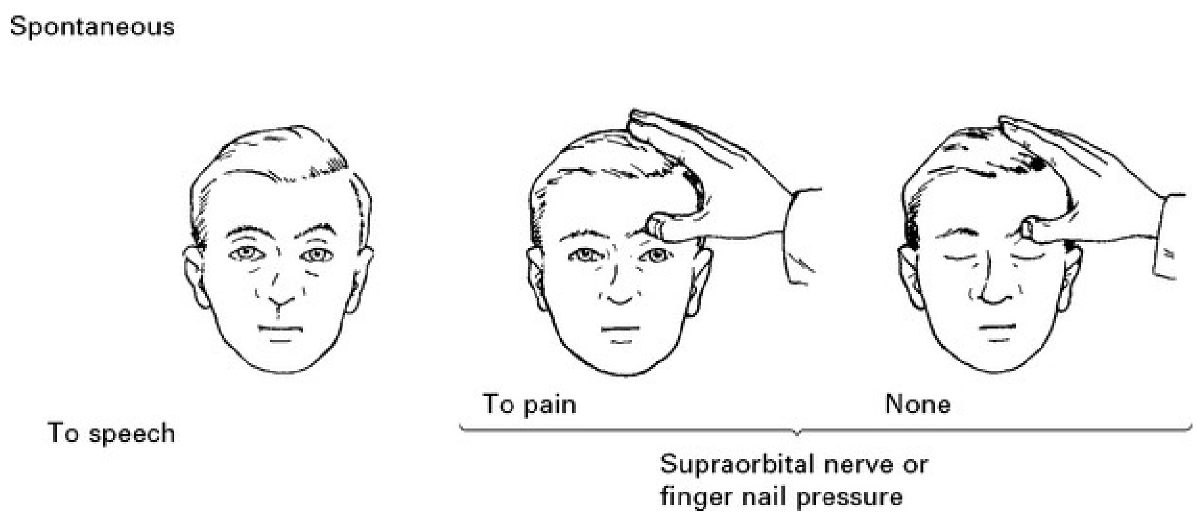


EXAMINATION – CONSCIOUS LEVEL ASSESSMENT

A wide variety of systemic and intracranial problems produce depression of conscious level. Accurate assessment and recording are essential to determine deterioration or improvement in a patient's condition. In 1974 Teasdale and Jennett, in Glasgow, developed a system for conscious level assessment. They discarded vague terms such as stupor, semicoma and deep coma, and described conscious level in terms of EYE opening, VERBAL response and MOTOR response.

The Glasgow coma scale is now used widely throughout the world. Results are reproducible irrespective of the status of the observer and can be carried out just as reliably by paramedics as by clinicians

EYE OPENING – 4 categories

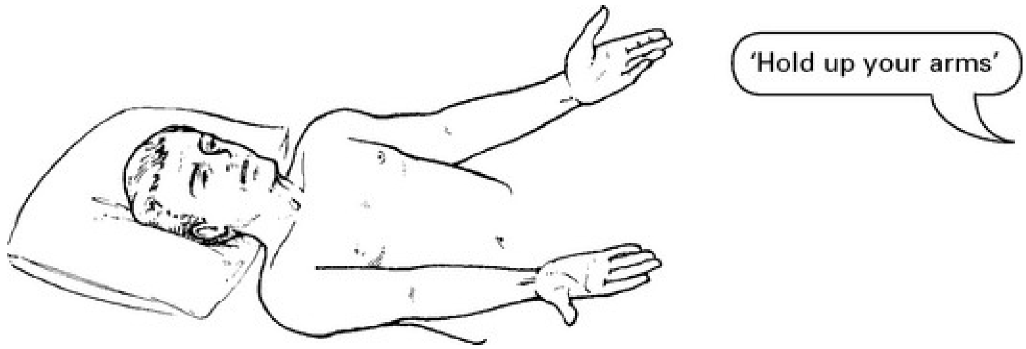


VERBAL RESPONSE – 5 categories

Orientated	– Knows place, *e.g.* Southern General Hospital and time, *e.g.* day, month and year
Confused	– Talking in sentences but disorientated in time and place
Words	– Utters occasional words rather than sentences
Sounds	– Groans or grunts, but no words
None	

MOTOR RESPONSE – 5 categories

Obeys commands



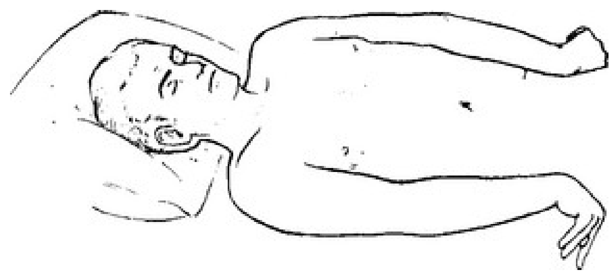
Localising to pain

Apply a painful stimulus to the supraorbital nerve, *e.g.* rub thumb nail in the supraorbital groove, increasing pressure until a response is obtained. If the patient responds by bringing the hand up beyond the chin = 'localising to pain'. (Pressure to nail beds or sternum at this stage may not differentiate 'localising' from 'flexing'.)

Flexing to pain

If the patient does not localise to supraorbital pressure, apply pressure with a pen or hard object to the nail bed. Record elbow flexion as 'flexing to pain'. Spastic wrist flexion may or may not accompany this response.

Extending to pain



If in response to the same stimulus elbow extension occurs, record as 'extending to pain'. This is always accompanied by spastic flexion of the wrist.

None

Before recording a patient at this level, ensure that the painful stimulus is adequate.

During examination the motor response may vary. Supraorbital pain may produce an extension response, whereas fingernail pressure produces flexion. Alternatively one arm may localise to pain; the other may flex. When this occurs record the best response during the period of examination (this correlates best with final outcome). For the purpose of conscious level assessment use only the arm response. Leg response to pain gives less consistent results, often producing movements arising from spinal rather than cerebral origin.

EXAMINATION – HIGHER CEREBRAL FUNCTION

COGNITIVE SKILL

Dominant hemisphere disorders

Listen to language pattern	– hesitant	Expressive dysphasia
	– fluent	Receptive dysphasia
Does the patient understand simple/complex spoken commands? *e.g.* 'Hold up both arms, touch the right ear with the left fifth finger.'		Receptive dysphasia
Ask the patient to name objects.		Nominal dysphasia
Does the patient read correctly?		Dyslexia
Does the patient write correctly?		Dysgraphia
Ask the patient to perform a numerical calculation, *e.g.* serial 7 test, where 7 is subtracted serially from 100.		Dyscalculia
Can the patient recognise objects? *e.g.* ask patient to select an object from a group.		Agnosia

Non-dominant hemisphere disorders

Note patient's ability to find his way around the ward or his home.	Geographical agnosia
Can the patient dress himself?	Dressing apraxia
Note the patient's ability to copy a geometric pattern, *e.g.* ask patient to form a star with matches or copy a drawing of a cube.	Constructional apraxia

Mini Mental Status Examination (MMSE) is used in the assessment of DEMENTIA (page 127).

MEMORY TEST

Testing requires alertness and is not possible in a confused or dysphasic patient.

IMMEDIATE memory –	Digit span – ask patient to repeat a sequence of 5, 6, or 7 random numbers.
RECENT memory –	Ask patient to describe present illness, duration of hospital stay or recent events in the news.
REMOTE memory –	Ask about events and circumstances occurring more than 5 years previously.
VERBAL memory –	Ask patient to remember a sentence or a short story and test after 15 minutes.
VISUAL memory –	Ask patient to remember objects on a tray and test after 15 minutes.

REASONING AND PROBLEM SOLVING

Test patient with two-step calculations, *e.g.* ‘I wish to buy 12 articles at 7 pence each. How much change will I receive from £1?’

Ask patient to reverse 3 or 4 random numbers.

Ask patient to explain proverbs.

Ask patient to sort playing cards into suits.

The examiner must compare patient’s present reasoning ability with expected abilities based on job history and/or school work.

EMOTIONAL STATE

Note:

Anxiety or excitement

Depression or apathy

Emotional behaviour

Uninhibited behaviour

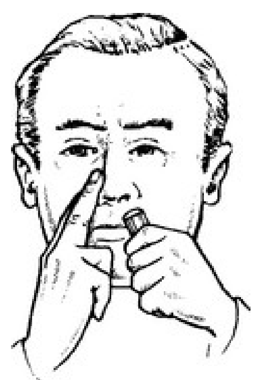
Slowness of movement or responses

Personality type or change.

CRANIAL NERVE EXAMINATION

OLFACTORY NERVE (I)

Test both perception and identification using aromatic non-irritant materials that avoid stimulation of trigeminal nerve fibres in the nasal mucosa, *e.g.* soap, tobacco.



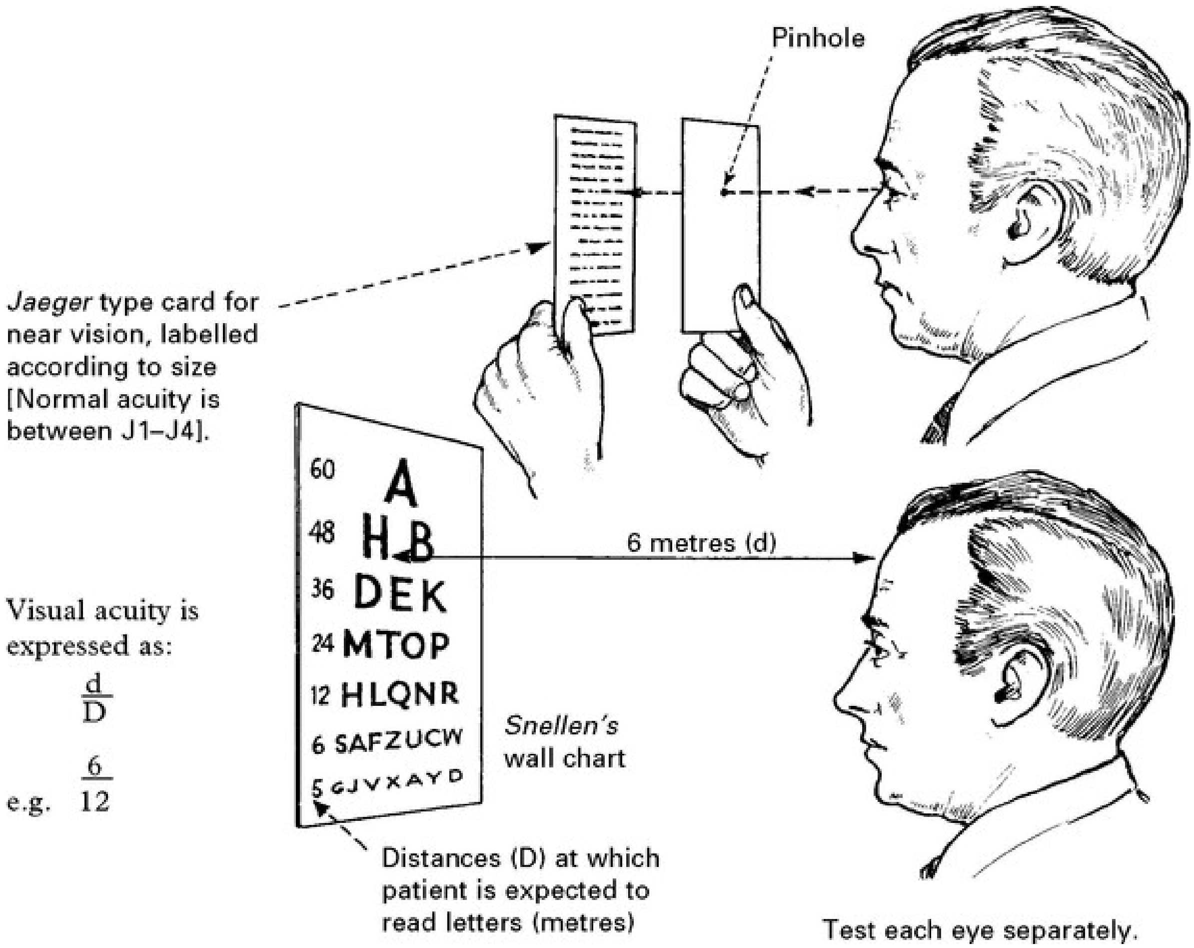
One nostril is closed while the patient sniffs with the other.

OPTIC NERVE (II)

Visual acuity

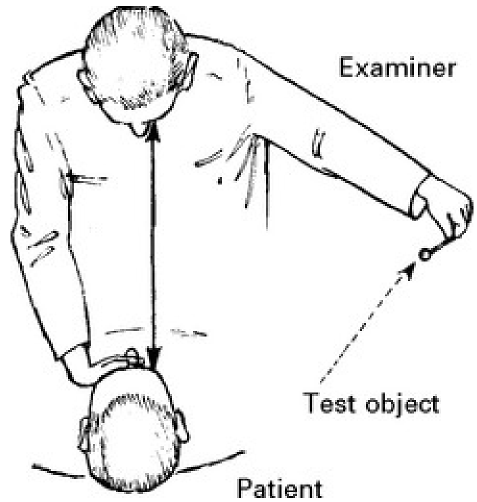
- *severe* deficit – Can patient see
  - light?
  - movement?
- Can patient count fingers?
- *mild* deficit – Record reading acuity with wall or hand chart.

N.B. *Refractive error* (i.e. inadequate focussing on the retina, *e.g.* hypermetropia, myopia) can be overcome by testing reading acuity through a pinhole. This concentrates a thin beam of vision on the macula.



Visual fields

Gross testing by CONFRONTATION. Compare the patient's fields of vision by advancing a moving finger or, more accurately, a red 5 mm pin from the extreme periphery towards the fixation point. This maps out 'cone' vision. A 2 mm pin will define central field defects which may only manifest as a loss of colour perception.

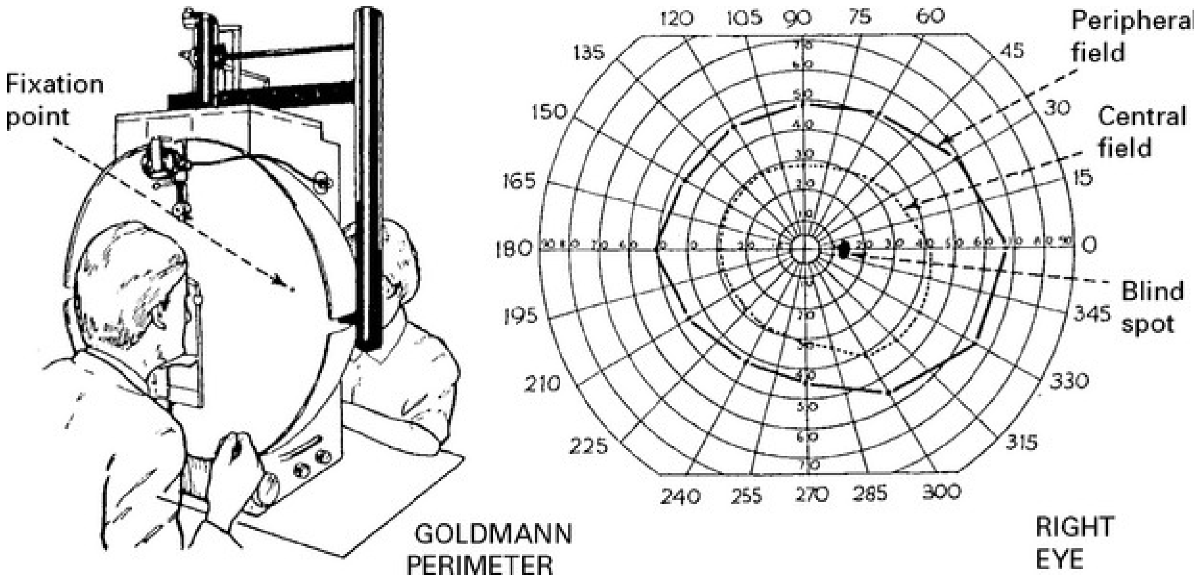


In the temporal portion of the visual field the physiological blind spot may be detected. A 2 mm object should disappear here.

The patient must fixate on the examiner's pupil.

Peripheral visual fields are more sensitive to a moving target and are tested with a GOLDMANN PERIMETER.

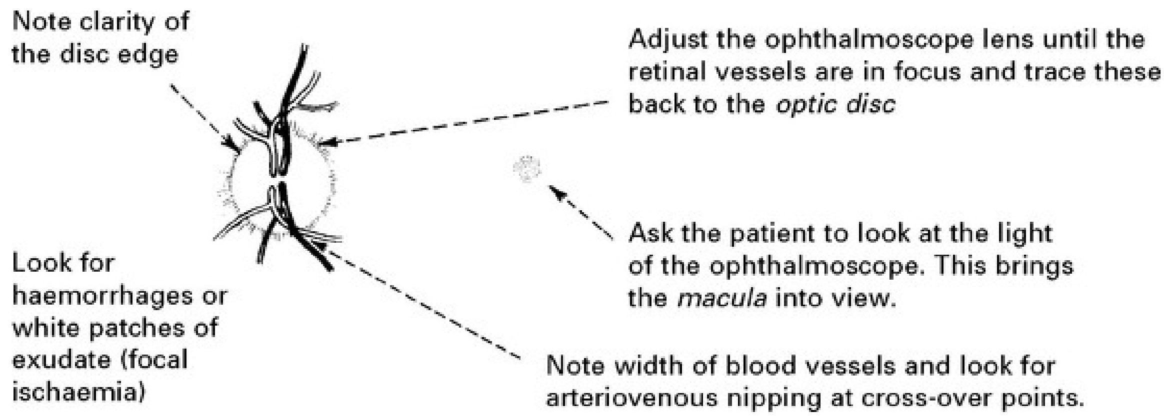
The patient fixes on a central point. A point of light is moved centrally from the extreme periphery. The position at which the patient observes the target is marked on a chart. Repeated testing from multiple directions provides an accurate record of visual fields.



Central fields are charted with either a Goldmann perimeter using a small light source of lesser intensity or a TANGENT (BJERRUM) SCREEN. The HUMPHREY FIELD ANALYSER provides an alternative and particularly sensitive method of testing central fields. This records the *threshold* at which the patient observes a static light source of increasing intensity.

Optic fundus (*Ophthalmoscopy*)

Ask the patient to fixate on a distant object away from any bright light. Use the right eye to examine the patient's right eye and the left eye to examine the patient's left eye.



If small pupil size prevents fundal examination, then dilate pupil with a quick acting mydriatic (homatropine). This is contraindicated if either an acute expanding lesion or glaucoma is suspected.

Pupils

Note:

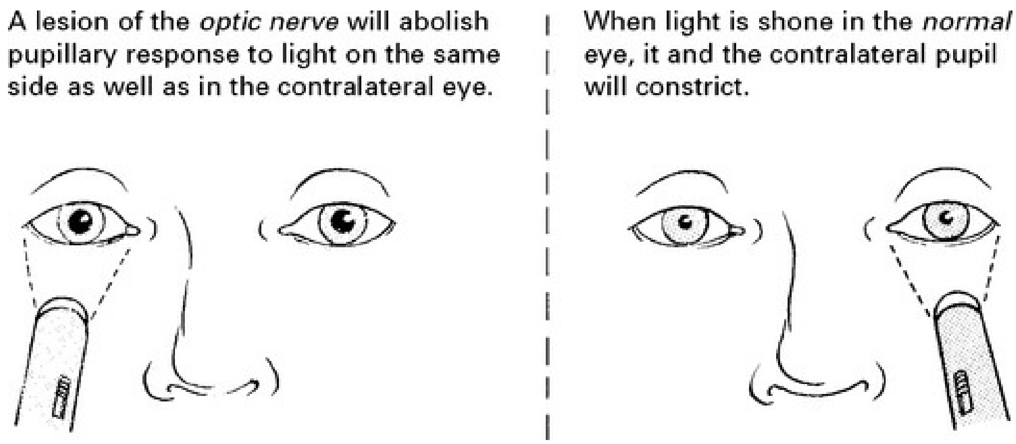
Size (small = miosis/large = mydriasis)

Shape

Equality

Reaction to light: both pupils constrict when light is shone in either eye

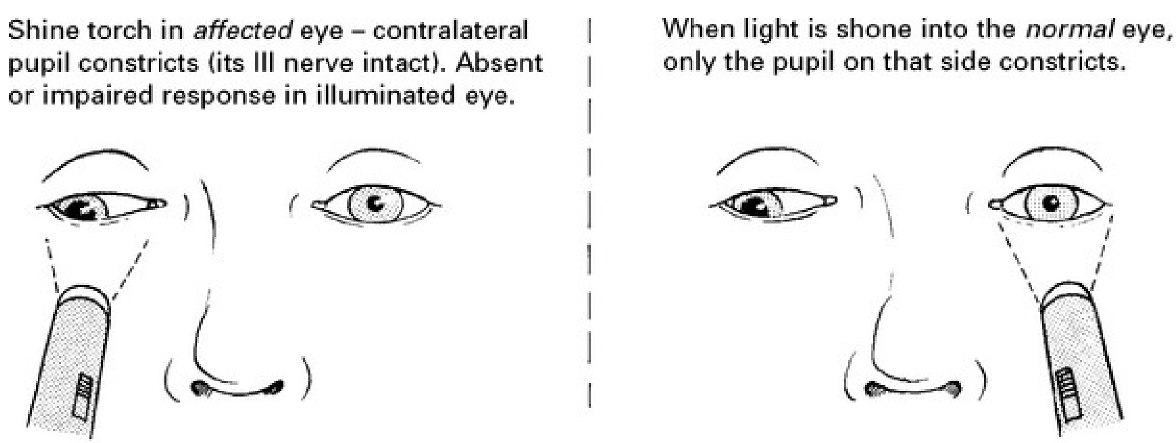
Reaction to accommodation and convergence: pupil constriction occurs when gaze is transferred to a near point object.

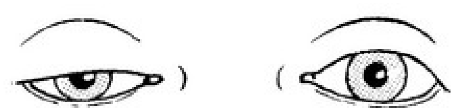


OCULOMOTOR (III), TROCHLEAR (IV) AND ABDUCENS (VI) NERVES

A lesion of the III nerve produces impairment of eye and lid movement as well as disturbance of pupillary response.

Pupil: The pupil dilates and becomes 'fixed' to light.





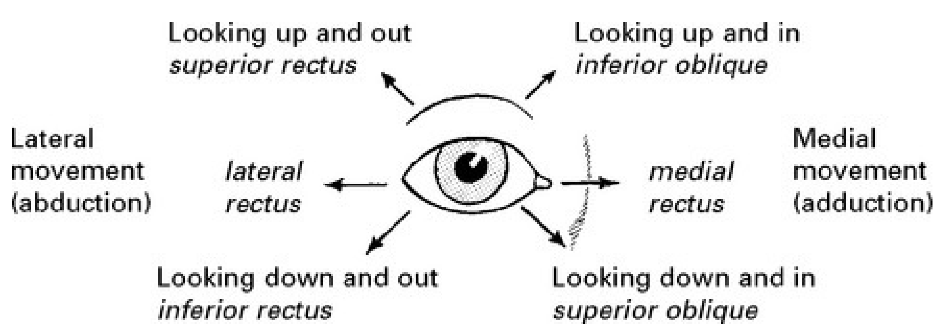
Ptosis: Ptosis is present if the eyelid droops over the pupil when the eyes are fully open. Since the levator palpebrae muscle contains both skeletal and smooth muscle, ptosis signifies either a III nerve palsy or a sympathetic lesion and is more prominent with the former.

Ocular movement

Steady the patient's head and ask him to follow an object held at arm's length. Observe the full range of horizontal and vertical eye movements.

Note any *malalignment or limitation of range.*

Examine eye movements in the six different directions of gaze representing maximal individual muscle strength.

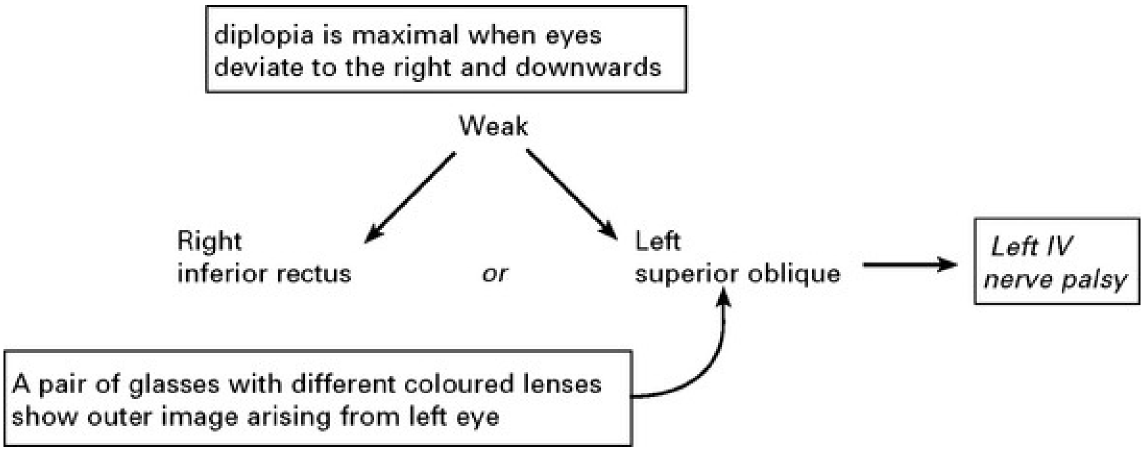


Question patient about *diplopia*; the patient is more likely to notice this before the examiner can detect impairment of eye movement. If present:

– note the *direction of maximum displacement* of the images and determine the pair of muscles involved

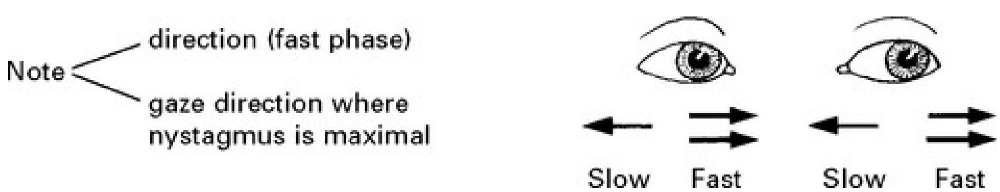
– identify the source of the *outer image* (from the defective eye) using a transparent coloured lens.

e.g.



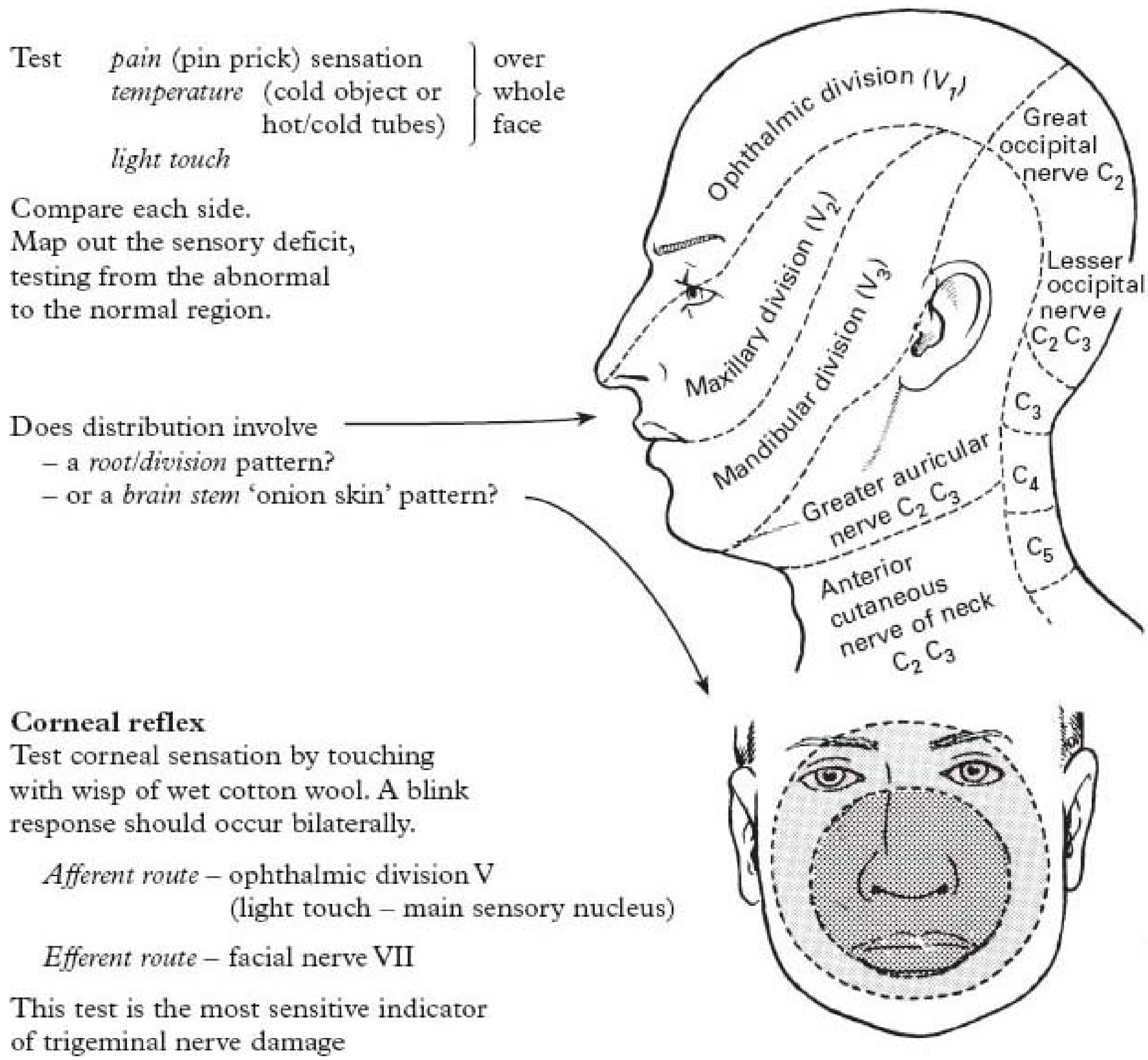
Conjugate movement: Note the ability of the eyes to move together (conjugately) in horizontal or vertical direction or tendency for gaze to fix in one particular direction.

Nystagmus: This is an upset in the normal balance of eye control. A slow drift in one direction is followed by a fast corrective movement. Nystagmus is maximal when the eyes are turned in the direction of the fast phase. Nystagmus 'direction' is usually described in terms of the fast phase and may be horizontal or vertical. Test as for other eye movements, but remember that 'physiological' nystagmus can occur when the eyes deviate to the endpoint of gaze.

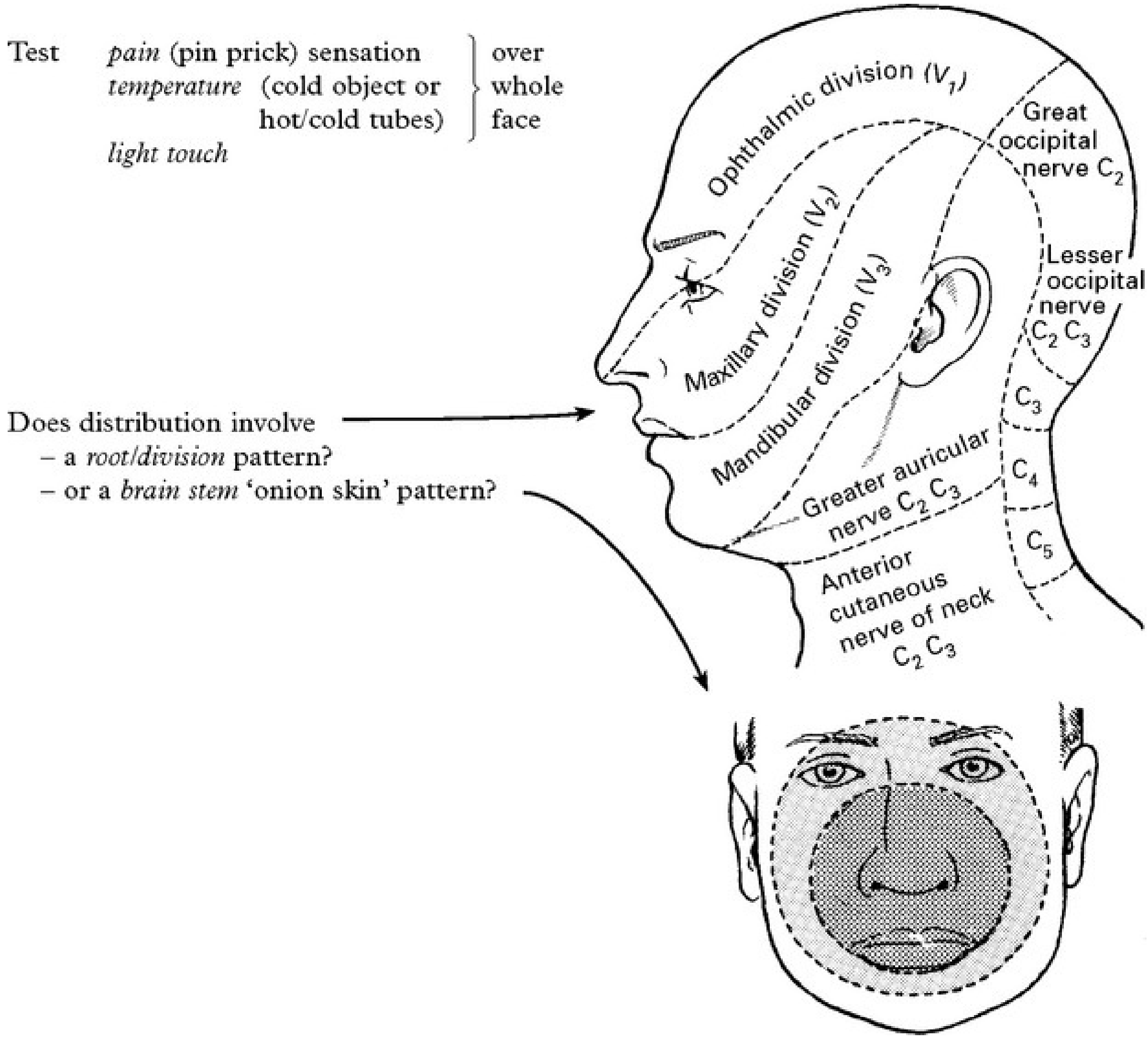


e.g. Nystagmus to the left maximal on left lateral gaze.

TRIGEMINAL NERVE (V)



Compare each side. Map out the sensory deficit, testing from the abnormal to the normal region.



Corneal reflex

Test corneal sensation by touching with wisp of wet cotton wool. A blink response should occur bilaterally.

Afferent route – ophthalmic division V (light touch – main sensory nucleus)

Efferent route – facial nerve VII

This test is the most sensitive indicator of trigeminal nerve damage

Motor examination

Observe for wasting and thinning of temporalis muscle – 'hollowing out' the temporalis fossa.

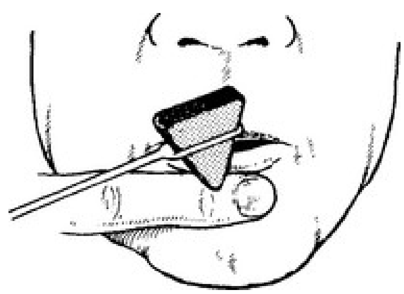
Ask the patient to clamp jaws together. Feel temporalis and masseter muscles. Attempt to open patient's jaws by applying pressure to chin. Ask patient to open mouth. If pterygoid muscles are weak the jaw will deviate to the weak side, being pushed over by the unopposed pterygoid muscles of the good side.

Jaw jerk

Ask patient to open mouth and relax jaw. Place finger on the chin and tap with hammer:

Slight jerk – normal

Increased jerk – bilateral upper neuron lesion.



FACIAL NERVE (VII)

Observe patient as he talks and smiles, watching for:

– eye closure

– asymmetrical elevation of one corner of mouth

– flattening of nasolabial fold.

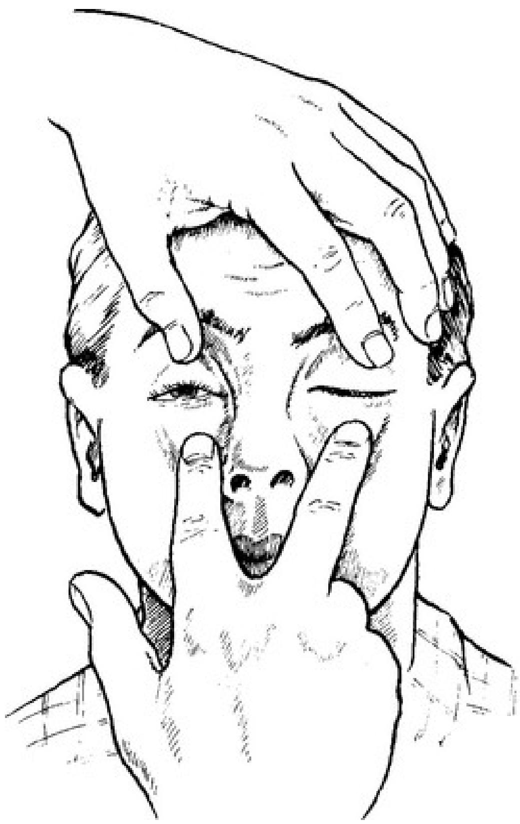
Patient is then instructed to:

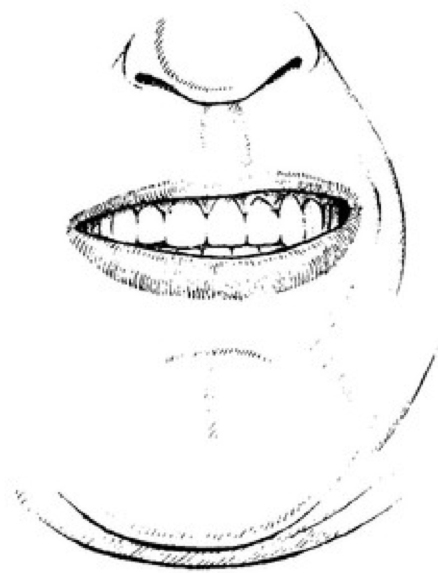
– wrinkle forehead (frontalis) (by looking upwards)

– close eyes while examiner attempts to open them (orbicularis oculi)

– purse lips while examiner presses cheeks (buccinator)

– show teeth (orbicularis oris)





Taste may be tested by using sugar, tartaric acid or sodium chloride. A small quantity of each substance is placed anteriorly on the appropriate side of the protruded tongue.

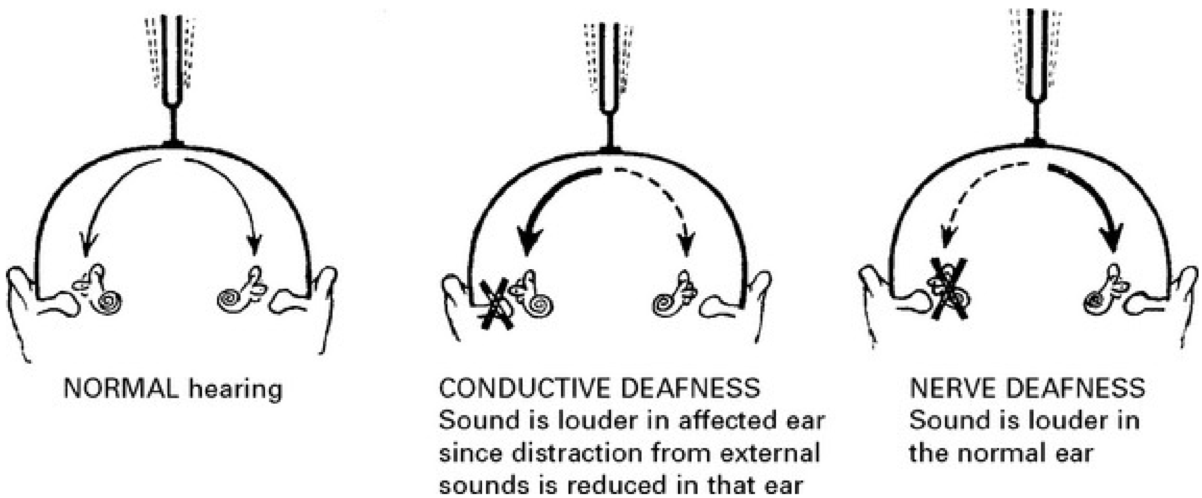
AUDITORY NERVE (VIII)

Cochlear component

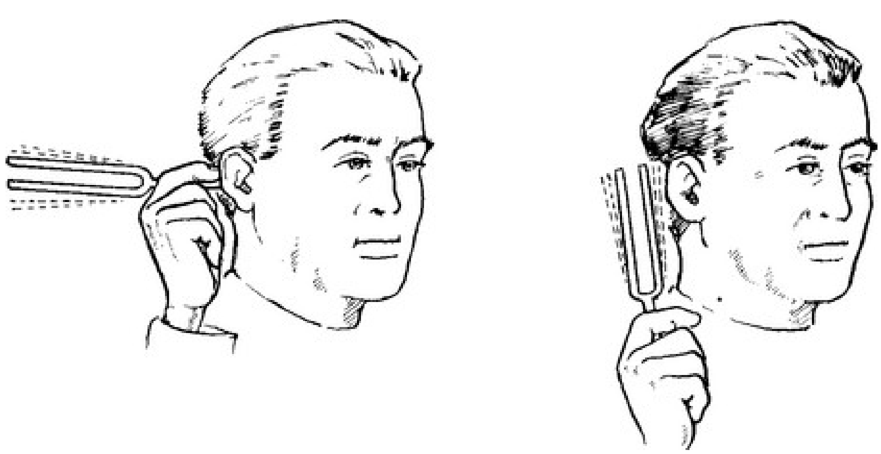
Test by whispering numbers into one ear while masking hearing in the other ear by occluding and rubbing the external meatus. If hearing is impaired, examine external meatus and the tympanic membrane with auroscope to exclude wax or infection.

Differentiate conductive (middle ear) deafness from perceptive (nerve) deafness by:

1. ***Weber's test:*** Hold base of tuning fork (512 Hz) against the vertex. Ask patient if sound is heard more loudly in one ear.



2. ***Rinne's test:*** Hold the base of a vibrating tuning fork against the mastoid bone. Ask the patient if note is heard. When note disappears – hold tuning fork near the external meatus. Patient should hear sound again since air conduction via the ossicles is better than bone conduction.



In *conductive deafness*, bone conduction is better than air conduction.

In *nerve deafness*, both bone and air conduction are impaired.

Further auditory testing and examination of the **vestibular component** requires specialised investigation (see pages 62–65).

GLOSSOPHARYNGEAL NERVE (IX): VAGUS NERVE (X)

These nerves are considered jointly since they are examined together and their actions are seldom individually impaired.

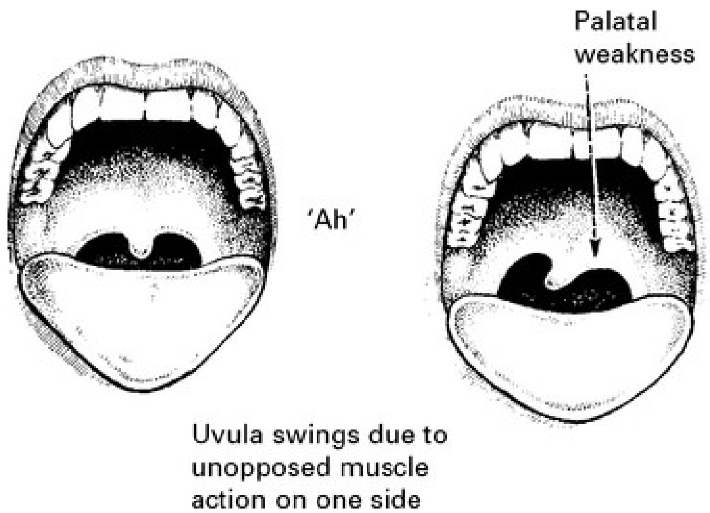
Note patient's *voice* – if there is vocal cord paresis (X nerve palsy), voice may be high pitched. (Vocal cord examination is best left to an ENT specialist.)

Note any *swallowing* difficulty or nasal regurgitation of fluids.

Ask patient to open mouth and say *'Ah'*. Note any asymmetry of palatal movements (X nerve palsy).

Gag reflex

Depress patient's tongue and touch palate, pharynx or tonsil on one side until the patient 'gags'. Compare sensitivity on each side (*afferent* route – IX nerve) and observe symmetry of palatal contraction (*efferent route* – X nerve).



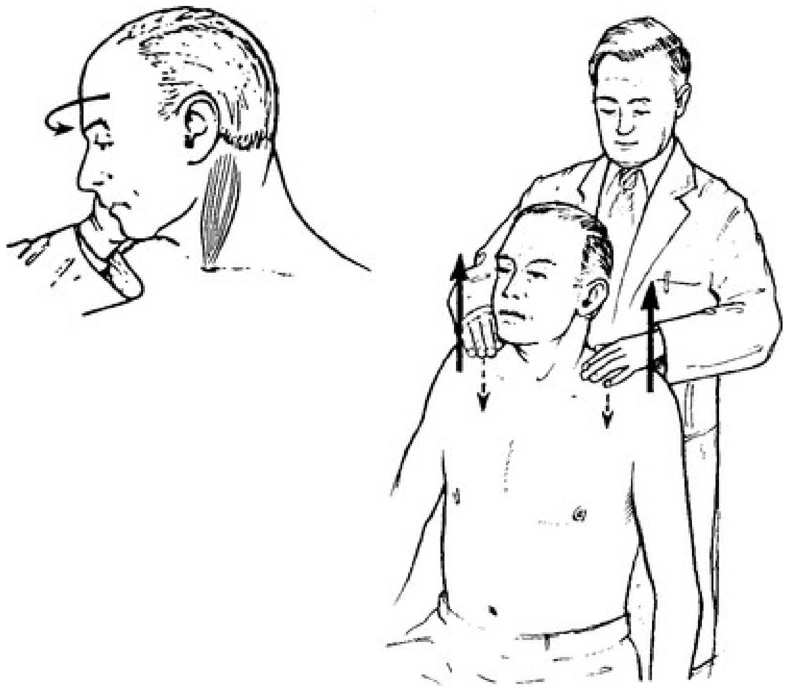
Absent gag reflex = loss of sensation and/or loss of motor power. (Taste in the posterior third of the tongue (IX) is impractical to test).

ACCESSORY NERVE (XI)

Sternomastoid

Ask patient to rotate head against resistance. Compare power and muscle bulk on each side. Also compare each side with the patient pulling head forward against resistance.

N.B. The left sternomastoid turns the head to the right and *vice versa*.



Trapezius

Ask patient to ‘shrug’ shoulders and to hold them in this position against resistance. Compare power on each side. Patient should manage to resist any effort to depress shoulders.

HYPOGLOSSAL NERVE (XII)

Ask patient to open mouth; inspect tongue.

Look for

– evidence of atrophy (increased folds, wasting)

– fibrillation (small wriggling movements).

Ask patient to protrude tongue. Note any difficulty or deviation. (N.B. apparent deviation may occur with facial weakness – if present, assess tongue in relation to teeth.)

Protruded tongue deviates towards side of weakness.

Non protruded tongue cannot move to the opposite side.

Dysarthria and dysphagia are minimal.

EXAMINATION – UPPER LIMBS

MOTOR SYSTEM

Appearance

Appearance

Note:	– any *asymmetry* or *deformity*	
	– muscle *wasting* – muscle *hypertrophy*	If in doubt, measure circumference at fixed distance above/below joint. Note muscle group involved.
	– muscle *fasciculation*	irregular, non-rhythmical contraction of muscle fascicules, increased after exercise and on tapping muscle surface.
	– muscle *myokimia*	a rapid flickering of muscle fibres, particularly in orbicularis oculi but occasionally in large muscles, after exercise or with fatigue – 'Benign Fasciculation'.

Tone

Ensure that the patient is relaxed, and assess tone by alternately flexing and extending the elbow or wrist.

Note: – decrease in tone

– increase in tone
- *'Clasp-knife':* the initial resistance to the movement is suddenly overcome (upper motor neuron lesion).
- *'Lead-pipe':* a steady increase in resistance throughout the movement (extrapyramidal lesion).
- *'Cog-wheel':* ratchet-like increase in resistance (extrapyramidal lesion).

Power

Muscle weakness. The degree of weakness is 'scored' using the MRC (Medical Research Council) scale.

Score 0 – No contraction

Score 1 – Flicker

Score 2 – Active movement/gravity eliminated

Score 3 – Active movement against gravity

Score 4 – Active movement against gravity and resistance

Score 5 – Normal power

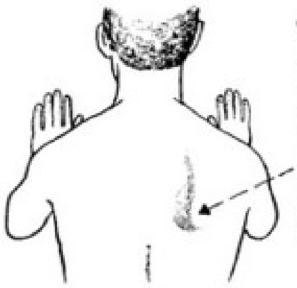
If a pyramidal weakness is suspect (i.e. a weakness arising from damage to the motor cortex or descending motor tracts (see pages 193–198) the following test is simple, quick, yet sensitive.

Ask the patient to hold arms outstretched with the hands supinated for up to one minute. The eyes are closed (otherwise visual compensation occurs). The weak arm gradually pronates and drifts downwards.

With possible involvement at the spinal root or nerve level (lower motor neuron), it is essential to test individual muscle groups to help localise the lesion.

When testing muscle groups, think of *root* and *nerve* supply.

Test for *Serratus anterior:*

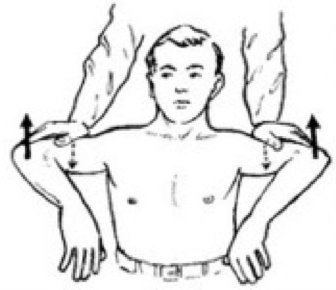


C5, C6, C7 roots
Long thoracic nerve

Patient presses arms against wall

Look for winging of scapula i.e. rises from chest wall

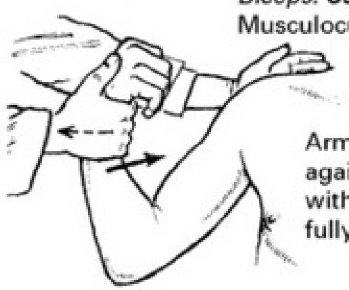
Shoulder abduction



Deltoid:
C5, C6 roots
Axillary nerve

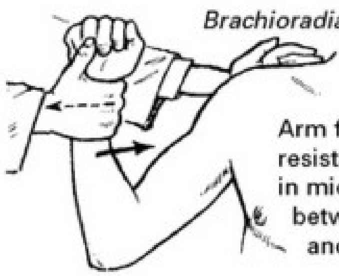
Arm (at more than 15° from the vertical) abducts against resistance

Elbow flexion



Biceps: **C5, C6** roots
Musculocutaneous nerve

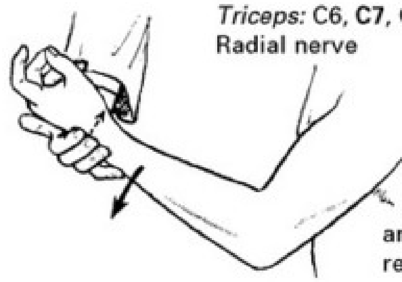
Arm flexed against resistance with the hand fully supinated



Brachioradialis: C5, **C6** roots.
Radial nerve

Arm flexed against resistance with hand in mid-position between pronation and supination

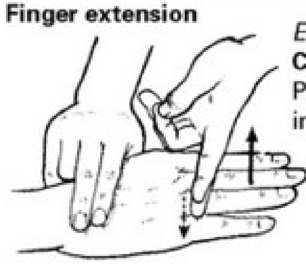
Elbow extension



Triceps: C6, **C7**, C8 roots
Radial nerve

Patient extends arm against resistance

Finger extension

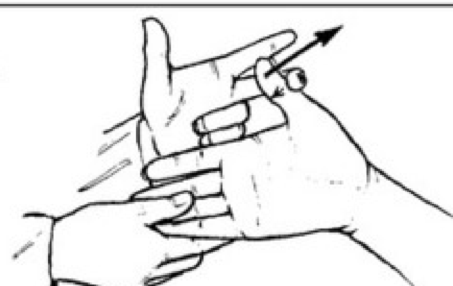


Extensor digitorum:
C7, C8 roots
Posterior interosseous nerve

Patient extends fingers against resistance

Thumb extension – terminal phalanx

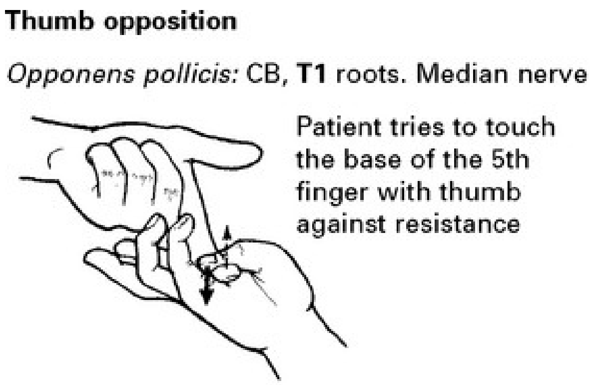
Extensor pollicis longus and brevis: **C7**, C8 roots
Posterior interosseous nerve
Thumb is extended against resistance

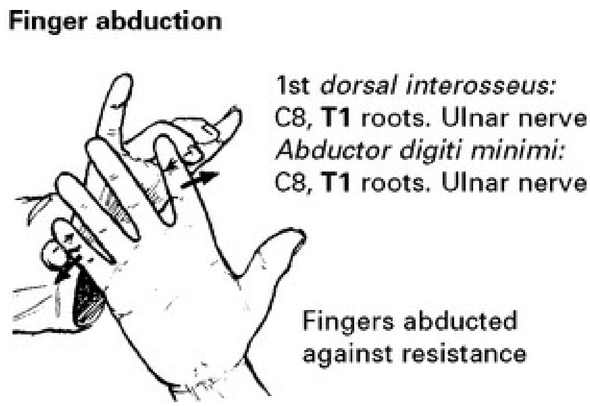


Finger flexion – terminal phalanx

Flexor digitorum profundus I and II: C7, **C8** roots
Median nerve
Flexor digitorum profundus III and IV: C7, **C8** roots
Ulnar nerve

Examiner tries to extend patient's flexed terminal phalanges



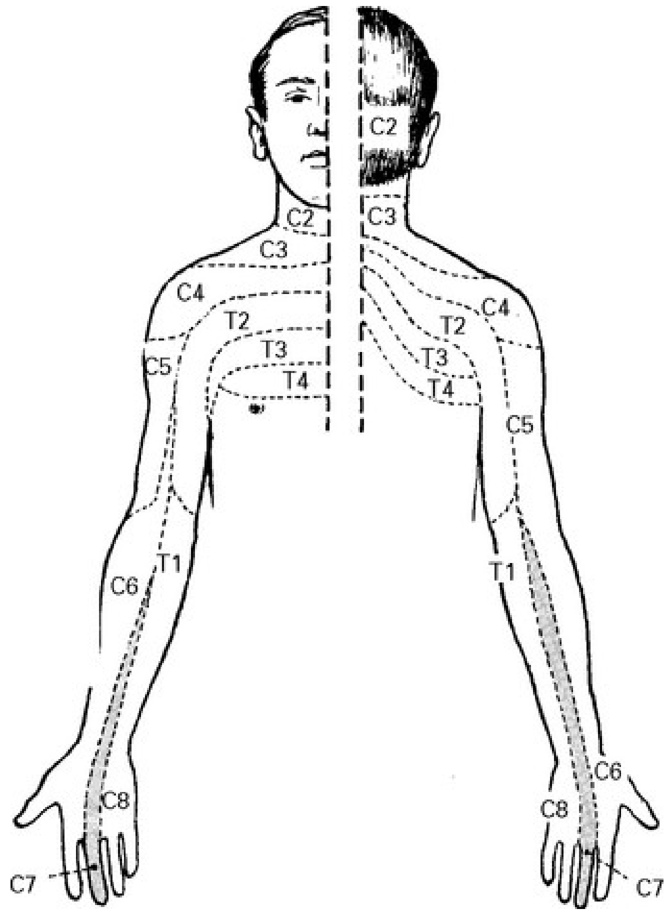


[Note: not all muscle groups are included in the foregoing, but only those required to identify and differentiate nerve and root lesions.]

SENSATION

Pain

Pin prick with a sterile pin provides a simple method of testing this important modality. Firstly, check that the patient detects the pin as 'sharp', *i.e.* painful, then rapidly test each dermatome in turn.



Memorising the dermatome distribution is simplified by noting that 'C7' extends down the middle finger.

If pin prick is impaired, then more carefully map out the extent of the abnormality, moving from the abnormal to the normal areas.

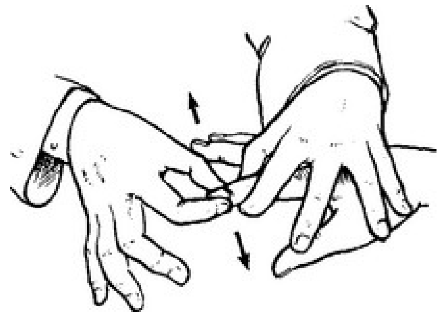
Light touch

This is tested in a similar manner, using a wisp of cotton wool.

Temperature

Temperature testing seldom provides any additional information. If required, use a cold object or hot and cold test tubes.

Joint position sense



Hold the sides of the patient's finger or thumb and demonstrate 'up and down' movements.

Repeat with the patient's eyes closed. Ask patient to specify the direction of movement.

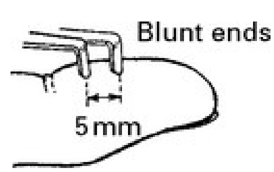
Ask the patient, with eyes closed, to touch his nose with his forefinger or to bring forefingers together with the arms outstretched.

Vibration

Place a vibrating tuning fork (usually 128 c/s) on a bony prominence, *e.g.* radius. Ask the patient to indicate when the vibration, if felt, ceases. If impaired, move more proximally and repeat. Vibration testing is of value in the early detection of demyelinating disease and peripheral neuropathy, but otherwise is of limited benefit.

If the above sensory functions are normal and a cortical lesion is suspected, it is useful to test for the following:

Two point discrimination: the ability to discriminate two blunt points when simultaneously applied to the finger, 5 mm apart (cf, 4 cm in the legs).

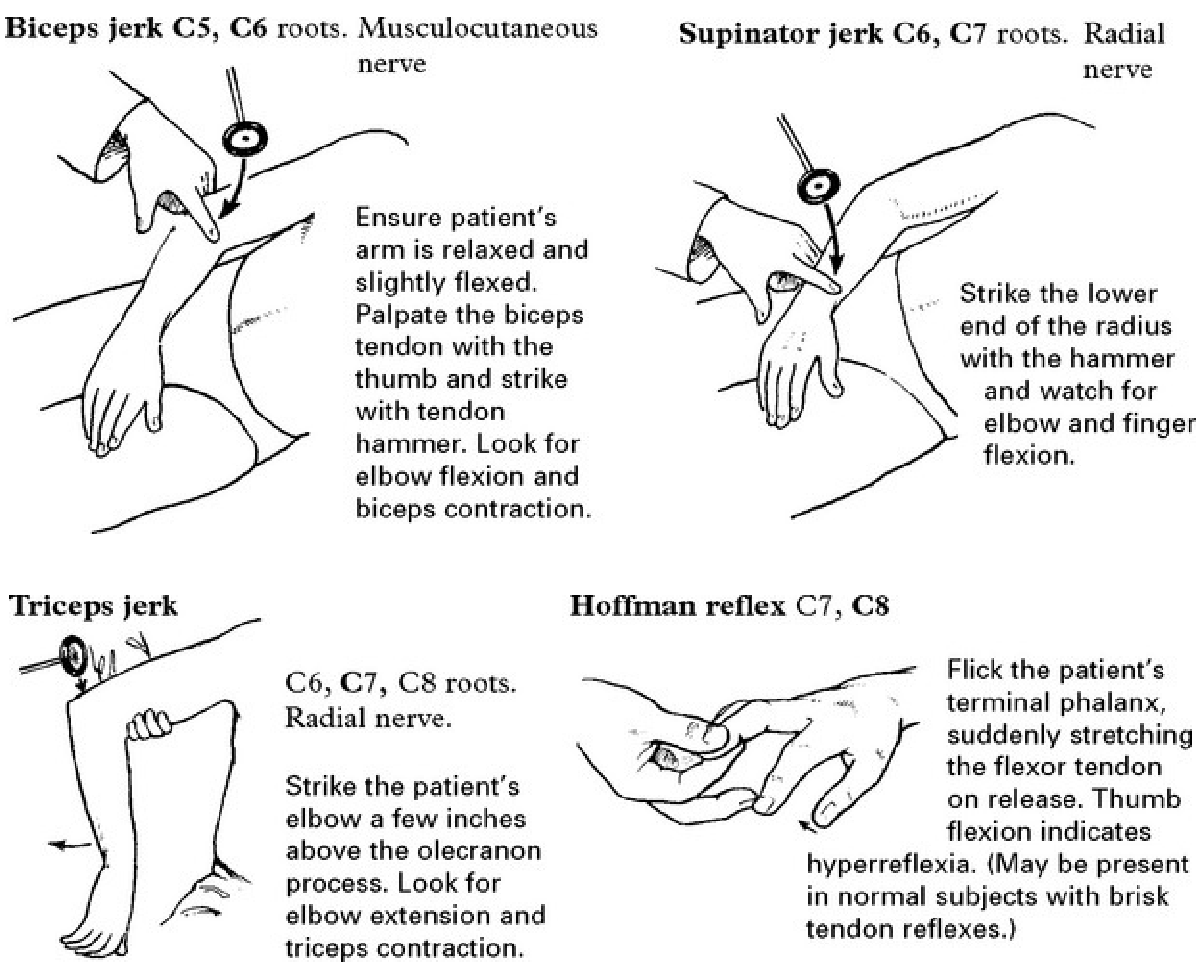


Sensory inattention (perceptual rivalry): the ability to detect stimuli (pin prick or touch) in both limbs, when applied to both limbs simultaneously.

Stereognosis: the ability to recognise objects placed in the hand.

Graphaesthesia: the ability to recognise numbers or letters traced out on the palm.

REFLEXES



Reflex enhancement

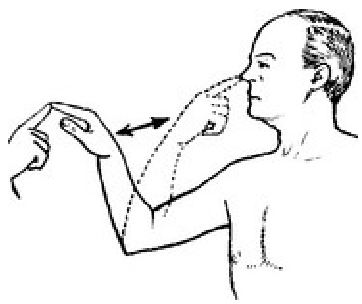
When reflexes are difficult to elicit, enhancement occurs if the patient is asked to 'clench the teeth'.

CO-ORDINATION

Inco-ordination (ataxia) is often a prominent feature of cerebellar disease (see page 182). Prior to testing, ensure that power and proprioception are normal.

Inco-ordination

Finger – nose testing

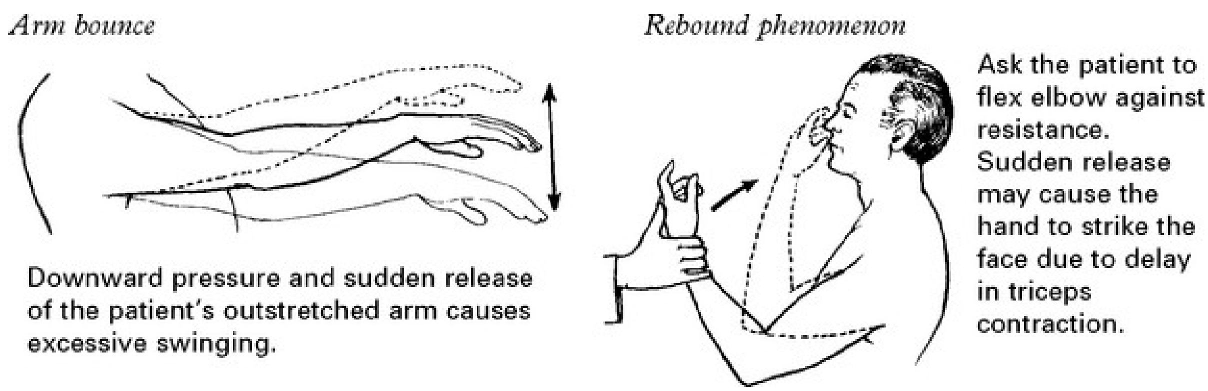


Ask patient to touch his nose with finger (eyes open).

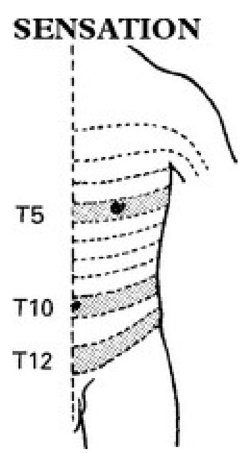
Look for jerky movements – DYSMETRIA or an INTENTION TREMOR (tremor only occurring on voluntary movement).

Ask patient to alternately touch his own nose then the examiner's finger as fast as he can. This may exaggerate the intention tremor and may demonstrate DYSDIADOCHOKINESIA – an inability to perform rapidly alternating movements.

This may also be shown by asking the patient to rapidly supinate and pronate the forearms or to perform rapid and repeated tapping movements.



EXAMINATION – TRUNK



Test pin prick and light touch in dermatome distribution as for the upper limbs.

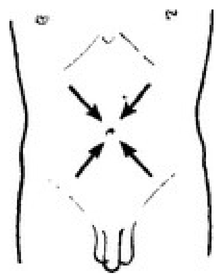
Levels to remember:

T5 – at *nipple*

T10 – at *umbilicus*

T12 – at *inguinal ligament.*

Abdominal reflexes: T7 – T12 roots. Stroke or lightly scratch the skin towards the umbilicus in each quadrant in turn. Look for abdominal muscle contraction and note if absent or impaired. (N.B. Reflexes may normally be absent in obesity, after pregnancy, or after abdominal operations.)



Cremasteric reflex: L1, L2 root. Scratch inner thigh. Observe contraction of cremasteric muscle causing testicular elevation.

SPHINCTERS

Examine abdomen for distended bladder.

Note evidence of urinary or faecal incontinence.

Note tone of anal sphincter during rectal examination.

Anal reflex: S4, S5 roots. A scratch on the skin beside the anus causes a reflex contraction of the anal sphincter.

EXAMINATION – LOWER LIMBS

MOTOR SYSTEM

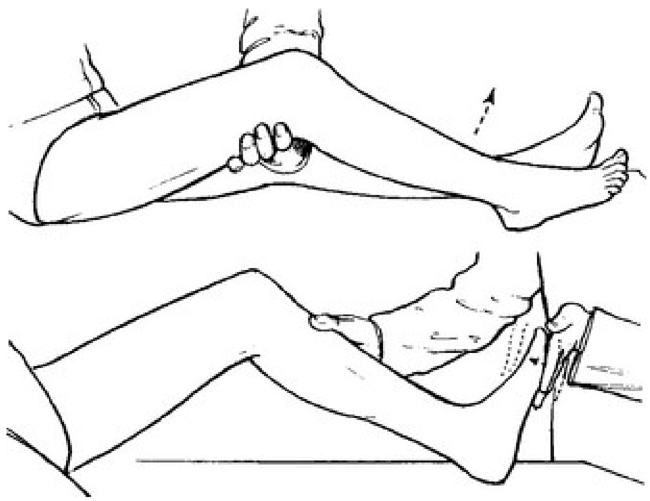
Appearance: Note:
- *asymmetry* or *deformity*
- muscle *wasting*
- muscle *hypertrophy*
- muscle *fasciculation*
- muscle *myokimia*

as in the upper limbs

Tone

Try to relax the patient and alternately flex and extend the knee joint. Note the resistance.

Roll the patient's legs from side to side. Suddenly lift the thigh and note the response in the lower leg. With increased tone the leg kicks upwards.



Clonus

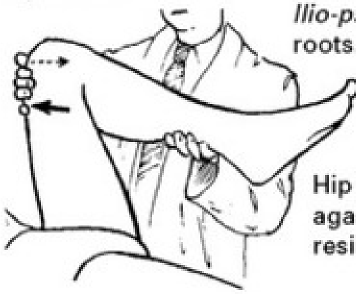
Ensure that the patient is relaxed. Apply sudden and sustained flexion to the ankle. A few oscillatory beats may occur in the normal subject, but when this persists it indicates increased tone.

Power

When testing each muscle group, think of *root* and *nerve* supply.

Hip flexion

Ilio-psoas: **L1, L2,** L3 roots. Femoral nerve

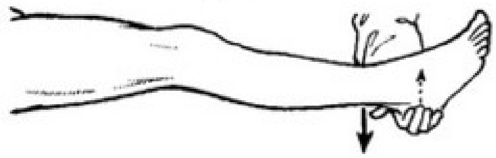


Hip flexed against resistance

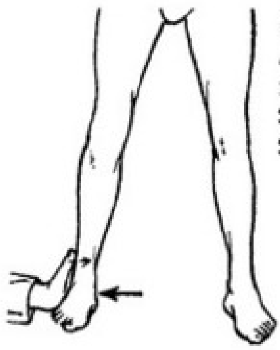
Hip extension

Gluteus maximus: **L5, S1,** S2 roots. Inferior gluteal nerve

Patient attempts to keep heel on bed against resistance



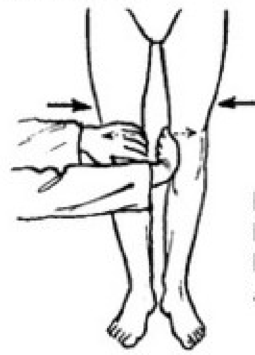
Hip abduction



Gluteus medius and minimus and tensor fasciae latae: **L4, L5,** S1 roots. Superior gluteal nerve

Patient lying on back tries to abduct the leg against resistance

Hip adduction



Adductors: **L2, L3,** L4 roots. Obturator nerve

Patient lying on back tries to pull knees together against resistance

Knee flexion

Hamstrings L5, **S1,** S2 roots. Sciatic nerve

Patient pulls heel towards the buttock and tries to maintain this position against resistance

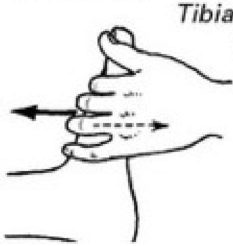
Knee extension

Quadriceps: L2, **L3, L4** roots. Femoral nerve

Patient tries to extend knee against resistance

Dorsiflexion

Tibialis anterior: **L4,** L5 roots. Deep peroneal nerve



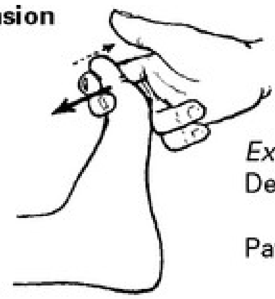
Patient dorsiflexes the ankle against resistance. May have difficulty in walking on heels

Plantarflexion

Gastrocnemius, soleus: **S1, S2,** roots. Tibial nerve.

Patient plantarflexes the ankle against resistance. May have difficulty in walking on toes before weakness can be directly detected

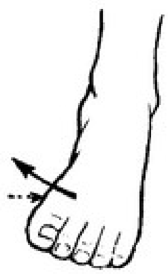
Toe extension



Extensor hallucis longus, extensor digitorum longus: **L5,** S1 roots.
Deep peroneal nerve

Patient dorsiflexes the toes against resistance

Inversion

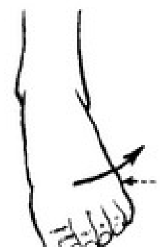


Tibialis posterior: **L4, L5** root.
Tibial nerve

Patient inverts foot against resistance

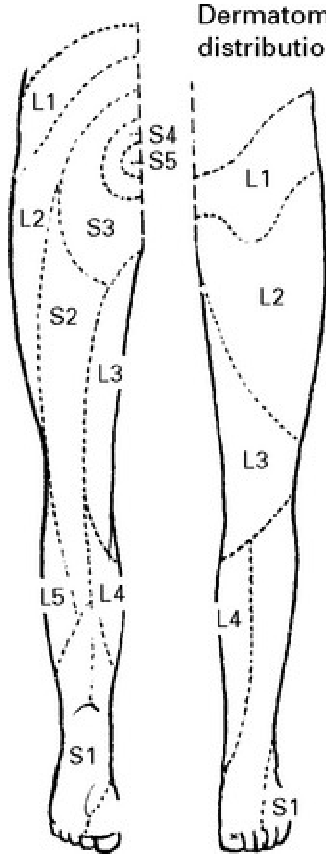
Eversion

Peroneus longus and brevis: **L5, S1** roots.
Superficial peroneal nerve

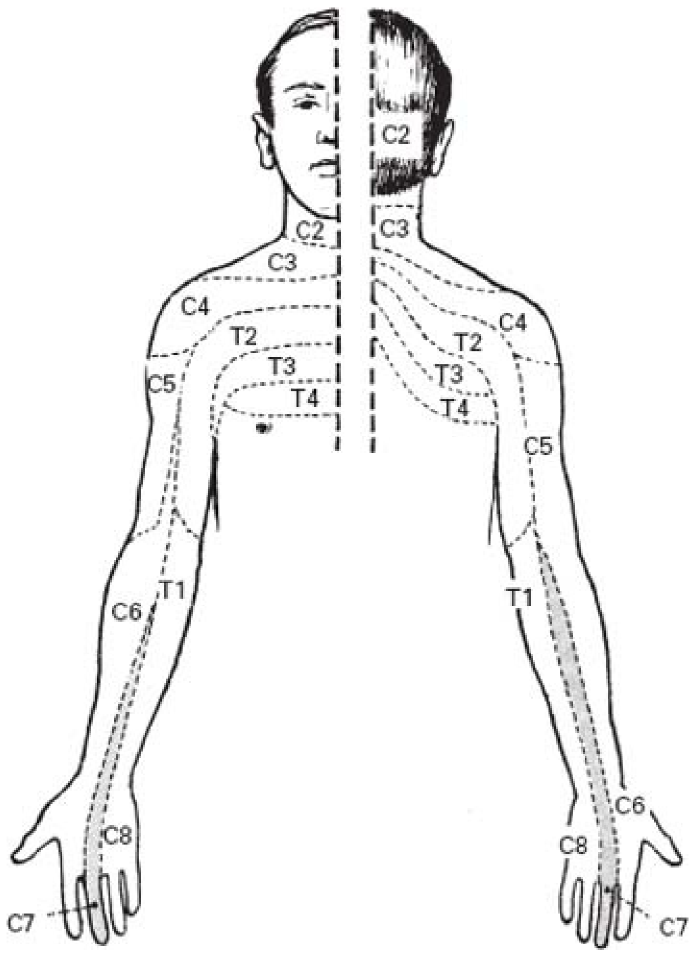


Patient everts foot against resistance

SENSATION



Dermatome distribution



Joint position sense

Firstly, demonstrate flexion and extension movements of the big toe. Then ask patient to specify the direction with the eyes closed.

If deficient, test ankle joint sense in the same way.

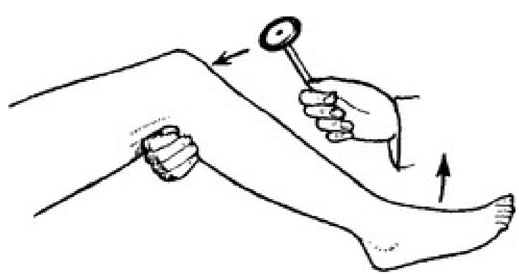
Vibration

Test vibration perception by placing a tuning fork on the malleolus. If deficient, move up to the head of the fibula or to the anterior superior iliac spine.

REFLEXES

Knee jerk: L2, L3, **L4** roots.

Ensure that the patient's leg is relaxed by resting it over examiner's arm or by hanging it over the edge of the bed. Tap the patellar tendon with the hammer and observe quadriceps contraction. Note impairment or exaggeration.



Ankle jerk: S1, S2 roots.

Externally rotate the patient’s leg. Hold the foot in slight dorsiflexion. Ensure the foot is relaxed by palpating the tendon of tibialis anterior. If this is taut, then no ankle jerk will be elicited.

Tap the Achilles tendon and watch for calf muscle contraction and plantarflexion.

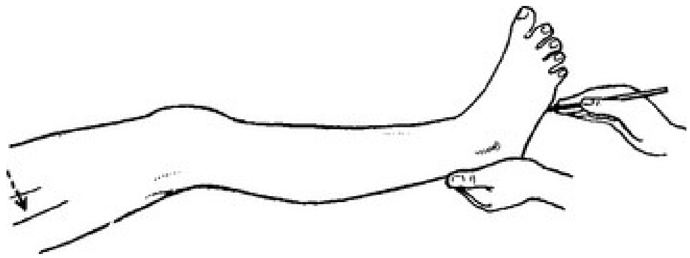
Reflex enhancement

When reflexes are difficult to elicit, they may be enhanced by asking the patient to clench the teeth or to try to pull clasped hands apart (Jendrassik’s manoeuvre).

Plantar response

Check that the big toe is relaxed. Stroke the lateral aspect of the sole and across the ball of the foot. Note the first movement of the big toe. Flexion should occur. Extension due to contraction of extensor hallucis longus (a ‘Babinski’ reflex) indicates an upper motor neuron lesion. This is usually accompanied by synchronous contraction of the knee flexors and tensor fasciae latae.

Elicit Chaddock’s sign by stimulating the lateral border of the foot. The big toe extends with upper motor neuron lesions.

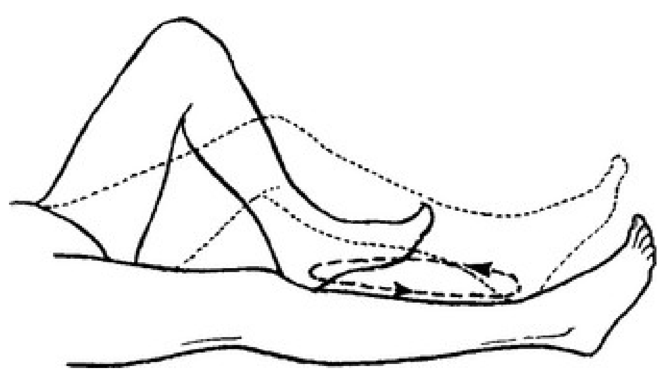


To avoid ambiguity do not touch the innermost aspect of the sole or the toes themselves.

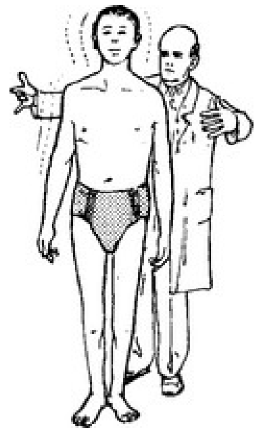
EXAMINATION – POSTURE AND GAIT

CO-ORDINATION

Ask patient to repeatedly run the heel from the opposite knee down the shin to the big toe. Look for ATAXIA (inco-ordination). Ask patient to repeatedly tap the floor with the foot. Note any DYSDIADOCHOKINESIA (difficulty with rapidly alternating movement)



Romberg's test



Ask patient to stand with the heels together, first with the eyes open, then with the eyes closed.

Note any excessive postural swaying or loss of balance

- Present when eyes open or closed = cerebellar deficit (cerebellar ataxia)
- Present only when eyes are closed ('positive' Romberg's) = proprioceptive deficit (sensory ataxia)

GAIT

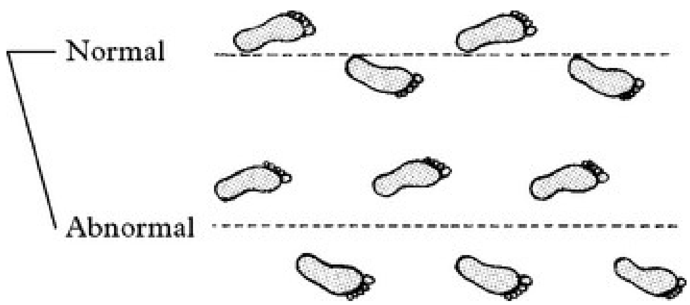
Note:

– Length of step and width of base

– Abnormal leg movements (e.g. excessively high step)

– Instability (gait ataxia)

– Associated postural movements (e.g. pelvic swinging)



If normal, repeat with *tandem* walking, *i.e.* heel to toe. This will exaggerate any instability.

EXAMINATION OF THE UNCONSCIOUS PATIENT

HISTORY

Questioning relatives, friends or the ambulance team is an essential part of the assessment of the unconscious or the unco-operative patient.

Has the patient sustained a head injury – leading to admission, or in the preceding weeks?

Did the patient collapse suddenly?

Did limb twitching occur?

Have symptoms occurred in the preceding weeks?

Has the patient suffered a previous illness?

Does the patient take medication?

GENERAL EXAMINATION

Lack of patient co-operation does not limit general examination and this may reveal important diagnostic signs. In addition to those features described on page 4, also look for signs of head injury, needle marks on the arm and evidence of tongue biting. Also note the smell of alcohol, but beware of attributing the patient's clinical state solely to alcohol excess.

NEUROLOGICAL EXAMINATION

Conscious level: This assessment is of major importance. It not only serves as an immediate prognostic guide, but also provides a baseline with which future examinations may be compared. Assess conscious level as described previously (page 5) in terms of eye opening, verbal response and motor response.

For research purposes, a score was applied for each response, with 'flexion' subdivided into 'normal' and 'spastic flexion', giving a total coma score of '15 points'. Many coma observation charts (page 31) still use a '14 point scale' with 5 points on the motor score. The '14 point' scale records less observer variability, but most guidelines for head injury management use the '15 point' scale.

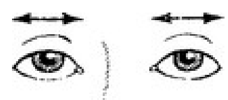
Eye opening		**Verbal response**		**Motor response**	
Spontaneous	4	Orientated	5	Obeying commands	6
To speech	3	Confused	4	Localising	5
To pain	2	Words	3	Normal flexion	4
None	1	Sounds	2	Spastic flexion	3
		None	1	Extension	2
				None	1

It is important to avoid the tendency to simply quote the patient's total score. This can be misleading. Describing the conscious level in terms of the actual responses *i.e.* 'no eye opening, no verbal response and extending', avoids any confusion over numbers.

Pupil response Fundi Corneal reflex – tone Limb – reflexes – plantar response	Lack of patient co-operation does not prevent objective assessment of these features described before, but elucidation of other relevant neurological signs requires a different approach.

Eye movements

Observe any **spontaneous** eye movements.



Elicit the **oculocephalic (doll's eye) reflex.**

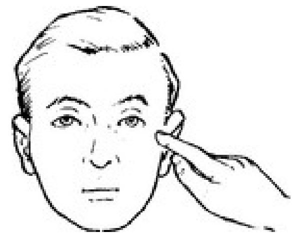
Rotation or flexion/extension of the head in a comatose patient produces transient eye movements in a direction opposite to that of the movement.

Note whether the movements, if present, are *conjugate* (i.e. the eyes move in parallel) or *dysconjugate* (i.e. the eyes do not move in parallel). These ocular movements assess midbrain and pontine function.

Elicit the **oculovestibular reflex** (caloric testing, see page 65).

Visual fields

In the unco-operative patient, the examiner may detect a hemianopic field defect when 'menacing' from one side fails to produce a 'blink'.

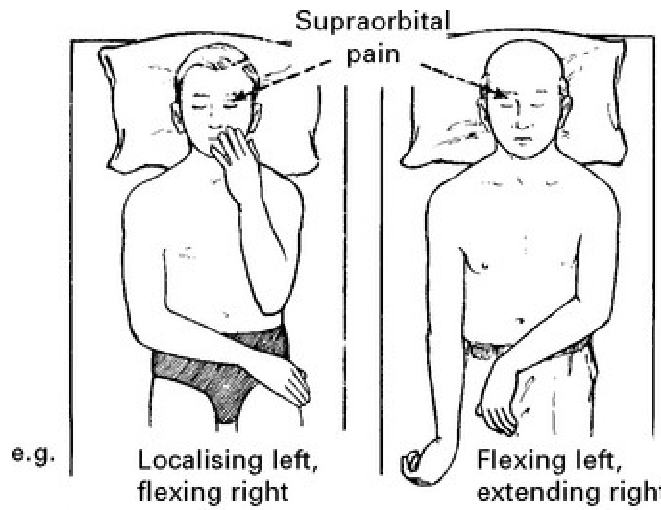


Facial weakness

Failure to 'grimace' on one side in response to bilateral supraorbital pain indicates a facial weakness.

Limb weakness

Detect by comparing the response in the limbs to painful stimuli. If pain produces an *asymmetric* response, then limb weakness is present. (If the patient 'localises' with one arm, hold this down and retest to ensure that a similar response cannot be elicited from the other limb.)

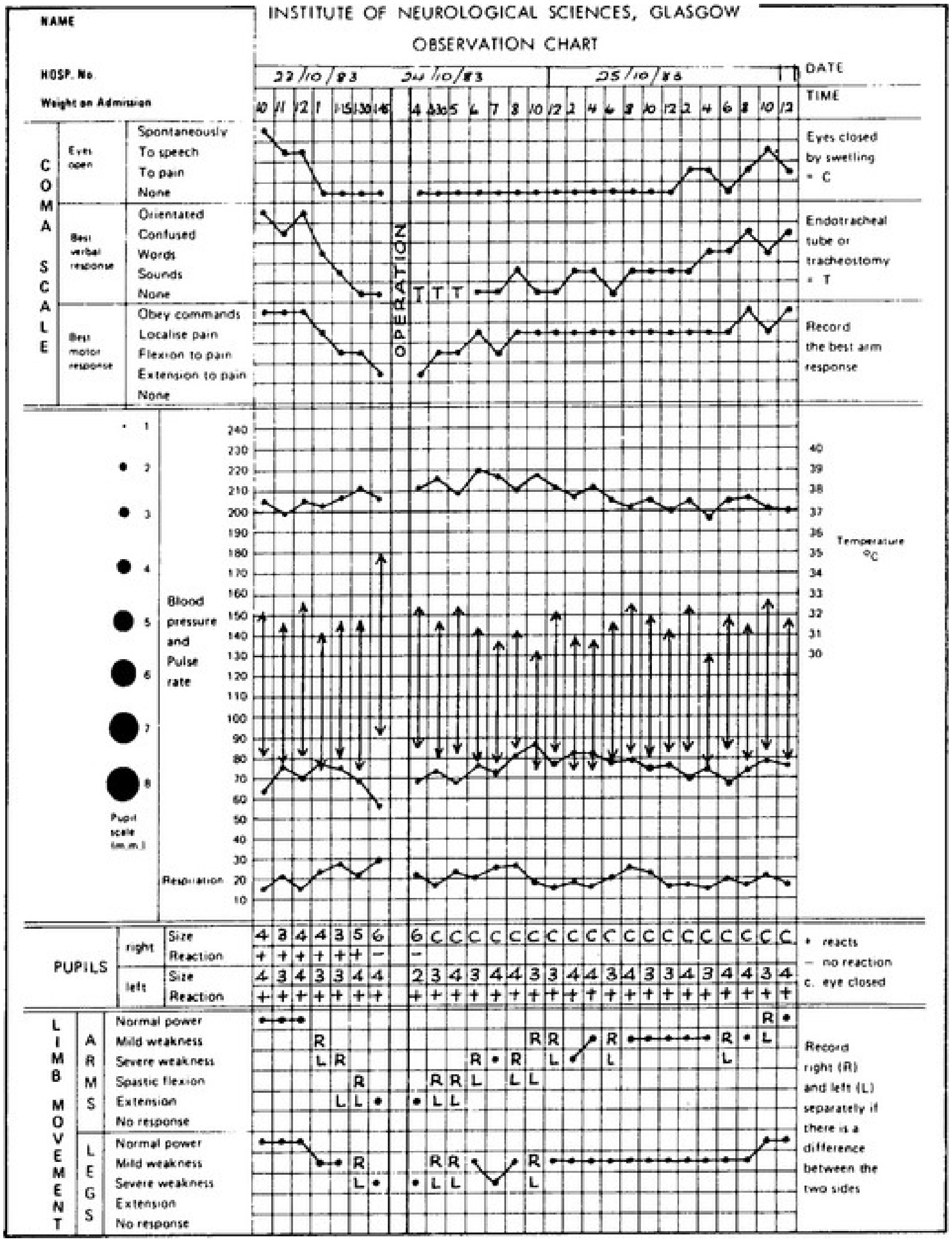


Both patients are in coma; both have an asymmetric response to pain indicating a right arm weakness and focal brain damage.

Pain stimulus applied to the toe nails or Achilles tendon similarly tests power in the lower limbs. Variation in tone, reflexes or plantar responses between each side also indicates a focal deficit. In practice, if the examiner fails to detect a difference in response to painful stimuli, these additional features seldom provide convincing evidence.

THE NEUROLOGICAL OBSERVATION CHART

Despite major advances in intracranial investigative techniques, none has replaced clinical assessment for monitoring the patient's neurological state. The neurological observation chart produced by Teasdale and Jennett incorporates the most relevant clinical features, *i.e. coma scale (eye opening, verbal and motor response), pupil size* and *reaction to light, limb responses* and *vital signs*. The frequency of observation (normally 2-hourly) depends on the individual patient's needs. The chart enables immediate evaluation of the trend in the patient's clinical state.



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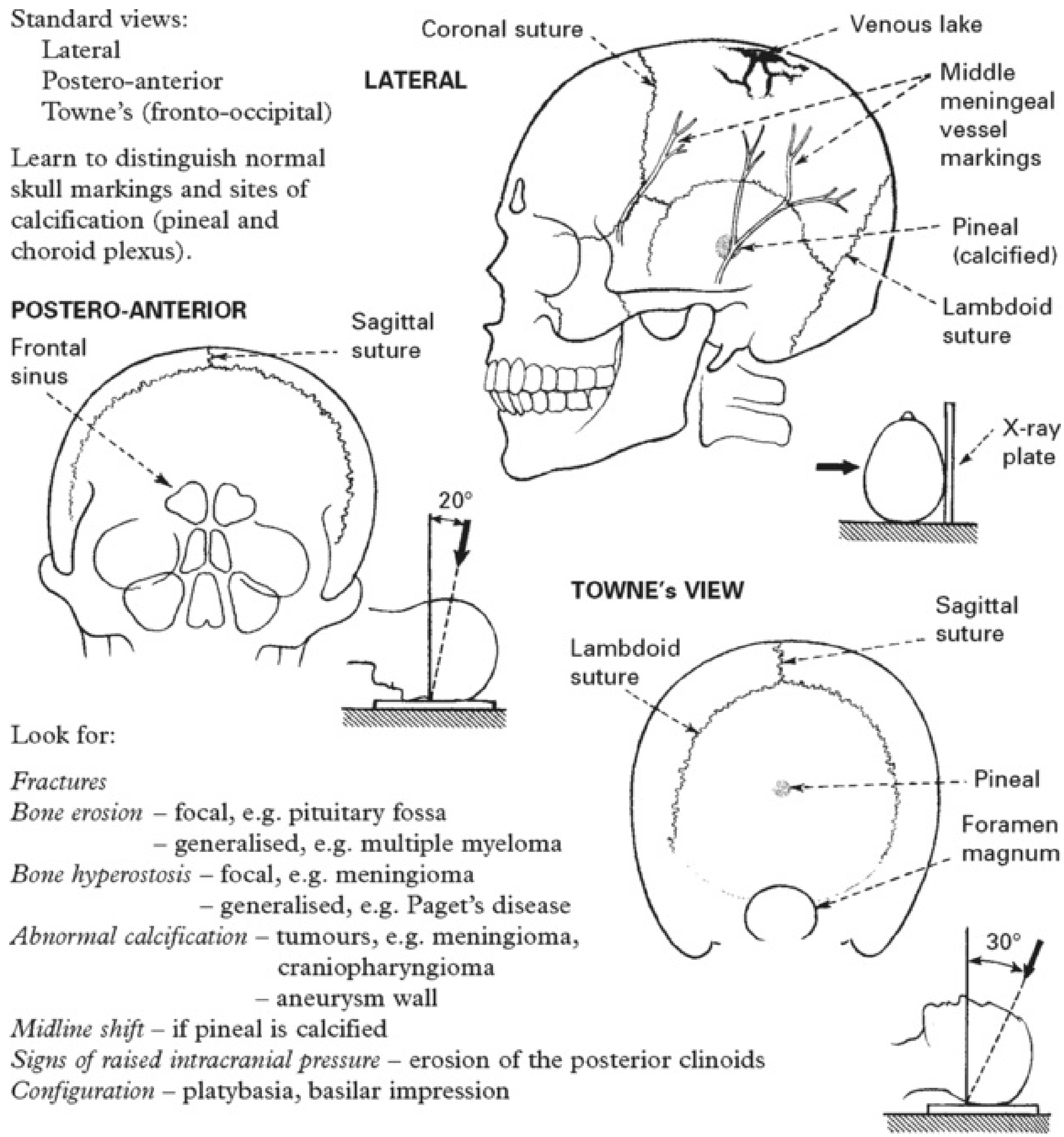
SECTION II

INVESTIGATIONS OF THE CENTRAL AND PERIPHERAL NERVOUS SYSTEMS

SKULL X-RAY

With the development of more advanced imaging techniques, skull X-ray is now less often used, but may still provide useful information.



More specific views are available, but in practice have been replaced by other imaging techniques, *e.g.*

Base of skull (submentovertical) – cranial nerve palsies

Optic foramina – progressive blindness

Sella turcica – visual field defects

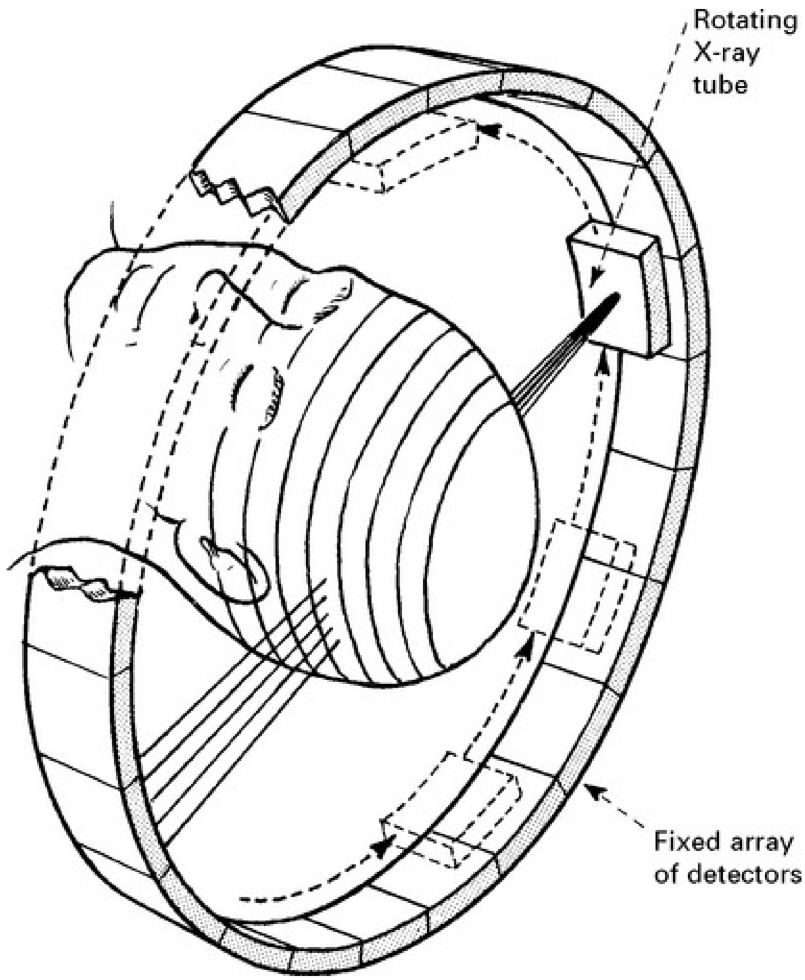
Petrous/internal auditory meatus – sensorineural deafness.

COMPUTERISED TOMOGRAPHY (CT) SCANNING

The development of this non-invasive technique in the 1970s revolutionised the investigative approach to intracranial pathology. A pencil beam of X-ray traverses the patient's head and a diametrically opposed detector measures the extent of its absorption. Computer processing, multiple rotating beams and detectors arranged in a complete circle around the patient's head enable determination of absorption values for multiple small blocks of tissue (voxels). Reconstruction of these areas on a two-dimensional display (pixels) provides the characteristic CT scan appearance. For routine scanning, slices are 3–5 mm wide. The latest 'spiral' or 'helical' CT scanners use a large bank of detectors (multislice) and the patient moves through the field during scanning so that the X-ray beams describe a helical path. This considerably reduces scanning time and is of particular value when slices of 1–2 mm thickness provide greater detail. These 'high definition' views permit coronal and sagittal reconstructions and allow detailed examination of certain areas *e.g.* the orbit, pituitary fossa and cerebellopontine angle.

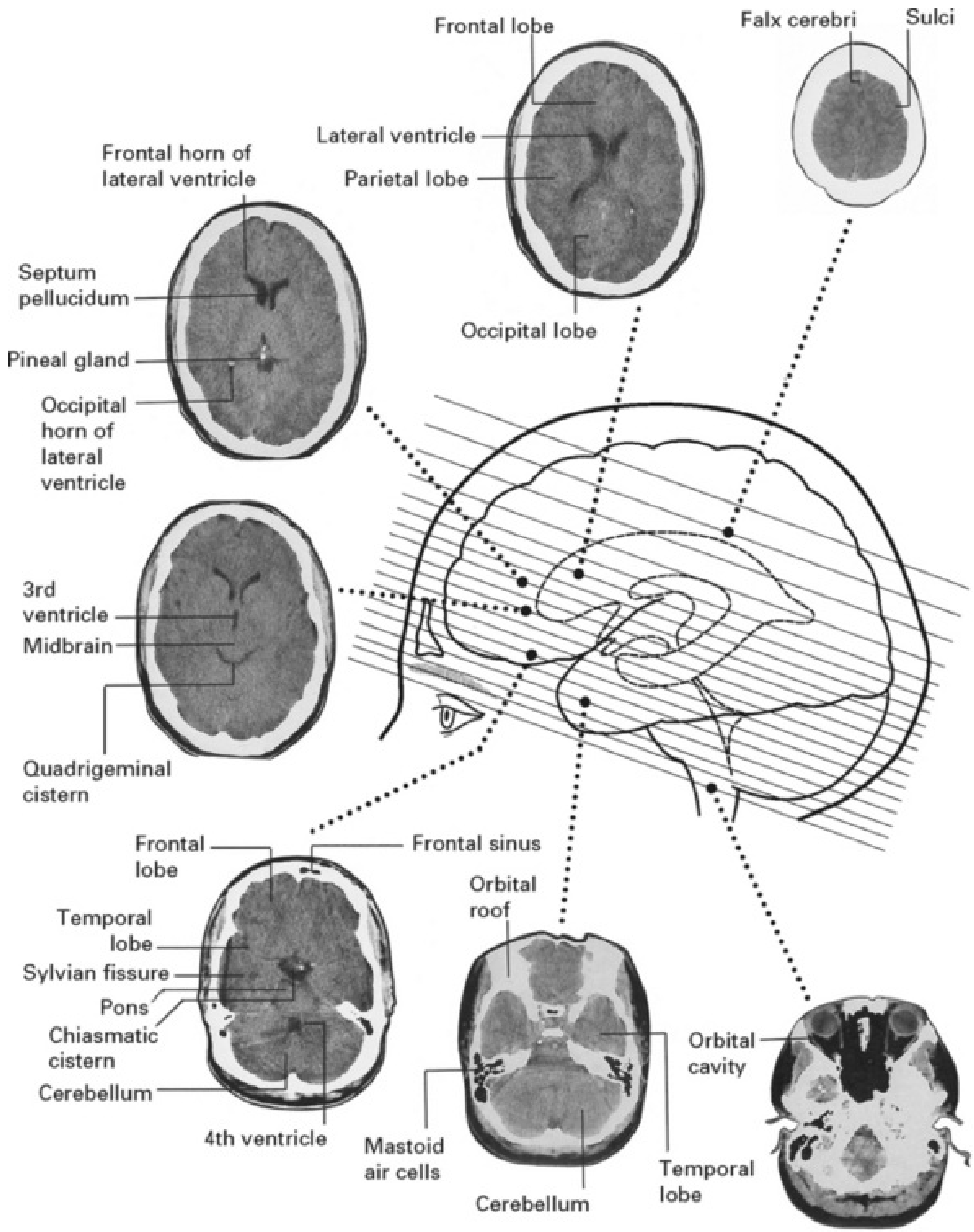
Selecting different window levels displays tissues of different X-ray density more clearly. Most centres routinely provide two images for each scanned level of the lumbar spine, one to demonstrate bone structures, the other to show soft tissue within and outwith the spinal canal.

An intravenous iodinated water-soluble contrast medium is administered when the plain scan reveals an abnormality or if specific clinical indications exist, *e.g.* suspected arteriovenous malformation, acoustic schwannoma or intracerebral abscess. Intravenous contrast shows areas with increased vascularity or with impairment of the blood–brain barrier.



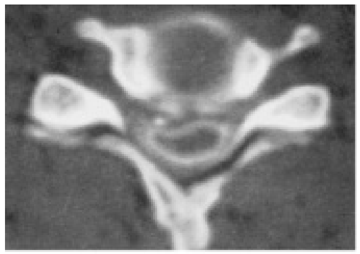
Note: diagram illustrates individual slices. In the latest generation scanners, the beam describes a helical pathway around the head.

NORMAL SCAN



Spinal CT scanning

If MRI is unavailable, CT of the spine can demonstrate the bony canal, intervertebral foramen and disc protrusion. After instilling some intrathecal contrast, CT scanning clearly demonstrates lesions compressing the spinal cord or the cervicomedullary junction.



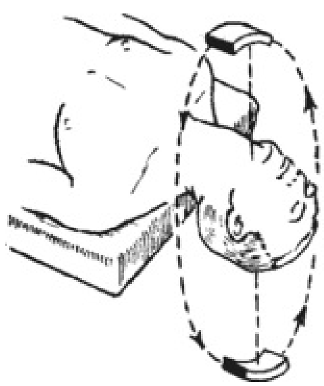
Cervical disc compressing one side of the spinal cord.

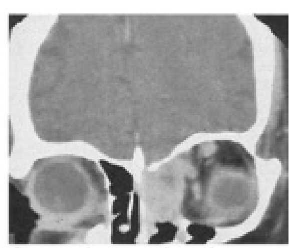
Coronal and sagittal reconstruction

CT imaging in the coronal plane is difficult and in the sagittal plane, virtually impossible. Two dimensional reconstruction of a selected plane may provide more information, but requires CT slices of narrow width *e.g.* 1–2 mm.

Reconstruction showing orbital tumour and relationships in the coronal plane

Coronal CT scanning





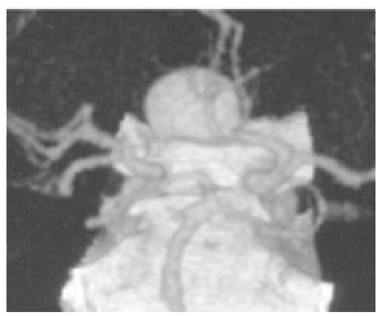
Coronal scan showing a tumour of the ethmoidal sinus

In the absence of the latest scanners and good reconstruction, full neck extension combined with maximal angulation of the CT gantry permits direct coronal scanning.

CT angiography

Helical scanning during infusion of intravenous contrast provides a non-invasive method of demonstrating intracranial vessels in 2 and 3-D format. The ability to rotate the image through 360° more clearly demonstrates vessels and any abnormalities. Many reports claim that 3-D CT angiography is as accurate as conventional angiography in detecting small aneurysms.

3-D CT angiogram showing an anterior communicating artery aneurysm



CT perfusion imaging

Following the infusion of contrast it is possible to construct brain perfusion maps. Ischaemic regions receive less contrast and appear as low density areas. This technique can be of value in predicting outcome from acute stroke.

Xenon-enhanced computed tomography (XE-CT)

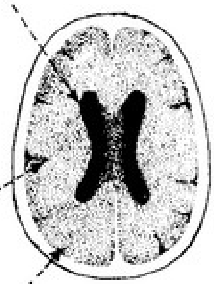
Inhaled stable xenon mixed with O_2 crosses the intact blood–brain barrier. CT scanning detects changes in tissue density as xenon accumulates producing quantitative maps of regional blood flow. This technique determines the degree and extent of cerebral ischaemia.

Interpretation of the cranial CT scan

Before contrast enhancement note:

VENTRICULAR SYSTEM

Size
Position
Compression of one or more horns, i.e. frontal, temporal or occipital



WIDTH OF CORTICAL SULCI AND THE SYLVIAN FISSURES

SKULL BASE AND VAULT

Hyperostosis
Osteolytic lesion
Remodelling
Depressed fracture

MULTIPLE LESIONS

may result from:

Tumour – metastases
– lymphoma
Abscesses
Granuloma
Infarction
Trauma

ABNORMAL TISSUE DENSITY

Identify the site, and whether the lesion lies within or without the brain substance.
Note the 'MASS EFFECT':
– midline shift
– ventricular compression
– obliteration of the basal cisterns, sulci

High density
Blood
Calcification – tumour
– arteriovenous malformation/aneurysm
– hamartoma
(Calcification of the pineal gland, choroid plexus, basal ganglia and falx may occur in normal scans.)

Low density
Infarction (arterial/venous)
Tumour
Abscess
Oedema
Encephalitis
Resolving haematoma

Mixed density
Tumour
Abscess
Arteriovenous malformation
Contusion
Haemorrhagic infarct

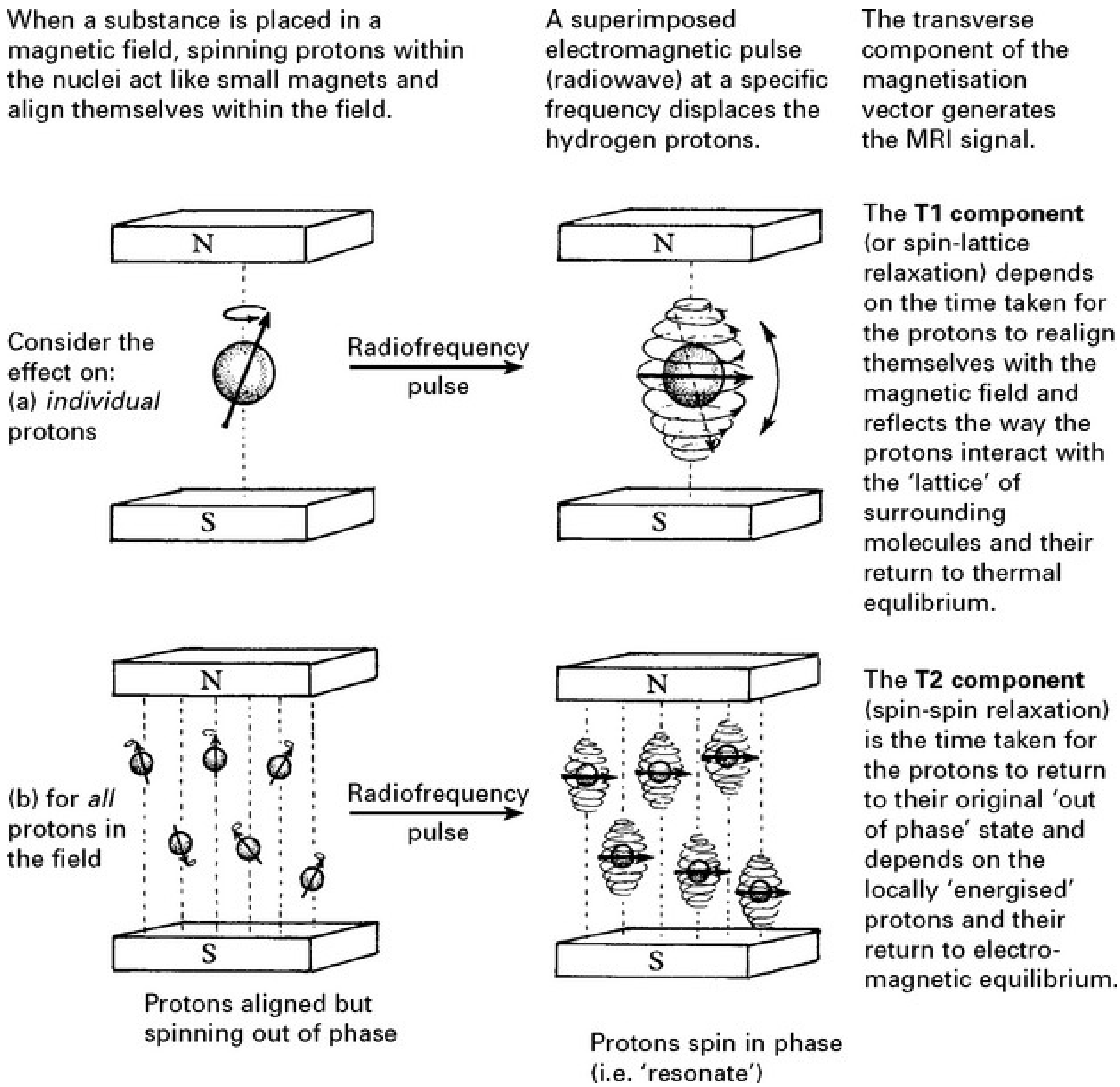
After contrast enhancement:

Vessels in the circle of Willis appear in the basal slices. Look at the extent and pattern of contrast uptake in any abnormal region. Some lesions may only appear after contrast enhancement.

MAGNETIC RESONANCE IMAGING (MRI)

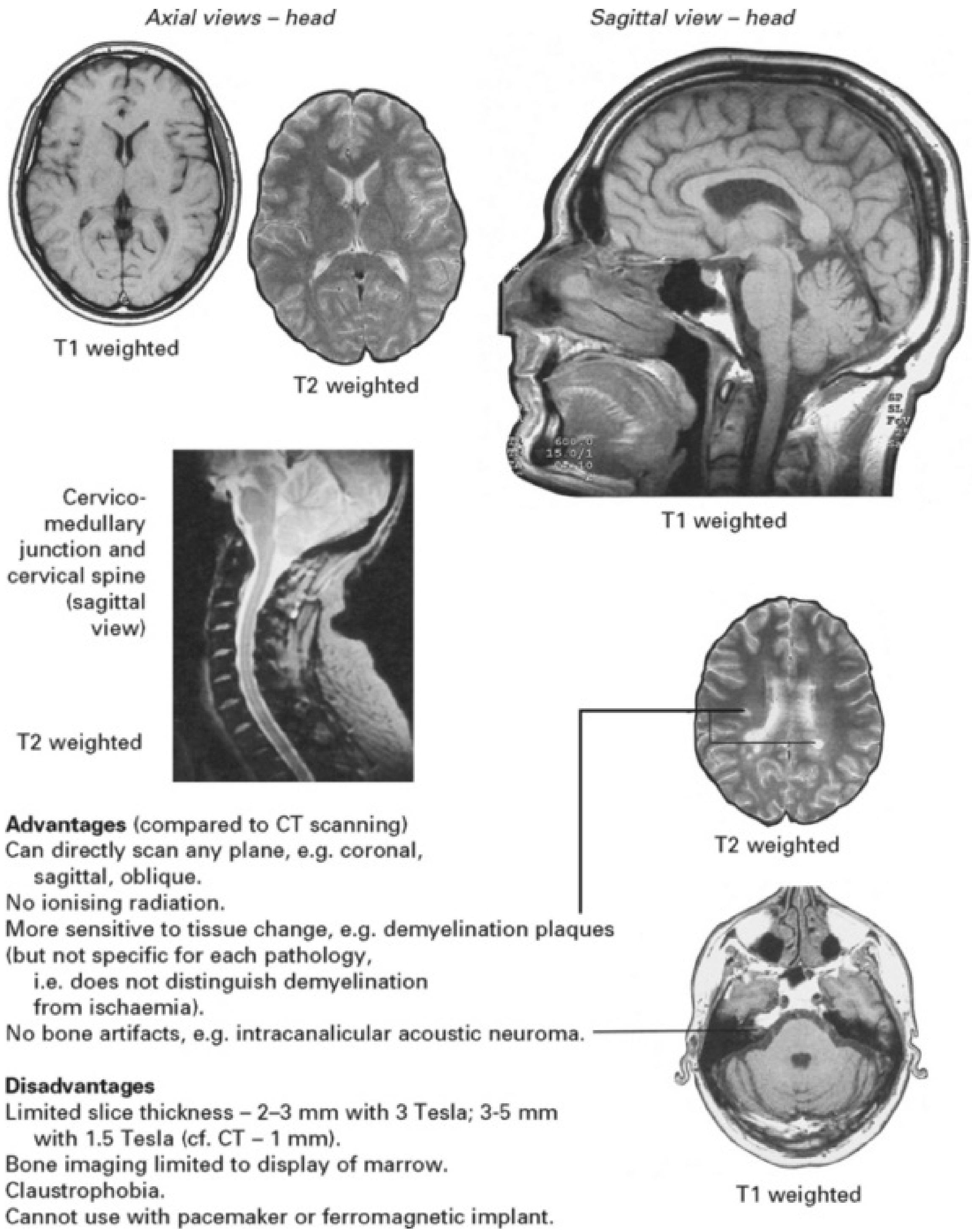
For many years, magnetic resonance techniques aided chemical analysis in the food and petrochemical industries. The development of large-bore homogeneous magnets and computer assisted imaging (as in CT scanning) extended its use to the mapping of hydrogen nuclei (i.e. water) densities and their effect on surrounding molecules in vivo. Since these vary from tissue to tissue, MRI can provide a detailed image of both head and body structures. The latest echo-planer MR imaging permits rapid image acquisition.

Physical basis



A variety of different radiofrequency pulse sequences (saturation recovery (SR), inversion recovery (IR) and spin echo (SE)) combined with computerised imaging produce an image of either proton density or of T1 or T2 weighting depending on the sequence employed.

Normal MRI images (T1/T2 weighting in relation to normal grey/white matter)



Interpretation of abnormal MRI image

Look for structural abnormalities and abnormal intensities indicating a change in tissue T1 or T2 weighting *in relation to normal grey and white* *matter*. (A prolonged T1 relaxation time gives hypointensity, *i.e.* more black; a prolonged T2 relaxation time gives hyperintensity, *i.e.* more white).

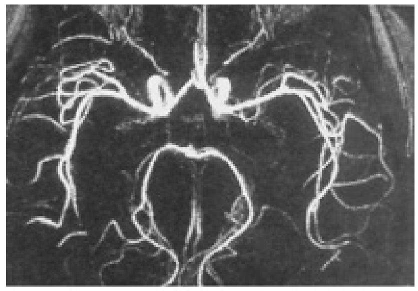
Tissue/lesion			T1 weighting Intensity –	T2 weighting Intensity –
CSF, cyst, hygroma, cerebromalacia			↓↓	↑
Ischaemia, oedema, demyelination, most malignant tumours			↓	↑
Fat e.g. dermoid tumour, lipoma, some metastasis, atheroma			↑	↑
Meningioma (usually identified from structural change or surrounding oedema)			=	=
Evolution of haemorrhage				
Hyperacute	0–2 hours	Intracellular Oxy-Hb	=, slight ↓	↑
Acute	2 hours – 5 days	Intracellular Deoxy-Hb	=, slight ↓	↓↓
Early subacute	5 – 10 days	Intracellular Met-Hb	↑↑	↓↓
Late subacute	10 days – weeks	Free Met-Hb	↑↑	↑↑
Chronic	Months – years	Haemosiderin, Ferritin	=, ↓	↓↓

Paramagnetic enhancement

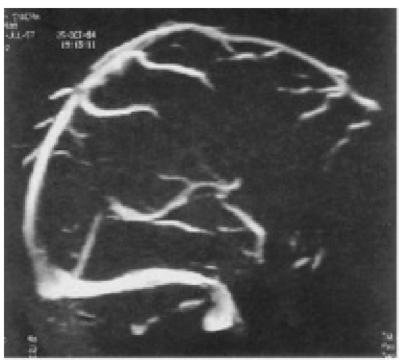
Some substances *e.g.* gadolinium, induce strong local magnetic fields – particularly shortening the T1 component. After intravenous administration, leakage of gadolinium through regions of damaged blood–brain barrier produces marked enhancement of the MRI signal, *e.g.* in ischaemia, infection, tumours and demyelination. Gadolinium may also help differentiate tumour tissue from surrounding oedema.

MR Angiography (MRA)

Rapidly flowing protons can create different intensities from stationary protons and the resultant signals obtained by special sequences can demonstrate vessels, aneurysms and arteriovenous malformations. Vessels displayed simultaneously, may make interpretation difficult, but selection of a specific MR section can demonstrate a single vessel or bifurcation. By selecting a specific flow velocity, MRA will show either arteries or veins. The resolution has improved with 3 Tesla MRA, but may still miss aneurysms seen on intra-arterial DSA (see page 45).

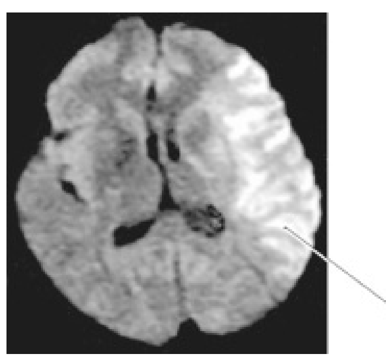


Cerebral arteries from below



Cerebral veins – oblique view showing sagittal sinus

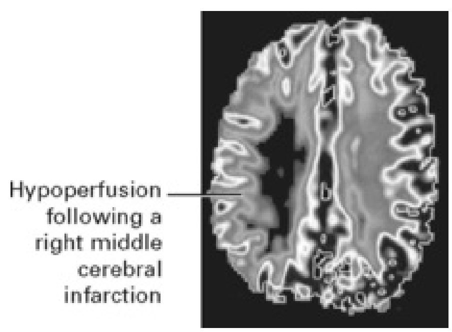
Diffusion-weighted MRI (DWI)



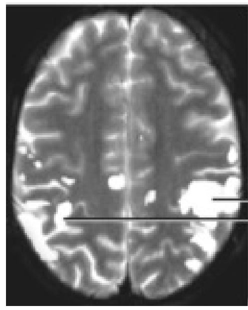
Images are based on an assessment of thermally driven translational movement of water and other small molecules within the brain. In acute ischaemia, cytotoxic oedema restrains diffusion. The degree of restricted diffusion is quantified with a parameter termed the apparent diffusion coefficient (ADC). ADC values fall initially, then normalise and prolong as ischaemic tissues become necrotic and are replaced by extracellular fluid. DWI shows size, site and age of ischaemic change. Whilst the volume normally increases within the first few days, the initial lesion size correlates best with the final outcome. This image shows restricted diffusion in a left middle cerebral artery infarct 3 hours from the onset.

Perfusion-weighted MRI (PWI)

Images are obtained by ‘bolus tracking’ after rapid contrast injection. A delay in contrast arrival and reduced concentration signifies hypoperfusion of that brain region. Soon after onset, ischaemic changes on PWI appear larger than on DWI. *The difference between PWI and DWI may reflect dysfunctional salvagable tissue* (ischaemic penumbra see page 245). Early resolution of the PWI abnormality indicates recanalisation of an occluded vessel, whereas in those who do not recanalise the DWI volume expands to fill a large part of the original PWI lesion.



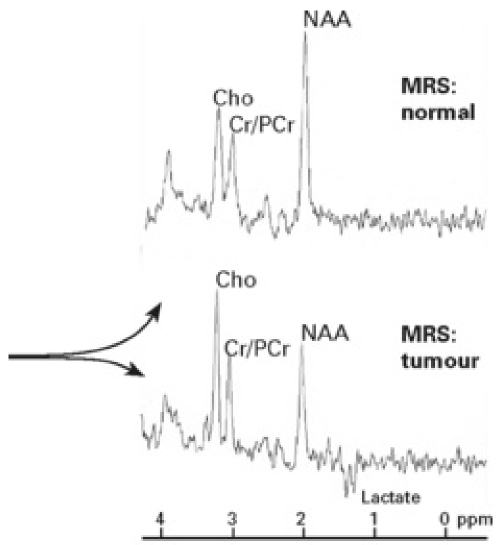
Functional MRI (fMRI)



The oxygenated state of haemoglobin influences the T2 relaxation time of perfused brain. A mismatch between the supply of oxygenated blood and oxygen utilisation in activated areas, produces an increase in venous oxygen content within post capillary venules causing signal change due to blood oxygenation level dependent (BOLD) contrast. Improved spatial and temporal resolution has increased the scope of functional imaging, leading to greater understanding of normal and abnormal brain function. A demonstration of the exact proximity of eloquent regions to areas of proposed resection, helps minimise damage.

Magnetic resonance spectroscopy (MRS)

Spectroscopic techniques generate information on in vivo biochemical changes in response to disease. Concentrations of chemicals of biological interest are minute but measurement can be undertaken in single or multiple regions of interest of around $1.5cm^3$. N-acetylaspartate (a neuronal marker) and lactate are studied by ^1H-MRS, whilst adenosine triphosphate phosphocreatine and inorganic phosphate are measured by ^{31}P-MRS. MRS is gradually emerging from being a research tool to play a role in tumour characterisation, the confirmation of metabolic brain lesions and the study of degenerative disease.

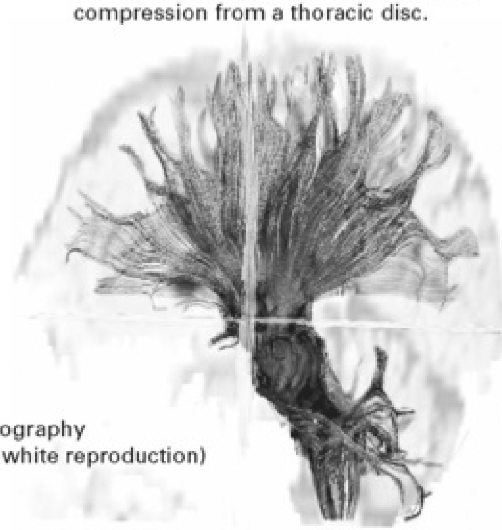


Cho – Choline, Cr/PCr – Creatine/ phosphocreatine, NAA – N-acetylaspartate.

^{1}H-MRS from both regions of normal brain and from a grade II astrocytoma. The tumour trace shows a high choline peak, due to high membrane turnover, a grossly reduced peak of N-acetylaspartate and the presence of lactate, confirming anaerobic metabolism.

Diffusion tensor imaging (DTI) - tractography

As with diffusion-weighted MRI, diffusion tensor imaging utilises the movement of water. Water diffuses more rapidly in the direction aligned with the internal structure. Each MR voxel has a rate of diffusion and a preferred direction. The structure of the white matter tracts facilitates movement of water through the brain in the direction of the tract (anisotropic diffusion). *Tractography* is the technique which makes use of this directional information. For each voxel a colour-coded tensor (or vector) can be created which reflects 3-dimensional orientation of diffusion, the colour reflecting the direction. By this means neural tracts can be demonstrated along their whole length.



This imaging technique has demonstrated interruption of white matter tracts in patients who have suffered a traumatic diffuse axonal injury. Of even more clinical value is the technique's ability to show whether intrinsic tumours infiltrate or deflect crucial structures such as the corticospinal tracts, potentially of value in pre-operative planning and performing tumour resection. The accuracy of DTI tractography and its use as an operative guide still requires validation.

ULTRASOUND

Extracranial

When the probe (i.e. a transducer) – frequency 5–10 MHz, is applied to the skin surface, a proportion of the ultrasonic waves emitted are reflected back from structures of varying acoustic impedance and are detected by the same probe. These reflected waves are reconverted into electrical energy and displayed as a two-dimensional image (β-mode).

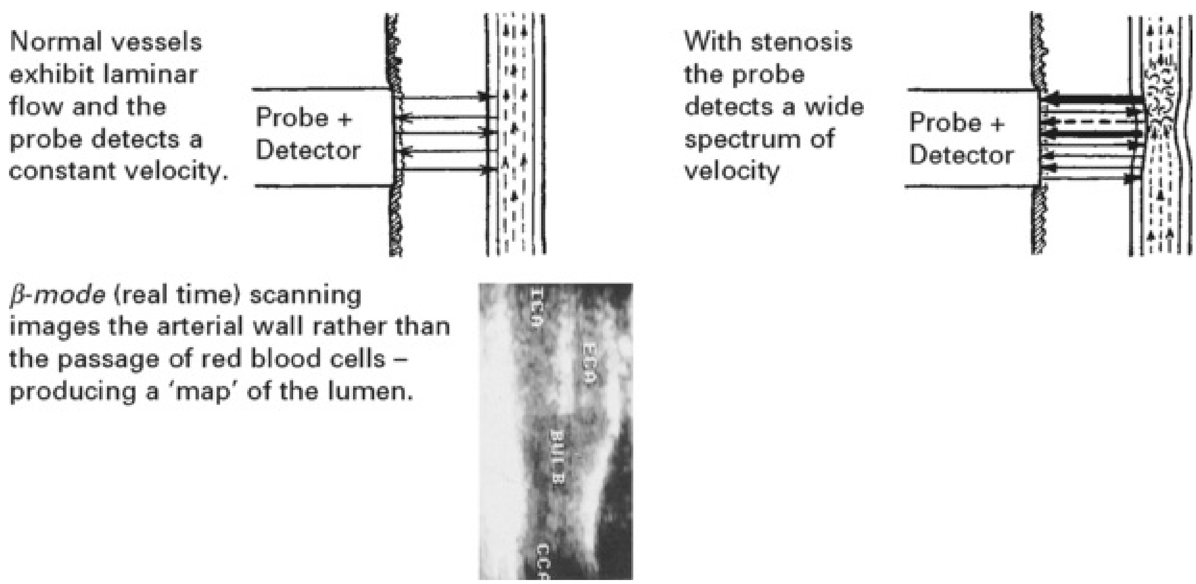
When the probe is directed at moving structures, such as red blood cells within a blood vessel lumen, frequency shift of the reflected waves occurs (the Doppler effect) proportional to the velocity of flowing blood.

Doppler ultrasound uses *continuous wave* (CW) or *pulsed wave* (PW). The former measures frequency shift anywhere along the path of the probe. Pulsed ultrasound records frequency shift at a specific depth.

Duplex scanning combines β-mode with doppler, simultaneously providing images from the vessels from which the velocity is recorded.

Colour Coded Duplex (CCD) uses colour coding to superimpose flow velocities on a two dimensional ultrasound image.

Applications: assessment of extracranial carotid and vertebral arteries.



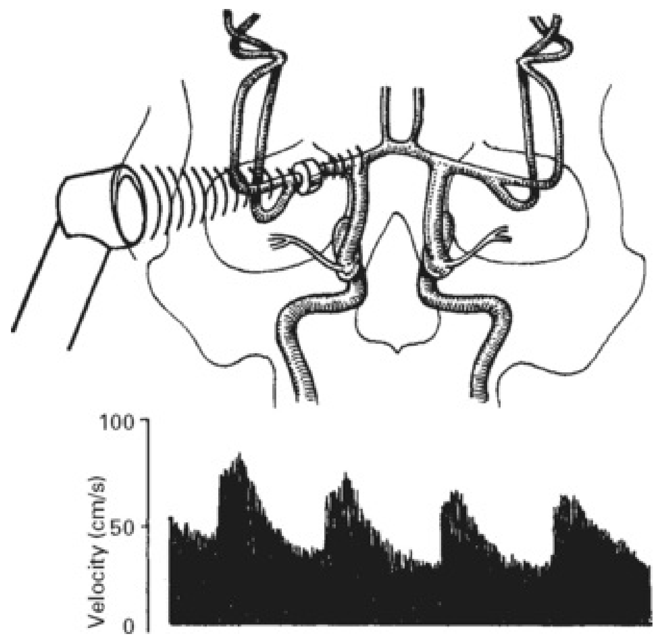
Intracranial – transcranial Doppler ultrasound

By selecting lower frequencies (2 MHz), ultrasound is able to penetrate the thinner parts of the skull bone.

Combining this with a pulsed system gives reliable measurements of flow velocity in the anterior, middle and posterior cerebral arteries and in the basilar artery.

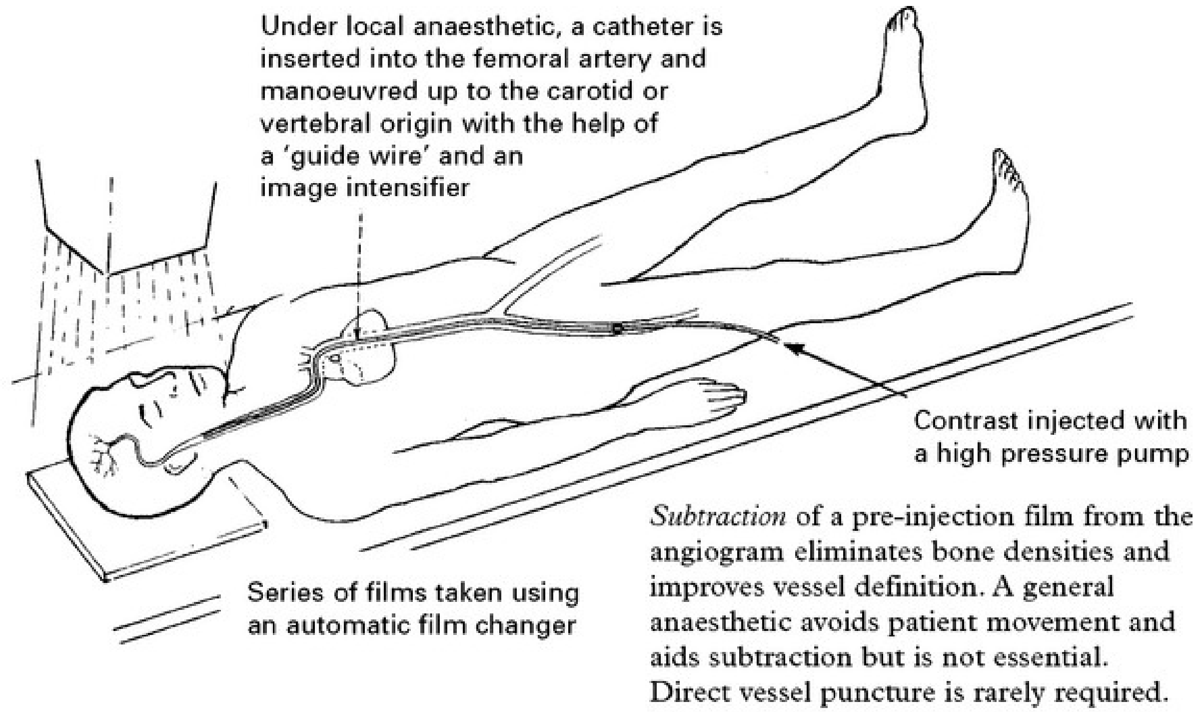
Applications:

Assessment of intracranial haemodynamics in extracranial occlusive/stenotic vascular disease. Detection of vasospasm in subarachnoid haemorrhage.



ANGIOGRAPHY

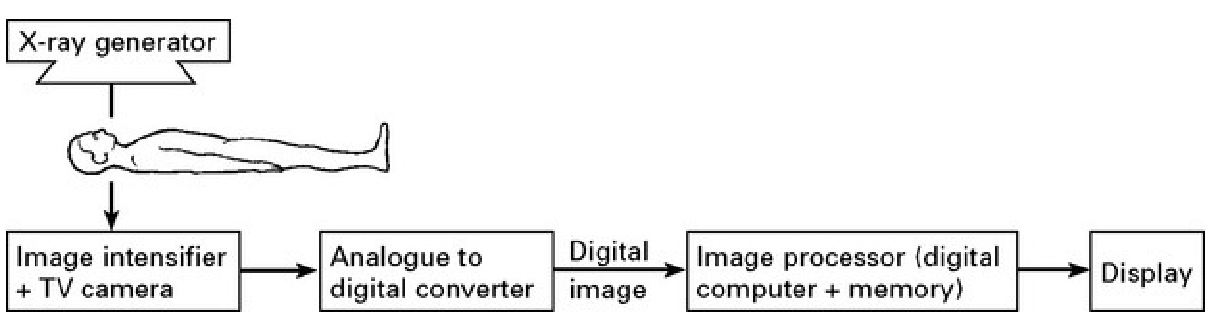
Many neurological and neurosurgical conditions require accurate delineation of both intra-and extracranial vessels. Intra-arterial injection of contrast, imaged by digital subtraction (DSA), remains the gold standard for imaging intracranial vessels.



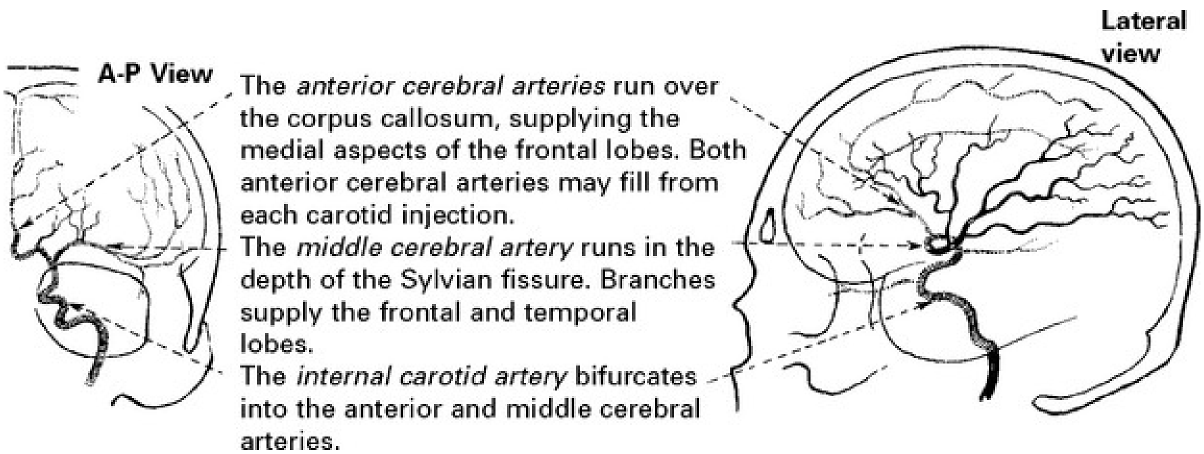
Phase – arterial
– capillary
– venous

Most information is now derived from the arterial phase. Prior to the availability of CT scanning, the position of the cerebral vessels helped localise intracranial structures.

Digital subtraction angiography (DSA) depends upon high-speed digital computing. Exposures taken before and after the administration of contrast agents are instantly subtracted 'pixel by pixel'. With the latest equipment, data processing provides 3D imaging of vessels and permits magnification of specific areas and rotation of the 3D image in any plane.

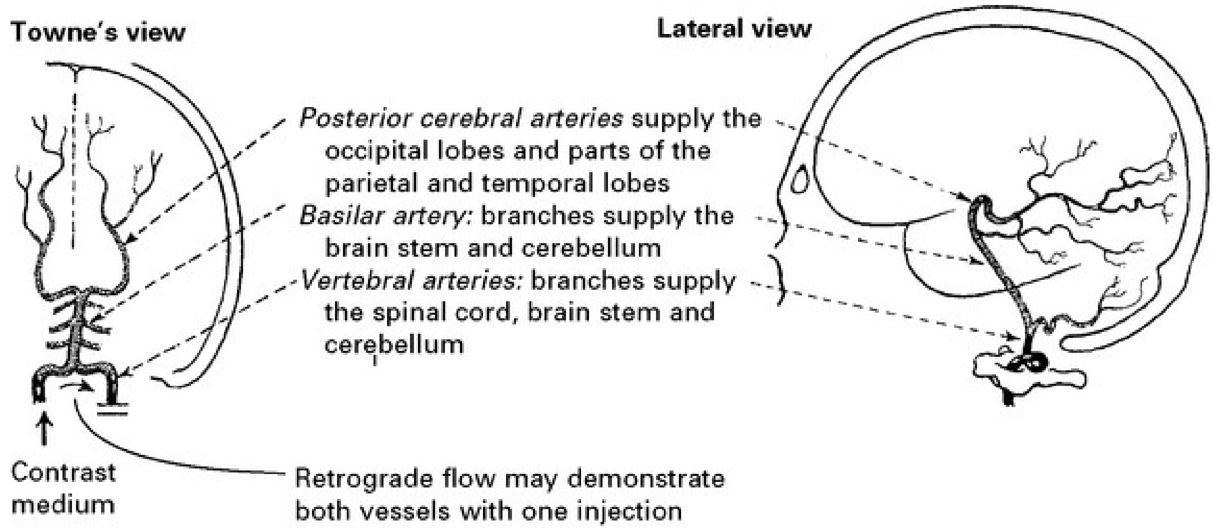


CAROTID ANGIOGRAPHY



In the absence of the ability to rotate the image, **oblique views** may aid identification of some lesions, *e.g.* aneurysms.

VERTEBRAL ANGIOGRAPHY



In *carotid* and *vertebral* angiography look for:

- Vessel *occlusion*, *stenosis*, *plaque formation* or *dissection*
- *Aneurysms*
- *Arterio-venous malformations*
- Abnormal *tumour circulation*
- Vessel *displacement* or *compression*.

Although superseded by the CT scan in tumour detection, angiography may give useful information about feeding vessels and the extent of vessel involvement with the tumour.

Complications

The development of non-ionic contrast mediums, *e.g.* iohexol, iopamidol, has considerably reduced the risk of complications during or following angiography.

Cerebral ischaemia: caused by emboli from an arteriosclerotic plaque broken off by the catheter tip, hypotension or vessel spasm following contrast injection. The small amount of contrast used for intra-arterial DSA carries low risk. In the hands of experienced radiologists, permanent neurological deficit occurs in only one in every 1000 investigations (one in 100 in arteriopaths).

Contrast sensitivity: mild sensitivity to the contrast occasionally develops, but this rarely causes severe problems.

CT angiography (see page 37), **Magnetic Resonance Angiography (MRA)** (see page 41)

INTERVENTIONAL ANGIOGRAPHY

With recent advances, endovascular techniques now play an important role in neurosurgical management.

Embolisation: *Particles* (e.g. Ivalon sponge) injected through the arterial catheter will occlude small vessels; *e.g.* those feeding meningioma or glomus jugulare tumours, thus minimising operative haemorrhage.

'Glue' (isobutyl-2-cyanocrylate) can be injected into both high and low flow arteriovenous malformations. Operative excision is greatly facilitated; if the lesion is completely obliterated, this may even serve as a definitive treatment.

Balloons inflated, then detached from the catheter tip will occlude high flow systems involving large vessels, *e.g.* carotico-cavernous fistula, high flow arteriovenous malformations.

Platinum coils inserted into the aneurysm fundus through a special catheter can produce complete or partial obliteration. Many centres now use this technique as a first line treatment for intracranial aneurysms, particularly those at the basilar bifurcation (see page 288). Temporary inflation of a balloon within the parent vessel during coiling can help prevent occlusion of the parent vessel in wide necked aneurysms *(balloon remodelling)* (see page 289).

Stents are now available for use in intracranial vessels and can prevent prolapse of platinum coils into the vessel lumen.

All techniques carry some risk of cerebral (or spinal) infarction from inadvertent distal embolisation when used in the internal carotid or spinal systems.

Angioplasty: Inflation of an intravascular balloon within a vasospastic segment of a major vessel may reverse cerebral ischaemia, but the technique is not without risk. No large trials of effectiveness exist.

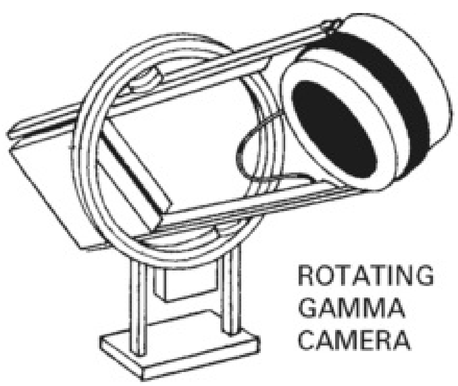
RADIONUCLIDE IMAGING

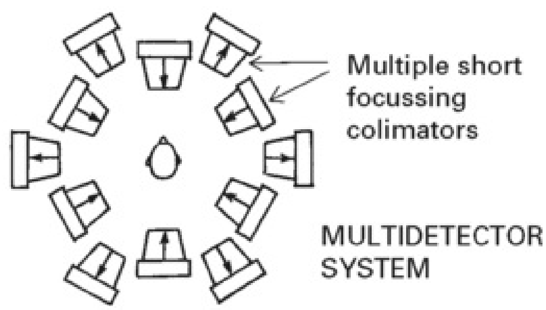
Single photon emission computed tomography (SPECT)

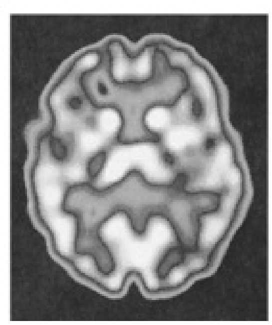
There are two components to imaging with radioactive tracers – the detecting system and the labelled chemical. Each has become increasingly sophisticated in recent years. SPECT uses compounds labelled with gamma-emitting tracers (ligands), but unlike conventional scanning, acquires data from multiple sites around the head. Similar computing to CT scanning provides a two-dimensional image depicting the radioactivity emitted from each 'pixel'. This gives improved definition and localisation. Various ligands have been developed but a $^{99}Tc^{m}$ labelled derivative of propylamine oxime (HMPAO) is the most frequently used. This tracer represents *cerebral blood flow* since it rapidly diffuses across the blood–brain barrier, becomes trapped within the cells, and remains long enough to allow time for scanning. Of the total injected dose, 5% is taken up by the brain and 86% of this activity remains in the brain at least 24 hours.

Ligands for SPECT scanning	Purpose
HMPAO	Cerebral blood flow
^{123}I–FP–CIT	Dopamine presynaptic receptors
^{123}I–IBZM	Dopamine postsynaptic receptors
^{123}I–Iomazenil	Benzodiazepine receptors
^{123}I–CNB	Cholinergic receptors
^{123}I–MK801	Glutamate receptors
^{201}Thallium–chloride	High grade tumour/breakdown blood–brain barrier
^{123}I–tyrosine	Low grade tumour component

A rotating gamma camera is often used for detection, although fixed multidetector systems will produce higher quality images. Data are normally reconstructed to give axial images but coronal and sagittal can also be produced.





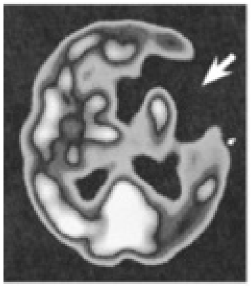


The normal scan – HMPAO (10 mm resolution)

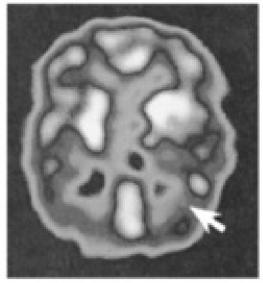
The tomogram can be co-registered with structural imaging (CT or MRI) to aid interpretation.

Clinical applications

– Detection of early ischaemia in
OCCLUSIVE and HAEMORRHAGIC CEREBROVASCULAR DISEASE



Absence of blood flow corresponds with area of infarction and tissue loss seen on structural imaging.



– Assessment of blood flow changes in DEMENTIA

Blood flow is generally reduced, especially in temporal and parietal lobes

– Evaluation of patients with intractable EPILEPSY of temporal lobe origin

Normal subject

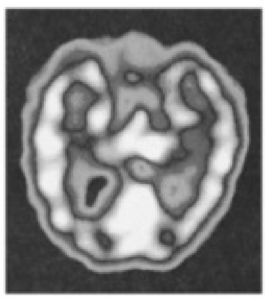
Scan of temporal lobe showing symmetrical pattern of blood flow more prominent in grey matter

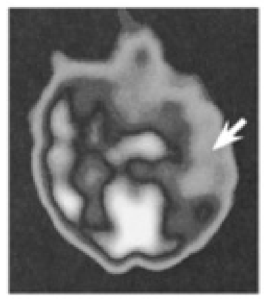
Patient with temporal lobe epilepsy

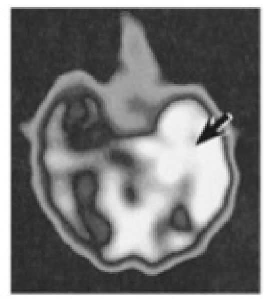
An *interictal* scan shows reduced flow throughout the temporal lobe

An *ictal* scan (i.e. HMPAO injected during the seizure) shows a marked hyperperfusion of the temporal lobe

The plane of scan lies in the same axis as the temporal lobe

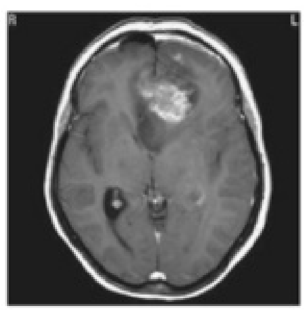






Such findings aid localisation of the epileptic focus and selection of patients for surgical treatment.

Thallium SPECT: a high uptake of thallium indicates rapidly dividing cells and can help differentiate low and high grade TUMOURS.

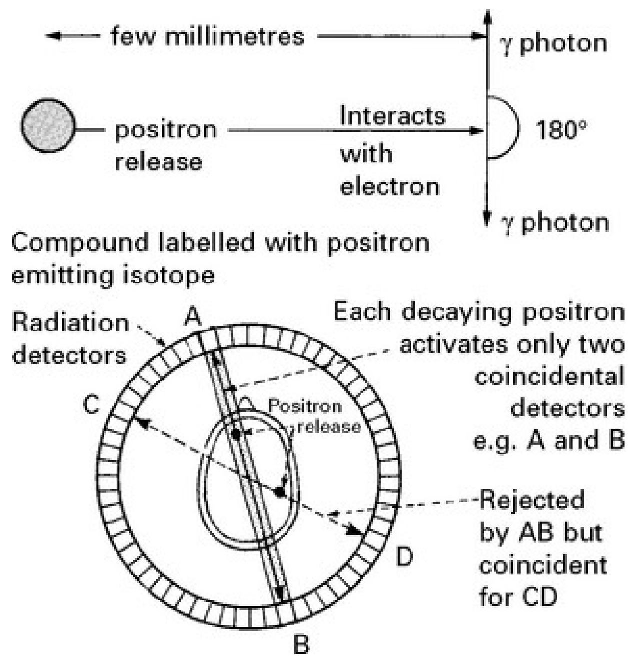


MRI showing apparent high grade tumour

^{201}Thallium scan showing high uptake

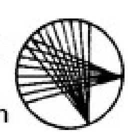
Positron emission tomography (PET)

PET uses positron-emitting isotopes (radionuclides) bound to compounds of biological interest to study specific physiological processes quantitatively. Positron-emitting isotopes depend on a cyclotron for production and their half-life is short; PET scanners only exist on adjacent sites which limits availability for routine clinical use.



Each decaying positron results in the release of two photons in diametric opposition; these activate two coincidental detectors. Multiple pairs of detectors and computer processing techniques enable quantitative determination of local radioactivity (and density of the labelled compound) for each 'voxel' (a cube of tissue) within the imaged field. Reconstruction using similar techniques to CT scanning produces the PET image.

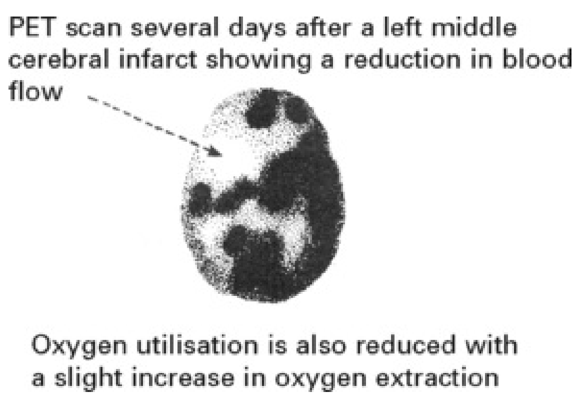
Each detector is linked to several others in a fan shaped distribution



Isotope	Binding compound		Measurement under study
^{15}Oxygen	Carbon monoxide	– inhalation	*Cerebral blood volume (CBV)*
^{15}Oxygen	Water	– i.v. bolus	*Cerebral blood flow (CBF)*
^{18}Fluorine	Fluorodeoxyglucose	– i.v. bolus	*Cerebral glucose metabolism (CMRgl)*
^{15}Oxygen	Oxygen	– inhalation	*Cerebral oxygen utilisation ($CMRO_2$)*
			Oxygen extraction factor (OEF)
^{11}Carbon	Drug, e.g. phenytoin	– i.v. bolus	*Drug receptor site*
^{11}Carbon	Methyl spiperone	– i.v. bolus	*Dopamine binding site*

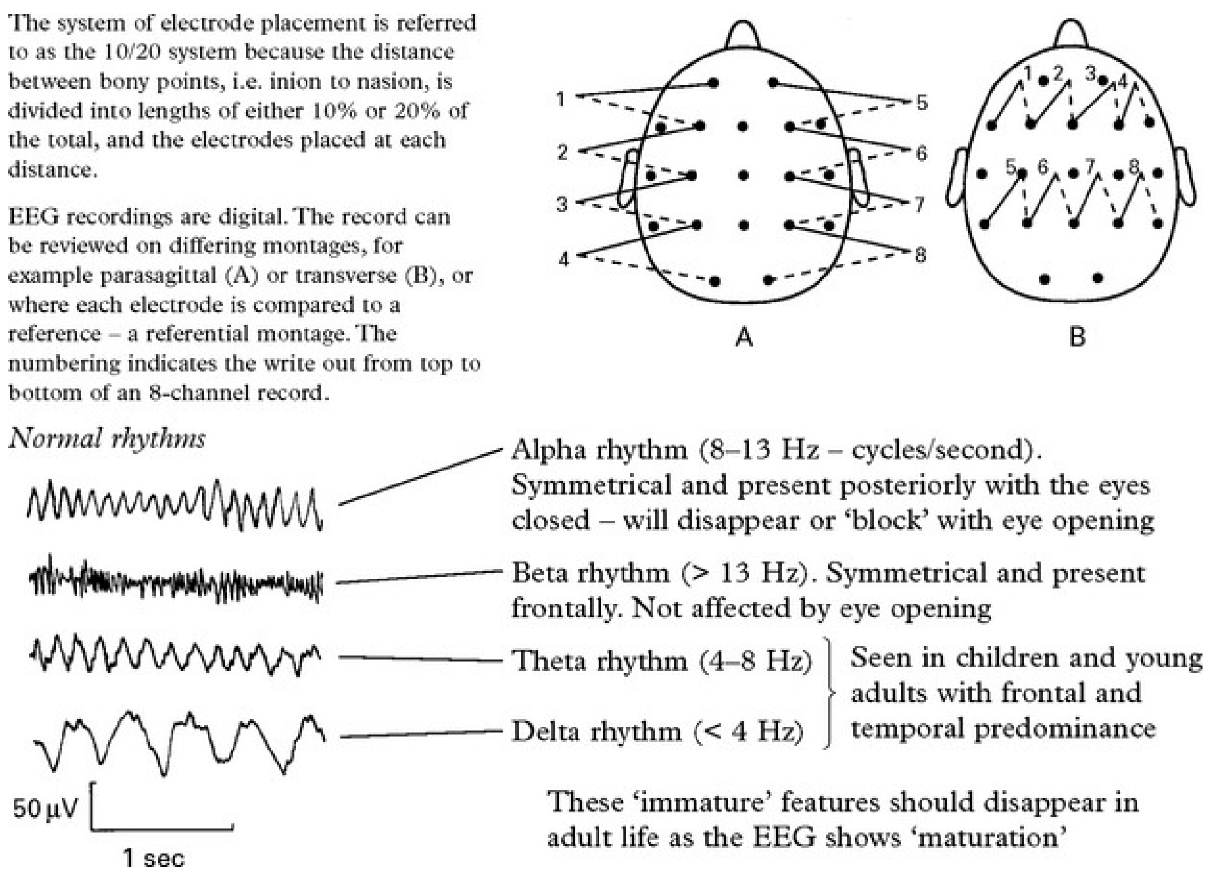
Clinical and research uses

PET scanning is used primarily as a research tool to elucidate the relationships between cerebral blood flow, oxygen utilisation and extraction in focal areas of ischaemia or infarction (page 245) in patients with dementia, epilepsy and brain tumours. Identification of neurotransmitter and drug receptor sites aids the understanding and management of psychiatric (schizophrenia) and movement disorders. Whole body PET scans can also identify occult tumour in patients with paraneoplastic syndromes (page 549).



ELECTROENCEPHALOGRAPHY (EEG)

Electroencephalography examines by means of scalp electrodes the spontaneous electrical activity of the brain. Tiny electrical potentials, which measure millionths of volts, are recorded, amplified and displayed on either 8 or 16 channels of a pen recorder. Low and high frequency filters remove unwanted signals such as muscle artefact and mains interference.



As well as recording a resting EEG stressing the patient by hyperventilation and photic stimulation (a flashing strobe light) may result in an electrical discharge supporting a diagnosis of epilepsy.

More advanced methods of telemetry and foramen ovale recording may be necessary

– to establish the diagnosis of 'epilepsy' if doubt remains

– to determine the exact frequency and site of origin of the attacks

– to aid classification of seizure type.

Telemetry: utilises a continuous 24–48 hour recording of EEG, often combined with a videotape recording of the patient. Increasing availability of this and ambulatory recording has greatly improved diagnostic accuracy and reliability of seizure classification.

Foramen ovale recording: a needle electrode is passed percutaneously through the foramen ovale to record activity from the adjacent temporal lobe.

INTRACRANIAL PRESSURE MONITORING

Although CSF pressure may be measured during lumbar puncture, this method is of limited value in intracranial pressure measurement:

An isolated pressure reading does not indicate the trend or detect pressure waves.

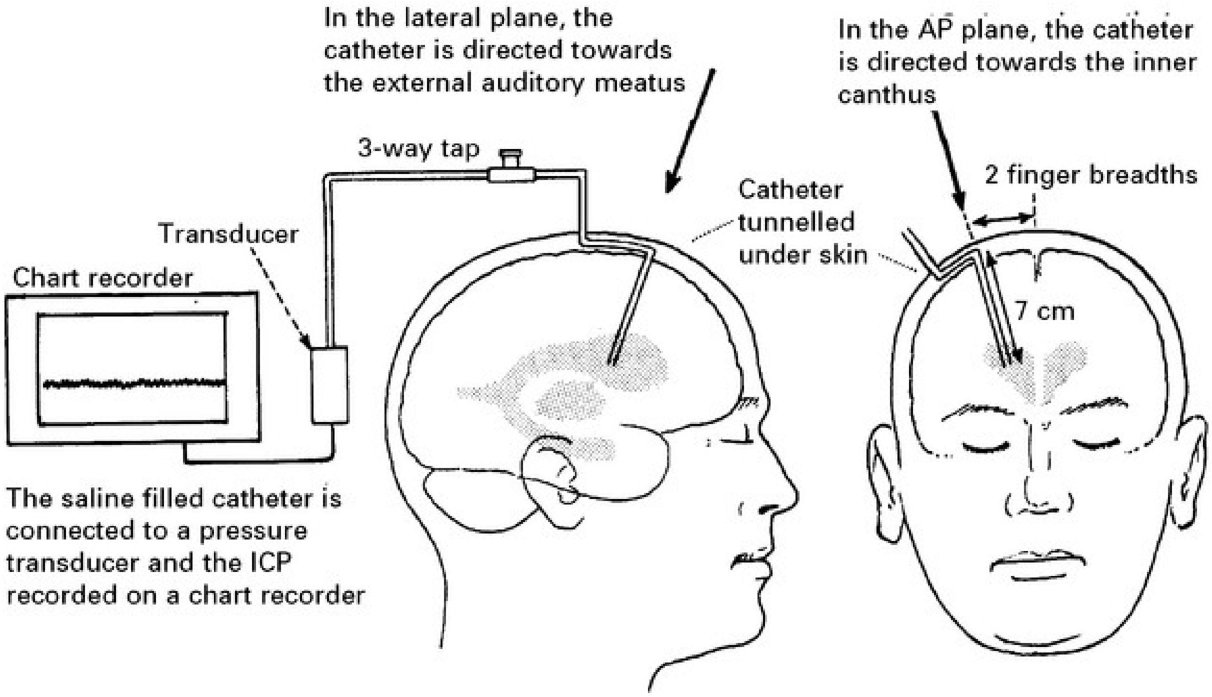
Lumbar puncture is contraindicated in the presence of an intracranial mass.

Pressure gradients exist between different intracranial and spinal compartments, especially in the presence of brain shift.

Many techniques are now available to measure intracranial pressure. In most instances a transducer either lying on the brain surface or inserted a few millimetres into the brain substance suffices, but a catheter inserted into the lateral ventricle remains the 'gold' standard by which other methods are compared.

Ventricular catheter insertion

A ventricular catheter is inserted into the frontal horn of the lateral ventricle through a frontal burr hole or small drill hole situated two finger breadths from the midline, behind the hairline and anterior to the coronal suture.

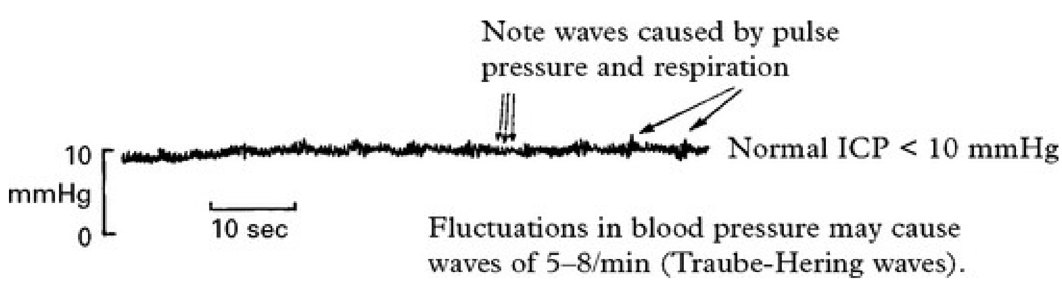


Complications

Intracerebral haemorrhage following catheter insertion rarely occurs.

Ventriculitis occurs in from 10–17%. Minimise this risk by tunnelling catheter under the skin and removing as soon as is practicable.

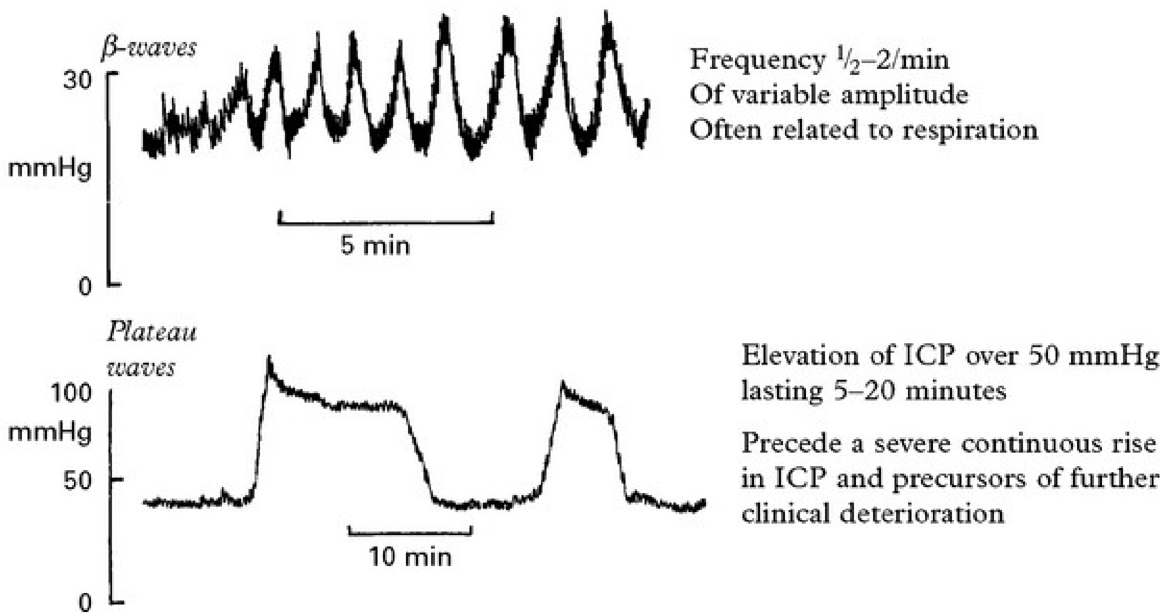
NORMAL PRESSURE TRACE



ABNORMAL PRESSURE TRACE

Look for: *Increase in the mean pressure* – > 20 mmHg – moderate elevation
> 40 mmHg – severe increase in pressure

N.B. As ICP increases, the amplitude of the pulse pressure wave increases.



CLINICAL USES OF ICP MONITORING

– Investigation of normal pressure hydrocephalus – the presence of β waves for > 5% of a 24-hour period suggests impaired CSF absorption and the need for a drainage operation.

– Postoperative monitoring – a rise in ICP may precede clinical evidence of haematoma formation or cerebral swelling.

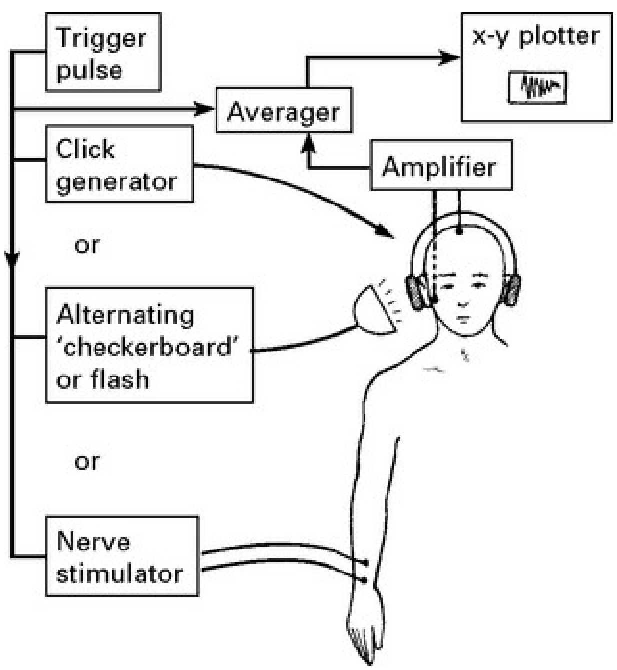
– Small traumatic haematomas – ICP monitoring may guide management and indicate the need for operative removal.

– ICP monitoring is required during treatment aimed at reducing a raised ICP and maintaining cerebral perfusion pressure.

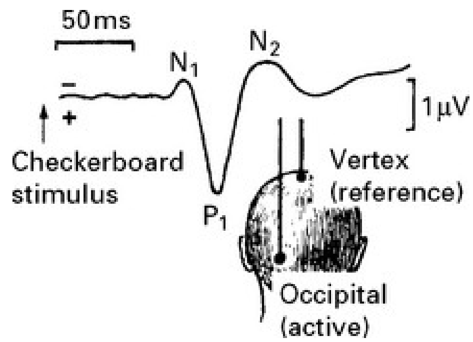
EVOKED POTENTIALS – VISUAL, AUDITORY AND SOMATOSENSORY

RECORDING METHODS

Stimulation of any sensory receptor evokes a minute electrical signal (i.e. microvolts) in the appropriate region of the cerebral cortex. Averaging techniques permit recording and analysis of this signal normally lost within the background electrical activity. When sensitive apparatus is triggered to record cortical activity at a specific time after the stimulus, the background electrical 'noise' averages out, *i.e.* random positive activity subtracts from random negative activity, leaving the signal evoked from the specific stimulus.



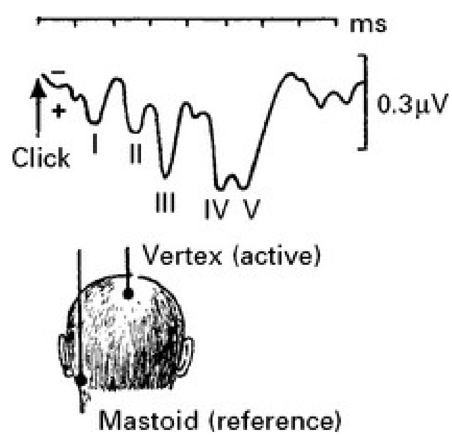
Visual evoked potential (VEP)



A stroboscopic flash diffusely stimulates the retina; alternatively an alternating checkerboard pattern stimulates the macula and produces more consistent results. The evoked visual signal is recorded over the occipital cortex. The first large positive wave (PC_1) provides a useful point for measuring conduction through the visual pathways.

Uses: *Multiple sclerosis detection* – 30% with normal ophthalmological examination have abnormal VEP. *Peroperative monitoring* – pituitary surgery.

Brain stem auditory evoked potential (BAEP)



Electrical activity evoked in the first 10 milliseconds after a ‘click’ stimulus provides a wave pattern related to conduction through the auditory pathways in the VIII nerve and nucleus (waves I and II) and in the pons and midbrain (waves III–V). Longer latency potentials (up to

500 ms), recorded from the auditory cortex in response to a 'tone' stimulus, are of less clinical value.

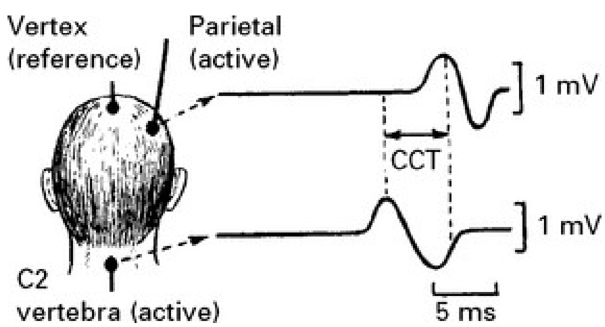
Uses: *Hearing assessment* – especially in children.

Detection of intrinsic and extrinsic brain stem and cerebellopontine angle lesions, e.g. vestibular schwannoma.

Peroperative recording during cerebellopontine angle tumour operations.

Assessment of brain stem function in coma.

Somatosensory evoked potentials (SEP)



The sensory evoked potential is recorded over the parietal cortex in response to stimulation of a peripheral nerve (e.g. median nerve). Other electrodes sited at different points along the sensory pathway record the ascending activity. Subtraction of the latencies between peaks provides conduction time between these sites.

Central conduction time (CCT): sensory conduction time from the dorsal columns (or nuclei) to the parietal cortex.

Uses: *Detection of lesions in the sensory pathways*

– brachial plexus injury

– demyelination.

Peroperative recording – straightening of scoliosis
– removal of spinal tumours/AVM } spinal conduction
– aneurysm operation with temporary vessel occlusion – CCT.

Motor Evoked Potential (MEP)

Subtraction of the latencies between motor evoked potentials elicited by applying a brief magnetic stimulus to either the motor cortex, the spinal cord or the peripheral nerves gives *peripheral* and *central motor conduction velocities*.

MYELOGRAPHY

Now rarely used due to availability of MRI and CT scanning. Injection of water-soluble contrast into the lumbar theca and imaging flow up to the cervicomedullary junction provides a rapid (although invasive) method of screening the whole spinal cord and cauda equina for compressive lesions (e.g. disc disease or spondylosis, tumours, abscesses or cysts). For suspected lumbosacral disc disease, contrast is screened up to the level of the conus *i.e.* RADICULOGRAPHY (but a normal study does not exclude the possibility of a laterally situated disc). CT scanning and MRI have gradually replaced the need for myelography, but the introduction of a low dose of water-soluble contrast considerably enhances axial CT scan images of the spinal cord and nerve roots.

Problems

Headache occurs in 30%, *nausea* and *vomiting* in 20% and *seizures* in 0.5%.

Arachnoiditis – previously a major complication with oil based contrast MYODIL, but rarely occurs with water soluble contrast.

Haematoma – occurs rarely at the injection site.

Impaction of spinal tumour – may follow CSF escape and aggravate the *effects of cord compression, leading to clinical deterioration.*

LUMBAR PUNCTURE (LP)

Lumbar puncture is used to obtain cerebrospinal fluid for analysis and to drain CSF and reduce intracranial pressure, for example in patients with idiopathic intracranial hypertension, communicating hydrocephalus or CSF fistula.

TECHNIQUE

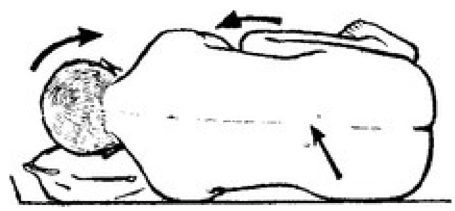
Use the smallest gauge possible to reduce post LP headaches (PLPH), preferably 22G or 20G. Using ‘atraumatic’ needles rather than standard cutting needles reduces the frequency of PLPH, for 22G needles from ~20% to ~5%.

1. *Correct positioning of the patient is essential.* Open the vertebral laminae by drawing the knees up to the chest and flexing the neck. Ensure the back is parallel to the bed to avoid rotation of the spinal column.

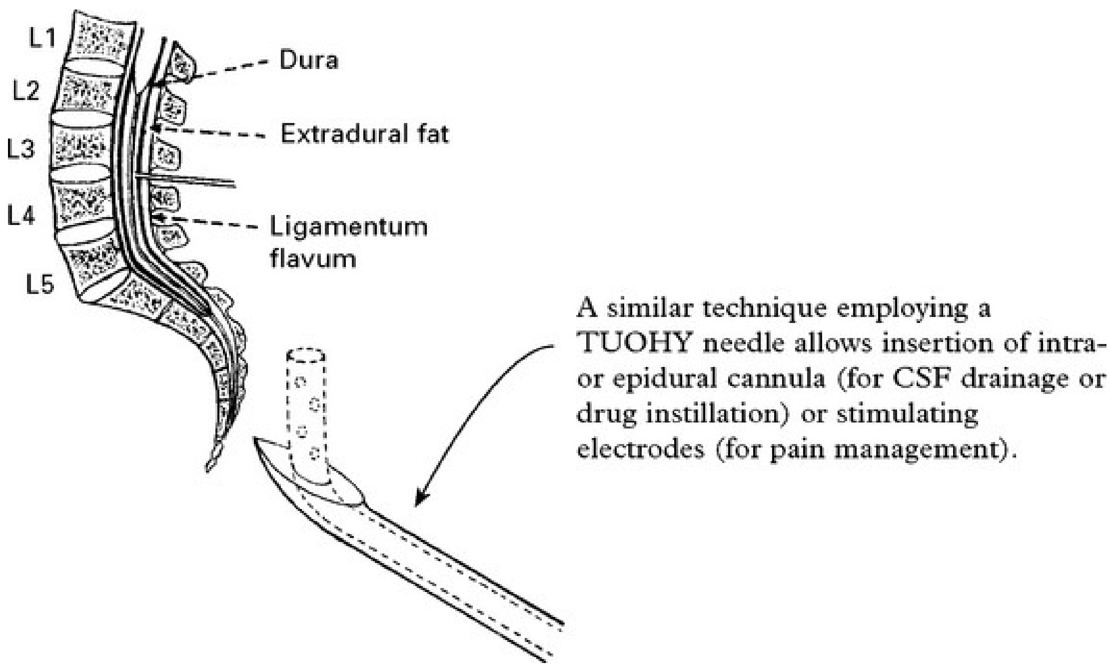
2. Identify the site. Usually aim for the L3/4 space at iliac crest level, but since the spinal cord ends at L1 any space from L2/L3 to L5/S1 is safe.

3. Clean the area and insert a few millilitres of local anaesthetic.

4. Ensure the stylet of the LP needle is fully home and insert at a slight angle towards the head, so that it parallels the spinous processes. Some resistance is felt as the needle passes through the ligamentum flavum, the dura and arachnoid layers.



5. Withdraw the stylet and collect the CSF. If bone is encountered, withdraw the needle and reinsert at a different angle. If the position appears correct yet no CSF appears, rotate the needle to free obstructive nerve roots.



Avoid lumbar puncture

– if raised intracranial pressure is suspected. Even a fine needle leaves a hole through which CSF will leak. In the presence of a space-occupying lesion, especially in the posterior fossa, CSF withdrawal creates a pressure gradient which may precipitate tentorial herniation.

– if platelet count is less than 40000 and prothrombin time is less than

50% of control.

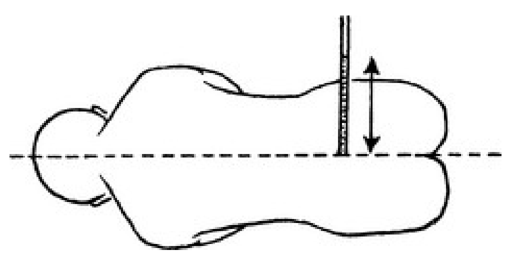
CEREBROSPINAL FLUID

CSF COLLECTION

Subarachnoid haemorrhage (SAH), or puncture of a blood vessel by the needle, may account for blood-stained CSF. To differentiate, collect CSF in three bottles.

CSF PRESSURE MEASUREMENT

Check that the patient's head (foramen of Munro) is level with the lumbar puncture. Connect a manometer via a 3-way tap to the needle and allow CSF to run up the column. Read off the height. Normal value: 100–200 mm CSF.



CSF ANALYSIS

Standard tests

1.	– RBC and differential WBC (normal = < 5 WBCs per mm^3)

Bacteriological	– Gram stain and culture – appearance of supernatant. Xanthochromia (yellow staining) results from subarachnoid haemorrhage with RBC breakdown, high CSF protein or jaundice.
2. Biochemical	– protein (normal = 0.15–0.45 g/l) – glucose (normal = 0.45–0.70 g/l) 40–60% of blood glucose simultaneously sampled.

Special tests

Suspected:	
Subarachnoid haemorrhage Malignant tumour	– spectrophotometry for blood breakdown products – cytology
Tuberculosis	– Ziehl-Neelson stain, Lowenstein-Jensen culture, polymerase chain reaction (PCR)
Non-bacterial infection Demyelinating disease	– virology, fungal and parasitic studies – oligoclonal bands
Neurosyphilis	– VDRL (Venereal Disease Research Laboratory) test – FTA-ABS (Fluorescent treponemal antibody absorption) test – *Treponema pallidum* immobilisation test (TPI)
Cryptococcus HIV	– culture and antigen detection – culture, antigen detection and antiviral antibodies (anti-HIV-IgG).

Complications

– tonsillar herniation (see page 83)

– transient headache (5–30% depending on needle type), radicular pain (10%), or ocular palsy (1%)

– epidural haemorrhage very rare.

ELECTROMYOGRAPHY/NERVE CONDUCTION STUDIES

Needle electromyography records the electrical activity occurring within a particular muscle.

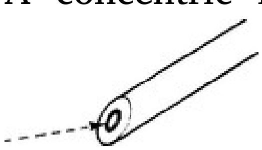
Nerve conduction studies measure conduction in nerves in response to an electrical stimulus.

Both are essential in the investigation of diseases of nerve (neuropathy) and muscle (myopathy).

Repetitive nerve stimulation tests are important in the evaluation of disorders of neuromuscular transmission, *e.g.* myasthenia gravis.

ELECTROMYOGRAPHY

A concentric needle electrode is inserted into muscle. The central



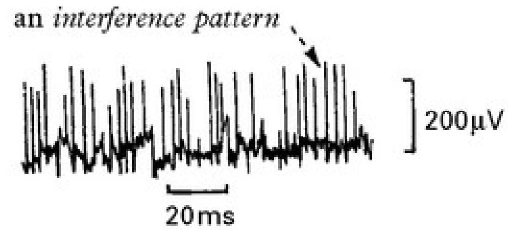
wire is the active electrode and the outer casing the reference electrode. This records from an area of 300μ radius.

The potential difference between the two electrodes is amplified and displayed on an oscilloscope. An audio monitor enables the investigator to 'hear' the pattern of electrical activity.

Normal muscle at rest is electrically 'silent' with a resting potential of 90 mV; as the muscle gradually contracts, *motor unit potentials* appear …

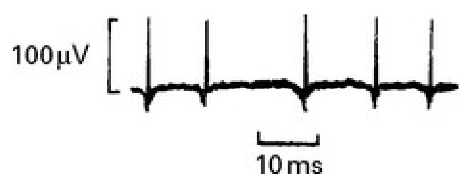
followed by the development of

Abnormalities take the form of:
Spontaneous activity in muscle when at rest.
Abnormalities of the motor unit potential.
Abnormalities of the interference pattern.
Special phenomena, e.g. myotonia.

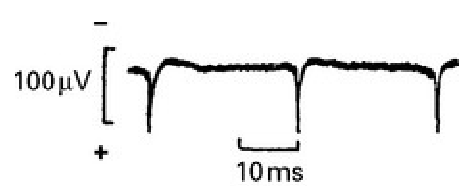


The recruitment of more and more motor units prevents identification of individual potentials

Spontaneous activity at rest



Fibrillation potentials are due to single muscle fibre contraction and indicate active denervation. They usually occur in neurogenic disorders, *e.g.* neuropathy.

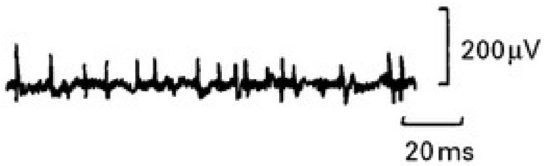


Positive sharp waves

Slow negative waves preceded by sharp positive spikes. Seen in chronically denervated muscle, *e.g.* motor neuron disease, but also in acute myopathy, *e.g.* polymyositis. These waves probably represent injury potentials.

Motor unit potential

In myopathies and muscular dystrophies, potentials are polyphasic and of small amplitude and short duration.



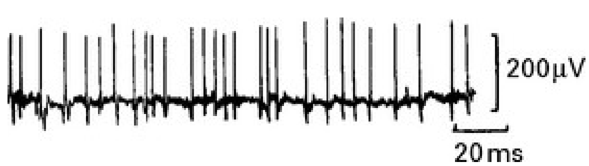
In neuropathy, the surviving motor unit potentials are also polyphasic but of large amplitude and long duration.

The enlarged potentials result from collateral reinnervation.

Interference pattern

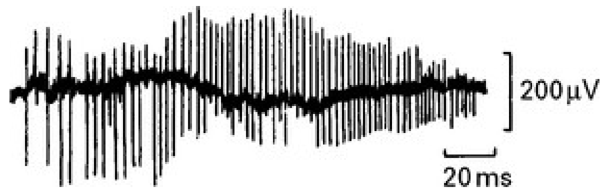
In myopathy, recruitment of motor units and the interference pattern remain normal. The interference pattern may even appear to increase due to fragmentation of motor units.

In neuropathy, there is a reduction in interference due to a loss of motor units under voluntary control.



Myotonia

High frequency repetitive discharge may occur after voluntary movement. The amplitude and frequency of the potentials wax and wane giving rise to the typical ‘dive bomber’ sound on the audio monitor.

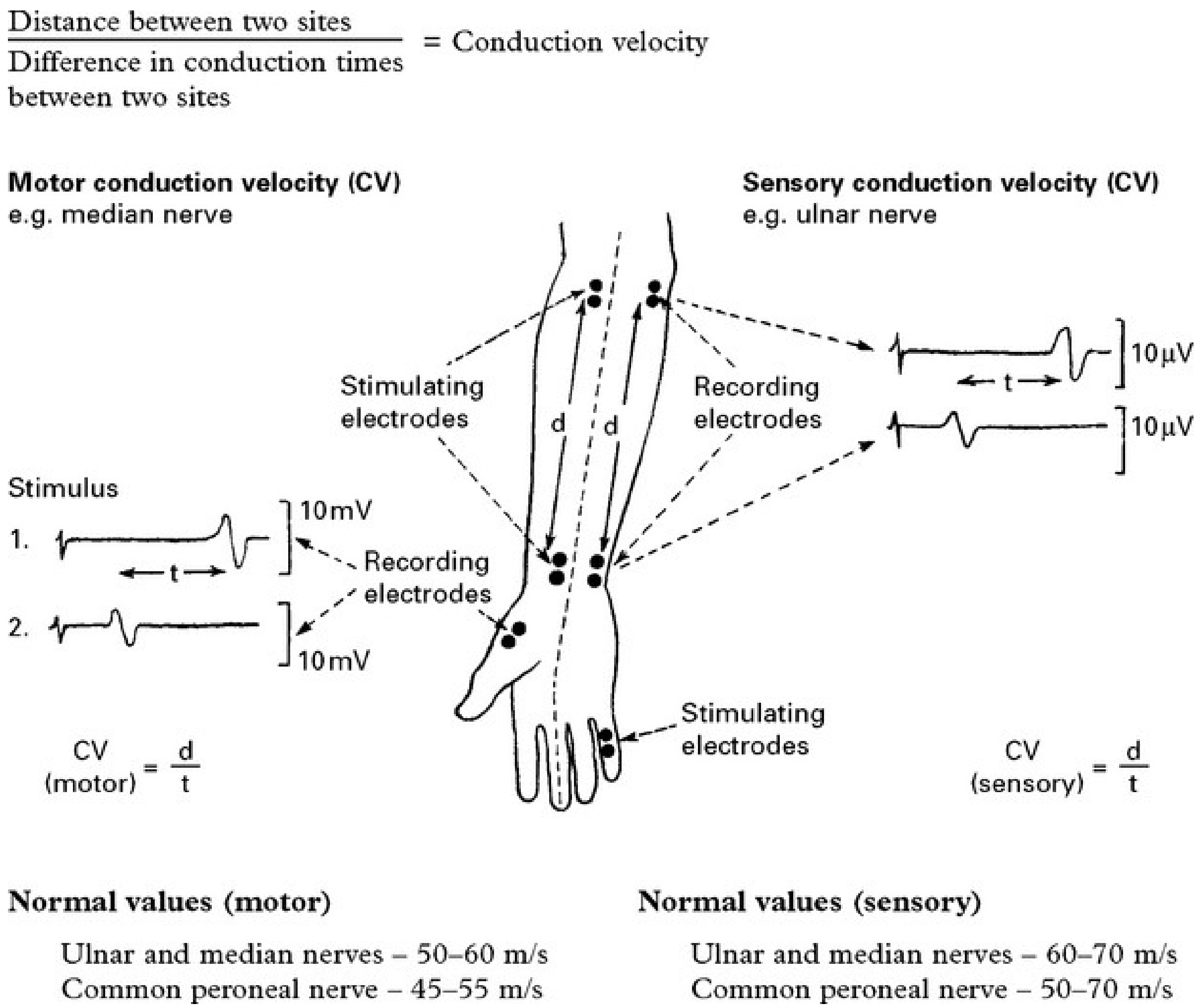


An abnormal myotonic discharge provoked by moving the needle electrode.

NERVE CONDUCTION STUDIES

Distal latency (latency from stimulus to recording electrodes), *amplitude* of the evoked response and *conduction velocity* all provide information on motor and sensory nerve function.

Conduction velocity: measurement made by stimulating or recording from two different sites along the course of a peripheral nerve.



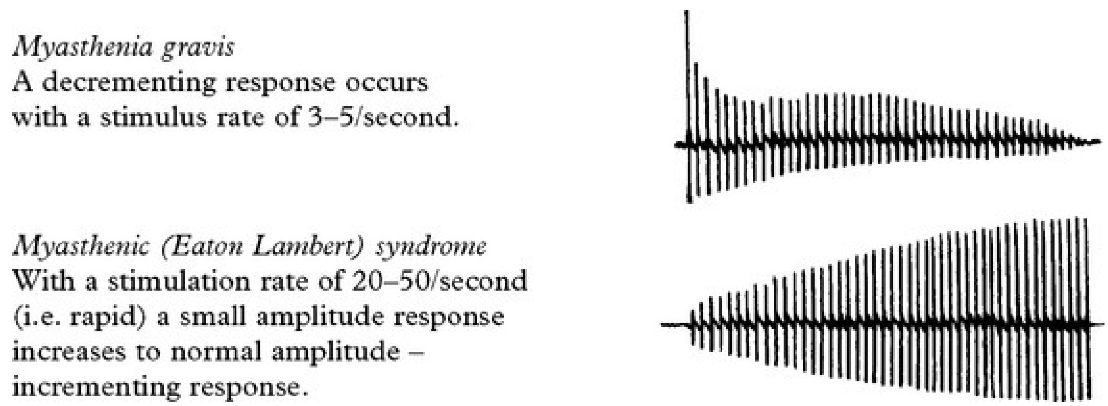
Motor conduction velocities slow with age.

Body temperature is important; a fall of 1°C slows conduction in motor nerves by approximately 2 metres per second.

Pathological delay occurs with nerve entrapments, demyelinating neuropathies (Guillain– Barré syndrome) and multifocal motor neuropathy.

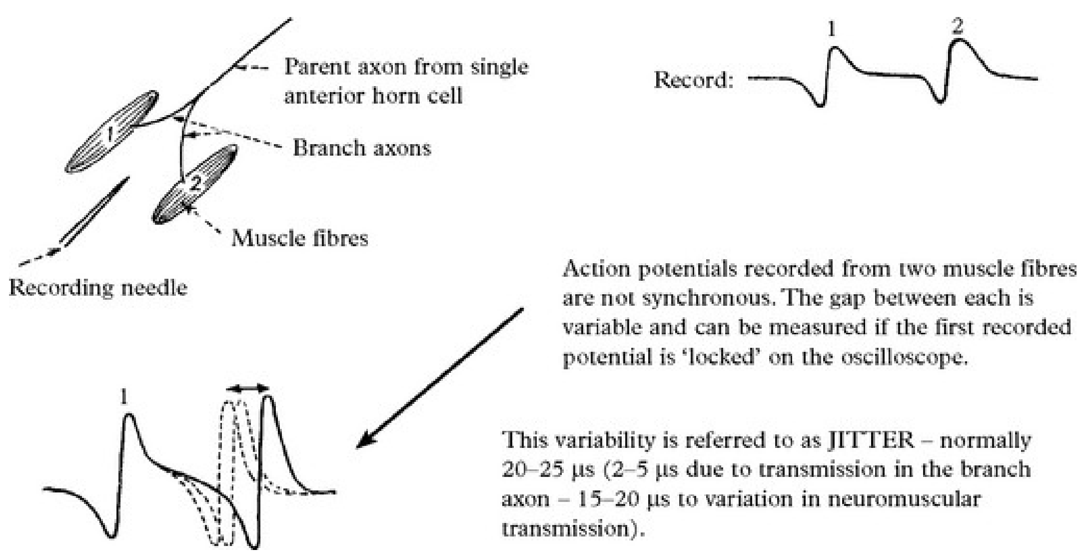
REPETITIVE STIMULATION

In the normal subject, repetitive stimulation of a motor nerve at a frequency of <30/second produces a muscle potential of constant form and amplitude. Increasing the stimulus frequency to >30/second results in fatigue manifest by a decline or 'decrement' in the amplitude. In patients with disorders of neuromuscular transmission, repetitive stimulation aids diagnosis:



SINGLE FIBRE ELECTROMYOGRAPHY

A standard concentric needle within muscle will record electrical activity 0.5–1 mm from its tip – sampling from up to 20 motor units. A 'single fibre' electromyography needle with a smaller recording surface detects electrical activity within 300 μ m of its tip – sampling 1–3 muscle fibres from a single motor unit.

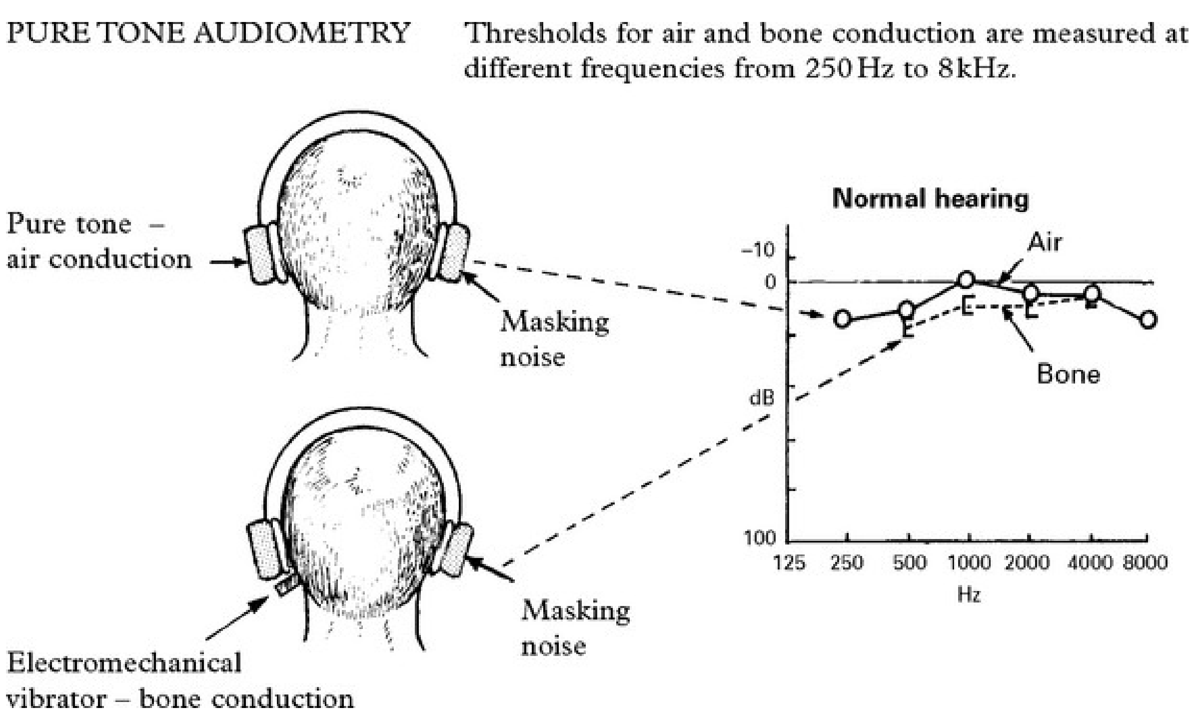


Single fibre electromyography is occasionally helpful in the investigation of disorders of neuromuscular transmission. In ocular myasthenia, the affected muscles are not accessible and frontalis is sampled instead.

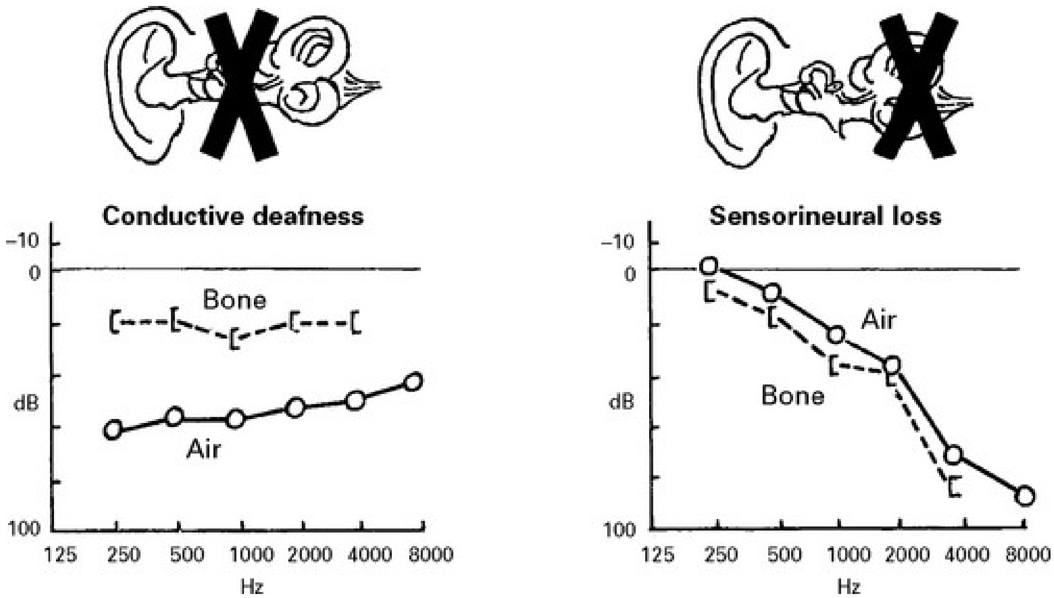
NEURO-OTOLOGICAL TESTS

AUDITORY SYSTEM

Neuro-otological tests help differentiate conductive, cochlear and retrocochlear causes of impaired hearing. They supplement Weber's and Rinne's test (page 16).

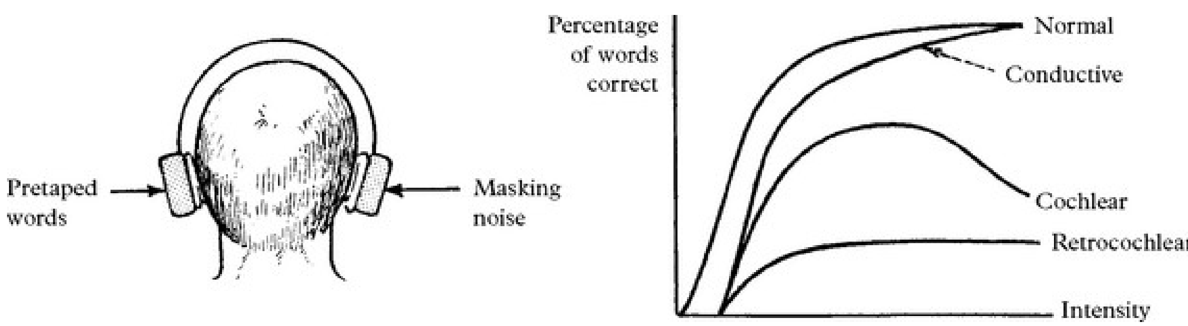


Sound conducted through air requires an intact ossicular system as well as a functioning cochlea and VIII nerve. Sound applied directly to the bone bypasses the ossicles.

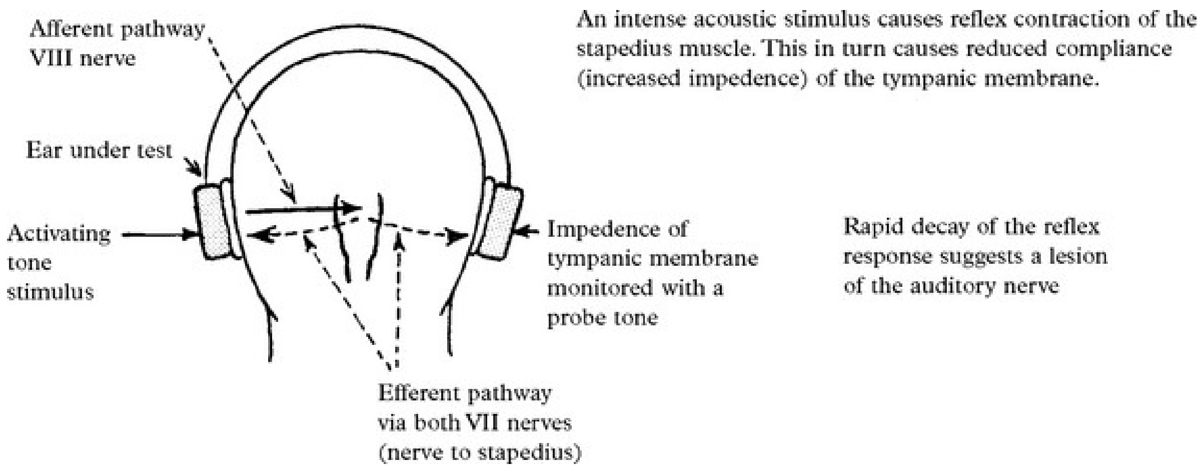


SPEECH AUDIOMETRY

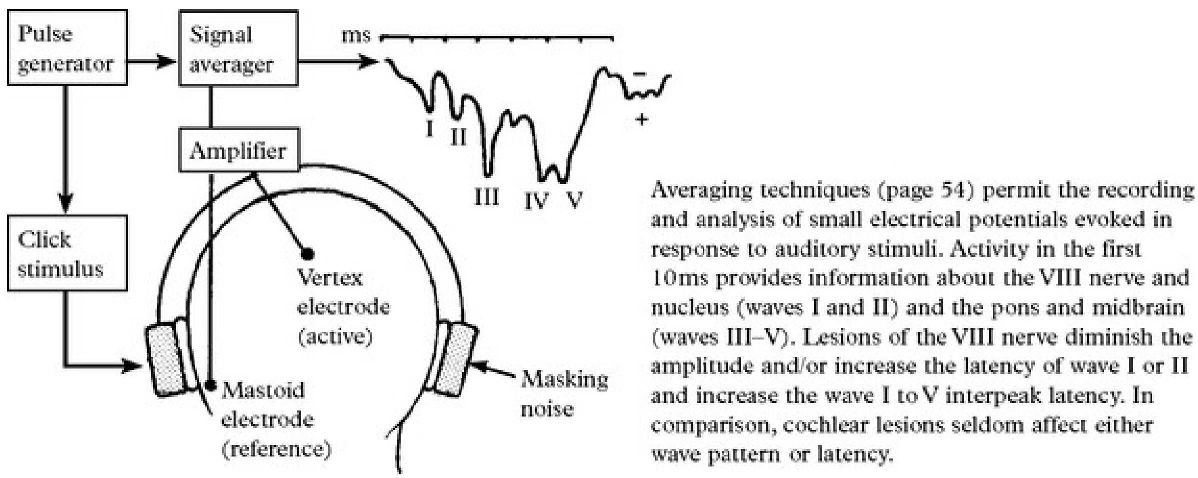
This test measures the percentage of words correctly interpreted as a function of the intensity of presentation and indicates the usefulness of hearing. The graph shows how different types of hearing loss can be differentiated.



STAPEDIAL REFLEX DECAY



AUDITORY BRAINSTEM EVOKED POTENTIAL



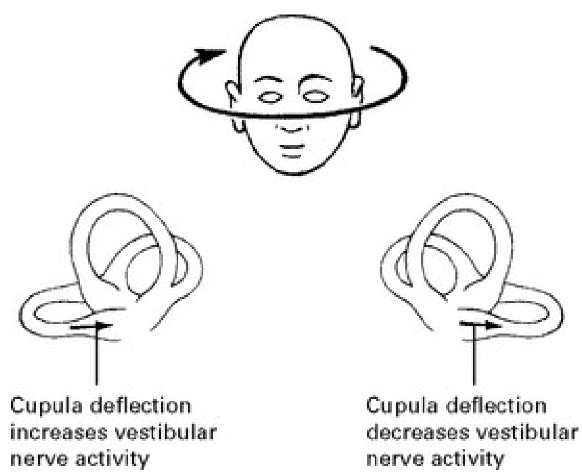
VESTIBULAR SYSTEM

Bedside vestibular function testing

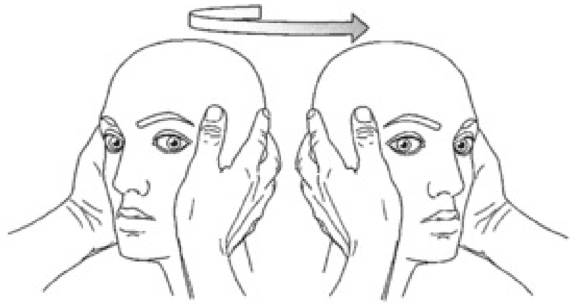
Hallpike's manoeuvre: see page 185.

Head thrust test

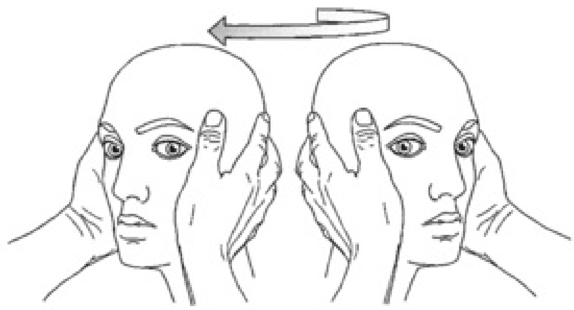
The semicircular canals detect rotational acceleration of the head. When the head is moved the endolymph stays in place relative to the skull and deflects the cupula within which the hair cells are imbedded. At rest the vestibular nerve from each semicircular canal has a background tonic firing rate. When the head is turned in one direction deflection of the hair cells increases the rate of firing from one canal and decreases the rate of firing from the paired contralateral canal (and vice versa). This activity acting through the III and VI nerves moves the eyes in a direction opposite to the rotation, tending to hold the eyes steady in space.



The *head thrust test* uses this to detect a peripheral unilateral vestibular lesion. The patient is asked to maintain gaze on the examiner's eyes. Slow rotation of the head (with minimal rotational acceleration) has no effect. With rapid head rotation in either direction, the gaze is maintained. In the presence of a unilateral vestibular lesion, if the head is turned rapidly *towards* the affected side, the firing rate does not increase in the vestibular nerve on this side and fails to maintain the position of gaze. The eyes move towards the affected side and this is followed by a catch up saccade. When the head is turned *away from* the affected side, increased activity in the normal ipsilateral vestibular nerve is sufficient to maintain the normal response.



Normal response in both eyes
– gaze is maintained



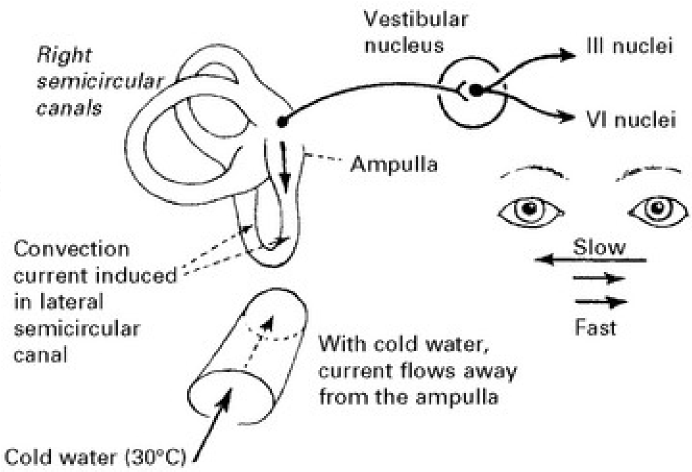
Rotation of the head to the affected side –
eyes drift to that side (right) followed by a catch up saccade

Caloric testing (vestibulo-ocular reflex)

Compensatory mechanisms may mask clinical evidence of vestibular damage – spontaneous and positional nystagmus. Caloric testing provides useful supplementary information and may reveal undetected vestibular dysfunction.

Method: Water at 30°C irrigated into the external auditory meatus. Nystagmus usually develops after a 20 second delay and lasts for more than a minute. The test is repeated after 5 minutes with water at 44°C.
Cold water effectively reduces the vestibular output from one side, creating an imbalance and producing eye drift towards the irrigated ear. Rapid corrective movements result in 'nystagmus' to the opposite ear. Hot water (44°) reverses the convection current, increases the vestibular output and changes the direction of nystagmus.

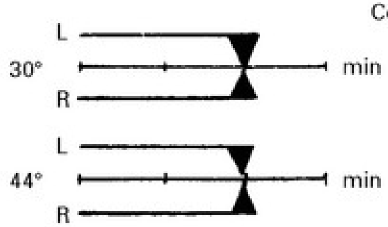
Stimulus is maximal with the head supported 30° from the horizontal (with the lateral semicircular canal in a vertical plane).



N.B. Ice water ensures a maximal stimulus when caloric testing for brain death or head injury prognostication.

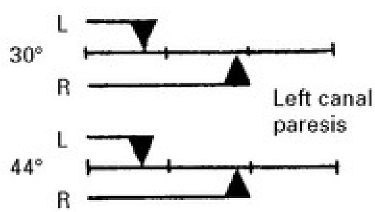
Time from onset of irrigation to the cessation of nystagmus is plotted for each ear, at each temperature.

Normal response

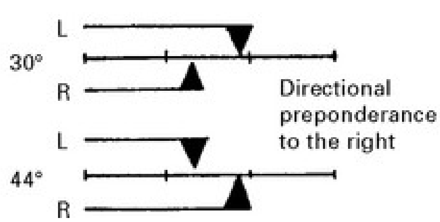


Damage to the labyrinth, vestibular nerve or nucleus results in one of two abnormal patterns, or a combination of both.

1. Canal paresis



2. Directional preponderance



Electronystagmography: The potential difference across the eye (the corneoretinal potential) permits recording of eye movements with laterally placed electrodes and enables detection of spontaneous or reflex induced nystagmus in darkness or with eyes closed.

This eliminates optical fixation which may reduce or even abolish nystagmus.

Canal paresis implies reduced duration of nystagmus on one side. It may result from either a peripheral or central (brain stem or cerebellum) lesion on that side.

Directional preponderance implies a more prolonged duration of nystagmus in one direction than the other. It may result from a central lesion on the side of the preponderance or from a peripheral lesion on the other side.

These tests combined with audiometry should differentiate a peripheral from a central lesion.

SECTION III

CLINICAL PRESENTATION, ANATOMICAL CONCEPTS AND DIAGNOSTIC APPROACH

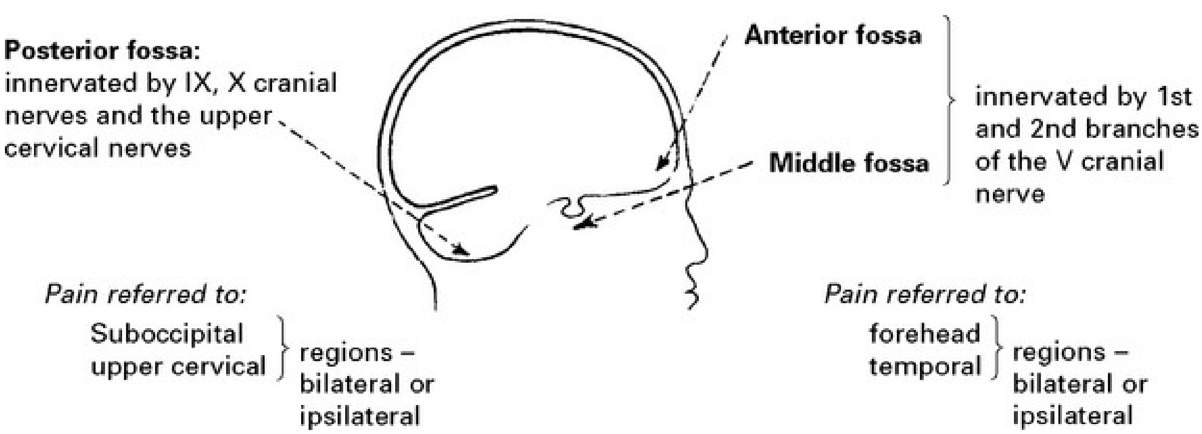
HEADACHE – GENERAL PRINCIPLES

Headache is a common symptom arising from psychological, otological, ophthalmological, neurological or systemic disease. In clinical practice tension-type headache is encountered most frequently.

Definition: Pain or discomfort between the orbits and occiput, arising from pain-sensitive structures.

Intracranial pain-sensitive structures are:

venous sinuses, cortical veins, basal arteries, dura of anterior, middle and posterior fossae.



Extracranial pain-sensitive structures are:

Scalp vessels and muscles, orbital contents, mucous membranes of nasal and paranasal spaces, external and middle ear, teeth and gums.

Estimated prevalence of headache in the general population

Type	Percentage
Tension type headache	50–70
Migraine	10–15
Medication overuse headache	4
Cluster headache	0.1
Raised intracranial pressure	< 0.01

Examination

Full general examination, including:

Ocular – acuity, tenderness, strabismus

Teeth and scalp

Percussion over frontal and maxillary sinuses
Full neurological examination.

HEADACHE – DIAGNOSTIC APPROACH

History: most information is derived from determining:

– the first attack or previous attacks

– whether onset is acute or gradual (days or weeks)

– whether attacks have recurred for many years (chronic)

– site of headache

– accompanying symptoms

– precipitating factors

The following table classifies causes in these categories:

(*) Indicates that attacks can be recurrent

Cause	Associated features which (if present) aid diagnosis	Further investigations (if required)
ACUTE		
*Sinusitis**	Preceding 'cold' nasal discharge	Imaging of nasal sinuses
*Migraine**	Visual/neurological aura, nausea, vomiting	
*Cluster headache**	Lacrimation, rhinorrhoea	
*Glaucoma**	'misting' of vision. 'haloes' around objects	Ophthalmological referral
Arterial dissection. carotid vertebral	Unilateral pain Horner's syndrome Symptoms of cerebral ischaemia	Vascular imaging: doppler, MR or CT angiogram
Retrobulbar neuritis	Loss of vision (unilateral)	Visual evoked response
Post-traumatic.	Following head injury	CT scan
Drugs/toxins	On vasodilator drugs	
Haemorrhage	Instantaneous onset vomiting, neck. stiffness, impaired conscious level	CT scan, lumbar puncture (see page 56)
Infection (meningitis, encephalitis)	As above but more gradual onset with pyrexia	CT scan, lumbar puncture (see page 56)
*Hydrocephalus**	Impaired conscious levels, leg weakness, impaired upward gaze	CT or MRI scan
SUBACUTE		
Infection (subacute, chronic meningitis, e.g. TB cerebral abscess)	Impaired conscious level, pyrexia, neck stiffness, focal neurological signs	CT or MRI scan lumbar puncture
*Intracranial tumour** *Chronic subdural haematoma** *Hydrocephalus**	Vomiting, papilloedema, impaired conscious level ± focal neurological signs	CT or MRI scan
*Idiopathic intracranial hypertension**	Papilloedema, visual obscurations. 6th nerve palsy	MRI and MR venogram Lumbar puncture
Temporal arteritis.	Thickened, tender, scalp arteries	ESR, temporal artery biopsy
Intracranial hypotension	Worse on standing.	MRI with gadolinium
CHRONIC		
*Tension-type headache**	Anxiety, depression	
*Transformed migraine**	Previous history of episodic migraine	
*Medication overuse headache**	Regular analgesic >15 days a month	
*Ocular 'eye strain'**	Impaired visual acuity	Refractive errors
*Drugs/toxins**	On vasodilator drugs	
*Cervical spondylosis**	Neck, shoulder, arm pain	X-ray cervical spine

Headache in children

Most causes of adult headache may occur in children. In this age group, the commonest type of headache is that accompanying any *febrile illness* or *infection of the nasal passages or sinuses*.

The clinician must not take a complaint of headache lightly; the younger the child, the more likely the presence of an underlying organic disease. Pyrexia may not only represent a mild 'constitutional' upset, but may result from *meningitis, encephalitis* or *cerebral abscess*. The presence of neck stiffness and/or impaired conscious level indicates the need for urgent investigation.

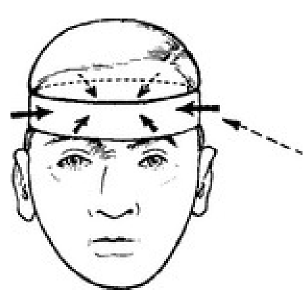
Although *intracranial tumours* are uncommon in childhood, when they occur they tend to lie in the *midline* (e.g. medulloblastoma, pineal region tumours). As a result, obstructive hydrocephalus often develops acutely with headache as a prominent initial symptom.

In a child with 'unexplained' headache, *CT or MRI scan* should be performed if the headache is acute or progressive or if there are other features (increase in head circumference, change in personality or decline in school performance) or in children under 5.

HEADACHE – SPECIFIC CAUSES

TENSION TYPE HEADACHE

This is the commonest form of headache experienced by 70% of males and 90% of females at some time in their lives.



Characteristics: Diffuse, dull, aching, 'band-like' headache, worse on touching the scalp and aggravated by noise; associated with 'tension' but not with other physical symptoms. Attacks may be chronic or episodic. Depression commonly co-exists.

Duration: Many hours–days.

Frequency: Infrequent or daily; worse towards the end of the day. May persist over many years.

Mechanism: 'Muscular' due to persistent contraction, *e.g.* clenching teeth, head posture, furrowing of brow. Some overlap with transformed migraine (see below).

Treatment: Reassurance. Attempt to reduce psychological stress and analgesic overuse (see medication-overuse headache). Amitriptyline and other tricyclic antidepressants or β-blockers.

MIGRAINE

Migraine is a common, often familial disorder characterised by *unilateral throbbing headache.*

Onset: Childhood or early adult life.

Incidence: Affects 5–10% of the population.

Female:male ratio: 2:1

Family history: Obtained in 70% of all sufferers.

Two recognisable forms exist:

Specific diagnostic criteria are required for migraine with and without aura.

MIGRAINE WITH AURA
An *aura* or warning of visual, sensory or motor type followed by headache – throbbing, unilateral, worsened by bright light, relieved by sleep, associated with nausea and, occasionally, vomiting.

MIGRAINE WITHOUT AURA (COMMON MIGRAINE)
The *aura* is absent. The headache has similar features, but it is often poorly localised and its description may merge with that of 'tension' headache.

The *aura* is absent. The headache has similar features, but it is often poorly localised and its description may merge with that of 'tension' headache.

The aura of migraine may take many forms. The visual forms comprise: flashing lights, zig-zags (fortifications), scintillating scotoma (central vision) and may precede visual field defects. Such auras are of visual (occipital) cortex origin.

The headache is recurrent, lasting from 2 to 48 hours and rarely occurring more frequently than twice weekly. In *migraine equivalents* the aura occurs without ensuing headache.

Specific types of migraine with aura

Basilar: Characterised by bilateral visual symptoms, unsteadiness, dysarthria, vertigo, limb paraesthesia, even tetraparesis. Loss of consciousness may ensue and precede the onset of headache. This form of migraine affects young women.

Hemiplegic: Characterised by an aura of unilateral paralysis (hemiplegia) which unusually persist for some days after the headache has settled. Often misdiagnosed as a 'stroke'. When familial, mendelian dominant inheritance is noted. Recovery is the rule.

Retinal

Unilateral (monocular) visual loss which is reversible and followed by headache. Ophthalmological examination between episodes is normal.

Precipitating factors in migraine

– Dietary: alcohol, chocolate and cheese (contain tyramine).

– Hormonal: often premenstrual or related to oral contraceptive (fluctuations in oestrogen).

– Stress, physical fatigue, exercise, sleep deprivation and minor head trauma.

Diagnosis

Clinical history with	– occasional positive family history – travel sickness or migraine variants (abdominal pains) in childhood – onset in childhood, adolescence, early adult life or menopause
Distinguish	– partial (focal) epilepsy (in hemiplegic or hemisensory migraine) – transient ischaemic attack (in hemiplegic or hemisensory migraine) – arteriovenous malformation – gives well localised but chronic headache) – hypoglycaemia

Management

(i) Identification and avoidance of precipitating factors

(ii) Treatment of acute attacks:

Simple *analgesics* (e.g. aspirin) with *metoclopramide* to enhance reduced absorption during an attack. If vomiting is prominent anti-emetic (domperidone or prochlorperazine) and analgesic can be helpful.

Sumatriptan (a selective $5HT_1$ agonist) and other triptans *e.g.* Naratriptan, Rizatriptan and Zolmitriptan – effectively reverse dilatation in extracranial vessels. Given orally or subcutaneously.

Ergotamine – widespread action on 5HT receptors reversing dilatation. Give orally or by inhalation, injection or by suppository.

Methylprednisolone i.m. or i.v. will halt the attack when prolonged (status migrainosus).

(iii) Prophylaxis: use only for frequent and severe attacks

Pizotifen ($5HT_2$ receptor blocker)

Propranolol (beta adrenergic receptor blocker)

Calcium antagonists (verapamil), antidepressants(amitriptyline) and *anticonvulsants*, (topiramate or sodium valproate).

Medication Overuse Headaches

Some patients with episodic tension headache or migraine find their headache pattern changes so that they have headaches most days. Many such patients take regular analgesics and/or triptans and this overuse (>14 days a month) can cause medication overuse headaches (MOH). These do not respond to prophylactic agents and will improve on stopping the regular analgesics; this can take some weeks and headaches can be worse in the short-term.

Transformed migraine

If patients with migraine go on to develop chronic daily headache without overusing medication this is ‘transformed migraine’. It usually responds to migraine prophylactic agents.

POST-TRAUMATIC HEADACHE

A ‘common migraine’ or ‘tension-like’ headache may arise after head injury and accompany other symptoms including lightheadedness, irritability, difficulty in concentration and in coping with work. This will often respond to amitriptyline or migraine prophylaxis.

CLUSTER HEADACHES (Histamine cephalgia or migrainous neuralgia)

‘Cluster headaches occur less frequently than migraine, and more often in men, with onset in middle age. Charecterised by episodes of severe unilateral pain, lasting 10 minutes to 2 hours, around one eye, associated with conjunctival injection, lacrimation, rhinorrhea and occasionally a transient Horner’s syndrome. The episodes occur between once and many times per day, often wakening from sleep at night. ‘Clusters’ of attacks separated by weeks or even many months. Alcohol may precipitate the attacks.’

Other Trigeminal Autonomic Cephalagias

Cluster headache is the most common form of trigeminal autonomic cephalalgia, where there is a combination of facial pain and autonomic dysfunction. Other rarer combinations of facial pain and antonomic symptoms include:

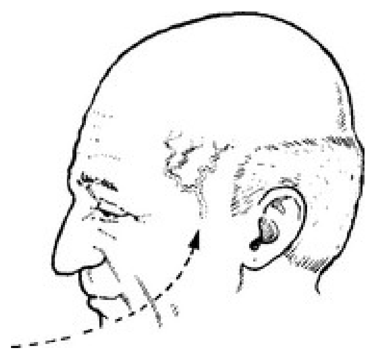
Hemicrania continua: continuous unilateral moderately severe head pain with exacerbations and variable tearing and partial Horner’s syndrome. More common in women than men (3:1). Responds dramatically to indometacin.

Paroxysmal hemicrania: Same pain but lasts 2-45 minutes multiple times a day. Responds to indometacin.

Short-lasting Unilateral Neuralgiform pain with Conjunctival injection and Tearing (SUNCT): brief pain lasting seconds to 3 minutes with associations described in its name. Women:men, 2:1. Does not respond to indometacin. Lamotrigine has some effect.

GIANT CELL (TEMPORAL) ARTERITIS

Giant cell arteritis, an autoimmune disease of unknown cause, presents with throbbing headache in patients over 60 often with general malaise. The involved vessel, usually the superficial temporal artery, may be tender, thickened, and but nonpulsatile.



Neurological symptoms: strokes, hearing loss, myelopathy and neuropathy.

Jaw claudication: pain when chewing or talking due to ischaemia of the masseter muscles is pathognomonic.

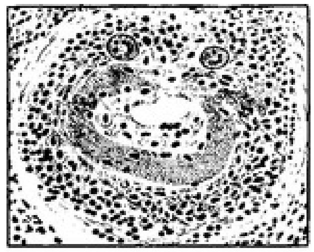
Visual symptoms are common with blindness (transient or permanent) or diplopia.

Associated systemic symptoms – weight loss, lassitude and generalised muscle aches – polymyalgia rheumatica in one-fifth of cases.

Duration: the headache is intractable, lasting until treated.

Mechanism:

Large and medium-sized arteries undergo intense ‘giant cell’ infiltration, with fragmentation of the lamina and narrowing of the lumen, resulting in distal ischaemia as well as stimulating pain sensitive fibres. Occlusion of important end arteries, *e.g.* the ophthalmic artery, may result in blindness; occlusion of the basilar artery may cause brain stem or bilateral occipital infarction.



Thickened wall with giant cell infiltrate

Diagnosis: ESR usually high. Blood film shows anaemia or thrombocytosis. C-reactive protein and hepatic alkaline phosphatase elevated. Biopsy of 1 cm length of temporal artery is often diagnostic.

Treatment: Urgent treatment, prednisolone 60 mg daily, prevents visual loss or brain-stem stroke, as well as relieving the headache. If complications have already occurred *e.g.* blindness, give parenteral high dose steroids. Monitoring the ESR allows gradual reduction in steroid dosage over several weeks to a maintenance level, *e.g.* 5 mg daily. Most patients eventually come off steroids; 25% require long-term treatment and if so, complications commonly occur.

HEADACHE FROM RAISED INTRACRANIAL PRESSURE

Characteristics:	**Associated features:**
– generalised. – aggravated by bending or coughing. – worse in the morning on awakening; may awaken patient from sleep.	– vomiting in later stages. – transient loss of vision (obscuration) with sudden change in posture. – eventual impairment of conscious level

– the severity of the headache gradually progresses.	**Management:** further investigations are essential – CT or MRI

Low pressure headache

Low pressure headache occurs most commonly after lumbar puncture but can arise spontaneously (Spontaneous intracranial hypotension). Headache is worse on standing and improves lying flat. After LP no investigation is needed. If no cause is apparent MRI will show downward displacement cerebellar tonsils and meningeal enhancement with contrast (Gd). Spontaneous improvement is usual. Occasionally a dural 'blood patch' at the site of CSF leak (post LP or epidural anaesthesia) is necessary.

HEADACHE DUE TO INTRACRANIAL HAEMORRHAGE

Characteristics:

– instantaneous onset.

– severe pain, spreading over the vertex to the occiput, or described as a 'sudden blow to the back of the head'.

– patient may drop to knees or lose consciousness.

Associated features:

– usually accompanied by vomiting.

– focal neurological signs suggest a haematoma.

Management: further investigation – CT scan/lumbar puncture (see Meningism, page 75).

N.B. Consider sudden severe headaches to be due to subarachnoid haemorrhage until proved otherwise.

NON-NEUROLOGICAL CAUSES OF HEADACHE

Local causes:

Sinuses: Well localised. Worse in morning. Affected by posture, *e.g.* bending.

X-ray – sinus opacified. Treatment – decongestants or drainage.

Ocular: Refraction errors may result in 'muscle contraction' headaches

– resolves when corrected with glasses.

Acute glaucoma can produce headache but is accompanied by other symptoms, *e.g.* misting of vision, 'haloes'.

Dental disease: Discomfort localised to teeth. Check for malocclusion.

Check temporomandibular joints.

Systemic causes:

Headache may accompany any febrile illness or may be the presenting feature of accelerated hypertension or metabolic disease, *e.g.* hypoglycaemia, hypercalcaemia.

Many drugs produce headache

– through vasodilatation, *e.g.* bronchodilators, antihistamines

– on withdrawal, *e.g.* amphetamines, benzodiazepines, caffeine.

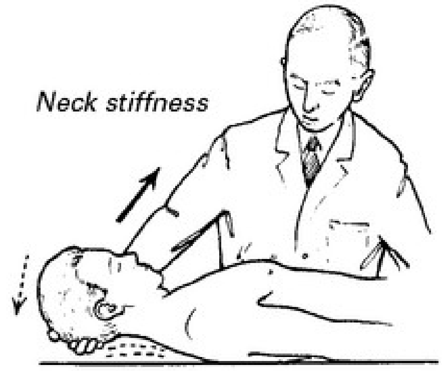
MENINGISM

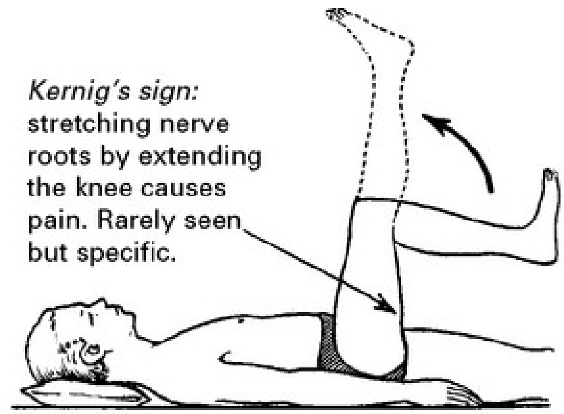
Evidence of meningeal irritation caused by infection or subarachnoid haemorrhage results in characteristic clinical features (though not all will necessarily be present):

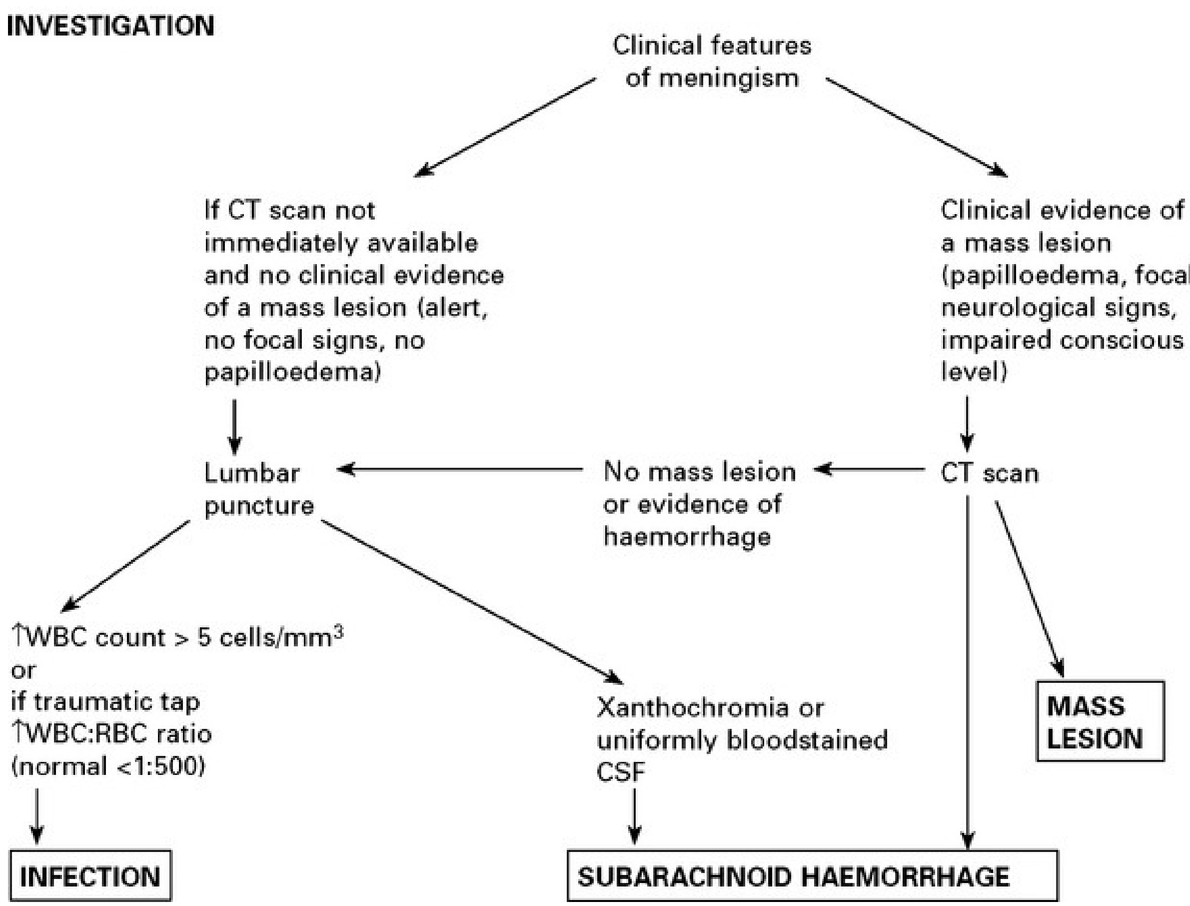
SYMPTOMS

1. Headache
2. Vomiting
3. Photophobia

SIGNS

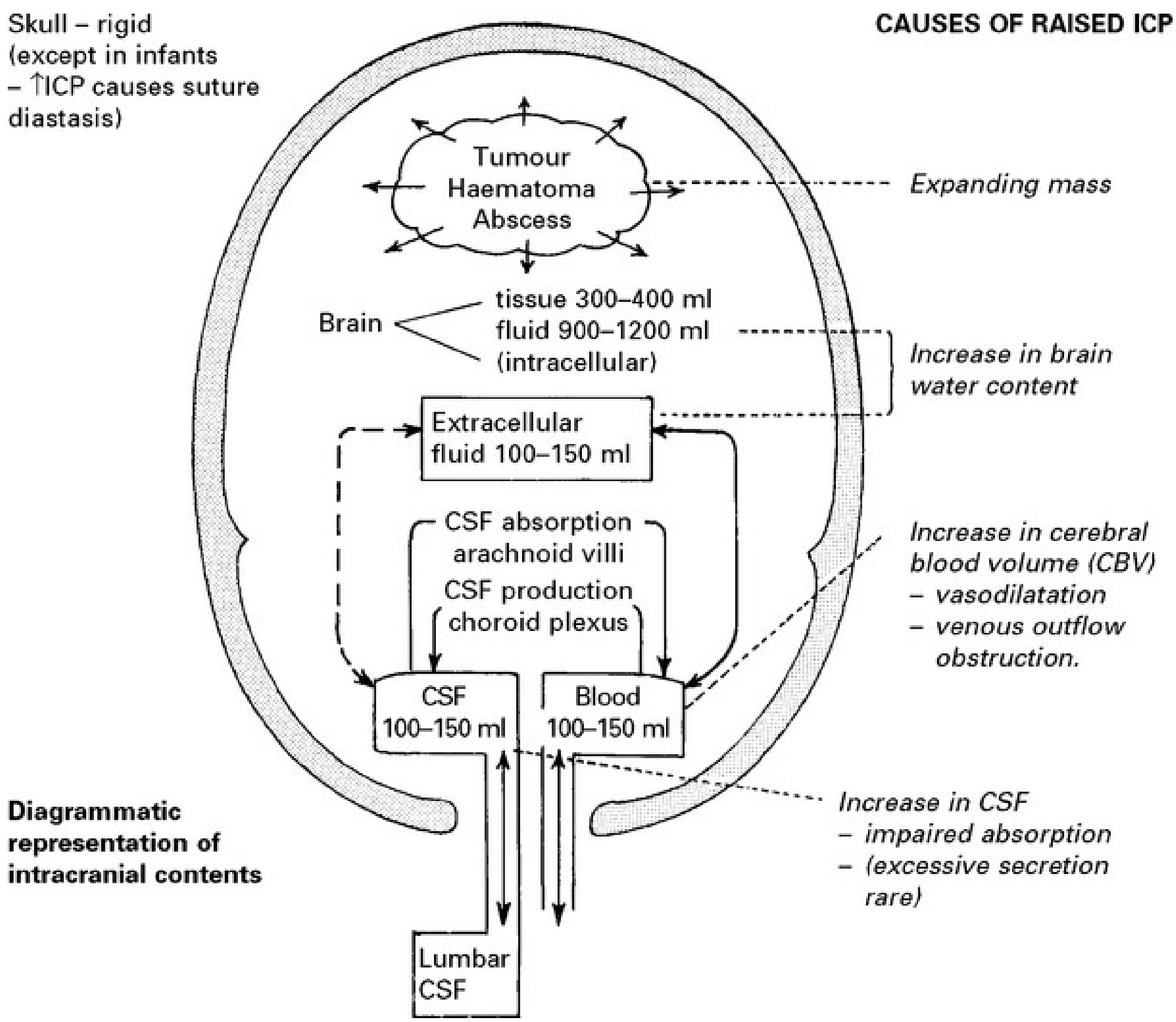






RAISED INTRACRANIAL PRESSURE

The skull is basically a rigid structure. Since its contents – brain, blood and cerebrospinal fluid (CSF) – are incompressible, an increase in one constituent or an expanding mass within the skull results in an increase in intracranial pressure (ICP) – the ‘Monro-Kellie doctrine’.



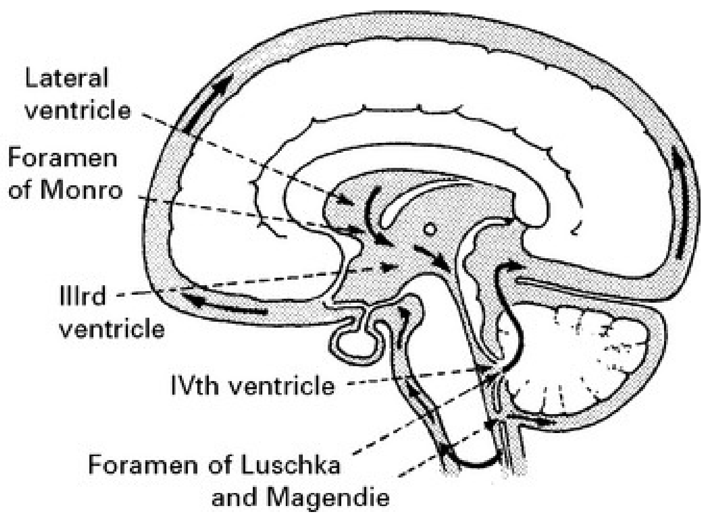
Compensatory mechanisms for an expanding intracranial mass lesion:

– Immediate { 1. ↓ CSF volume – CSF outflow to the lumbar theca
2. ↓ Cerebral blood volume
– Delayed – 3. ↓ Extracellular fluid

CEREBROSPINAL FLUID (CSF)

Secreted at a rate of 500 ml per day from the choroid plexus, CSF flows through the ventricular system and enters the subarachnoid space via the 4th ventricular foramina of Magendie and Luschka.

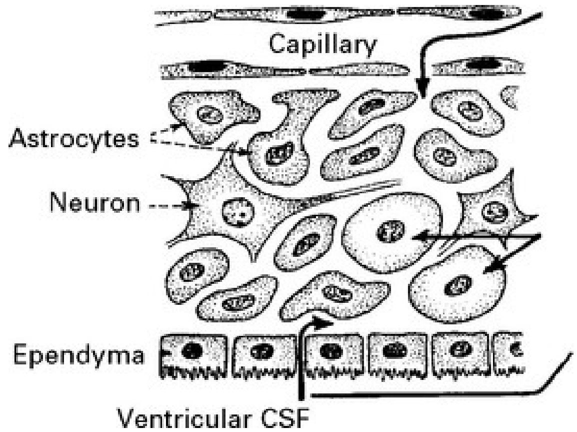
Under normal conditions, CSF flows freely through the subarachnoid space and is absorbed into the venous system through the arachnoid villi. If flow is obstructed at any point in the pathway, *hydrocephalus* with an associated rise in intracranial pressure develops, as a result of continued CSF production. With an expanding intracranial mass lesion, normal pressure is initially maintained by CSF expulsion to the expandable lumbar theca. Further expansion and subsequent brain shift may obstruct the free flow of CSF not only to the lumbar theca but also to the arachnoid villi, causing an acute rise in intracranial pressure.



BRAIN WATER/OEDEMA

Cerebral oedema – an excess of brain water – may develop around an intrinsic lesion within the brain tissue, *e.g.* tumour or abscess or in relation to traumatic or ischaemic brain damage, and contribute to the space-occupying effect.

Different forms of cerebral oedema exist:



Vasogenic: excess fluid (protein rich) passes through damaged vessel walls to the extracellular space – especially in the white matter. The extracellular fluid gradually infiltrates throughout normal brain tissue towards the ventricular CSF and this drainage route may aid clearance. *e.g.* adjacent to tumour.

Cytotoxic: fluid accumulates within cells – neurons and glia *i.e.* intracellular, *e.g.* toxic or metabolic states.

Interstitial: when obstructive hydrocephalus develops, CSF is forced through to the extracellular space especially in the periventricular white matter.

With *ischaemic* damage, as cell metabolism fails, intracellular Na^+ and Ca^{2+} increase and the cells swell *i.e.* cytotoxic oedema. Capillary damage follows and vasogenic oedema supervenes.

CEREBRAL BLOOD FLOW (CBF)/CEREBRAL BLOOD VOLUME (CBV)

Blood flow is dependent on blood pressure and the vascular resistance:

$$\text{Flow} = \frac{\text{Pressure}}{\text{Resistance}}$$

Inside the skull, intracranial pressure must be taken into account:

$$\text{Cerebral blood flow (CBF)} = \frac{\text{Cerebral perfusion pressure (CPP)} \atop \text{(i.e. systemic BP – intracranial pressure)}}{\text{Cerebral vascular resistance (CVR)}}$$

Under normal conditions the cerebral blood flow is coupled to the energy requirements of brain tissue. Various regulatory mechanisms acting on the arterioles maintain a cerebral blood flow sufficient to meet the metabolic demands.

FACTORS AFFECTING THE CEREBRAL VASCULATURE

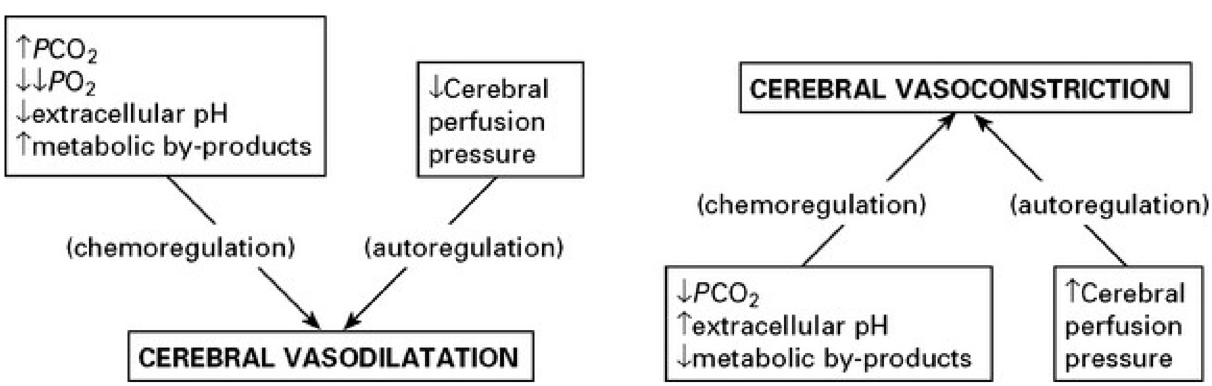
Chemoregulation

– Change in extracellular pH or an accumulation of metabolic by-products directly affects the vessel calibre.

– Any change in arteriolar PCO_2 has a direct effect on cerebral vessels, but only a reduction of PO_2 to < 50 mmHg has a significant effect.

Autoregulation

– A change in the cerebral perfusion pressure results in a compensatory change in vessel calibre.

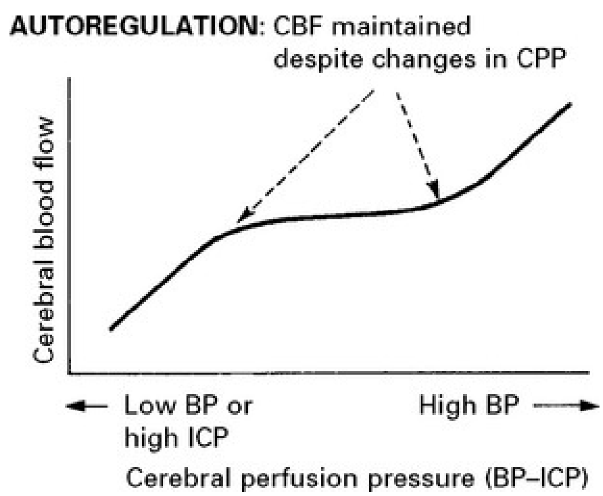


Any change in blood vessel diameter results in considerable variation in cerebral blood volume and this, in turn, directly affects intracranial pressure.

Energy requirements differ in different parts of the brain. To meet such needs in the white matter, flow is 20 ml/100 g/min, whereas in the grey matter flow is as high as 100ml/100g/min.

Autoregulation is a compensatory mechanism which permits fluctuation in the cerebral perfusion pressure within certain limits without significantly altering cerebral blood flow.

A drop in cerebral perfusion pressure produces vasodilation (probably due to a direct 'myogenic' effect on the vascular smooth muscle) thereby maintaining flow; a rise in the cerebral perfusion pressure causes vasoconstriction.



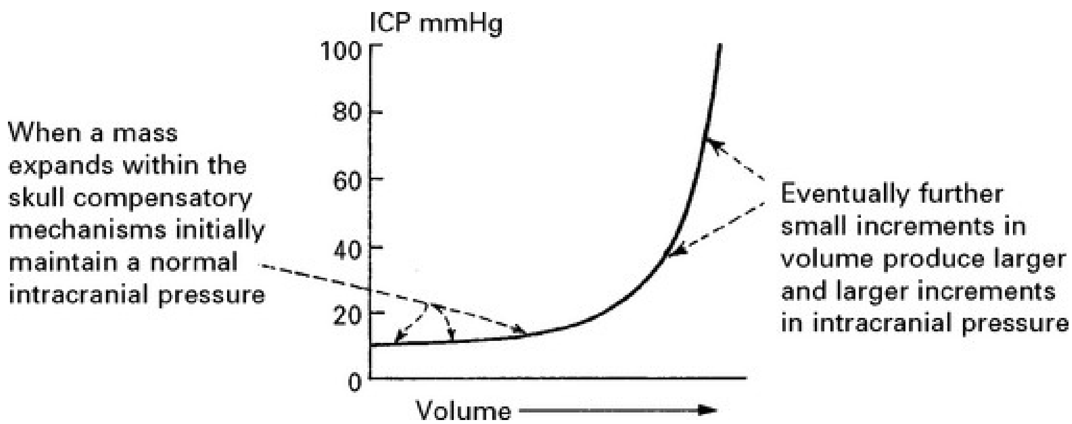
Neurogenic influences appear to have little direct effect on the cerebral vessels but they may alter the range of pressure changes over which autoregulation acts.

Autoregulation fails when the cerebral perfusion pressure falls below 60 mmHg or rises above 160 mmHg. At these extremes, cerebral blood flow is more directly related to the perfusion pressure.

In damaged brain (e.g. after head injury or subarachnoid haemorrhage), autoregulation is impaired; a drop in cerebral perfusion pressure is more likely to reduce cerebral blood flow and cause ischaemia. Conversely, a high cerebral perfusion may increase the cerebral blood flow, break down the blood–brain barrier and produce cerebral oedema as in hypertensive encephalopathy.

INTRACRANIAL PRESSURE (ICP)

Intracranial pressure, measured relative to the foramen of Monro, under normal conditions ranges from 0–135 mm CSF (0–10 mmHg) although very high pressure, *e.g.* 1000 mm CSF may occur transiently during coughing or straining.

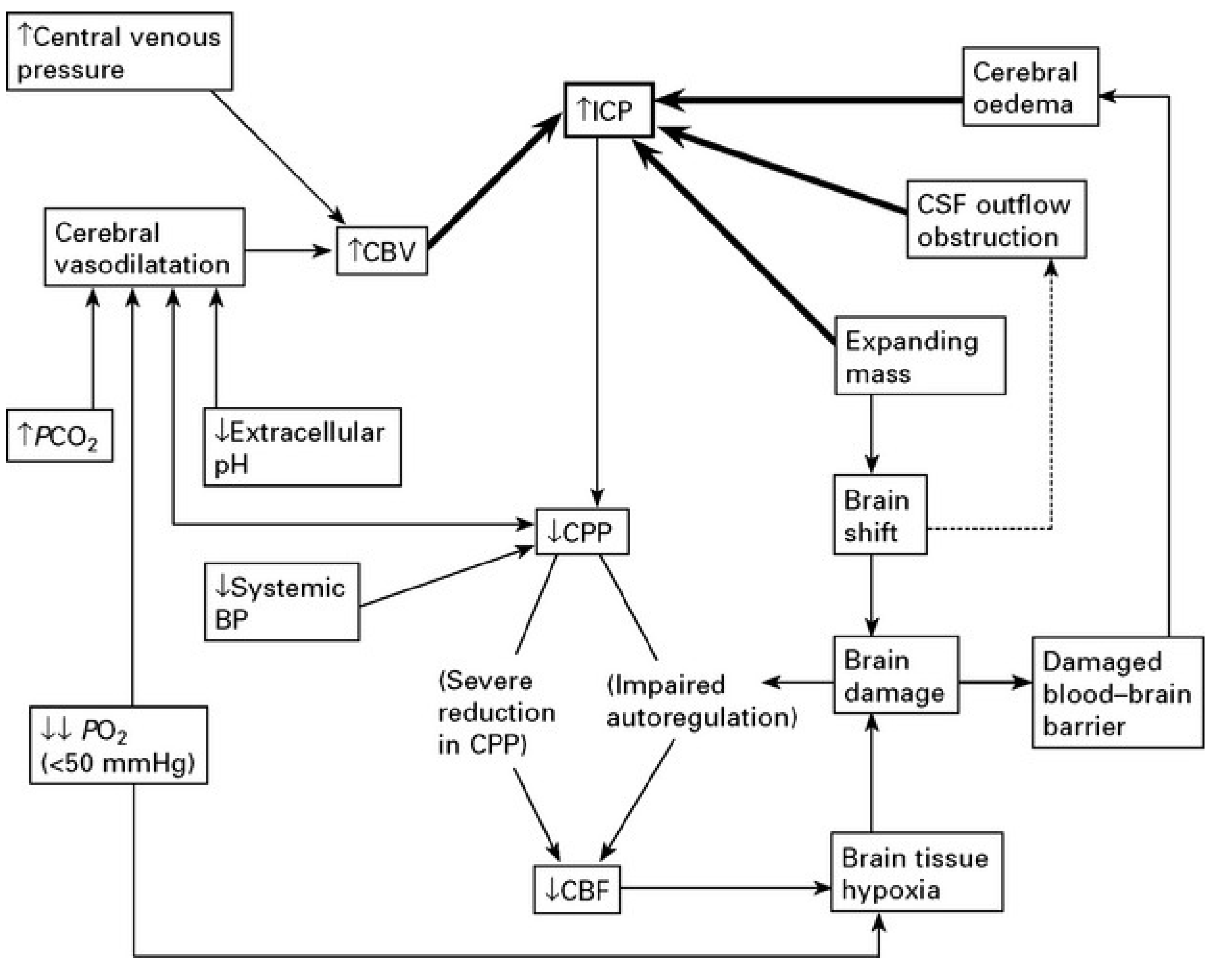


When intracranial pressure is monitored with a ventricular catheter, regular waves due to pulse and respiratory effects are recorded (page 53). As an intracranial mass expands and as the compensatory reserves diminish, transient pressure elevations (pressure waves) are superimposed. These become more frequent and more prominent as the mean pressure rises.

Eventually the rise in intracranial pressure and resultant fall in cerebral perfusion pressure reach a critical level and a significant reduction in cerebral blood flow occurs. Electrical activity in the cortex fails at flow rates about 20 ml/100 g/min. If autoregulation is already impaired these effects develop even earlier. When intracranial pressure reaches the mean arterial blood pressure, cerebral blood flow ceases.

INTERRELATIONSHIPS

Many factors affect intracranial pressure and these should not be considered in isolation. Inter-relationships are complex and feedback pathways may merely serve to compound the brain damage.



CLINICAL EFFECTS OF RAISED INTRACRANIAL PRESSURE

A raised ICP will produce symptoms and signs but does not cause neuronal damage provided cerebral blood flow is maintained. Damage does, however, result from brain shift – tentorial or tonsillar herniation.

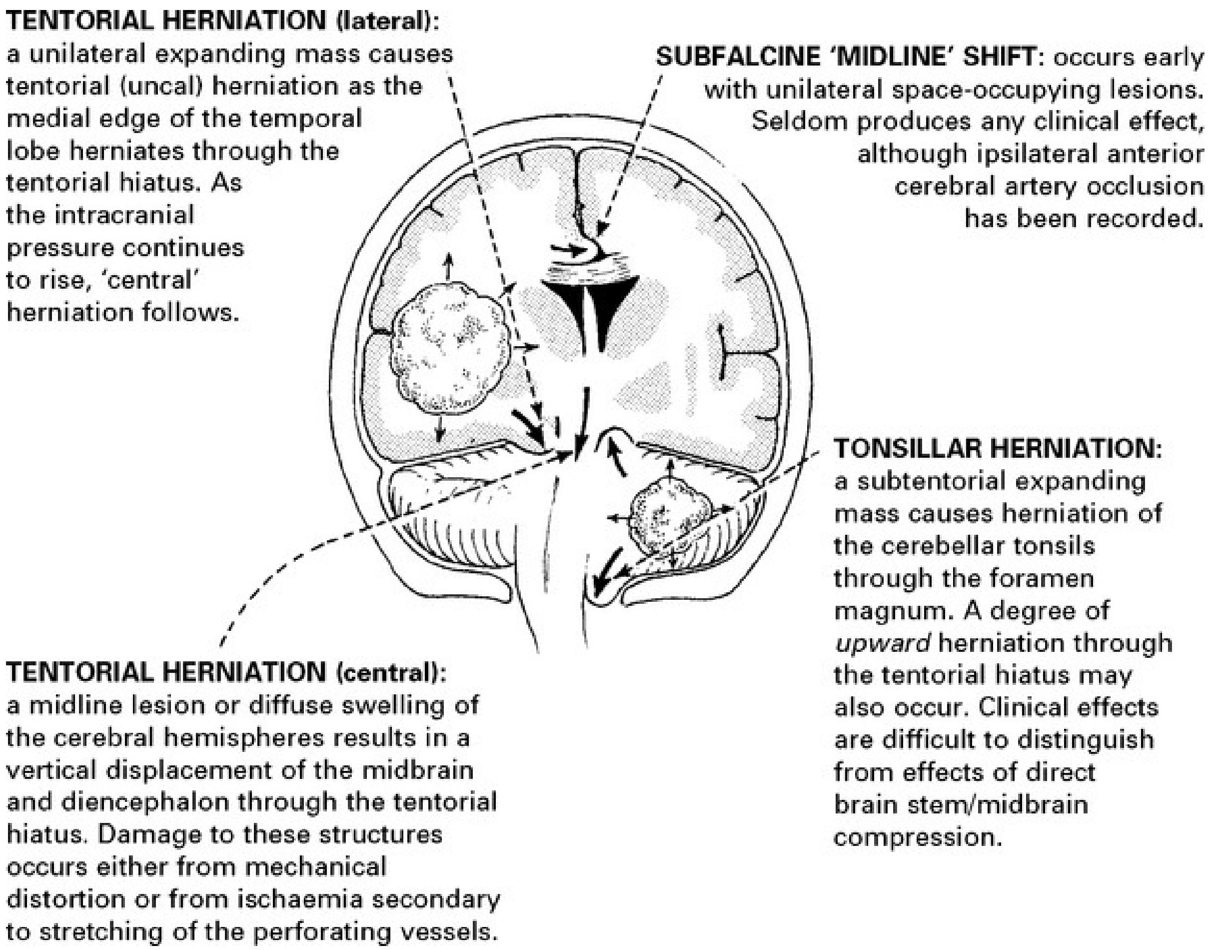
Clinical features due to ↑ICP:

1. *Headache* – worse in the mornings, aggravated by stooping and bending.

2. *Vomiting* – occurs with an acute rise in ICP.

3. *Papilloedema* – occurs in a proportion of patients with ICP. It is related to CSF obstruction and does not necessarily occur with brain shift alone. Increased CSF pressure in the optic nerve sheath impedes venous drainage and axoplasmic flow in optic neurons. Swelling of the optic disc and retinal and disc haemorrhages result. Vision is only at risk when papilloedema is both severe and prolonged.

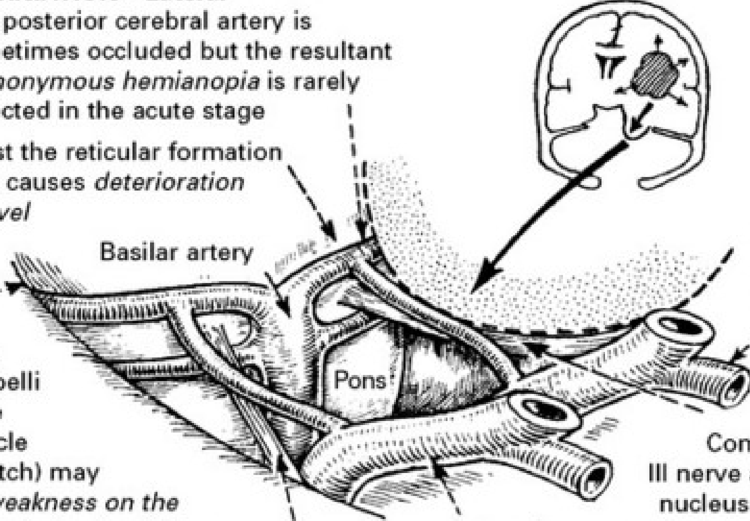
BRAIN SHIFT – TYPES



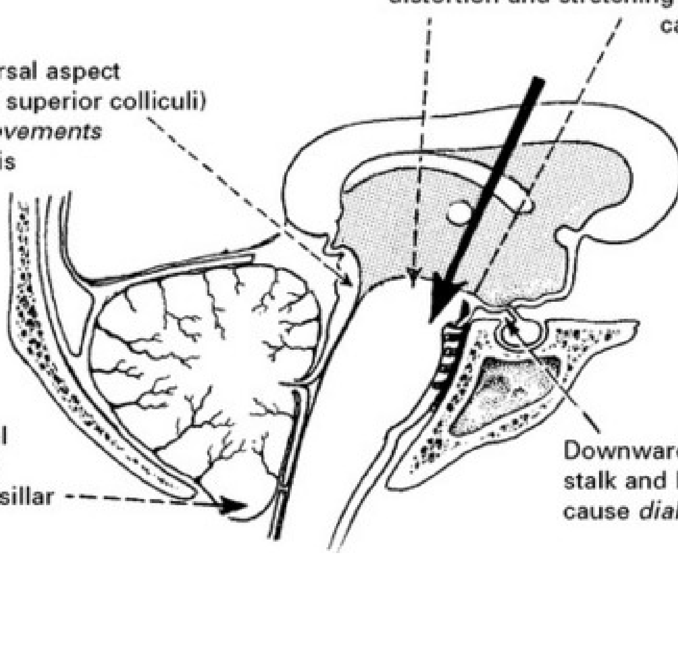
Unchecked lateral tentorial herniation leads to central tentorial and tonsillar herniation, associated with progressive brain stem dysfunction from midbrain to medulla.

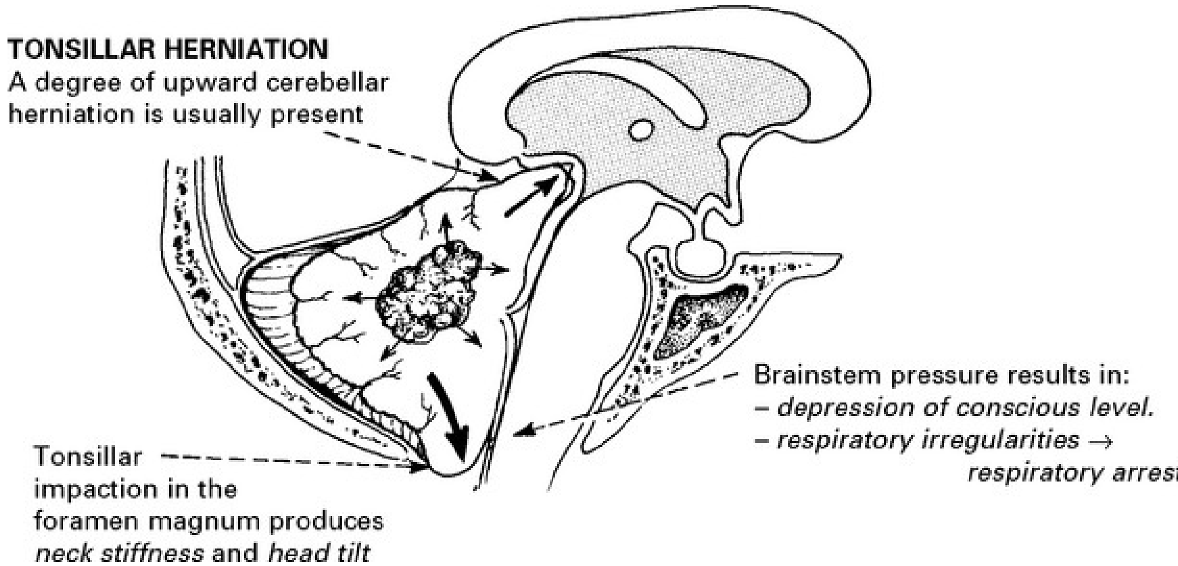
CLINICAL EFFECTS OF BRAIN SHIFT

TENTORIAL HERNIATION – Lateral



TENTORIAL HERNIATION – Central





An injudicious lumbar puncture in the presence of a subtentorial mass may create a pressure gradient sufficient to induce tonsillar herniation.

N.B. Harvey Cushing described cardiovascular changes – an increase in blood pressure and a fall in pulse rate, associated with an expanding intracranial mass, and probably resulting from direct medullary compression. The clinical value of these observations is often overemphasised. They are often absent; when present they are invariably preceded by a deterioration in conscious level.

INVESTIGATIONS

Patients with suspected raised intracranial pressure require an urgent CT/MRI scan. Intracranial pressure monitoring where appropriate (see page 53).

TREATMENT OF RAISED INTRACRANIAL PRESSURE

When a rising intracranial pressure is caused by an expanding mass, or is compounded by respiratory problems, treatment is clear-cut; the mass must be removed and blood gases restored to normal levels – by ventilation if necessary.

In some patients, despite the above measures, cerebral swelling may produce a marked increase in intracranial pressure. This may follow removal of a tumour or haematoma or may complicate a diffuse head injury. Artificial methods of lowering intracranial pressure may prevent brain damage and death from brain shift, but some methods lead to reduced cerebral blood flow, which in itself may cause brain damage (see page 84).

Intracranial pressure is monitored with a ventricular catheter or surface pressure recording device (see page 52). Treatment may be instituted when the mean ICP is $>$ 25 mmHg. Ensure cause is not due to constriction of neck veins.

Methods of reducing intracranial pressure

Mannitol infusion: An i.v. bolus of 100 ml of 20% mannitol infused over 15 minutes reduces intracranial pressure by establishing an osmotic gradient between the plasma and brain tissue. *This method 'buys' time prior to craniotomy in a patient deteriorating from a mass lesion.* Mannitol is also used 6 hourly for a 24–48 hour period in an attempt to reduce raised ICP. Repeated infusions, however, lead to equilibration and a high intracellular osmotic pressure, thus counteracting further treatment. In addition, repeated doses may precipitate lethal rises in arterial blood pressure and acute tubular necrosis. Its use is therefore best reserved for emergency situations.

CSF withdrawal: Removal of a few ml of CSF from the ventricle immediately reduces the intracranial pressure. Within minutes, however, the pressure will rise and further CSF withdrawal will be required. In practice, this method is of limited value, since CSF outflow to the lumbar theca results in a diminished intracranial CSF volume and the lateral ventricles are often collapsed. Continuous CSF drainage may make most advantage of this method.

Sedatives: If intracranial pressure fails to respond to standard measures then sedation may help under carefully controlled conditions.

Propofol, a short acting anaesthetic agent, reduces intracranial pressure but causes systemic vasodilatation. If this occurs pressor agents may be required to prevent a fall in blood pressure and a reduction in cerebral perfusion. Avoid high doses of Propofol; rhabdomyolysis may result and carries a 70% mortality.

Barbiturates (thiopentone) reduce neuronal activity and depress cerebral metabolism; a fall in energy requirements theoretically protects ischaemic areas. Associated vasoconstriction can reduce cerebral blood volume and intracranial pressure but systemic hypotension and myocardial depression also occur. *Clinical trials of barbiturate therapy have not demonstrated any improvement in outcome.*

Controlled hyperventilation: Bringing the $P\mathrm{CO_2}$ down to 3.5kPa by hyperventilating the sedated or paralysed patient causes vasoconstriction. Although this reduces intracranial pressure, the resultant reduction in cerebral blood flow may aggravate ischaemic brain damage and do more harm than good (see page 232). Maintaining the blood pressure and the *cerebral perfusion pressure* (CPP) ($>$60 mmHg) appears to be as important as lowering intracranial pressure.

Decompressive craniectomy: This technique is gaining renewed interest in treating raised ICP unresponsive to other methods. The principal concern is that although reducing mortality, unacceptable levels of morbidity may result. A randomised trial of decompressive craniectomy in head injury is currently underway.

Hypothermia: Cooling to 34°C lowers ICP. Although hypothermia after cardiac arrest with slow rewarming has been reported to improve outcome, trials in head injured patients have failed to demonstrate significant benefit.

Steroids: By stabilising cell membranes, steroids play an important role in treating patients with oedema surrounding intracranial tumours. Trials have found no evidence of benefit after traumatic or ischaemic damage.

COMA AND IMPAIRED CONSCIOUS LEVEL

Consciousness is regarded as a state of awareness of self and surroundings. Impaired consciousness is due to disturbed arousal or content of mental function.

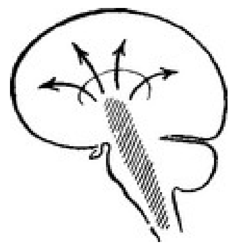
Many pathological processes may impair conscious level and numerous terms have been employed to describe the various clinical states which result, including obtundation, stupor, semicoma and deep-coma. These terms result in ambiguity and inconsistency when used by different observers. Recording conscious level with the *Glasgow coma scale* (page 5) avoids these difficulties and clearly describes the level of arousal. With this scale:

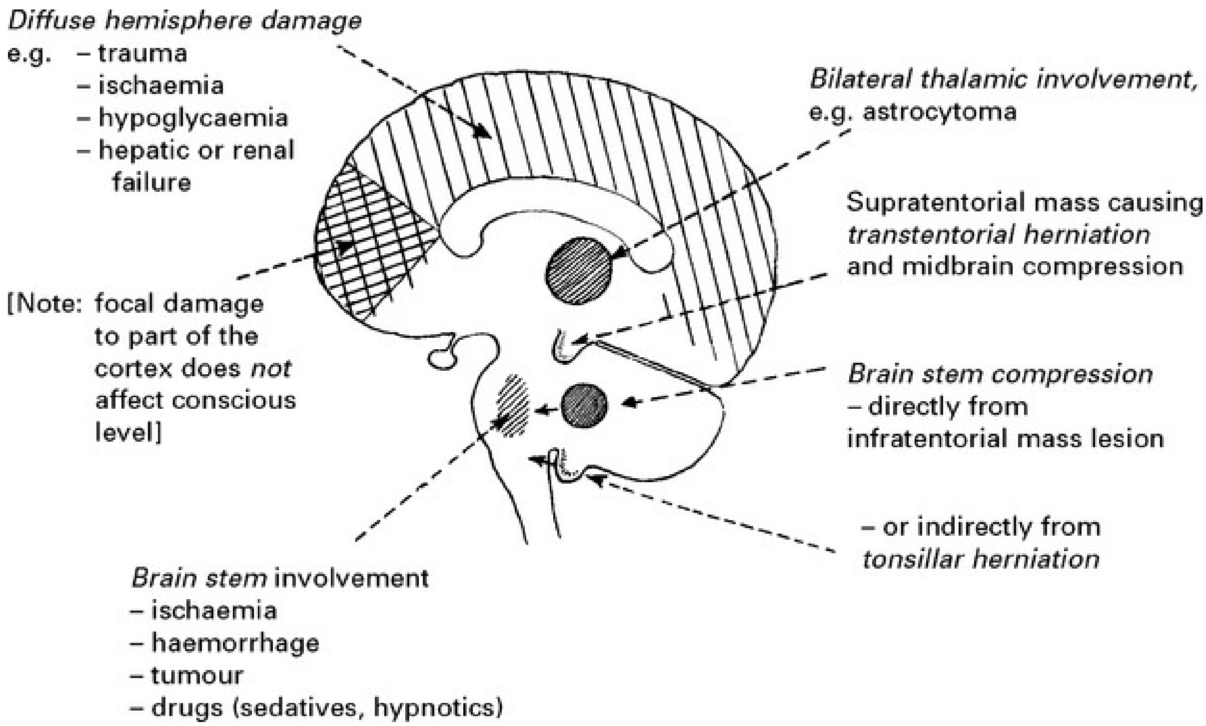
COMA = NO SPEECH, NO EYE OPENING, NO MOTOR RESPONSE

In this section we describe conditions which may present with, or lead to, coma. Patients experiencing ‘transient disturbance of conscious level’ require a different approach.

Pathophysiology of coma

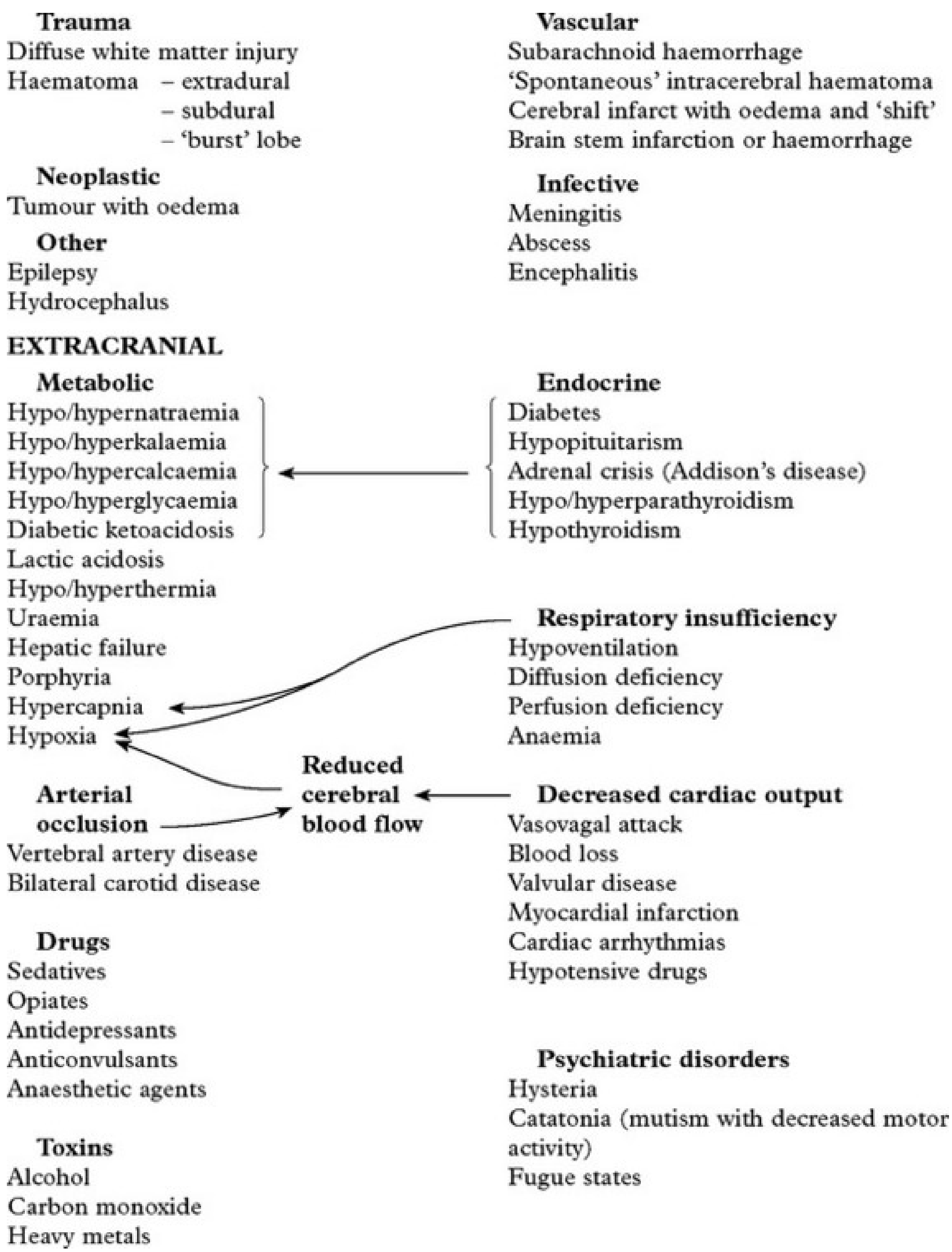
A ‘conscious’ state depends on intact cerebral hemispheres, interacting with the ascending reticular activating system in the brain stem, midbrain, hypothalamus and thalamus. Lesions diffusely affecting the cerebral hemispheres, or directly affecting the reticular activating system cause impairment of conscious level:





CAUSES

INTRACRANIAL



Examination of the unconscious patient (see pages 29, 30)

DIAGNOSTIC APPROACH

Questioning friends, relatives or the ambulance team, followed by general and neurological examination all provide important diagnostic information.

History	Possible cause of coma/impaired conscious level
Head injury leading to admission	*Diffuse shearing injury and/or intracranial haematoma*
Previous head injury (e.g. 6 weeks)	*Chronic subdural haematoma*
Sudden collapse	*Intracerebral haemorrhage* *Subarachnoid haemorrhage*
Limb twitching, incontinence	*Epilepsy/postictal state*
Gradual development of symptoms	*Mass lesion, metabolic or infective cause*
Previous illness – diabetes	*Hypo- or (less likely) hyperglycaemia*
– epilepsy	*Postictal state*
– psychiatric illness	*Drug overdose*
– alcoholism or drug abuse	*Drug toxicity*
– viral infection	*Encephalitis*
– malignancy	*Intracranial metastasis*

General examination

Note the presence of:

Laceration, bruising, CSF leak	*Head injury*
Internal auditory meatus – bleeding	*Head injury*
pus	*Cerebral abscess/meningitis*
Enlarged head, Tense anterior fontanelle } In infant	*Raised intracranial pressure*
Neck stiffness, retraction	*Tonsillar herniation* *Meningitis*
Positive Kernig's sign	*Meningitis*
Tongue biting	*Epilepsy/postictal state*
Emaciation, hepatomegaly, lymphadenopathy	*Intracranial metastasis*
Infection source (ears, sinus, lungs, valvular disease)	*Cerebral abscess, meningitis*
Pyrexia	*Cerebral abscess, meningitis* *Subarachnoid, intracerebral, pontine haemorrhage*

Possible cause of coma/impaired conscious level

Hypotensíon/blood loss → Reduced cardiac output → *Cerebral ischaemia*
Cardiac arrhythmias → Reduced cardiac output; → Emboli
Valvular disease → Emboli → *Cerebral ischaemia*
Respiratory insufficiency → Anoxia → *Cerebral ischaemia*
Smell of alcohol → *Alcohol abuse*
Needle marks on limbs → *Drug abuse*
'Snout' rash → *Solvent abuse*

Neurological examination

Signs of raised intracranial pressure (ICP)
- papilloedema
- tense anterior fontanelle (in infants)

} → *Intracranial mass lesion*

Neurological signs
- unilateral, dilated, fixed pupil → *Intracranial mass lesion*; *Hydrocephalus*
- bilateral dilated, fixed pupils → *Hydrocephalus*; *Diffuse cerebral swelling, e.g. anoxia*; *Drugs – anticholinergics, sympathomimetics* } *overdose*
- pinpoint pupils → *Drugs – opiates, parasympathomimetics*; → *Pontine haemorrhage*
- eye movements absent (spontaneous or reflex)
  - → pupils fixed → *Severe – trauma – ischaemia – haemorrhage*; → *Drugs (transient effect)*
  - → pupils usually reacting → *Hypoxic/hepatic encephalopathy*
- asymmetric limb response (i.e. hemi/monoparesis) → *Focal brain damage, e.g.*
  - *tumour*
  - *trauma*
  - *haematoma*
  - *encephalitis*

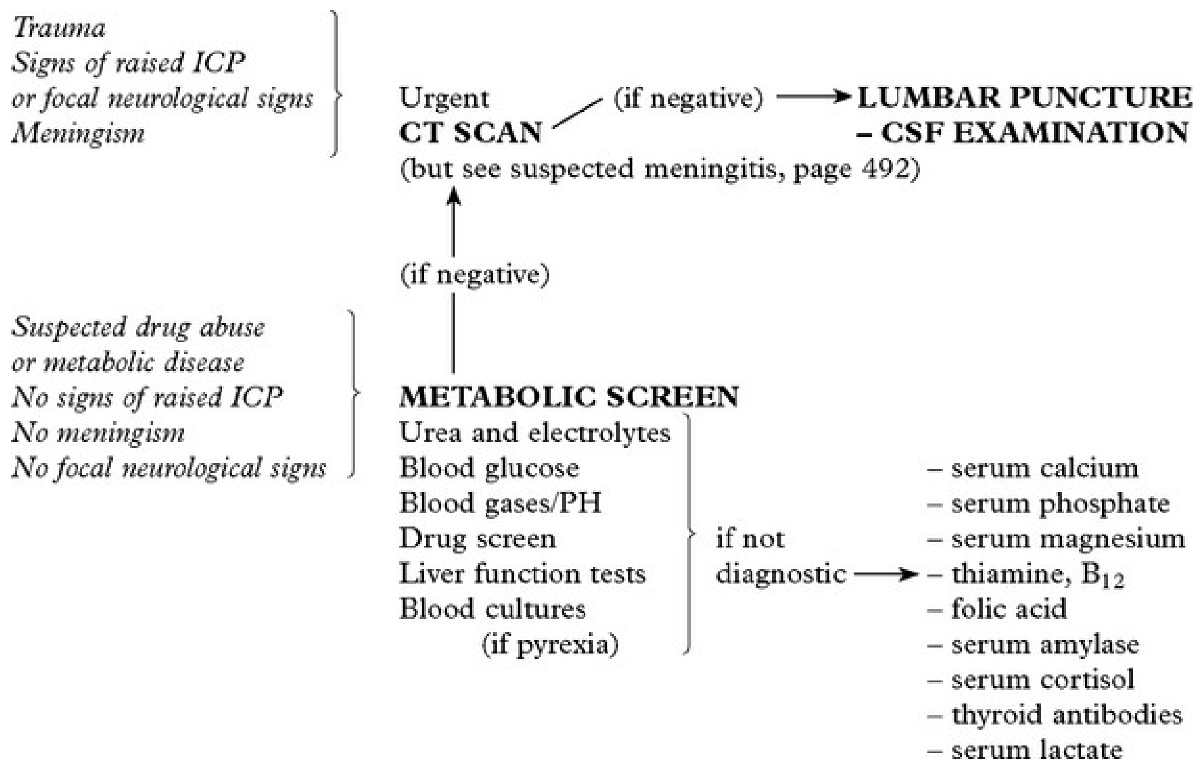
[N.B. – *hepatic encephalopathy*
– *hypoglycaemia*
– *uraemia*
} → occasionally produce asymmetrical responses]

– Symmetrical limb responses
– Reacting pupils
– Full eye movements
} → suggest a *metabolic encephalopathy or drug toxicity*

– Subhyaloid/vitreous haemorrhage (on fundoscopy) → *Subarachnoid haemorhage*

Investigations

The sequence of investigations depends on clinical suspicion:



In addition

CHEST X-RAY – may reveal a bronchial carcinoma.

ELECTROENCEPHALOGRAPHY – may provide evidence of – subclinical epilepsy

– herpes simplex encephalitis

– metabolic encephalopathy.

MRI – has a limited role in the investigation of coma. More sensitive than CT scan in demonstrating small ischaemic changes and early encephalitis.

Prognosis

Although *conscious level examination* does not aid diagnosis, it plays an essential role in patient management and along with the *duration of coma, pupil response* and *eye movements* provides valuable prognostic information. Non-traumatic coma tends to carry a better prognosis (see page 214).

TRANSIENT LOSS OF CONSCIOUSNESS

Many conditions causing coma may also transiently affect a patient's conscious level. This results from:

Syncope:

Reduction in cerebral arterial oxygen supply can be caused by *cardiac arrhythmias*, *cardiac outflow obstruction* or *vasovagal attack.*

Seizure:

Pseudoseizure (non-epileptic attack disorder) – see below

Acute toxic or metabolic coma:

– Drug abuse – *alcohol, solvents or barbiturates* – may cause transient, intermittent confusion.

– *Hypoglycaemia*

DIAGNOSTIC APPROACH

History

Try to obtain a history from eyewitness as well as from the patient themselves.

History from the patient:

Context: may suggest likely cause – a collapse when having blood taken suggests syncope; an episode arising from sleep suggests a seizure.

Prodrome: a brief sensation of déjà vu before the episode indicates a focal onset seizure; a feeling of lightheadness, sweatiness and visual fading suggests syncope.

Recovery: a rapid recovery suggest syncope; waking in the ambulance suggest seizure.

History from witness (find them; phone them):

How long the patient was out for; – syncope is typically less than 1 minute; seizures usually longer.

What they did; brief asynchronous jerking movements occur in syncope; more prolonged synchronous tonic clonic movements occur in seizures.

Any colour change; 'ashen' suggests syncope; cyanosed suggests seizure.

How quickly they recovered; rapid recovery suggest syncope.

Silent witnesses:

Incontinence is common in all forms of loss of consciousness and does not distinguish between a seizure and syncope. Tongue biting strongly suggests a seizure as do other much less common injuries – posterior dislocation of the shoulder or vertebral fracture.

Investigation is directed by the clinical history:

Electroencephalography (EEG) may reveal a focal or generalized disturbance – *epilepsy*.

Electrocardiography (ECG) and 24 hour ECG may reveal a *cardiac arrythmia*.

Head up tilt-table testing may reveal *neurocardiogenic syncope* or *orthostatic hypotension.*

Echocardiography may reveal *cardiomyopathy*.

Blood glucose may indicate *hypoglycaemia*.

EEG telemetry is occassionally needed.

Often attacks of unconsciousness remain unexplained and possibly have a psychological basis. The circumstances of the attack (e.g. during an argument), the non-stereotyped nature of the episode suggest a non-organic explanation. Such attacks are often mistaken for a seizure and are referred to as *pseudoseizures or non-epileptic attacks* (see page 99).

CONFUSIONAL STATES AND DELIRIUM

Of all acute medical admissions, 5–10% present with a **confused verbal response**, *i.e.* disorientation in time and/or place. Most patients are easily distracted, have slowed thought processes and a limited concentration span. Some may lose interest in the examination to the point of drifting off to sleep.

Perceptual disorders (illusions and hallucinations) may accompany the confused state – **delirium**. This is often associated with withdrawal and lack of awareness or with restlessness and hyperactivity.

Primary neurological disorders contribute to only 10% of those patients presenting with an acute confusional state. In the elderly, postoperative disorientation is particularly common and multiple factors probably apply; in these patients the prognosis is good.

The Confusion Assessment Method (CAM) is used to confirm delirium.

Feature 1 – Acute onset and fluctuating course.

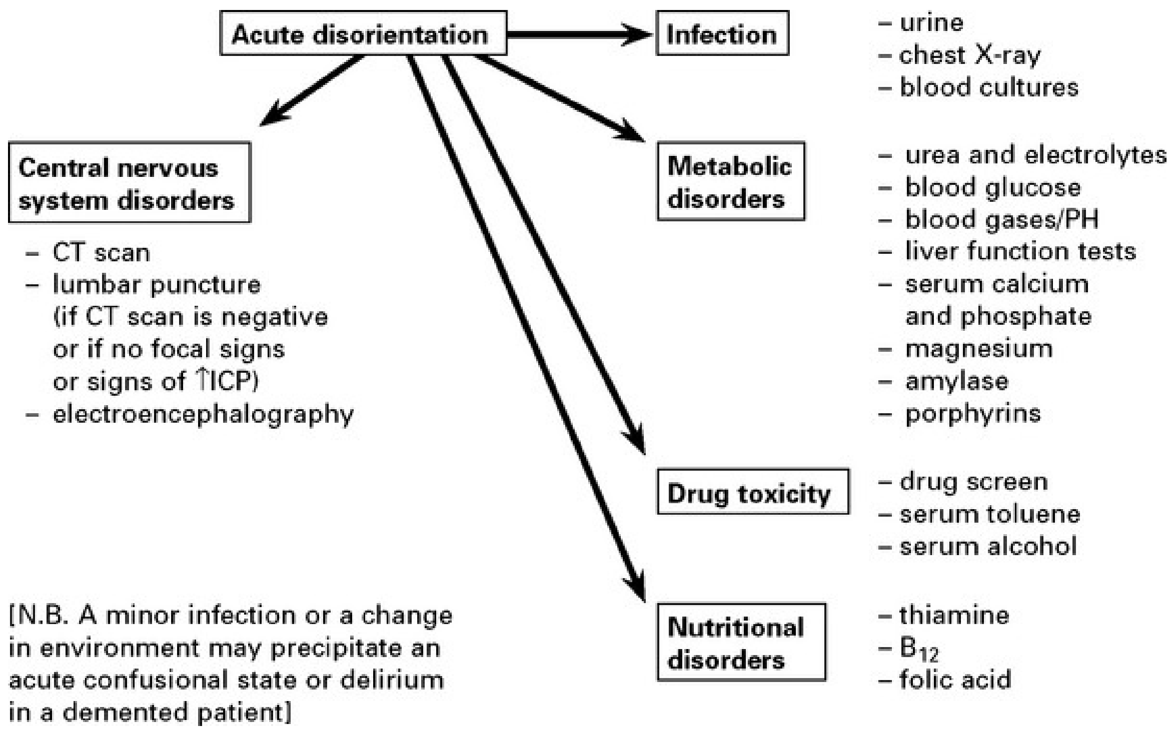
Feature 2 – Inattention.

Feature 3 – Disorganised thinking.

Feature 4 – Altered level of consciousness.

The presence of features 1 and 2 and either 3 or 4 are diagnostic.

DIAGNOSTIC APPROACH



EPILEPSY

Definitions

A seizure or epileptic attack is the consequence of a paroxysmal uncontrolled discharge of neurons within the central nervous system. The clinical manifestations range from a major motor convulsion to a brief period of lack of awareness.

The *prodrome* refers to mood or behavioural changes which may precede the attack by some hours.

The *aura* refers to the symptom immediately before a seizure and will localise the attack to its point of origin within the nervous system.

The *ictus* refers to the attack or seizure itself.

The *postictal period* refers to the time immediately after the ictus during which the patient may be confused, disorientated and demonstrate automatic behaviours.

The stereotyped and uncontrollable nature of the attack is characteristic of epilepsy.

A patient is said to have *epilepsy* when they have had more than one seizure. It is important to remember that epilepsy is not a single condition; epilepsy can be the symptom of other disorders and there are numerous different epilepsy syndromes.

Pathogenesis

Epilepsy has been described since ancient times. The 19th century neurologist Hughlings-Jackson suggested 'a sudden excessive disorderly discharge of cerebral neurons' as the causation of the attack. Berger (1929) recorded the first electroencephalogram (EEG) and not long after, it was appreciated that certain seizures were characterised by particular EEG abnormalities.

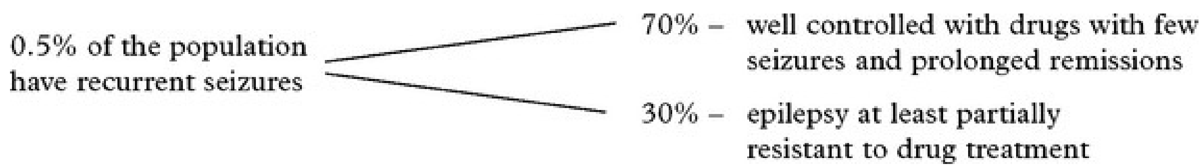
Recent studies in animal models of *focal epilepsy* suggest a central role for the excitatory neurotransmitter glutamate. This produces a depolarisation shift by activating receptors which in turn facilitate cellular influx of Na^+, K^+ and Ca^{2+}. Gamma amino butyric acid (GABA) has an important inhibitory influence in containing abnormal cortical discharges and preventing the development of generalised seizures.

Epilepsies have complex inheritance; molecular genetics studies in rarer syndromes with autosomal dominant features have identified genes that code for ion channel subunits, either ligand or voltage gated (Channelopathies).

Incidence and course

Epilepsy presents most commonly in childhood and adolescence or in those over 65, but may occur for the first time at any age.

5% of the population suffer a single seizure at some time.



Though there is considerable variability depending on seizure type, 6 years after diagnosis 40% of patients have had a substantial remission; after 20 years – 75%.

SEIZURE CLASSIFICATION

The classification of epilepsy involves two steps:

1. The classification of the seizure types

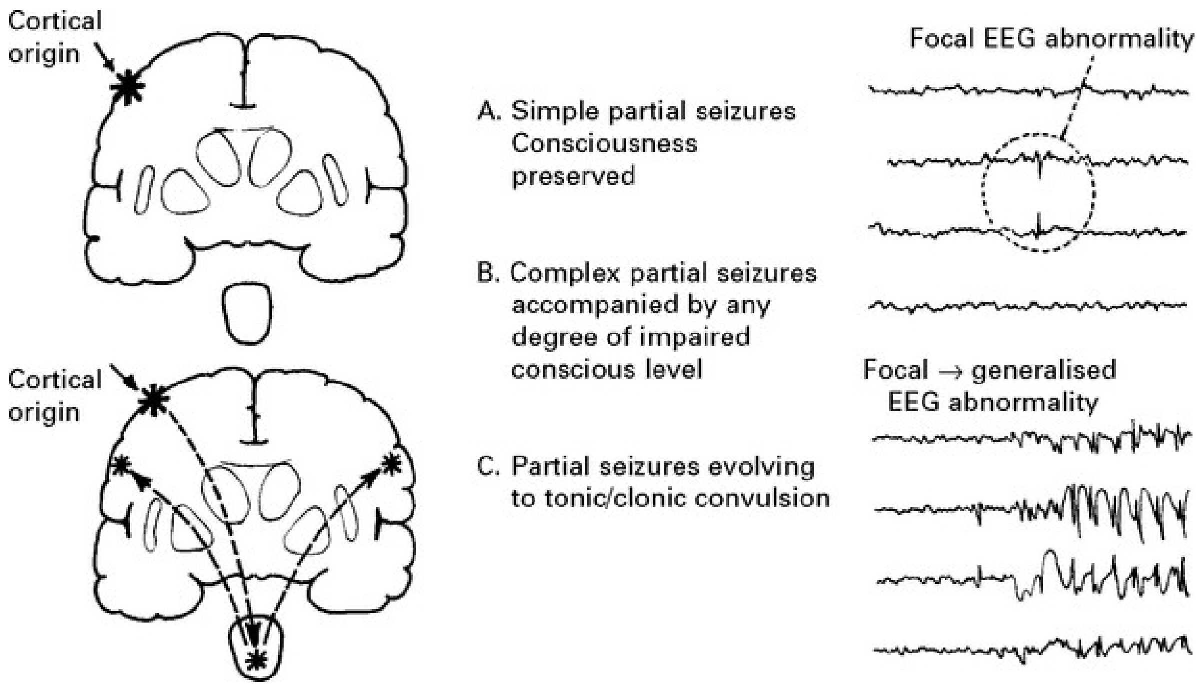
2. Integration of seizure type with history, family history, EEG and imaging (as needed)

CLASSIFICATION OF SEIZURE TYPE

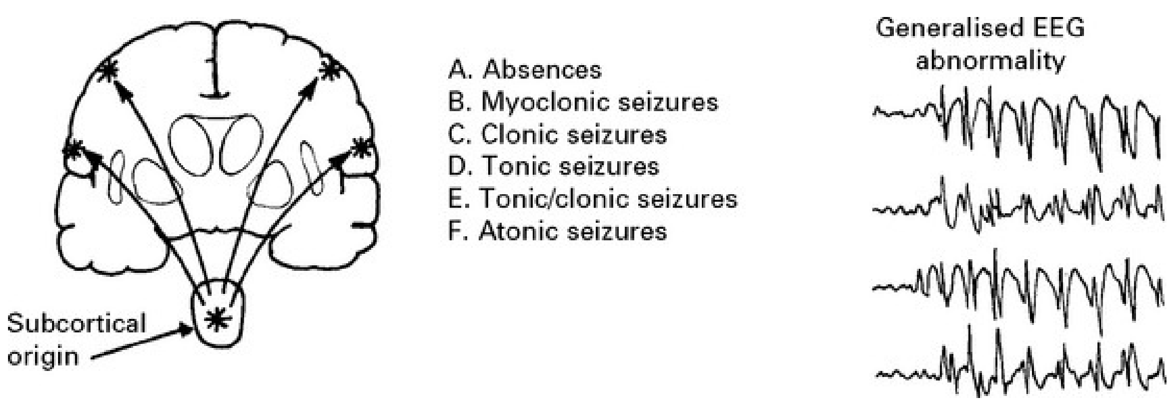
Attacks which begin **focally** from a single location within one hemisphere are distinguished from those of a **generalised** nature which probably commence in deeper midline structures and project to both hemispheres simultaneously.

1. PARTIAL (focal, localisation related) SEIZURE

Classified by site of onset (frontal, temporal, parietal or occipital lobe) and by severity:



2. GENERALISED SEIZURES (convulsive or non-convulsive)



3. UNCLASSIFIED SEIZURES, There may be insufficient information to classify a seizure.

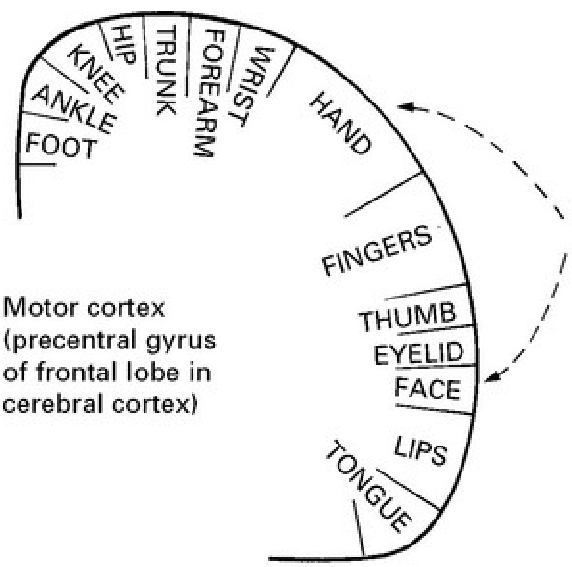
THE PARTIAL SEIZURES

Partial seizures are classified according to both their:

Severity – simple; complex partial; evolving to tonic/clonic convulsion

Semiology – what happens during the seizure, which reflects the site of origin, in order of frequency: temporal, frontal, parietal and occipital lobes.

FRONTAL LOBE SEIZURES



There are a number of seizure types:

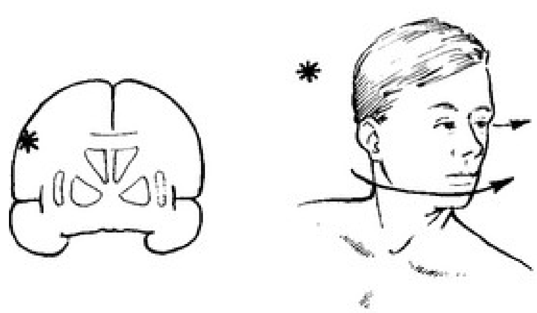
Jacksonian motor seizures consists of a 'march' of involuntary movement from one muscle group to the next.

Movement is clonic (shaking) and usually begins in hand or face – these having the largest representative cortical area.

Motor seizures with the above 'march' are quite rare, usually they are less localised, involving many muscle groups simultaneously and are tonic (rigid) or clonic.

After a motor seizure the affected limb(s) may remain weak for some hours before return of function occurs – **Todd's paralysis.**

Adversive seizures



The patient is aware of movement of the head. Attacks often progress to loss of consciousness and tonic/clonic epilepsy. The patient's eyes and head turn away from the site of the focal origin.

Supplementary motor area seizures can result in more complicated stereotyped movements often arising from sleep – for example cycling movement.

PARIETAL LOBE SEIZURES

These arise in the sensory cortex (parietal lobe), the patient describing paraesthesia or tingling in an extremity or on the face sometimes associated with a sensation of distortion of body image. A 'march' similar to the Jacksonian motor seizure may occur. Motor symptoms occur concurrently – the limb appears weak without involuntary movement.

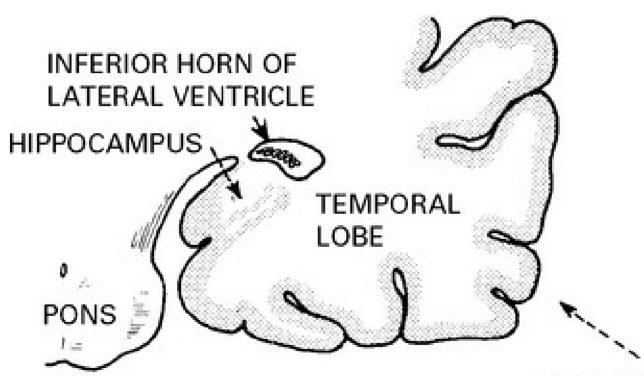
The representation of limbs, trunk, *etc.* in the post-Rolandic sensory cortex is similar to that of the motor cortex.

VISUAL, AUDITORY and AUTONOMIC simple partial seizures occur, but are rare.

Frontal and Parietal seizures indicate structural brain disease, the focal onset localising the lesion. Full investigation is mandatory.

TEMPORAL LOBE SEIZURES

These attacks are characterised by a complex aura (initial symptom) often with some impairment of consciousness.



The most common seizure origin lies in the medial part of the temporal lobe, hippocampus or lateral surface of the lobe.

Coronal section through the pons showing medial aspect of the temporal lobe and hippocampus

The nature of the attack

The content of attacks may vary in an individual patient. Commonly encountered symptoms include:

Visceral disturbance: Gustatory (taste) and olfactory (smell) hallucinations, lip smacking, epigastric fullness, choking sensation, nausea, pallor, pupillary changes (dilatation), tachycardia.

Memory disturbance: Déjà vu ('something has happened before'), jamais vu ('feeling of unfamiliarity'), depersonalisation, derealisation, flashbacks, formed visual or auditory hallucinations.

Motor disturbance: Fumbling movement, rubbing, chewing, semi-purposeful limb movements.

Affective disturbance: Displeasure, pleasure, depression, elation, fear.

A constellation of these symptoms associated with subtle clouding of consciousness characterises a temporal lobe onset seizure.

AUTOMATISM occurs during the state of clouding of consciousness either during or after the attack (postictal) and takes the form of involuntary, often complicated, motor activity. In ambulatory automatism, subjects may 'wander off'.

Confusion and headache after an attack are common. The whole episode may last for seconds but occasionally may be prolonged and a rapid succession or cluster of attacks may occur. Attacks show an increased incidence in adolescence and early adult life. A history of birth trauma or febrile convulsions in infancy may be obtained. Lesions in the hippocampus occur as a result of anoxia or from the convulsion itself and act as a source of further epilepsy. When surgery is carried out, hippocampal sclerosis is often found. Occasionally other pathologies are identified, such as dysembryoplastic neuroepithelial tumours (DNET), vascular malformations and low-grade astrocytomas.

OCCIPITAL LOBE SEIZURES

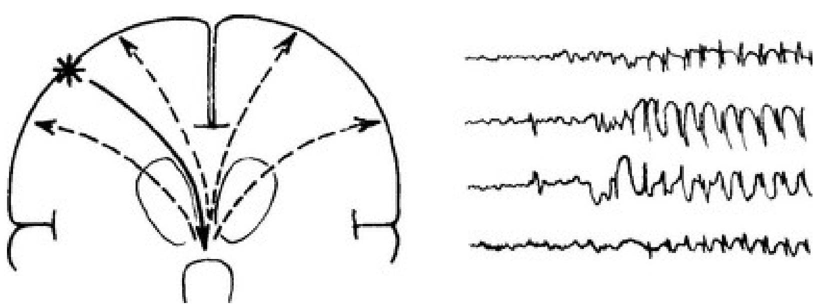
These are uncommon. Typically there is an elementary visual hallucination – a line or flash – prior to a tonic-clonic seizure.

PARTIAL SEIZURES EVOLVING TO TONIC/CLONIC CONVULSION

Seizure discharges have the capacity to spread from their point of origin and excite other structures. When spread occurs to the subcortical structures (thalamus and upper reticular formation) their excitation releases a discharge which spreads back to the cerebral cortex of both hemispheres, resulting in a tonic/clonic seizure. This chain of events is reflected in the electroencephalogram (EEG).

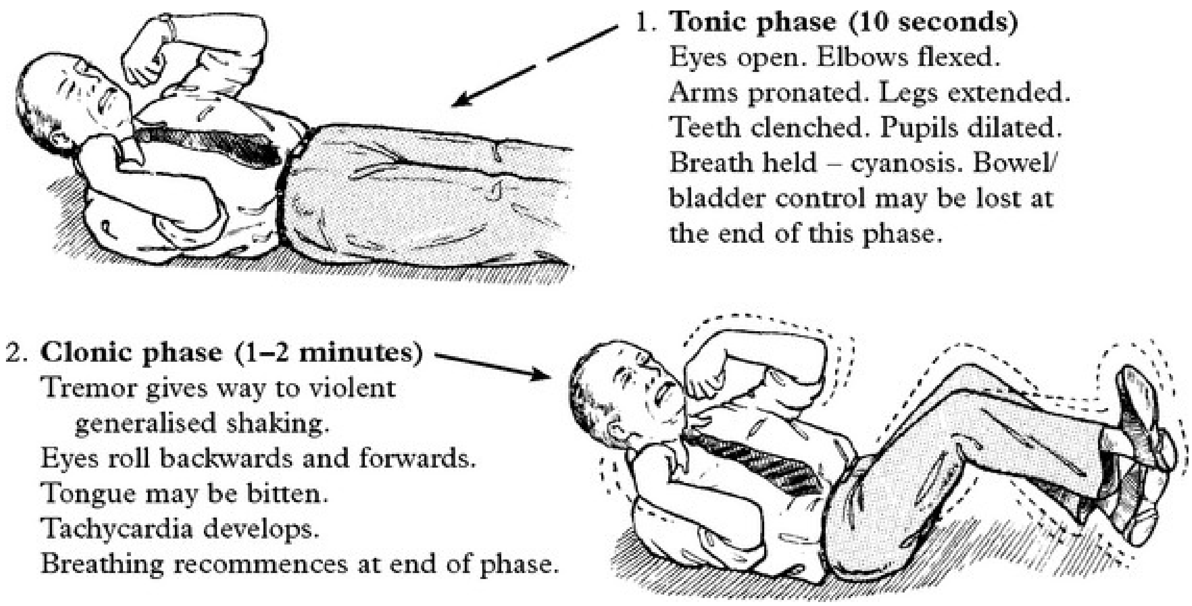
The symptoms before the tonic/clonic convulsion give a clue to the site of the initial discharge (simple partial or complex partial).

An eyewitness account is important because retrograde amnesia may prevent recall of the onset.



TONIC/CLONIC ATTACKS

Loss of consciousness; falls to the ground.



The patient then sleeps with stertorous respiration and cannot be roused. On regaining consciousness, confusion and headache are present. He may feel exhausted for hours or even days afterwards. Muscles may ache as a result of violent movement and muscle damage occurs with elevation of the muscle enzyme creatinine phosphokinase (CPK). Trauma occurs frequently, either as a result of the fall, or as a result of the movements, *e.g.* posterior dislocation of the shoulder. Very rarely sudden death may occur from inhalation or an associated cardiac arrhythmia.

The differentiation of these attacks from pseudoseizures will be discussed later.

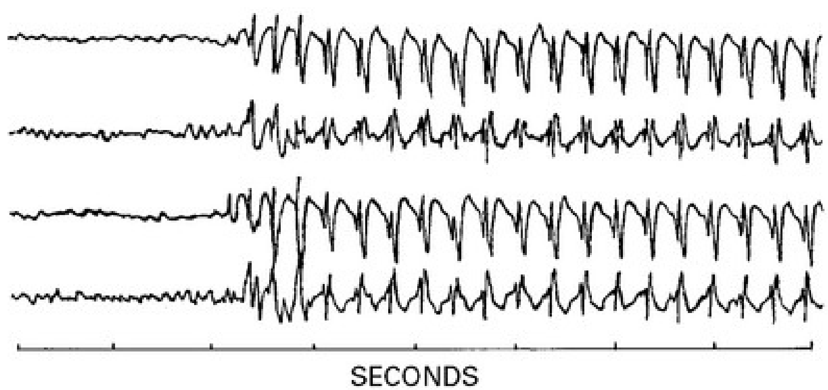
GENERALISED SEIZURES

Generalised seizure attacks arise from subcortical structures and involve both hemispheres. Consciousness may be impaired and motor manifestations are bilateral.

ABSENCES (previously called Petit mal)

The patient (usually a child) stares vacantly, eyes may blink. The absence may occur many times a day with a duration of 5–15 seconds and may be induced by hyperventilation.

The *ELECTROENCEPHALOGRAM (EEG)* is diagnostic.



3 per second spike and wave activity occurs in all leads, persisting as long as the seizure. Hyperventilation evokes this appearance during recording.

Similarly, photic stimulation – flashing a light in both eyes – may produce spike and wave discharge.

ABSENCE STATUS

Long periods of clouding of consciousness with continuing ‘spike and wave’ activity on the EEG.

MYOCLONIC SEIZURES

Sudden, brief, generalised muscle contractions. They often occur in the morning and are occasionally associated with tonic/clonic seizures. The commonest disorder is benign juvenile myoclonic epilepsy (JME) with onset after puberty. Myoclonus on the edge of sleep is normal. Myoclonus also occurs in degenerative and metabolic disease (see page 190).

TONIC SEIZURES

Sudden sustained muscular contraction associated with immediate loss of consciousness.

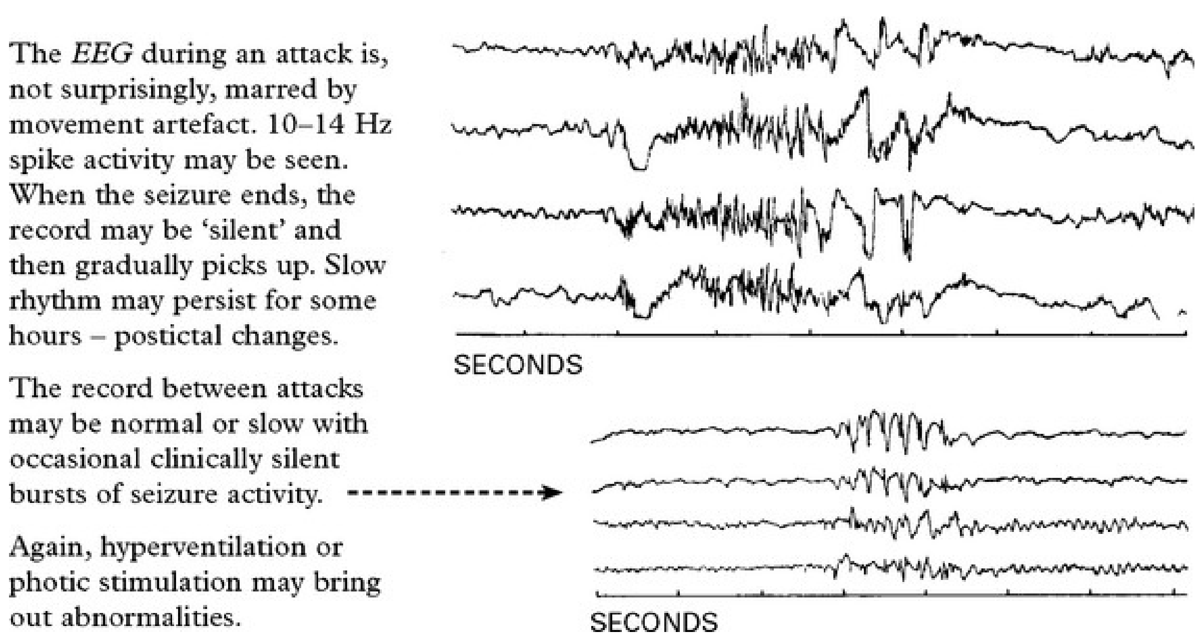
Tonic episodes occur as frequently as tonic/clonic episodes in children and should alert the physician to a possible anoxic aetiology.

In adults, tonic attacks are rare.

GENERALISED SEIZURES

TONIC/CLONIC SEIZURES (previously called Grand mal)

Primary tonic/clonic seizures occur without warning or aura. The *epileptic cry* at onset results from tonic contraction of respiratory muscles with partial closure of vocal cords. The tonic phase is associated with rapid neuronal discharge. The clonic phase begins as neuronal discharge slows.



ATONIC SEIZURES

These are rare and almost always occur in patients with other types of seizure. They are characterised by a loss of muscle tone and a sudden fall. Consciousness may only be lost briefly. The EEG shows polyspike activity or low voltage fast activity.

SYMPTOMATIC SEIZURES

Seizures can be symptoms of acute brain pathology. If the patient goes on to develop recurrent seizures this is symptomatic epilepsy (see later).

The age of onset gives a clue to the causation.

Newborn	Infancy and Childhood	Adult
Asphyxia	Febrile convulsions	Trauma
Intracranial haemorrhage	CNS infection	Drugs and alcohol
Hypocalcaemia	Trauma	CNS infection
Hypoglycaemia	Congenital defects	Intracranial haemorrhage
Hyperbilirubinaemia	Inborn errors of metabolism	Tumours
Water intoxication	Tumours	Vascular disease
Inborn errors of metabolism		Hypoglycaemia

Seizures occur in about 5% of patients following stroke and in 5% of patients with multiple sclerosis.

SEIZURES – DIFFERENTIAL DIAGNOSIS

The following should be considered in the differential diagnosis of seizures – **SYNCOPE (VASOVAGAL) ATTACKS**

Syncope usually occurs when the patient is standing and result from a global reduction of cerebral blood flow.

Prodromal pallor, nausea and sweating occur associated with a feeling of lightheadness and often fading of vision. If the patient sits down, the attack may pass off or proceed to a brief loss of consciousness.

Brief asynchronous jerks are common as is urinary incontinence. Tonic and clonic movements may develop if impaired cerebral blood flow is prolonged (‘anoxic’ seizures).

Mechanism: Peripheral vasodilatation with drop in blood pressure followed by vagal overactivity with fall in heart rate.

Syncopal attacks occur in hot, crowded rooms (e.g. classroom) or in response to pain or emotional disturbance.

'Reflex' syncope from cardiac slowing may occur with carotid sinus compression. Similarly, cough syncope may result from vigorous coughing.

CARDIAC ARRHYTHMIAS

Seen in situations such as complete heart block (Adams-Stokes attacks).

Prolonged arrest of cardiac rate or critical reduction will progressively lead to loss of consciousness – tonic jerks – cyanosis/stertorous respiration – fixed pupils and extensor plantar responses.

On recovery of normal cardiac rhythm, the degree of persisting neurological damage depends upon the duration of the episode and the presence of pre-existing cerebrovascular disease. In suspected patients, electrocardiography is mandatory. Continuous (24 hours) ECG monitoring may be necessary.

HYPOGLYCAEMIA

Amongst other neuroglycopenic manifestations, seizures or intermittent behavioural disturbances may occur. A rapid fall of blood sugar is associated with symptoms of catecholamine release, *e.g.* palpitations, sweating, *etc.* In 'atypical' seizures exclude a metabolic cause by blood sugar estimation when symptomatic.

EPISODIC CONFUSION

Intermittent confusional episodes caused by drugs (e.g. barbiturates) or toxins (e.g. solvents).

PANIC ATTACKS Hyperventilation can induce focal motor and sensory symptoms.

NARCOLEPSY

Inappropriate sudden sleep episodes may easily be confused with epilepsy (see page 107).

DISSOCIATIVE SEIZURES (pseudoseizures, non-epileptic attack disorder, NEAD)

A difficult distinction lies between epileptic seizures and dissociative seizures. The latter are heterogeneous comprising episodes in which shaking/thrashing and apparent loss of consciousness occur. The episodes are often variable (rather than stereotyped), prolonged, with a rapid recovery. Often patients with epilepsy will also manifest such attacks. Patients may have a history of other functional illness and have an increased frequency of preceding sexual or physical trauma (about 30%). Dissociative seizures are usually thought to be a subconscious disorder. Rarely some patients do have insight and the episodes are part of a facticious disorder or malingering. EEG studies, particularly with video telemetry, may help discriminate. Management depends on helping the patient understand and manage the episodes, for example with cognitive behavioural therapy, managing any associated depression or anxiety and stopping unnecessary anticonvulsants.

EPILEPSY – CLASSIFICATION

The classification of epilepsy brings together the seizure semiology and other aspects of the history and investigations. The International League Against Epilepsy classified epilepsies as:

- **Idiopathic** – thought to be primarily genetic with generalised seizures, sometimes grouped as more specific syndromes (see below). Account for about 10–20% of cases.

• **Symptomatic** – partial onset seizures associated with a structural lesion, such as tumour, cortical dysplasia, infection, head injury or trauma – about 30–40% of cases. The combination of the site of seizure onset and the underlying pathology leads to the diagnosis: for example 'post traumatic frontal lobe epilepsy' or 'temporal lobe epilepsy due to mesial temporal sclerosis' or 'symptomatic occipital lobe epilepsy secondary to an arteriovenous malformation'.

• **Cryptogenic** – partial onset seizures for which no cause has been found. Account for about 50% of patients.

With developments in understanding, particularly in genetics, limitations with this generally practical classification have arisen – for example familial frontal onset epilepsy (associated with a mutation in the gene encoding the neuronal nicotinic acetylcholine receptor (nAChR) alpha-4 subunit) is an idiopathic yet partial onset epilepsy. Newer proposals under consideration suggest the classification should move to 'genetic', 'structural/metabolic' or 'of unknown cause' rather than the groups given above.

Selected Idiopathic Epilepsy Syndromes (by age of onset)

Childhood absence epilepsy (common)

Absence seizures begin between 4 and 12 years of age. Family history in 40% of patients. The absence may occur many times a day with a duration of 5–15 seconds.

Frequent episodes lead to falling off in scholastic performance.

Attacks rarely present beyond adolescence.

In 30% of children, adolescence may bring tonic/clonic seizures.

Distinction of absences from complex partial seizures is straightforward; the latter are longer – 30 seconds or more – and followed by headache, lethargy, confusion and automatism.

EEG finds 3 Hz spike and wave (page 97)

Juvenile myoclonic epilepsy (common)

Myoclonic jerks begin in teenage years, typically in the morning. Develop tonic/clonic seizures, often with sleep deprivation, in late teens. Occasionally have absence seizures. EEG frequently finds 4-5 polyspike and wave discharges.

West Syndrome (rare)

Infants present with diffusely abnormal EEGs, tonic clonic convulsions, myoclonic jerks and mental retardation following perinatal trauma or asphyxia. The seizures are sometimes called infantile spasms and the abnormal EEG pattern between events – hypsarrhythmia. Mortality or severe disability is high.

Lennox-Gastaut Syndrome (rare)

This similar syndrome presents later between 1–7 years of age. The response to anticonvulsant treatment and the degree of retardation is variable. The condition is associated with a large number of disorders including hypoxia, intracranial haemorrhage, toxoplasmosis, cytomegalovirus infection and tuberous sclerosis.

The **REFLEX EPILEPSIES** are a rare group of seizure disorders in which tonic/clonic or complex partial seizures are evoked by sensory stimuli. These stimuli can be certain pieces of music (**Musicogenic epilepsy**), reading (**reading epilepsy**) or performing calculations (**arithmetical epilepsy**).

EPILEPSY – INVESTIGATION

Investigations are directed at:

- corroborating the diagnosis of epilepsy
- classifying the type of epilepsy
- looking for an underlying cause
- eliminating alternative diagnoses

The relative emphasis of these elements will depend on the clinical situation.

For most patients the clinical diagnosis of a seizure is secure and the emphasis is to seek the cause and to classify the epilepsy to direct treatment. In others the main concern is whether the episodes are seizures or an alternative diagnosis.

Neuroimaging

All adults and all with focal onset seizures should be scanned. MRI brain imaging is more sensitive than CT and many lesions, for example small tumours, cortical dysplasia or hippocampal sclerosis will be missed on CT.

EEG

Standard interictal EEG is relatively insensitive – though this varies according to the type of epilepsy (it is very sensitive in childhood absence epilepsy). The interpretation of abnormalities requires caution; 0.5% of the normal population have interictal spikes or sharp waves (epileptic discharges) as compared to 30% of patients after their first seizure.

The pattern of abnormalites can point towards a focal or generalised onset and can supplement the clinical classification.

Sleep deprived EEG increases the yield but with the risk of provoking a seizure. EEG shortly after a seizure is more likely to find an abnormality.

Ambulatory EEG recording increases the chance of finding an abnormality and of recording a clinical event. The 'gold standard' investigation is simultaneous EEG monitoring and video monitoring (videotelemetry).

Eliminating alternatives

ECG should be done in all patients with seizures. This is a simple cheap test and a small number of epilepsy mimics can be identified this way, *e.g.* prolonged QT syndrome.

Prolonged ECG may be useful in patients with possible cardiac syncope – especially in patients with sleep associated events. Implantable loop recorders can be used when patients have infrequent events.

Head up tilt table testing is often helpful in the diagnosis of neurocardiogenic syncope.

Metabolic investigation to consider include fasting glucose for insulinoma and synacthen test for Addison's disease.

Advanced investigation

Volumetric MRI can identify hippocampal sclerosis not apparent on conventional imaging.

Functional imaging, using ictal and interictal SPECT may be helpful in identifying an epileptogenic focus when evaluating patients for surgery.

Advanced EEG techniques for example using sphenoidal electrodes or recording from surgically inserted intracranial grid or depth electrodes can help localise a focus before surgery.

EPILEPSY – TREATMENT

Basic principles: Most patients respond to anticonvulsant drug therapy. Drug treatment should be simple, preferably using one anticonvulsant (monotherapy). Polytherapy should be avoided to minimise adverse effects and drug interactions.

Treatment aims to prevent seizures without side effects though this is not always achieved. Surgery is an option in a small number on non-responders.

Teratogenicity: it is important to consider the teratogenetic risks when starting any anticonvulsant in a woman of childbearing age. Large prospective studies have established rates of major congenital malformations for widely used drugs: those on no medication, carbamazepine or lamotrigine had similar rates of around 3%; in valproate monotherapy the rate was significantly higher at 6%; polytherapy overall was about 6%, and 9% if valproate was one of the drugs.

Interactions: many anticonvulsants (especially carbamazepine, phenytoin, phenobarbitone) induce liver enzymes to increase metabolism of other drugs (notably the oral contraceptive, warfarin and other anticonvulsants); valproate inhibits liver enzymes.

Blood levels: monitoring levels is useful for phenytoin because of the difficult pharmacokinetics. Other blood levels can occasionally be useful to check the patient is taking the medication or for toxicity.

Drug choice:

Idiopathic generalised epilepsy: sodium valproate*; lamotrigine*; topiramate; levetiracetam; phenytoin.

Partial (focal) epilepsy: lamotrigine*; carbamazepine*; sodium valproate*; Phenytoin*; Phenobarbitone; Levetiracetam; Topiramate; Tiagabine; Zonisamide; Oxcarbazepine; Gabapentin; pregabalin;

lacosamide.

Those drugs asterisked are typically used for monotherapy others as ‘add-on’ therapy when control sub-optimal. The choice of anticonvulsant will be a balance between efficacy, adverse effects, teratogenicity and drug interactions and the patient should be involved in this decision.

Main adverse effects of main anticonvulsants:

Lamotrigine; rash – can produce Stevens–Johnson syndrome; drowsiness.

Carbamazepine and oxcarbazepine; rash; dose related drowsiness, ataxia, diplopia; hyponatraemia; thrombocytopenia.

Sodium valproate; abdominal pain, hair loss, weight gain, tremor, thrombocytopenia.

Phenytoin; gum hypertrophy, acne; ataxia, diplopia, skin thickening, neuropathy.

Phenobarbitone; sedation, behavioural changes, withdrawal seizures.

Gabapentin and pregabalin; drowsiness, ataxia, weight gain.

Topiramate and zonisamide; drowsiness, weight loss, renal stones, paraesthesiae.

Levetiracetam; irritability, weight loss.

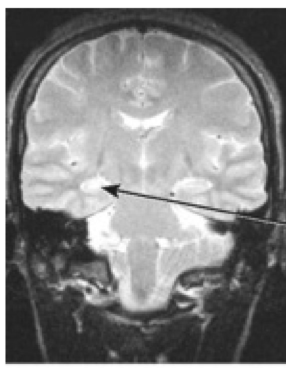
Lifestyle issues: Generally there should be as few restrictions as possible (see driving regulations). Patient should be made aware of potential triggers to avoid – sleep deprivation, excess alcohol, and, where relevant flashing lights (though most patients are not photosensitive). Sensible precautions – showering rather than taking a bath, avoiding heights – should be suggested.

EPILEPSY – SURGICAL TREATMENT

In some patients, particularly those with complex partial epilepsy, seizures remain intractable despite adequate drug administration and prevent a normal lifestyle; of those, a proportion will benefit from surgery.

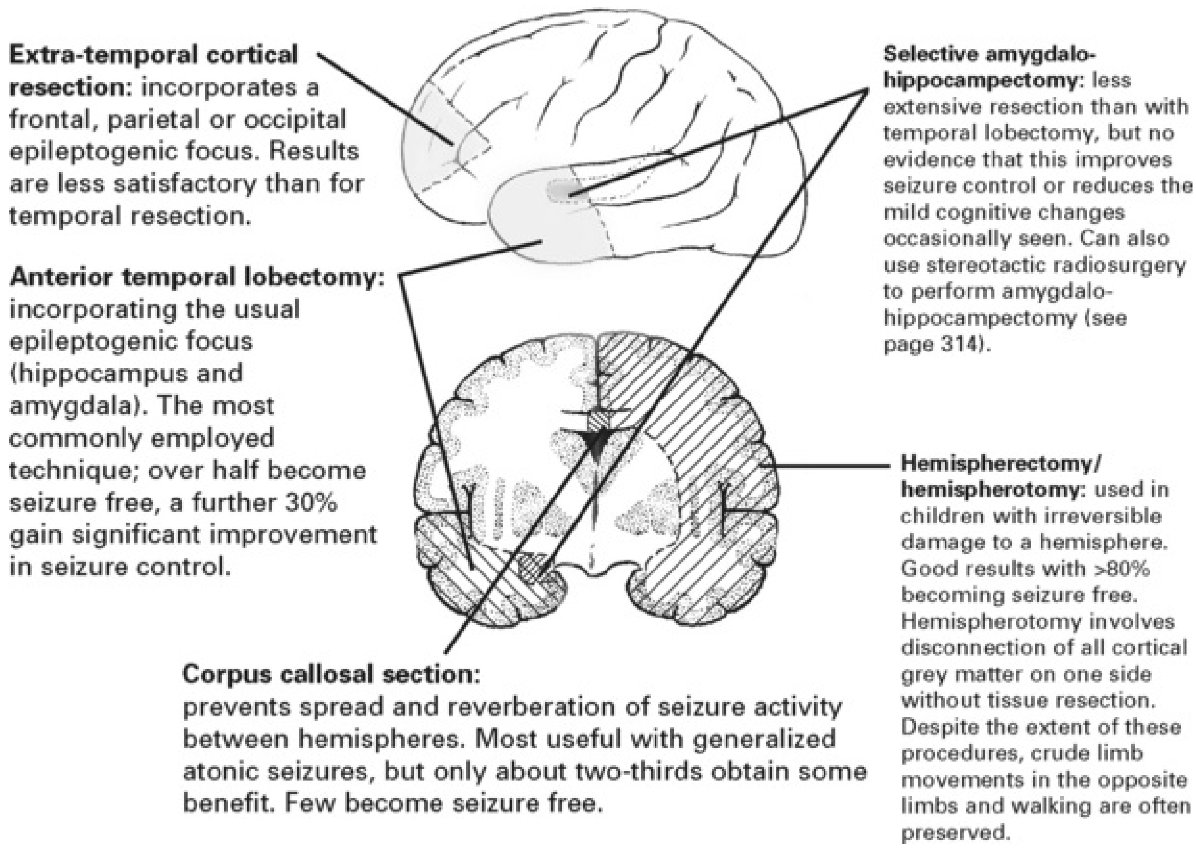
Operation is contraindicated in patients with severe mental retardation or with an underlying psychiatric problem.

Investigations: *Videotelemetry* (24–48 hr EEG), in some after electrode grid or depth electrode insertion and imaging with *MRI*, *SPECT* or *PET* scanning help identify the primary focus. Coronal MRI may show 'mesial temporal sclosis' or a structural abnormality (e.g. tumour, AVM, hamartoma or a neuronal migration disorder). The presence of such a lesion improves the chance of a good result with resective surgery.



T2 weighted MRI showing right mesial temporal sclerosis small, sclerotic (high T2 signal) hippocampus

Operative techniques



Vagal nerve stimulation (VNS): involves periodic stimulation of the left vagus nerve by an implanted stimulator. Considered in patients with intractable epilepsy not suited to the resective procedures. VNS appears to reduce neuronal excitability, but the exact mechanism remains obscure. About 30% of patients show a 50% seizure reduction within two years.

EPILEPSY – SPECIFIC ISSUES

WITHDRAWAL OF DRUG TREATMENT

Withdrawal of medication can be considered when the patient has been seizure free for 2 or more years. The decision to come off medication rests with the patient. There is a risk of recurrence (about 40% on average) with a temporary loss of driving licence (and risk of loss if seizures recur). The benefit depends on circumstances but will be greatest where there are drug side effects or in a woman planning pregnancy.

Several factors increase the likelihood of relapse of epilepsy after drug withdrawal:

– epilepsy associated with known cerebral disease

– response to starting treatment

– seizure type

– early childhood onset

EPILEPSY AND PREGNANCY

Seizures developing *during pregnancy*: The patient may present with the first seizure during pregnancy (when investigation is limited) or during the puerperium. Tumours and arteriovenous malformations can enlarge in pregnancy and produce such seizures; however, these causes are rare and most attacks are idiopathic. In late pregnancy seizures occur in association with hypertension and proteinuria as eclampsia. This is an emergency which needs to be managed in association with obsteticians. Intravenous magnesium sulphate and delivery is the recommended management.

When seizures present *post-partum* consider cortical venous thrombosis.

In patients with *established epilepsy* folic acid is recommended, preferably preconceptually, to reduce congenital malfomations (on little evidence). The risks of teratogenicity should be discussed with all women of childbearing ages *before they become pregnant*. Patients should be offered early detailed scans. Over 90% of pregnant women with epilepsy will deliver a normal child.

Strategies to best minimise the risk when nursing the baby need to be discussed, including any potential problems with breast feeding.

FEBRILE CONVULSIONS

Febrile convulsions occur in the immature brain as a response to high fever, probably as a result of water and electrolyte disturbance.

Usually occurs between 6 months and 3 years of age.

Long-term follow up suggests a liability to develop seizures in later life (unassociated with fever) especially in males, when seizures are prolonged and have focal features.

Treatment is aimed at preventing a prolonged seizure by sponging the patient and using rectal diazepam. The role of prophylaxis after one seizures is debatable.

SUDDEN UNEXPLAINED DEATH IN EPILEPSY (SUDEP)

The Standardised Mortality Ratio (SMR) compares mortality in a group with a specific illness to age and sex matched controls. The SMR is increased 2–3 times in epilepsy. When accidental death and suicide are excluded it appears that some persons with epilepsy die abruptly of no clear cause. Such deaths could be seizure related (cardiac arrhythmias/suffocation); autopsy is usually uninformative. A community-based study suggests 1 SUDEP/year/370 persons with epilepsy. Patients and carers should be compassionately informed of this small risk.

DRIVING AND EPILEPSY (DVLA UK regulations for type 1 licence (cars))

Off treatment	Isolated (single) seizure: 1 year off driving; if MRI and EEG are normal DVLA will consider reducing this to 6 months.

	Withdrawal of treatment: 6 months off driving (excluding period of drug withdrawal)
On treatment	Patients must be free of attacks (whilst awake) for 1 year
	Patients must be free of attacks whilst asleep for 1 year unless they have a 3 year history of sleep related attacks alone.

STATUS EPILEPTICUS

A succession of tonic/clonic convulsions, one after the other with a gap between each, is referred to as **serial epilepsy**.

When consciousness does not return between attacks the condition is then termed **status epilepticus**. This state may be life-threatening with the development of pyrexia, deepening coma and circulatory collapse.

Status epilepticus may occur with frontal lobe lesions, following head injury, on reducing drug therapy (especially phenobarbitone), with alcohol or other sedation withdrawal, drug intoxications (tricyclic antidepressants), infections, metabolic disturbances (hyponatraemia) or pregnancy.

TREATMENT

There is no completely satisfactory approach.

Death occurs in 5–10%

NOTE: ALL DOSE REGIMES APPLY TO ADULTS AND NOT TO CHILDREN

General

Establish an airway.

O_2 inhalation 10 litres/minute.

I.V. infusion: 500 ml 5% dextrose/0.9N saline.

Vital signs recorded regularly – especially temperature.

Prevent hyperthermia (sponging, etc.).

Monitor and treat acidosis

During assessment consider:

Potential causes of status (i.e. infection, intracranial event, metabolic factors)

Potential complications (i.e. aspiration, rhabdomyolysis and renal failure)

Specific

Pre-hospital: *Diazepam* 10–20 mg rectally or *midazolam* 10 mg buccally Effective for 10–20 minutes then seizures may return.

Early status: *Lorazepam* 4 mg i.v.

Beware respiratory depression with repeated injections.

If not controlled then proceed to longer acting drug.

Established status: *Phenytoin* 15–18 mg/kg or *Fosphenytoin* 15–20 mg/kg intravenously. Needs to be given at 50 mg/minute with cardiac monitoring.

At this point status should be controlled and oral maintenance therapy re-established.

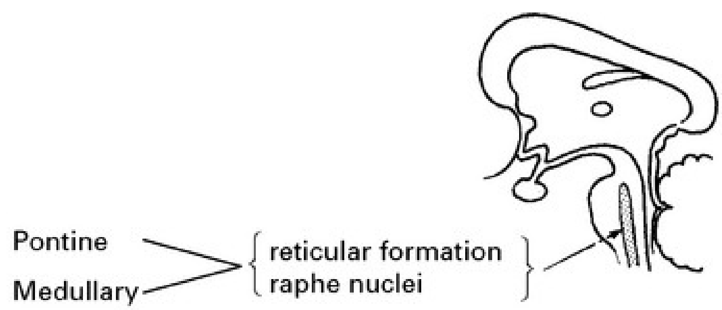
Refractory status: If control has not been achieved, the stage of refractory status is reached and general anaesthesia with Propofol should be commenced immediately (2 mg/kg i.v. bolus followed by continuous infusion of 5–10 mg/kg/h). Alternatively Thiopentone can be used (100–250 mg i.v. bolus over 20 sec with further 50 mg boluses every 2–3 min until control is achieved. This is then followed by continuous infusion.)

These treatments where possible should be used under EEG control to induce and maintain a 'burst suppression' pattern.

DISORDERS OF SLEEP

PHYSIOLOGY

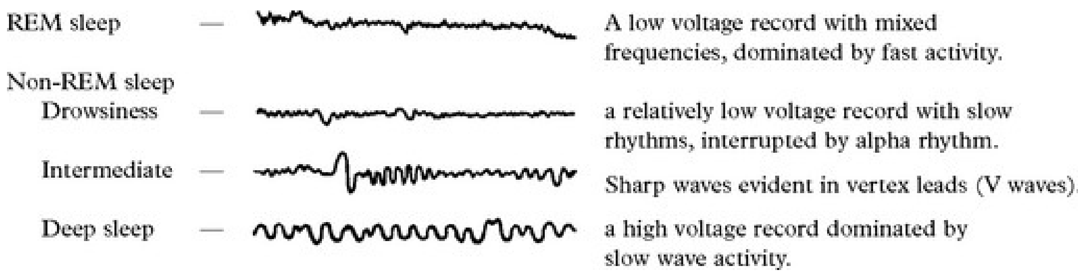
Sleep results from activity in certain sleep producing areas of the brain rather than from reduced sensory input to the cerebral cortex. Stimulation of these areas produces sleep; damage results in states of persistent wakefulness.



Two states of sleep are recognised:

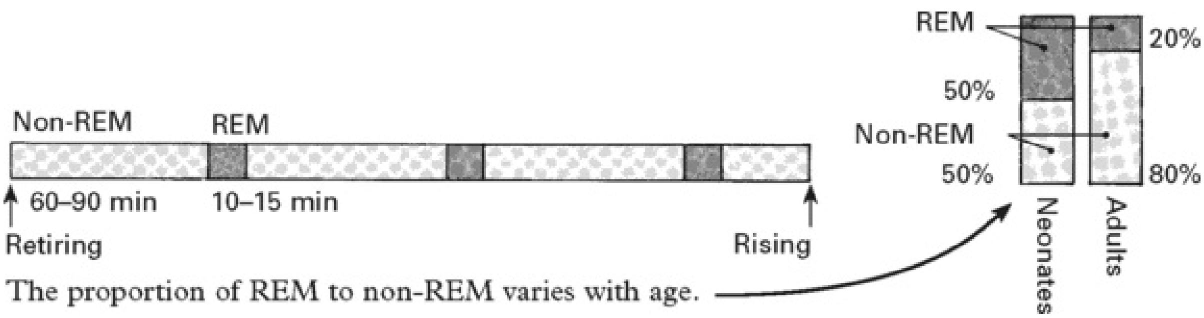
	1. **Rapid eye movement (REM) sleep**	2. **Non-rapid eye movement (non-REM) sleep**
Characterised by:	– Rapid conjugate eye movement – Fluctuation of temperature, BP, heart rate and respiration – Muscle twitching – Presence of dreams	– Absence of eye movement – Stability of temperature, BP, heart rate and respiration – Absence of muscle twitching – Absence of dreams
Originates in:	– Pontine reticular formation	– Midline pontine and medullary nuclei (raphe nuclei)
Mediated by:	– Noradrenaline (norepinephrine)	– Serotonin

The **electroencephalogram** shows characteristic patterns which correspond to the type and death of sleep.



The sleep pattern

In adults non-REM and REM sleep alternate throughout the night.



In view of the important role of serotonin and noradrenaline (norepinephrine) in sleep, it is understandable that drugs may affect the duration and/or content of sleep.

NARCOLEPSY AND CATAPLEXY

Narcolepsy: an irresistible desire to sleep in inappropriate circumstances and places. Attacks occur suddenly and are of brief duration unless patient remains undisturbed.

Cataplexy: sudden loss of postural tone. The patient crumples to the ground. Consciousness is preserved. Emotion – laughter or crying – can bring on an attack.

The narcolepsy/cataplexy tetrad

Only 10% of patients manifest the complex tetrad

Sleep paralysis: on awakening, the patient is unable to move. This may last for 2–3 minutes.

Hypnagogic hallucinations: vivid dreams or hallucinations occur as the patient falls asleep or occasionally when apparently awake.

Males are affected more than females. Prevalence 1:2000.

Onset is in adolescence/early adult life. The disorder is life long, but becomes less troublesome with age. It may have a familial incidence, or may occur after head injury, with multiple sclerosis, or with hypothalamic tumours. Pathological studies have found an early loss of hypothalamic neurons producing hypocretin/orexin, a wakefulness associated neurotransmitter.

Diagnosis

The suggestive history is supported by EEG studies. The multiple sleep latency test (MSLT) is diagnostic in showing onset of REM within 15 min of sleep onset in 2 of 4 naps (short sleeps).

Treatment

The non-amphetamine stimulant Modafinil, a wake promoting agent, reduces daytime sleepiness. Amphetamines are more potent but carry the risk of habituation. Sodium oxybate is a newer agent that improves night-time sleep and reduces cataplexy. Selegilene, metabolised in part to amphetamine, has a stimulant effect and may help. Clomipramine and SSRIs are also worth trying. Occasionally modifying lifestyle alone by 'cat-napping' is sufficient.

OTHER SLEEP DISORDERS (PARASOMNIAS)

NIGHT TERRORS (pavor nocturnus)

These occur in children, shortly after falling asleep and during deep to intermediate non-REM sleep. The child awakes in a state of fright with a marked tachycardia, yet in the morning cannot recollect the attack. Such attacks are not associated with psychological disturbance, are self limiting and if necessary will respond to diazepam.

NIGHTMARES

These occur during REM sleep. Drug or alcohol withdrawal promotes REM sleep and is often associated with vivid dreams.

SOMNAMBULISM (sleep walking)

Sleep walking varies from just sitting up in bed to walking around the house with the eyes open, performing complex major tasks. Episodes occur during intermediate or deep non-REM sleep. In childhood, somnambulism is associated with night terrors and bed wetting, but not with psychological disturbance. In adults, there is an increased incidence of psychoneurosis. Prevention of injury is important.

In **REM sleep-behaviour disorder** patients physically act out their dreams sometimes hurting themselves or their sleeping partner. This is associated with Parkinson's disease and other dementias and may be the earliest symptom.

SLEEP STARTS (HYPNIC JERKS)

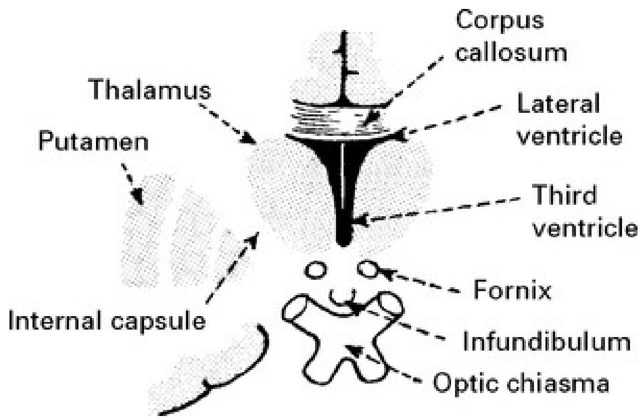
On entering sleep, sudden jerks of the arms or legs commonly occur and are especially frequent when a conscious effort is made to remain awake, *e.g.* during a lecture. This is a physiological form of myoclonus.

Other movement disorders in sleep: Restless legs, Dystonia, Bruxism (teeth grinding) and head banging.

HYPERSOMNIA

Rarely lesions (e.g. tumours or encephalitis) in the floor of the third ventricle may produce excessive sleepiness, often associated with diabetes insipidus.

Systemic disease such as hypothyroidism may result in hypersomnia, as may conditions which produce hypercapnia – chronic bronchitis, or primary muscle disease, *e.g.* dystrophia myotonica.



SLEEP APNOEA SYNDROMES

Respiratory rate fluctuates during REM sleep with occasional short episodes of apnoea. These are normal physiological events and are brief and infrequent.

Prolonged sleep apnoea results from central reduction of respiratory drive, a mechanical obstruction of the airway or a mixture of both.

Central causes:	**Mechanical causes:**
Brain stem medullary infarction or following cervical/foramen magnum surgery.	Obesity. Tonsillar enlargement.
	Myxoedema. Acromegaly.

When breathing ceases, the resultant hypercapnia and hypoxia eventually stimulate respiration.

Patients may present with daytime sleepiness, nocturnal insomnia and early morning headache. Snoring and restless movements are characteristic. In severe cases of sleep apnoea, hypertension may develop with right heart failure secondary to pulmonary arterial hypertension.

Polycythaemia and left heart failure may ensue.

Evaluation requires sleep oximetry and video recording with low level illumination. Fall in oxygen saturation may be as much as 50%.

Treatment depends on aetiology. Mechanical airway obstruction should be relieved; drugs such as theophylline are occasionally helpful. Continuous positive airway pressure (CPAP) applied to the nose may help. Surgical reconstruction of palate and oropharynx is offered in extreme cases.

The *Pickwickian syndrome*: sleep apnoea associated with obesity, named after the Dickens' fat boy who repeatedly fell asleep.

INSOMNIA

The most common sleep disorder, difficult to evaluate and of multiple causation including psychiatric, alcohol, drug related or due to systemic illness. Treatment depends on cause *e.g.* antidepressant.

HIGHER CORTICAL DYSFUNCTION

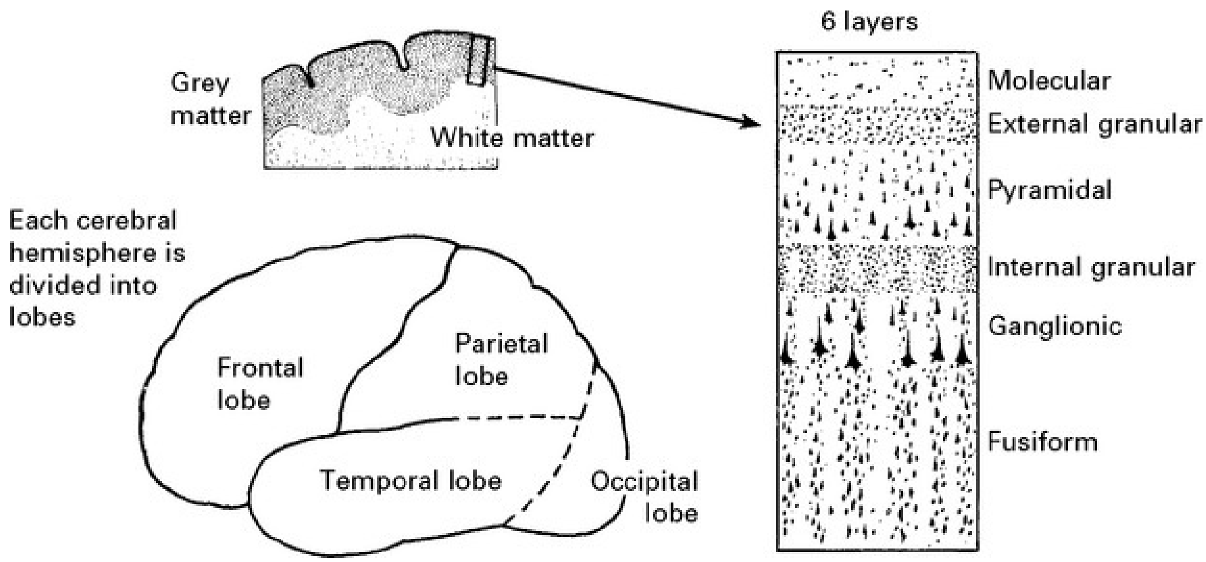
Specific parts of the cerebral hemispheres are responsible for a certain aspect of function. In normal circumstances these functions are integrated and the patient operates as a whole. Damage to part of the cortex will result in a characteristic disturbance of function. Interruption by disease of 'connections' between one part of the cortex and another will 'disconnect' function.

GENERAL ANATOMY

Brodmann, on the basis of histological differences, divided the cortex into 47 areas. Knowledge of these areas is not practical, though they are referred to often in some texts.

Six layers can be recognized in the cerebral cortex superficial to the junction with the underlying white matter.

The relative preponderance of each layer varies in different regions of the cortex and appears to be related to function.



The frontal motor cortex, dominated by pyramidal rather than granular layers, is termed the AGRANULAR CORTEX.

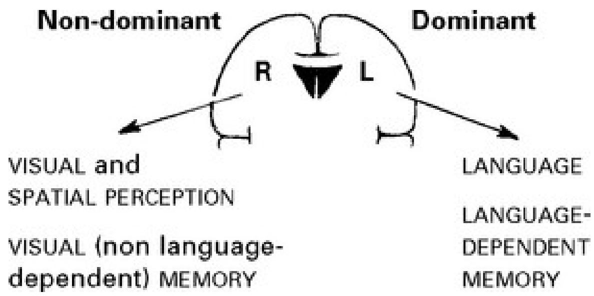
The parietal sensory cortex, dominated by granular layers, is termed the GRANULAR CORTEX.

The largest cells of the granular cortex are the giant cells of Betz. These give rise to some of the motor fibres of the corticospinal tract.

RIGHT AND LEFT HEMISPHERE FUNCTION

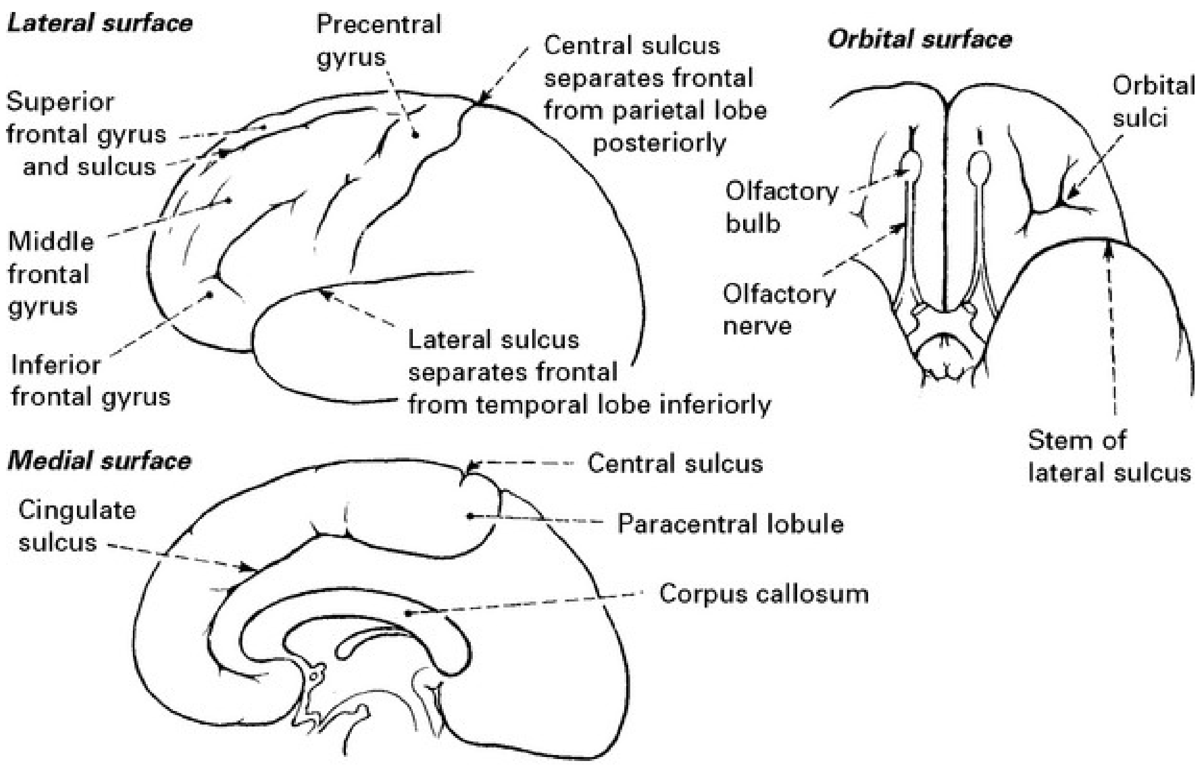
Unilateral brain damage reveals a difference in function between hemispheres. The left hemisphere is 'dominant' in right-handed people. In left-handed subjects the left hemisphere is dominant in the majority (up to 75%).

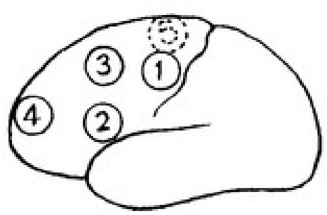
Hand preference may be hereditary, but in some cases disease of the left hemisphere in early life determines left-handedness.



Hemisphere dominance may be demonstrated by the injection of sodium amytal into the internal carotid artery. On the dominant side this will produce an arrest of speech for up to 30 seconds – the WADA TEST. Such a test may be important before temporal lobectomy for epilepsy when handedness/hemisphere dominance is in doubt.

FRONTAL LOBES





FRONTAL LOBE FUNCTION

1. Precentral gyrus – motor cortex contralateral movement – face, arm, leg, trunk.

2. Broca's area – dominant hemisphere – expressive centre for speech.

3. Supplementary motor area – contralateral head and eye turning.

4. Prefrontal areas – 'personality', initiative.

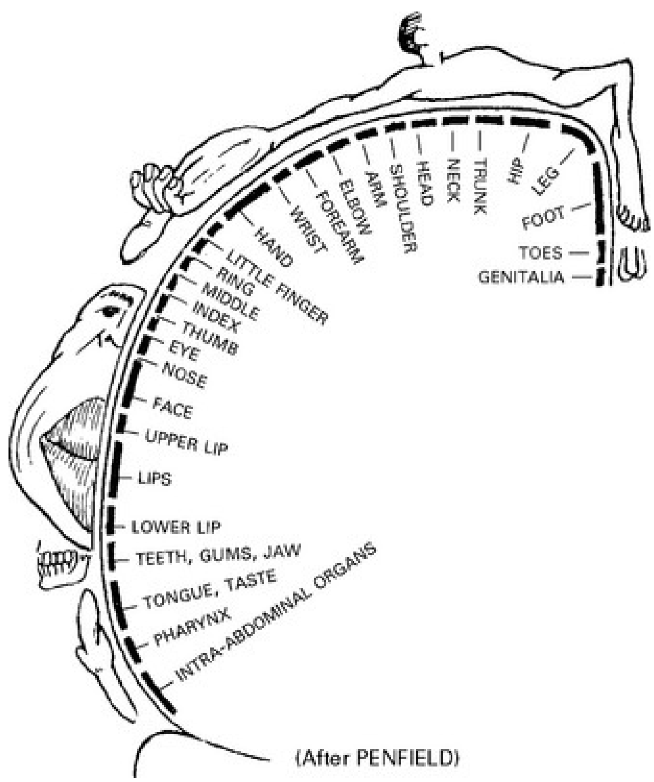
5. Paracentral lobule – cortical inhibition of bladder and bowel voiding.

IMPAIRMENT OF FRONTAL LOBE FUNCTION

1. **Precentral gyrus** Monoplegia or hemiplegia depending on extent of damage.

2. **Broca's area** (inferior part of dominant frontal lobe) Results in Broca's dysphasia (see page 124) (motor or expressive).

3. **Supplementary motor area** Paralysis of head and eye movement to opposite side. Head turns 'towards' diseased hemisphere and eyes look in the same direction.



4. **Prefrontal areas** (the vast part of the frontal lobes anterior to the motor cortex as well as undersurface – orbital – of frontal lobes)

Damage is often bilateral, *e.g.* infarction, following haemorrhage from anterior communicating artery aneurysm, neoplasm, trauma or anterior dementia, resulting in a change of personality with antisocial behaviour/loss of inhibitions.

Three pre-frontal syndromes are recognised

Orbitofrontal syndrome	*Frontal convexity syndrome*	*Medial frontal syndrome*
Disinhibition	Apathy	Akinetic
Poor judgement	Indifference	Incontinent
Emotional lability	Poor abstract thought	Sparse verbal output

Prefrontal lesions are also associated with:

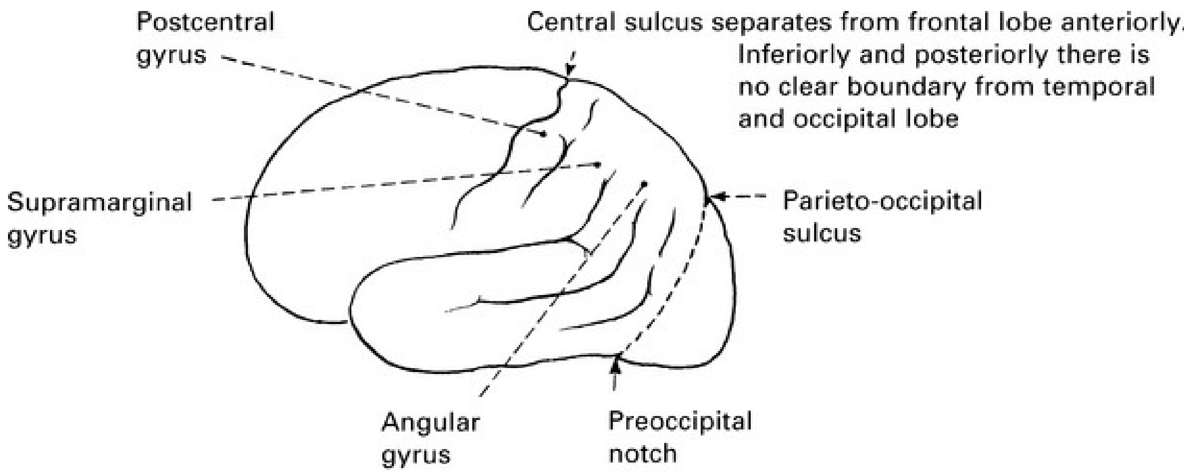
1. Primitive reflexes – grasp, pout, *etc.* (see page 127).
2. Disturbance of gait – ‘gait apraxia’.
3. Resistance to passive movements of the limbs – paratonia.

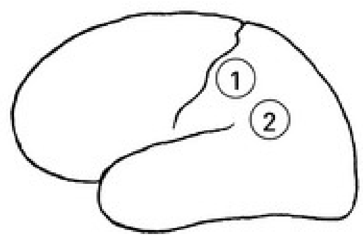
Unilateral lesions may show minor degrees of such change.

5. **Paracentral lobule**

Damage to the posterior part of the superior frontal gyrus results in incontinence of urine and faeces – ‘loss of cortical inhibition’. This is particularly likely with ventricular dilatation and is an important symptom of normal pressure hydrocephalus.

PARIETAL LOBES





PARIETAL LOBE FUNCTION

1. **Postcentral gyrus** (*granular cortex*)

The sensory cortex (representation similar to the motor cortex) receives afferent pathways for appreciation of posture, touch and passive movement.

2. **Supramarginal and angular gyri** (*dominant hemisphere*) make up part of Wernicke's language area.

This is the receptive area where auditory and visual aspects of comprehension are integrated.

The **non-dominant** parietal lobe is important in the concept of body image and the awareness of the external environment. The ability to construct shapes, *etc.* results from such visual/proprioceptive skills.

The **dominant** parietal lobe is implicated in the skills of handling numbers/calculation.

The **visual pathways** – the fibres of the optic radiation (lower visual field) – pass deep through the parietal lobe.

IMPAIRMENT OF PARIETAL LOBE FUNCTION

Disease of **either dominant or non-dominant** sensory cortex (postcentral gyrus) will result in contralateral disturbance of cortical sensation:

Postural sensation disturbed.

Sensation of passive movement disturbed.

Accurate localization of light touch may be disturbed.

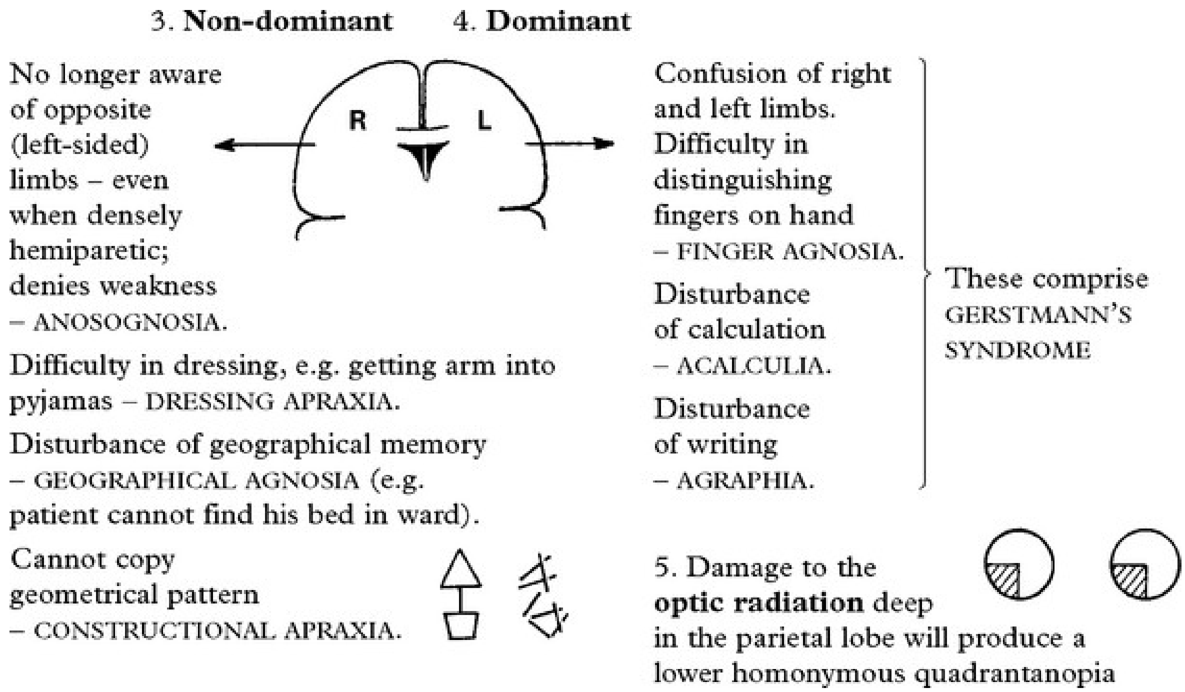
Discrimination between one and two points (normally 4 mm on finger tips) is lost.

Appreciation of size, shape, texture and weight may be affected, with difficulty in distinguishing coins placed in hand, *etc.* (astereognosis).

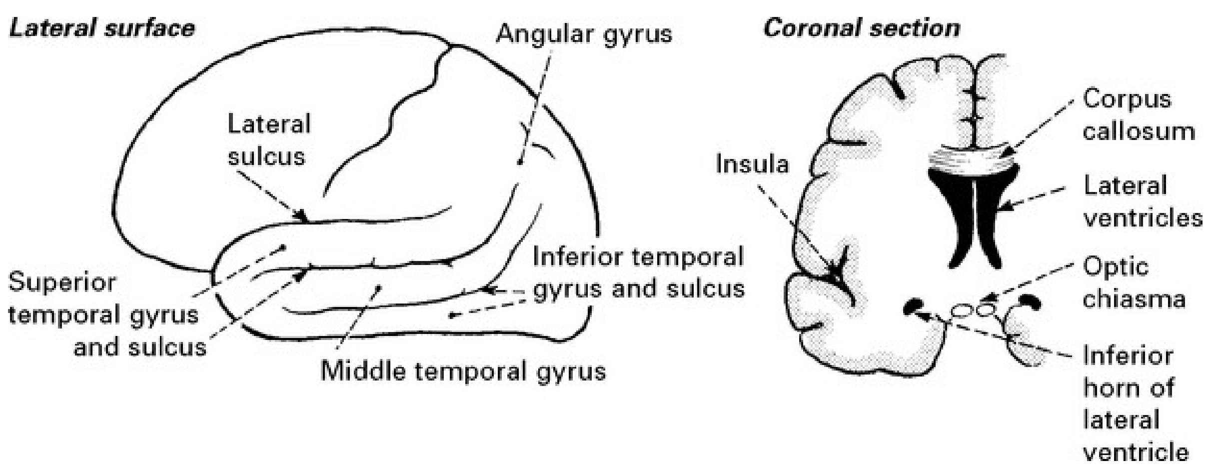
Perceptual rivalry (sensory inattention) is characteristic of parietal lobe disease. Presented with two stimuli, one applied to each side (e.g. light touch to the palm of the hand) simultaneously, the patient is only aware of that one contralateral to the normal parietal lobe.

As the gap between application of stimuli is increased (approaching 2–4 seconds) the patient becomes aware of both.

2. **Supramarginal and angular gyri** – Wernicke's dysphasia (see page 124).



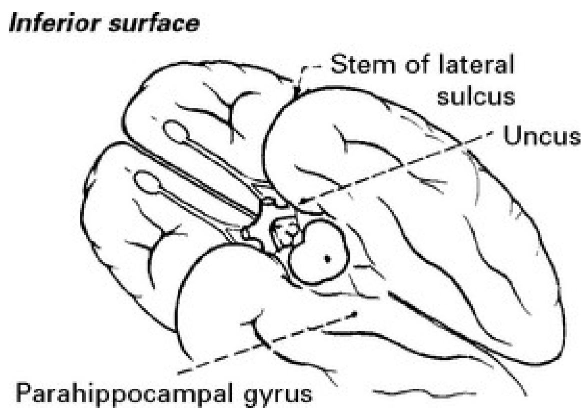
TEMPORAL LOBES



Anteriorly, the temporal lobe is separated from the frontal lobe by the lateral sulcus. Posteriorly and superiorly, separation from occipital and parietal lobes is less clearly defined.

The lateral sulcus is deep and contains 'buried' temporal lobe. The buried island of cortex is referred to as the INSULA.

The temporal lobe also has a considerable inferior and medial surface in contact with the middle fossa.



TEMPORAL LOBE FUNCTION

1. The **auditory cortex** lies on the upper surface of the superior temporal gyrus, buried in the lateral sulcus (Heschl's gyrus).

The **dominant** hemisphere is important in the hearing of language.

The **non-dominant** hemisphere is important in the hearing of sounds, rhythm and music. Close to the auditory cortex labyrinthine function is represented.

2. The **middle and inferior temporal gyri** are concerned with learning and memory (see later).

3. The **limbic lobe**: the inferior and medial portions of the temporal lobe, including the hippocampus and parahippocampal gyrus.

The sensation of olfaction is mediated through this structure as well as emotional/affective behaviour.

Olfactory fibres terminate in the uncus.

The limbic lobe or system also incorporates inferior frontal and medial parietal structures and will be discussed later.

4. The **visual pathways** pass deep in the temporal lobe around the posterior horn of the lateral ventricle.

IMPAIRMENT OF TEMPORAL LOBE FUNCTION

1. **Auditory cortex**

Cortical deafness: Bilateral lesions are rare but may result in complete deafness of which the patient may be unaware.

Lesions which involve surrounding association areas may result in difficulty in hearing spoken words (dominant) or difficulty in appreciating rhythm/music (non-dominant) – *AMUSIA*. Auditory hallucinations may occur in temporal lobe disease.

2. **Middle and inferior temporal gyri**

Disturbance or memory/learning will be discussed later.

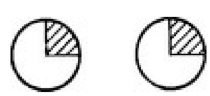
Disordered memory may occur in complex partial seizures either after the event – postictal amnesia – or in the event – déjà vu, jamais vu.

3. **Limbic lobe** damage may result in:

Olfactory hallucination with complex partial seizures.

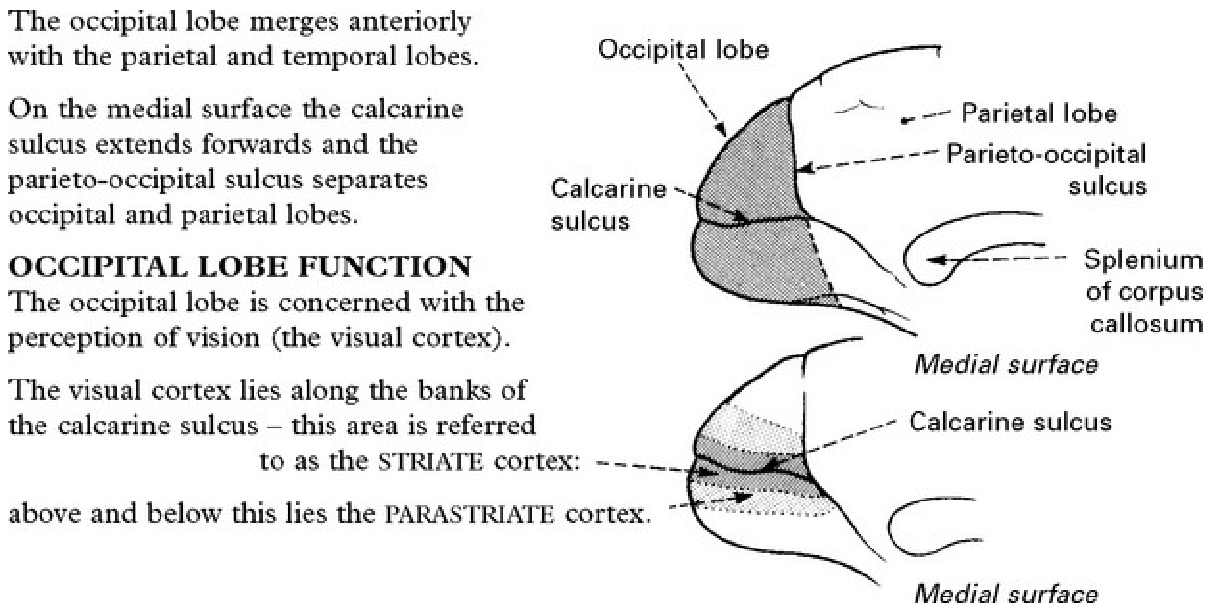
Aggressive or antisocial behaviour.

Inability to establish new memories (see later).



4. Damage to **optic radiation** will produce an upper homonymous quadrantanopia. Dominant hemisphere lesions are associated with Wernicke's dysphasia.

OCCIPITAL LOBE



The *striate* cortex is the primary visual cortex and when stimulated by visual input relays information to the *parastriate* – association visual cortex. This, in turn, connects with the parietal, temporal and frontal lobes both on the same side and on the opposite side (through the posterior part of the corpus callosum) so that the meaning of a visual image may be interpreted, remembered, *etc*.

The visual field is represented upon the cortex in a specific manner (page 140).

IMPAIRMENT OF OCCIPITAL LOBE FUNCTION

A cortical lesion will result in a homonymous hemianopia with or without involvement of the macula, depending on the posterior extent of the lesion.

When only the occipital pole is affected, a central hemianopia field defect involving the macula occurs with a normal peripheral field of vision.

Cortical blindness

Extensive bilateral cortical lesions of the striate cortex will result in cortical BLINDNESS. In this, the pupillary light reflex is normal despite the absence of conscious perception of the presence of illumination (light reflex fibres terminate in the midbrain).

Anton's syndrome

Involvement of both the striate and the parastriate cortices affects the interpretation of vision. The patient is unaware of his visual loss and denies its presence. This denial in the presence of obvious blindness characterizes Anton's syndrome.

Cortical blindness occurs mainly in vascular disease (posterior cerebral artery), but also following hypoxia and hypertensive encephalopathy or after surviving tentorial herniation.

Balint's syndrome

Inability to direct voluntary gaze, associated with visual agnosia (loss of visual recognition) due to bilateral parieto-occipital lesions.

Visual hallucinations are common in migraine when the occipital lobe is involved; also in epilepsy when the seizure source lies here.

Hallucinations of occipital origin are elementary – unformed – appearing as patterns (zig-zags, flashes) and fill the hemianopic field, whereas hallucinations of temporal lobe origin are formed, complex and fill the whole of the visual field.

Visual illusions also may occur as a consequence of occipital lobe disease. Objects appear smaller (MICROPSIA) or larger (MACROPSIA) than reality. Distortion of a shape may occur or disappearance of colour from vision.

These illusions are more common with non-dominant occipital lobe disease.

Prosopagnosia: the patient, though able to see a familiar face, *e.g.* a member of the family, cannot name it. This is usually associated with other disturbances of 'interpretation' and naming with intact vision such as colour agnosia (recognition of colours and matching of pairs of colours). Bilateral lesions at occipito-temporal junction are responsible.

APRAXIA

A loss of ability to carry out skilled movement despite adequate understanding of the task and normal motor power.

Constructional and dressing apraxia: See page 113, non-dominant parietal disease.

Gait apraxia: Difficulty in initiating walking – frontal lobe/anterior corpus callosum disease.

Oculomotor apraxia: Impaired voluntary eye movement – parieto-occipital disease.

Ideamotor apraxia: Separation of idea of movement from execution – cannot carry out motor command but can perform the required movement under different circumstances – dominant hemisphere (see later).

Ideational apraxia: Inability to carry out a sequence of movements each of which can be performed separately – frontal lobe disease.

HIGHER CORTICAL DYSFUNCTION – DISCONNECTION SYNDROMES

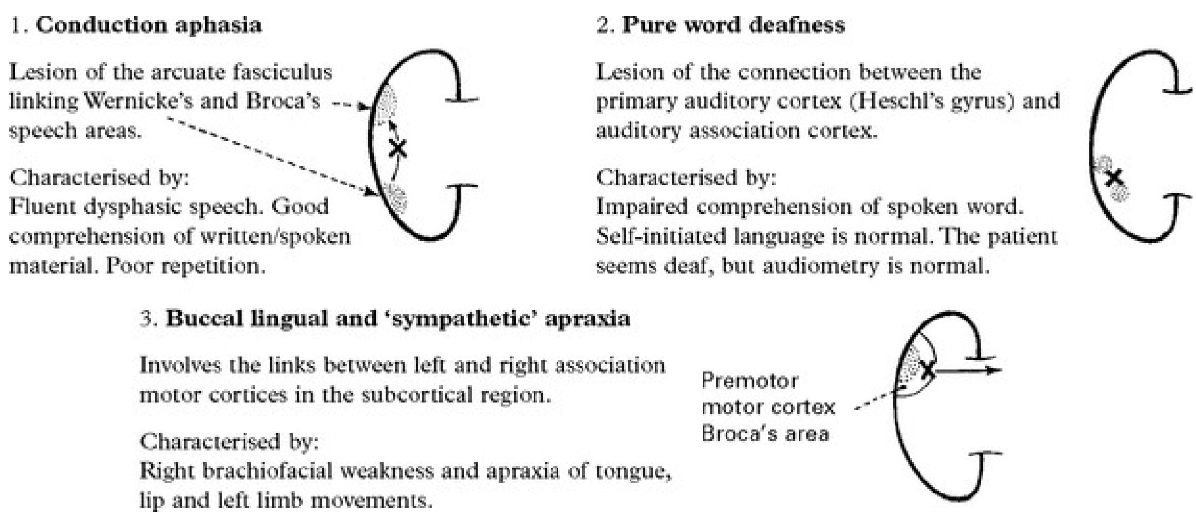
Cortical function is described, on the previous pages, ‘lobe by lobe’. These functions integrate by means of connections between hemispheres and lobes. Lesions of these connecting pathways disorganise normal function, resulting in recognizable syndromes – the disconnection syndromes. APRAXIA is a feature of some of these disorders.

The connecting pathways may be divided into:

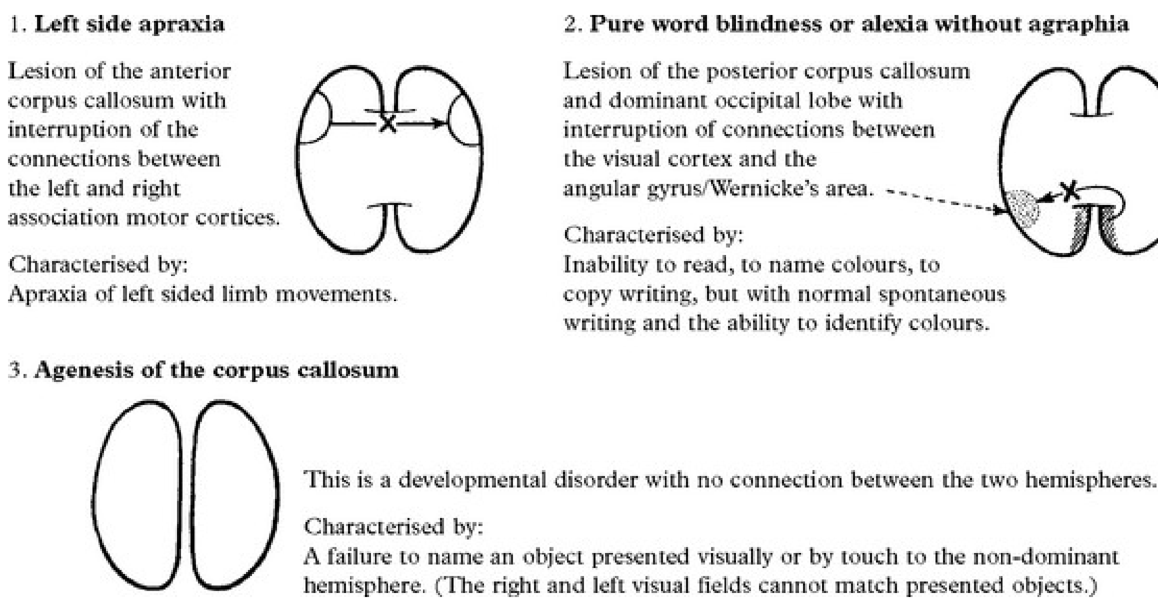
Intra hemispheric: lying in the subcortical white matter and linking parts of the same hemisphere.

Inter hemispheric: traversing the corpus callosum and linking related parts of the two hemispheres.

THE INTRAHEMISPHERIC DISCONNECTION SYNDROMES



THE INTERHEMISPHERIC DISCONNECTION SYNDROMES



HIGHER CORTICAL FUNCTION MEMORY

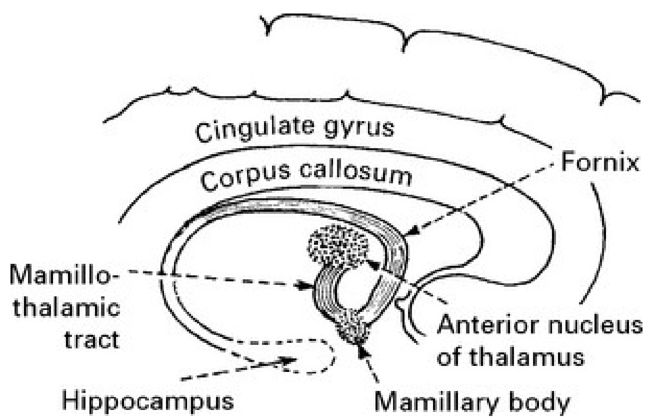
Normal memory involves the recognition, registering and cataloguing of a stimulus – *acquisition*, as well as the skill of appropriate recall – *retrieval*.

Verbal memory:	refers to material presented in the verbal form.
Visual memory:	denotes material presented without words or verbal mediation.
Episodic memory:	
Short term:	immediate recall of a short message.
Long term:	retrieval of recent or remote events.
Semantic memory:	refers to long established factual knowledge.

Disordered memory may be confused with disturbances of attention, motivation and concentration and requires detailed neuropsychological examination to properly assess.

THE ANATOMICAL BASIS OF MEMORY

The structures of the limbic system involved in the memory process are inferred from the pathological examination of diseases that disorder function. The *hippocampus*, a deep structure in the temporal lobe, ridges the floor of the lateral ventricle. Fimbriae of the hippocampus connect this structure to the *fornix*. There appears to be a loop from hippocampus → fornix → mamillary body → thalamus → cingulate gyrus → back to hippocampus.



	Mamillary bodies	Thalamus	Orbito-frontal cortex	Medial temporal cortex/ hippocampus	Fornix
Korsakoff 's	+	+			
Head trauma		+	+	+	+
Stroke		+		+	+
Encephalitis				+	+
Anoxia				+	+
Metabolic				+	+
Temporal lobectomy				+	
3rd ventricular operations					+

TESTS OF MEMORY (see examination, page 8)

These aim to distinguish loss of immediate, recent or remote memory.

Disorders may be further classified into those which affect memories established before the injury or damage – RETROGRADE AMNESIA – and those which affect memory of events following the injury or damage – ANTEROGRADE or POST-TRAUMATIC AMNESIA.

DISORDERS OF MEMORY

THE AMNESIC SYNDROME is characterised by –

Retrograde amnesia – impairment of memory for events that antedate illness or injury

Anterograde amnesia – inability to learn new verbal or non verbal information from onset of the illness or injury

Intact retrieval of old information

Intact intellectual function

Intact personality

Tendency to confabulate

CAUSES

Korsakoff's syndrome: results from – alcoholism, encephalitis, and head injury

Lesions occur within the thalamus and the mamillary bodies. Commonly associated with confabulation – a false rationalization of events and circumstances.

Post-traumatic amnesia: after trauma, retrograde amnesia may span several years, but with recovery, this gradually diminishes. The duration of post-traumatic amnesia on the other hand remains fixed and relates directly to the severity of the injury.

Amnesic stroke: bilateral medial temporal lobe infarction from a posterior circulation stroke is usually associated with hemiplegia and visual disturbance or loss *e.g.* Anton's or Balint's syndrome (page 115).

Amnesia with tumours: tumours that compress thalamic structures or the fornix may produce amnesia – *e.g.* colloid cyst of the 3rd ventricle.

Temporal lobectomy: amnesia will only occur if function in the unoperated temporal lobe is abnormal. Pre-operative assessment during a unilateral carotid injection of sodium amytal minimises this risk.

Transient global amnesia: typically a single episode lasting between 1 and 10 hours; the patient is bewildered, typically repeatedly asking the same questions, but with clear consciousness and often able to carry out complex tasks such as driving or cooking. Benign phenomenon probably associated with migraine. May be triggered by stress or exercise.

Transient epileptic amnesia: recurrent episodes of amnesia lasting 15 minutes to 1 hour, often on waking.

Psychogenic amnesia: affects overlearned and personally relevant aspects of memory *e.g.* 'What is my name?', while less well learned memory remains unaffected. Clinically evident acute mental stress may precipitate this. This inadequate defence mechanism suggests a serious underlying psychiatric or personality disorder.

DISORDERS OF MEMORY RETRIEVAL

Senescence – as part of normal aging, rapid retrieval of stored memory becomes defective.

Depression – impaired memory is a common complaint in depressive illness. The disorder is one of motivation and concentration.

Subcortical dementia – This will be described later (page 126). The major abnormality is that of a slowed (but correct) response rate to questions of memory function.

NB DEMENTIA, TUMOURS and CEREBROVASCULAR DISEASE are all often associated with memory loss but this is usually combined with evidence of more widespread disordered cognitive function.

DISORDERS OF SPEECH AND LANGUAGE

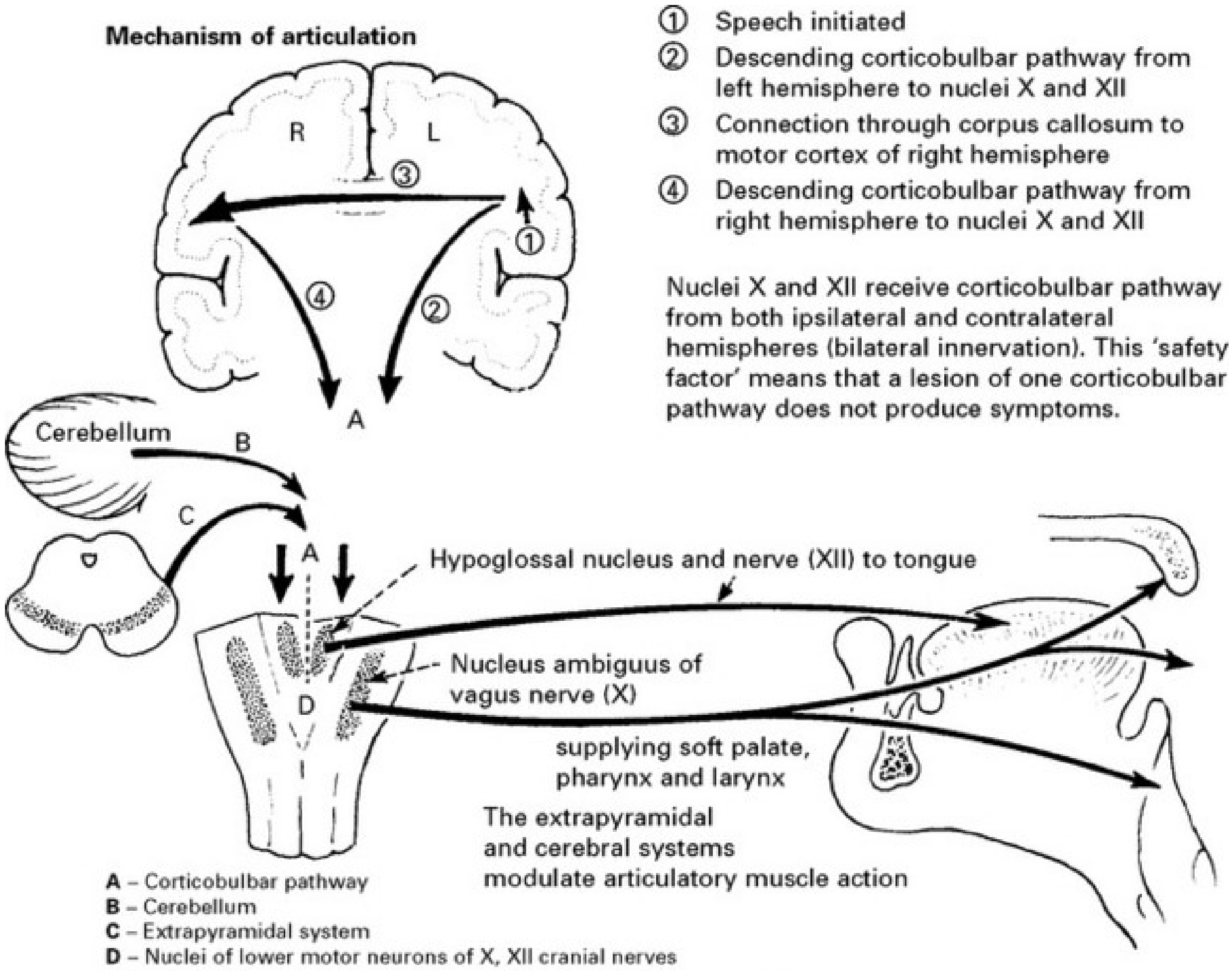
Introduction

Disturbed speech and language are important symptoms of neurological disease. The two are not synonymous. Language is a function of the dominant cerebral hemisphere and may be divided into (a) *emotional* – the instinctive expression of feelings representing the earliest forms of language acquired in infancy and (b) *symbolic or prepositional* – conveying thoughts, opinion and concepts. This language is acquired over a 20-year period and is dependent upon culture, education and normal cerebral development.

An understanding of disorders of speech and language is essential, not just to the clinical diagnosis but also to improve communication between patient and doctor. All too often patients with language disorders are labelled 'confused' as a consequence of superficial evaluation.

DYSARTHRIA

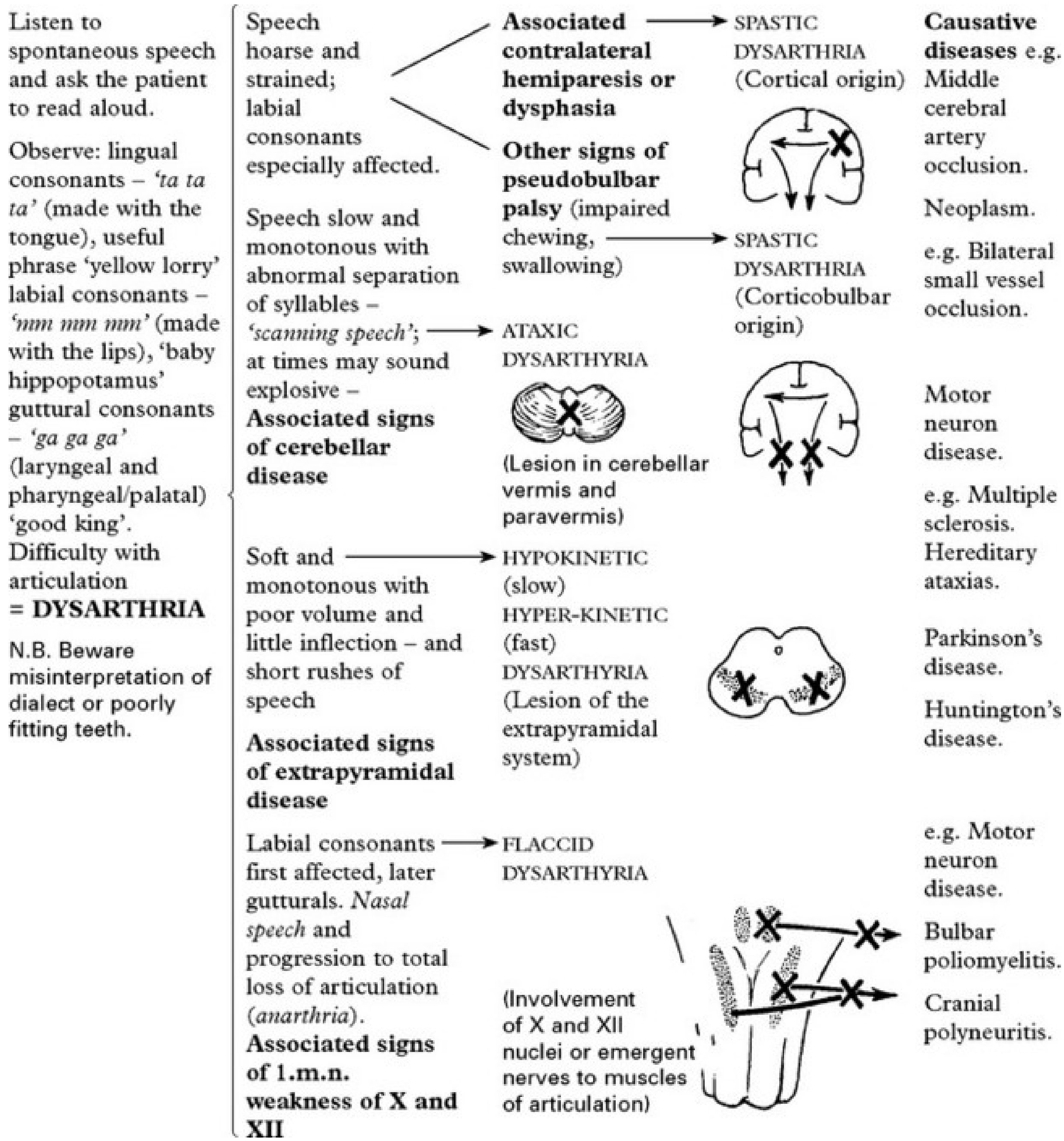
Dysarthria is a *disturbance of articulation* in which the content of speech – language – is unaffected.



Muscles of expression, innervated by the facial nerve, play an additional role in articulation and weakness also results in dysarthria.

DISORDERS OF SPEECH – DYSARTHRIA

DIAGNOSTIC APPROACH



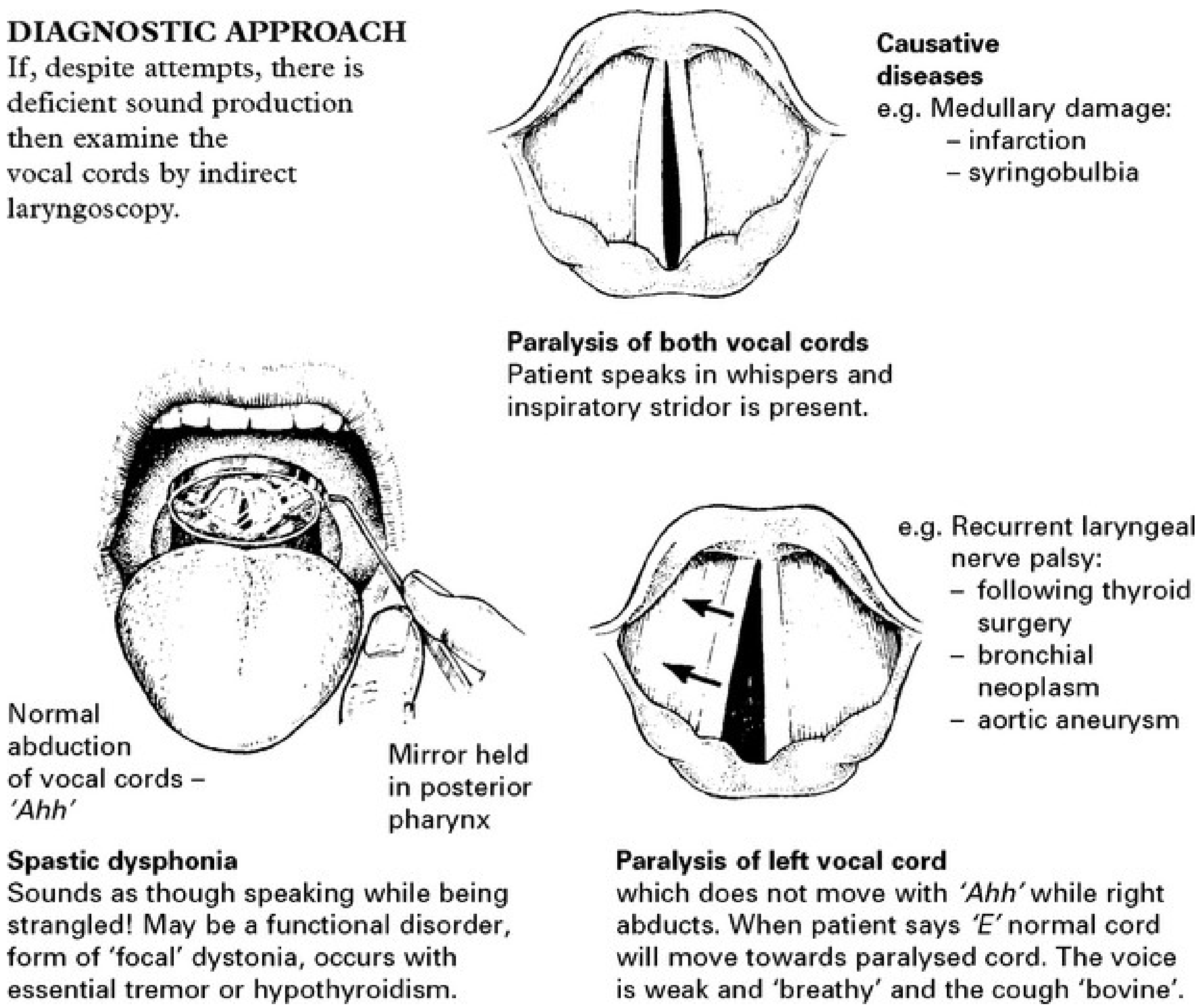
Many diseases affect multiple sites and a 'mixed' dysarthria occurs.

For example, multiple sclerosis with corticobulbar and cerebellar involvement will result in a mixed spastic/ataxic dysarthria.

DISORDERS OF SPEECH – DYSPHONIA

Sound is produced by the passage of air over the vocal cords.

Respiratory disease or vocal cord paralysis results in a disturbance of this facility – **dysphonia**. A complete inability to produce sound is referred to as aphonia. Dysarthria often co-exists.



OTHER DISORDERS OF SPEECH

Mutism: An absence of any attempt at oral communication. It may be associated with bilateral frontal lobe or third ventricular pathology (see Akinetic mutism).

Echolalia: Constant repetition of words or sentences heard in dementing illnesses.

Palilalia: Repetition of last word or words of patient's speech. Heard in extrapyramidal disease.

Logorrhoea: Prolonged speech monologues; associated with Wernicke's dysphasia.

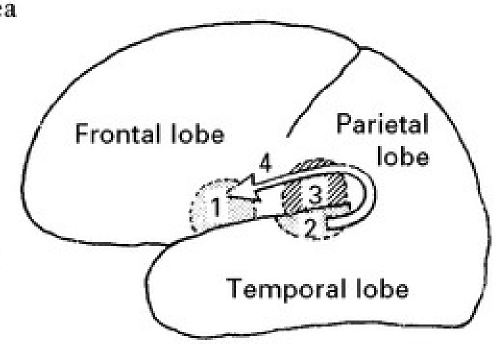
DISORDERS OF SPEECH – DYSPHASIA

Dysphasia is an acquired loss of production or comprehension of spoken and/or written language secondary to brain damage.

Hand preference is associated with 'hemisphere dominance' for language. In right-handed people the left hemisphere is dominant; in left-handed people the left hemisphere is dominant in most, though 25% have a dominant right hemisphere.

The cortical centres for language reside in the dominant hemisphere.

1. Broca's area
Executive or motor area for the production of language – lies in the inferior part of the frontal lobe on the lateral surface of the cerebral hemisphere abutting the mouth of the Sylvian fissure.



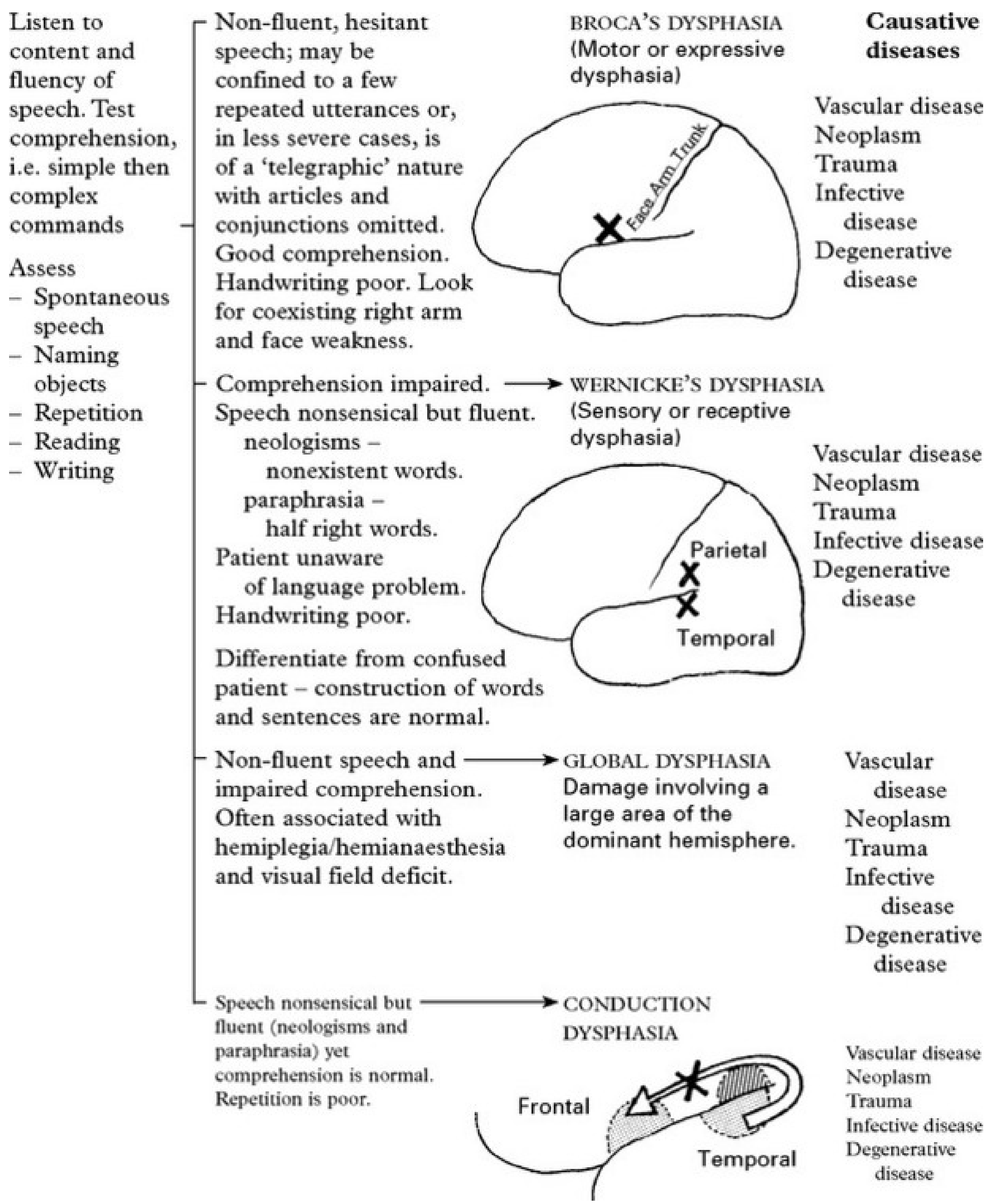
2 and **3. Receptive areas**
Here the spoken word is understood and the appropriatae reply or action initiated. These areas lie at the posterior end of the Sylvian fissure on the lateral surface of the hemisphere.

The temporal lobe receptive area (**2**) lies close to the auditory cortex of the transverse gyrus of the temporal lobe. The parietal lobe receptive area (**3**) lies within the angular gyrus.

Receptive and expressive areas must be linked in order to integrate function. The link is provided by (**4**), the **arcuate fasciculus**, a fibre tract which runs forward in the subcortical white matter.

Dysphasia may develop as a result of vascular, neoplastic, traumatic, infective or degenerative disease of the cerebrum when language areas are involved.

DIAGNOSTIC APPROACH



DEMENTIAS

Definition

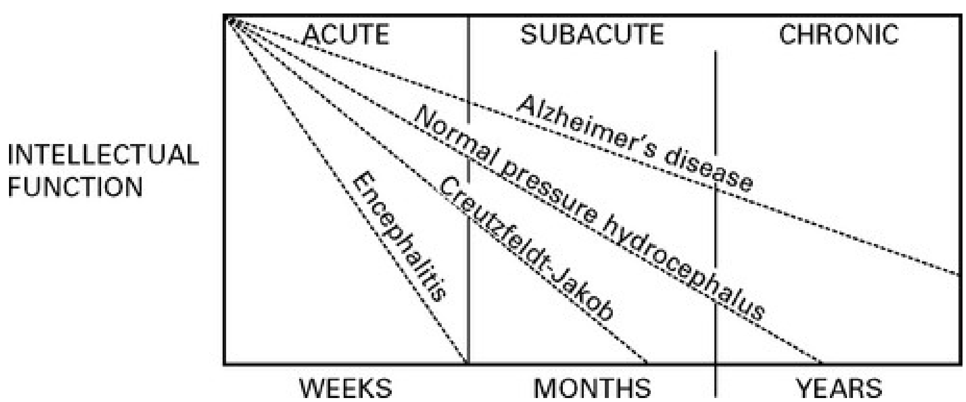
Progressive deterioration of intellect, behaviour and personality as a consequence of diffuse disease of the cerebral hemispheres, maximally affecting the cerebral cortex and hippocampus.

Distinguish from *delirium* which is an acute disturbance of cerebral function with impaired conscious level, hallucinations and autonomic overactivity as a consequence of toxic, metabolic or infective conditions.

Dementia may occur at any age but is more common in the elderly, increasing with age (approximate prevalence 1% in 60s, 5% in 70s, 15% in 80s). Dementia is a symptom of disease rather than a single disease entity. When occurring under the age of 65 years it is labelled 'presenile' dementia. This term is artificial and does not suggest a specific aetiology.

Clinical course:

The rate of progression depends upon the underlying cause.



The duration of history helps establish the cause of dementia; Alzheimer's disease is slowly progressive over years, whereas encephalitis may be rapid over weeks. Dementia due to cerebrovascular disease appears to occur 'stroke by stroke'.

All dementias show a tendency to be accelerated by change of environment, intercurrent infection or surgical procedures.

Development of symptoms

Introspective. Unsure of self. → Difficulty in coping with work and ordinary routine (retained insight). → Loss of insight, behavioural changes, Loss of inhibition.

↓

Long-term care. Cannot be left unattended. → Mutism, incontinence and DEATH

This initial phase of dementia may be inseparable from the *pseudodementia* of *depressive illness*.

DEMENTIAS – CLASSIFICATION

Based on cause

Degenerative
- *Pure dementia*
  - Alzheimer's disese (~60%)
  - Frontotemporal/Pick's disease (~5%)
- *Dementia plus syndromes*
  - Dementia with Lewy bodies (~10%)
  - Parkinson's disease with dementia
  - Corticobasal degeneration
  - Progressive supranuclear palsy
  - Huntington's disease

Cerebrovascular diseases (20%)
- Multiple infarct dementia
- Subcortical ischaemic vascular dementia
- Chronic subdural haematomas
- Cerebral amyloid angiopathy
- CADASIL (see later)

Structural disorders
- Normal pressure hydrocephalus

Infections
- Prion disesase (Creutzfeld-Jakob disease)
- Syphilis
- HIV
- Progressive multifocal leukoencephalopathy

Nutritional
- Wernicke Korsakoff (thiamine deficiency)
- B_{12} deficiency

Metabolic
- Hepatic disease
- Thyroid disease

Chronic inflammatory
- Multiple sclerosis
- Vasculitis

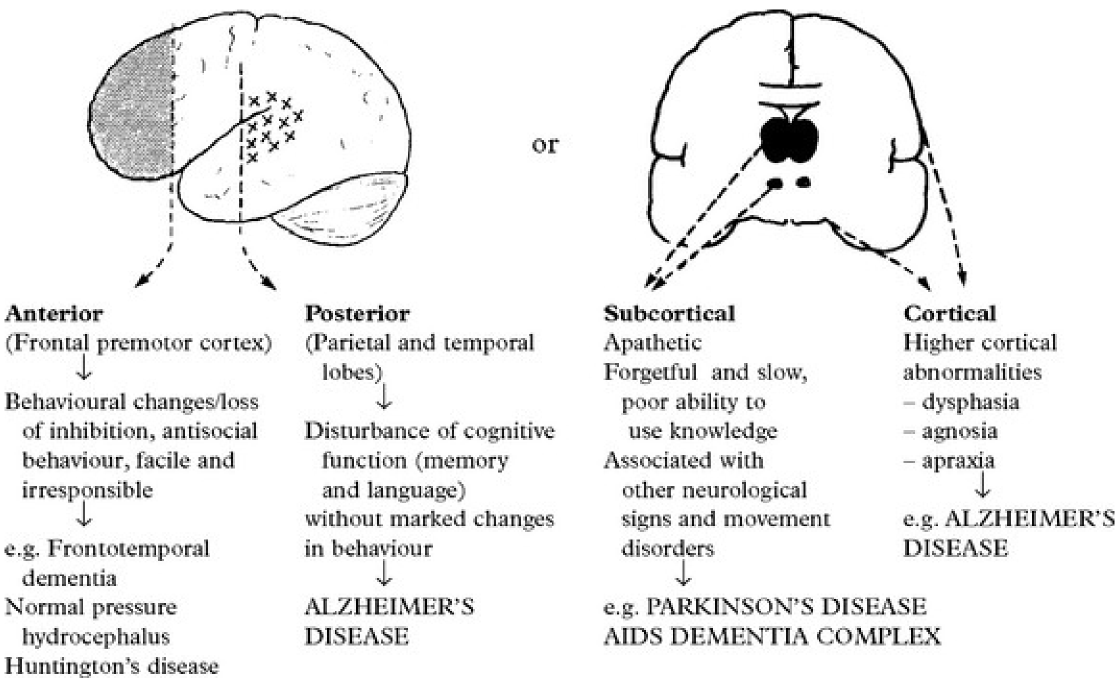
Trauma
- Head injury

Neoplasia and paraneoplasia
- Frontal tumour
- Limbic encephalitis

It is important to investigate all patients with dementia as many causes are treatable in practice 10–15% can be reversed.

Based on site

Subdividing dementia depending upon the site of predominant involvement is useful in clinical classification but has only limited value in predicting underlying pathology:



DEMENTIAS – HISTORY AND CLINICAL EXAMINATION

When obtaining a history from a patient with dementia and relative or carer, establish:

– Rate of intellectual decline – Nutrition status

– Impairment of social function – Drug history

– General health and relevant disorders, *e.g.* stroke, head injury

– Nutrition status

– Drug history

– Family history of dementia.

<table>
<tr><th>Tests to assess intellectual function are designed to check</th><th>The Mini Mental Status Examination (MMSE)</th><th></th></tr>
<tr><td rowspan="2">– memory

– abstract thought

– judgement

– specific focal cortical functions</td><td>Date orientation

Place orientation

Register 3 objects

Obeying verbal command</td><td>Serial sevens

Naming

Repeating</td></tr>
<tr><td colspan="2">Obeying written command

Writing/drawing

This is the standard tool of evaluation. Top Score = 30; score >24 normal; <24 suggests dementia

Folstein at el J. Psych Res 12:196–198 1975</td></tr>
</table>

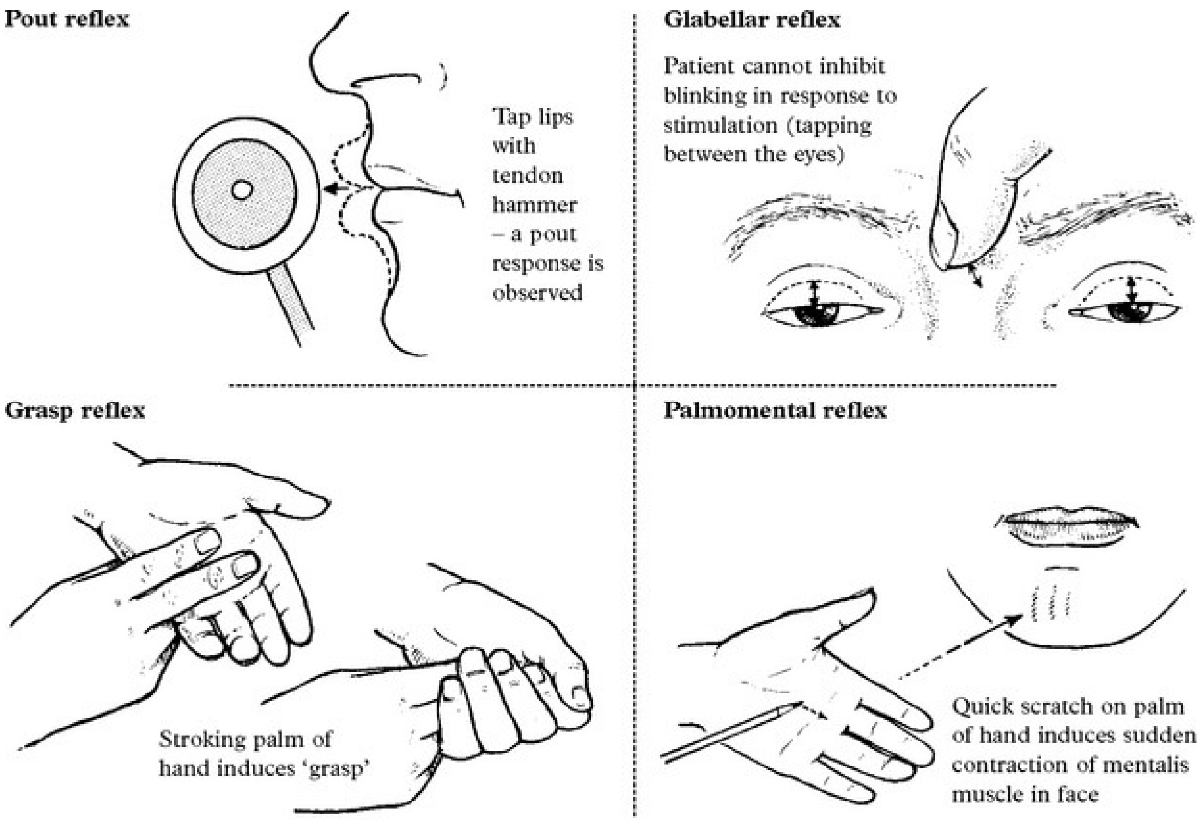
On neurological examination note:

– Focal signs

– Involuntary movements

– Pseudobulbar signs

– Primitive reflexes:



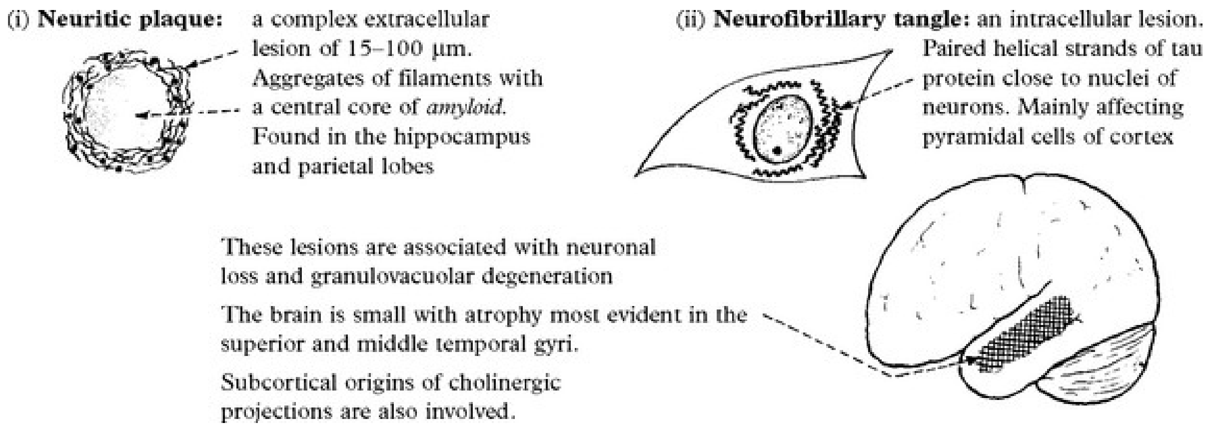
Primitive reflexes are present in infancy and in aged people, as well as in dementia.

DEMENTIAS – SPECIFIC DISEASES

ALZHEIMER'S DISEASE

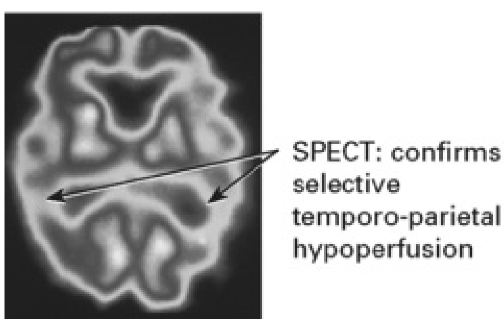
This is the commonest cause of dementia with an estimated half million sufferers in the UK. The disorder rarely occurs under the age of 45 years. The incidence increases with age. Up to 30% of cases are familial.

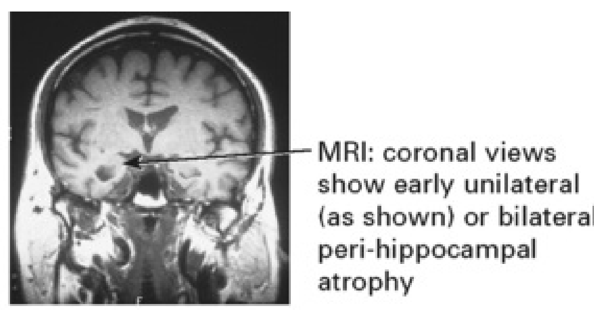
Pathology



Diagnosis This may be established during life by early memory failure, slow progression and exclusion of other causes. Specific clinical criteria have been established, mainly for research purposes. Whilst certain blood tests can identify populations at risk (i.e. APOE genotyping) these are of no diagnostic value in individual cases.

CT scanning: aids diagnosis by excluding multiple infarction or a mass lesion.





Causation

The cause of Alzheimer's disease is not known. Some genetic forms have been identified. The most common are mutations in the presenilin-1 gene (on chromosome 14). Patient's with Down's syndrome (trisomy 21) develop Alzheimer's pathology. The role of environmental toxins, especially aluminium, is uncertain. Early research suggested selective lesions of neurotransmitter pathways occurred and a disorder of cholinergic innervation was postulated. It is now known that many neurotransmitter pathways are defective.

Treatment

Centrally acting drugs such as acetylcholinesterase inhibitors (e.g. Donepezil, Rivastigmine, Galantamine) have been shown in trials to enhance cognitive performance in early disease. However they do not cure. Memantine is an NMDA antagonist that also provides some symptomatic relief.

DEMENTIAS – SPECIFIC DISEASES

MULTI-INFARCT (arteriosclerotic dementia)

This is an overdiagnosed condition which accounts for less than 10% of cases of dementia.

Dementia occurs 'stroke by stroke', with progressive focal loss of function. Clinical features of stroke profile – hypertension, diabetes, *etc.* – are present.

Diagnosis is obtained from the history and confirmed by CT scan.

Low density areas of infarction

These areas are not space-occupying and do not enhance after intravenous contrast

Treatment: Maintain adequate blood pressure control. Reduce cholesterol. Anti-platelet aggregants (aspirin).

FRONTOTEMPORAL DEMENTIA

This progressive condition accounts for 5% of all dementias, but about

20% of those under age of 65. There are three clinical patterns of presentation that are associated with differing areas of focal atrophy:

Behavioural variant – frontal lobe atrophy – change in personality; impaired judgement; apathy; stereotyped behaviours; loss of appropriate emotional response. Relatively preserved memory.

Progressive non-fluent aphasia – dominant temporal lobe atrophy – loss of verbal fluency, relatively preserved understanding.

Sematic dementia – bilateral temporal lobe atrophy – loss of knowledge of the meaning of words, and knowledge about the world.

The pathology is heterogeneous some have tau-positive inclusions (including Pick's disease) while others do not, some of whom have ubiquitin inclusions.

About 40% of patients have a family history of dementia and a number of the responsible genes have been identified, the most common being the progralulin *(PRGN)* mutation.

Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL)

This inherited disorder presents with migraine (often hemiplegic) in early adult life, progessing through TIAs and subcortical strokes to early dementia. The advent of MRI with its characteristic appearance has led to increasing recognition of what was previously only identified at autopsy. CADASIL has been mapped to the 'Notch 3' gene on chromosome 19 in many (though not all) cases allowing diagnostic testing. The role of the gene is uncertain and specific treatments not available.

AIDS DEMENTIA COMPLEX (see pages 515–516)

Approximately two-thirds of persons with AIDS develop dementia, mostly due to AIDS dementia complex. In some patients HIV is found in the CNS at postmortem. In others an immune mechanism or an unidentified pathogen is blamed. Dementia is initially of a ‘subcortical’ type. CT shows atrophy; MRI shows increased T2 signal from white matter. Imaging excludes other infections and neoplastic causes of intellectual decline.

Treatment with Zidovudine (AZT) halts and partially reverses neuropsychological deficit.

METABOLIC DEMENTIA

General medical examination is important in suggesting underlying systemic disease. B_{12} deficiency may produce dementia rather than subacute combined degeneration of the spinal cord. In alcoholics, consider not only Wernicke Korsakoff syndrome but also *chronic subdural haematoma*.

NORMAL PRESSURE HYDROCEPHALUS

Normal pressure hydrocephalus (NPH) is the term applied to the triad of:

Dementia	occurring in conjunction with
Gait disturbance	*hydrocephalus* and *normal*
Urinary incontinence	CSF pressure.

Two types occur:

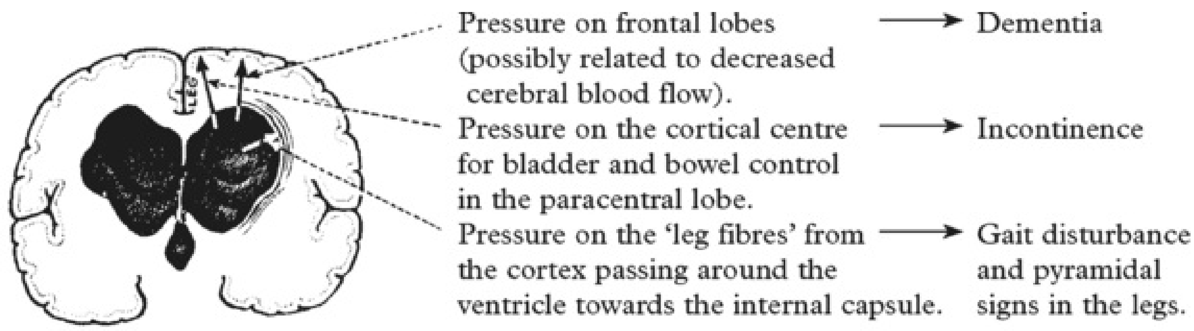
– NPH with a *preceding cause*	– subarachnoid haemorrhage – meningitis – trauma

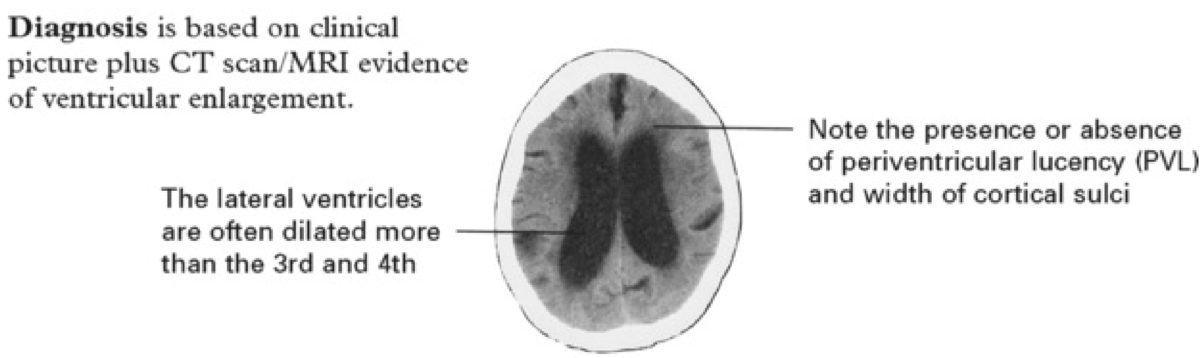
	– radiation-induced

(This must be distinguished from hydrocephalus with raised intracranial pressure associated with these causes.)

– NPH with no known preceding cause – *idiopathic* (50%).

Aetiology is unclear. It is presumed that at some preceding period, impedence to normal CSF flow causes raised intraventricular pressure and ventricular dilatation. Compensatory mechanisms permit a reduction in CSF pressure yet the ventricular dilatation persists and causes symptoms:





Normal pressure hydrocephalus must be differentiated from patients whose ventricular enlargement is merely the result of shrinkage of the surrounding brain, *e.g.* Alzheimer's disease. These patients do not respond to CSF shunting, whereas a proportion of patients with NPH (but not all) show a definitive improvement with shunting.

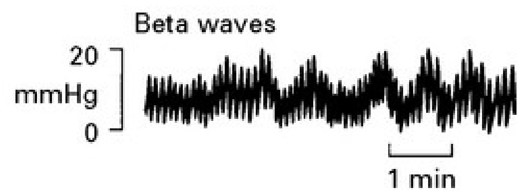
Investigations

Numerous tests have been assessed to predict those most likely to benefit from operation.

The most frequently used are –

(i) The presence of *beta waves* on *continuous intracranial pressure* monitoring for more than 5% of a 24 hour period.

(ii) Clinical improvement with *continuous lumbar CSF drainage* of 200 ml per day for three to five days.



Other tests include the presence of periventricular lucency or disproportionate sulcal width on CT scan, isotope cisternography and CSF infusion studies but predictive accuracy is low. Some believe that the risks of treatment are not warranted in the ‘ideopathic’ group.

Operation: Ventriculo-peritoneal shunting; although a small procedure, not without risk (see page 377).

Results: Improvement occurs in 50–70% of those patients with a known preceding cause *e.g.* subarachnoid haemorrhage. At best, 30% of the idiopathic group respond to shunting.

TRAUMA

Reduction of intellectual function is common after severe head injury. Chronic subdural haematoma can also present as progressive dementia, especially in the elderly. *Punch-drunk encephalopathy* (dementia pugilistica) is the cumulative result of repeated cerebral trauma. It occurs in both amateur and professional boxers and is manifest by dysarthria, ataxia and extrapyramidal signs associated with 'subcortical' dementia. There is no treatment for this progressive syndrome.

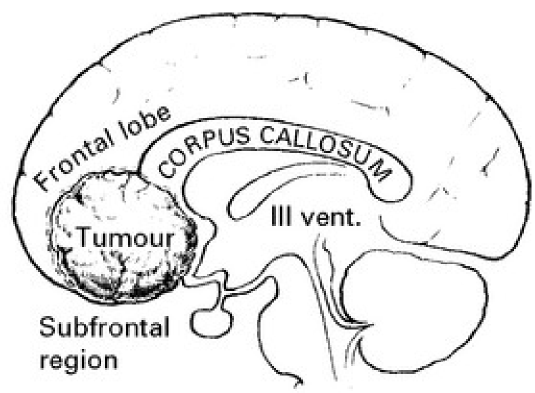
TUMOUR presenting as dementia

Concern is always expressed at the possibility of dementia being due to intracranial tumour. This is rare, but may happen when tumours occur in certain sites.

Mental or behavioural changes occur in 50–70% of all brain tumours as distinct from dementia which is associated with **frontal lobe tumours** (and subfrontal tumours), **III ventricle tumours** and **corpus callosum tumours**.

Suspect in recent onset dementia with focal signs, e.g. subfrontal lesions may be associated with loss of smell (I cranial nerve involvement) and optic atrophy (II cranial nerve involvement).

Cognitive impairment also occurs as a non metastatic complication of systemic malignancy (limbic encephalitis).



N.B. Dementia can occur as a symptom of a more widespread degenerative disorder

e.g. Parkinson's disease
Diffuse Lewy body disease
Progressive supranuclear palsy
Huntington's disease
Motor neuron disease

These will be considered later

DEMENTIA – DIAGNOSTIC APPROACH

It is neither practical nor essential to perform all the screening tests in every patient with dementia. The presenting features should guide investigations.

DEMENTIA	Suspected cause	Appropriate investigations
without neurological signs or systemic illness	– *Alzheimer's disease*	CT/MR scan Confirmation: pathology (post mortem)
	– *Frontotemporal dementia/Pick's disease*	
with neurological signs	– *Tumour*	CT/MR scan Confirmation: pathology (biopsy)
	– *Degenerative disease,* e.g. Huntington's disease	CT/MR scan Genetics Confirmation: pathology (biopsy or post mortem)
(gait disturbance and incontinence)	– *Normal pressure hydrocephalus* *Frontal lobe tumour*	CT/MR scan Confirmation: CSF pressure monitoring (tumour-biopsy)
with neurological signs and systemic symptoms and signs	– *Inflammatory disease,* e.g. Demyelinating disease (page 519) Vasculitis & collagen vascular disease	Serum autoantibodies Evoked responses CSF (immunology) CT/MR scan
	– *Infective disease,* e.g. AIDS Syphilis Meningitis	Serum antibodies (viral) VDRL, TPHA HIV status CSF examination CT/MR scan
with 'stroke risk factors' (page 519)	– *Multi-infarct state*	CT/MR scan)
with poor nutrition	– *Nutritional disease*	Serum B_1 (thiamine) Red cell transketolase (thiamine) Serum B_{12} Serum folate
with metabolic and endocrine symptoms and signs	– *Metabolic and endocrine disease*	Function tests: – thyroid – parathyroid – renal – hepatic – adrenal
with history of head trauma	– *Post-traumatic dementia*	CT/MR scan

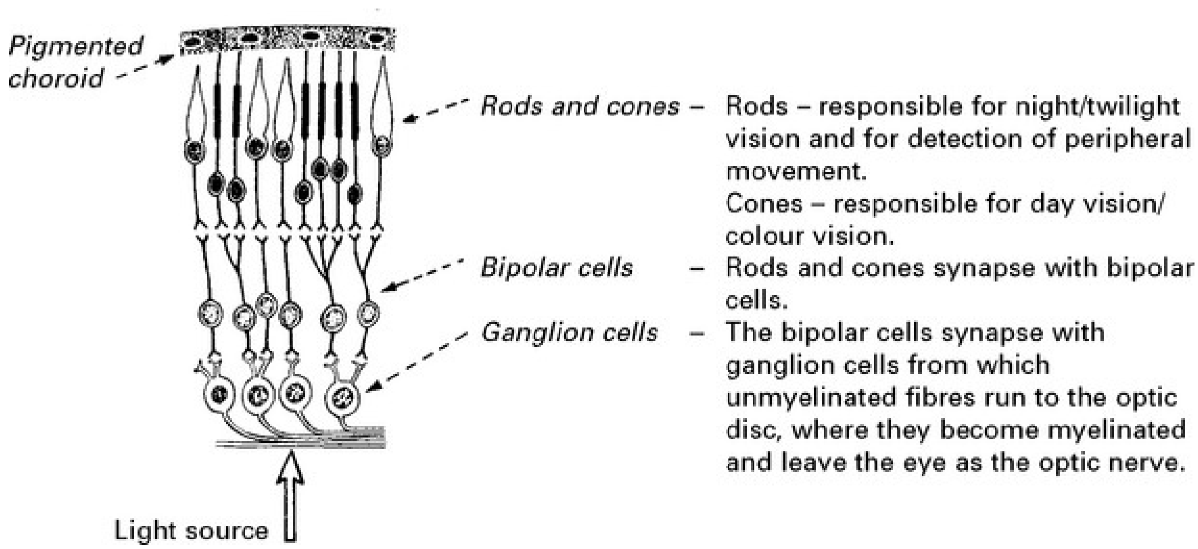
Neuropsychometric testing is performed
- to diagnose early dementia.
- evaluate atypical dementia.
- separate out depressive illness.
- monitor therapies.

When the reason for dementia is unclear, comprehensive investigation is essential to ensure that treatable nutritional, infective, metabolic and structural causes are not overlooked.

ANATOMY AND PHYSIOLOGY

Anatomically the visual system is contained in the supratentorial compartment. It is composed of peripheral receptors in the retina, central pathways and cortical centres. The control of ocular movement and pupillary responses are closely integrated.

The retina: three distinct layers of the retina are identified:

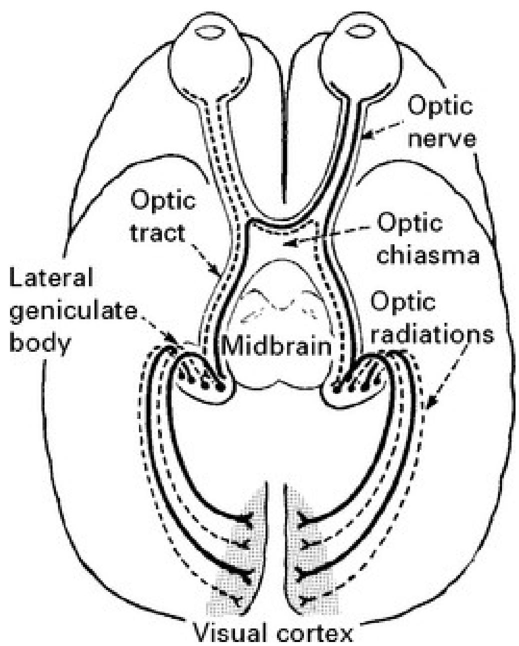


The *macular region* of the retina is its most important area for visual acuity. Here, cones lie in the greatest concentration whereas rods are more numerous in the surrounding retina.

The optic nerve leaves the orbit through the *optic foramen* and passes posteriorly to unite with the opposite optic nerve at the *optic chiasma*. Here, partial decussation occurs (axons from ganglion cells on the nasal side of the retina cross over to the opposite side).

The *optic tract* consisting of ipsilateral temporal and contralateral nasal fibres passes to the *lateral geniculate body*. A few fibres leave the tract before the lateral geniculate body and pass to the *superior colliculus* (fibres concerned with pupillary light reflex).

Axons of cell bodies in the lateral geniculate body make up the *optic radiation*. This enters the hemisphere in the most posterior part of the internal capsule, courses deep in parietal and temporal lobes and terminates in the *calcarine cortex* of the occipital lobe.

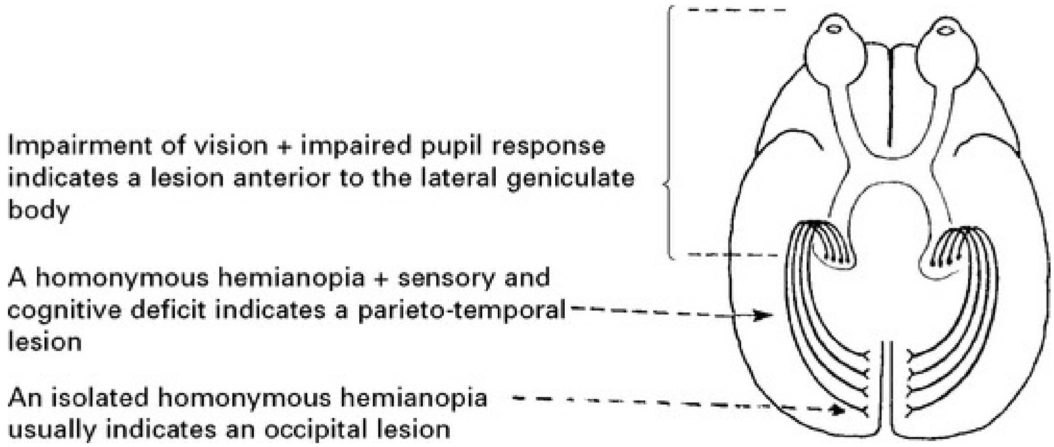


IMPAIRMENT OF VISION

CLINICAL APPROACH AND DIFFERENTIAL DIAGNOSIS

Patients presenting with visual impairment require a systematic examination, not only of *vision*, but also of the *pupillary response*, *eye movements*, and, unless the cause clearly lies within the globe, *a full neurological examination*.

The findings aid localisation of the lesion, *e.g.*



Refractive errors are excluded by testing visual acuity through a pinhole or by correcting a lens deformity (page 9).

Four types of refractive error exist:

PRESBYOPIA – failure of accommodation with age

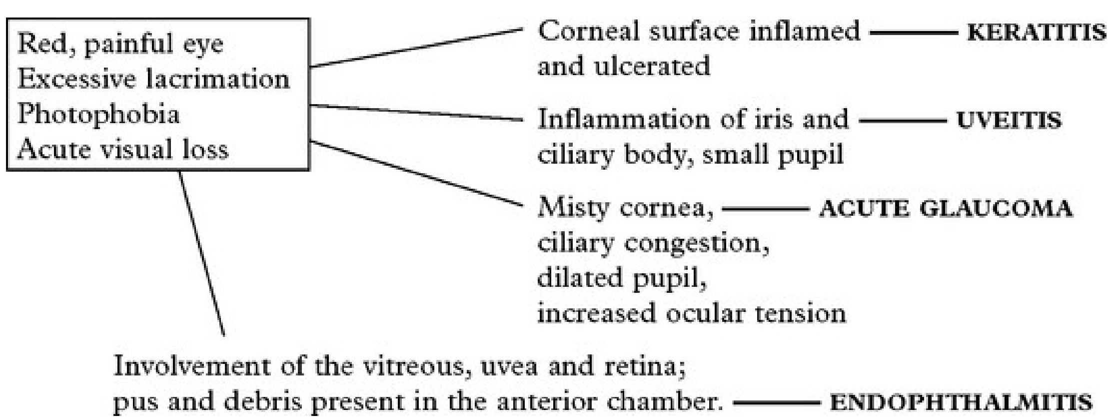
HYPERMETROPIA (long sightedness) – short eyeball

MYOPIA (short sightedness) – long eyeball

ASTIGMATISM – variation in corneal curvature

If this examination is normal, then the lesion lies in the retina, visual pathways or visual cortex.

Examine the globe and anterior chamber

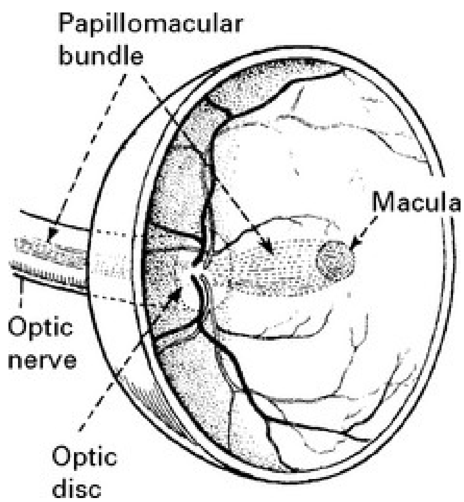


Examine the lens with an ophthalmoscope

Opacification indicates CATARACT.

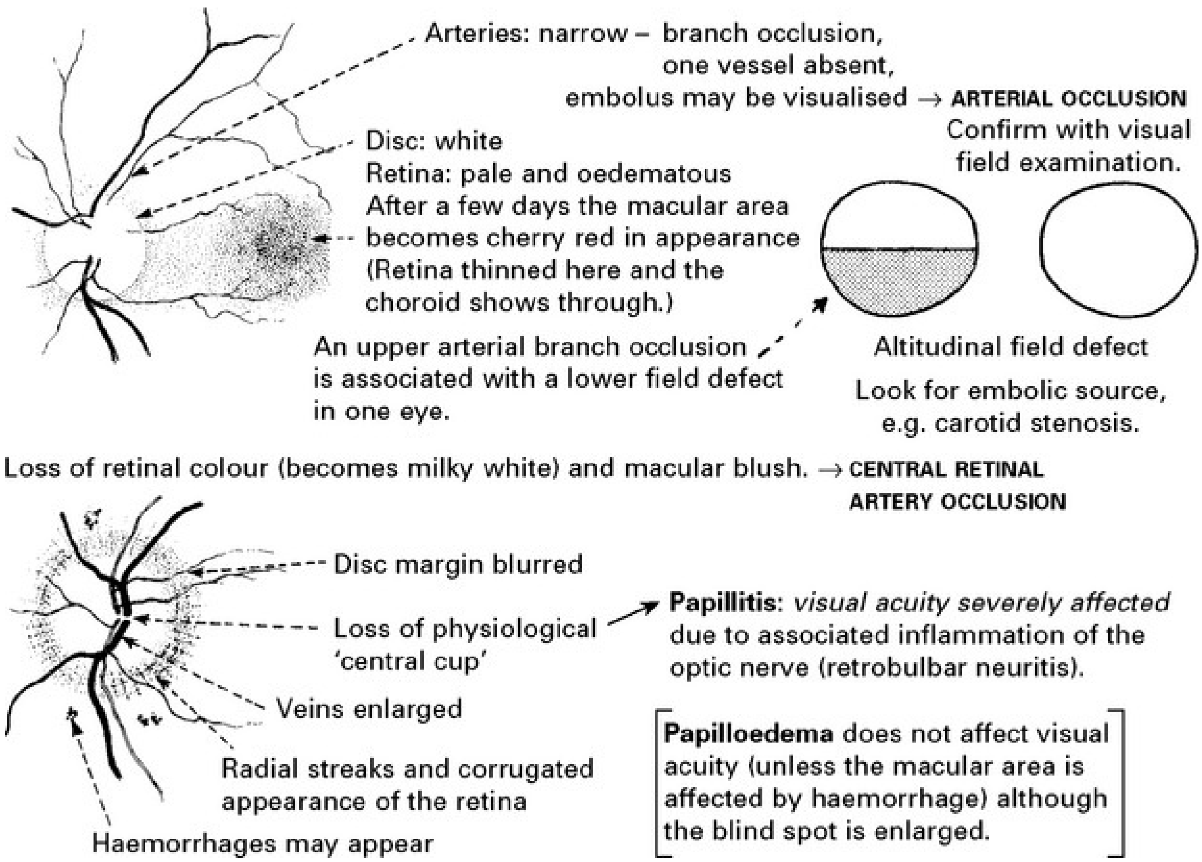
CLINICAL APPROACH AND DIFFERENTIAL DIAGNOSIS

Examine the posterior segment of the eye with an ophthalmoscope. Pupil dilatation may be required.



In the normal fundus, the *disc* is pale with a central cup and reddish-brown surrounding retina. Arteries and veins emerge from the optic disc. The *macula* is darker than the rest of the fundus and lies on the temporal side of the disc. One-third of all retinal fibres arise from the small macular region and pass to the optic nerve head (disc) as the *papillomacular bundle*. The macula is the region of sharpest vision (cone vision), whereas peripheral vision (rod vision) serves the purpose of perception of movement and directing central/macular vision. The optic nerve head contains no rods or cones and accounts for the physiological *blind spot* in normal vision. The macular fibres being so functionally active, are the most susceptible to damage and produce a specific defect in the visual field – a **scotoma**.

Retinal abnormality with acute impairment of vision



N.B. Distinguish:

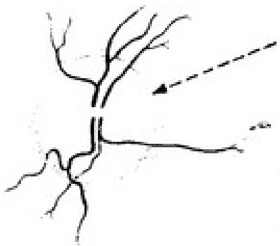
HYPERMETROPIC patients who have a pale indistinct disc often difficult to differentiate from early papilloedema.

HYPERTENSIVE RETINOPATHY – superficial haemorrhages and 'cotton wool' exudates.

PSEUDOPAPILLOEDEMA – 'DRUSEN' – hyaline bodies near the optic disc which raise the disc and blur the margin. This normal variant may be inherited.

Separation of the superficial retina from the pigment layer → **RETINAL DETACHMENT** (traumatic or spontaneous)

Retinal abnormalities with gradual impairment of vision



Disc white like a 'tennis ball' with 'punched out' margins: blood supply is less prominent and the number of arteries reduced → **OPTIC ATROPHY**

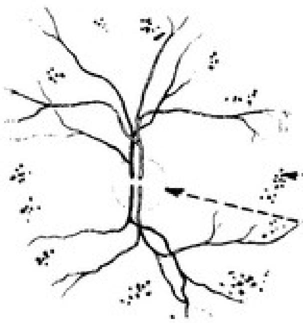
Primary (optic nerve disease): *compression, toxins, ischaemia, optic neuritis*

Secondary (following papilloedema):

↓

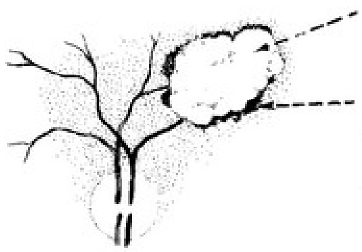
visual field charting (see later) may help differentiate cause

N.B. Any disease of the optic nerve or anterior visual pathway causing loss of vision will eventually result in optic atrophy.



Pigmentary deposits in the periphery of the retina → **RETINITIS PIGMENTOSA**

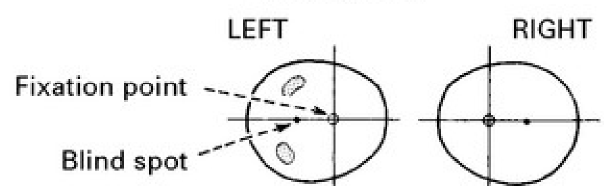
Progressive pallor of the optic disc



Areas of white sclera exposed along with areas of proliferation of retinal pigmentary epithelium – follows atrophy of the choroid → **CHOROIDITIS**

Occurs in *toxoplasmosis* and in *cytomegalovirus* infection

Field examination reveals a patchy loss.



Small deep haemorrhages and hard exudates in a long-standing diabetic → **DIABETIC RETINOPATHY**

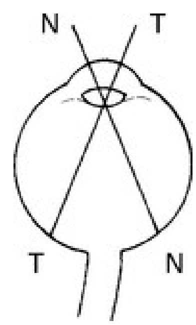
Dark oval mass – possibly related to secondary retinal detachment — in middle aged patient → **MALIGNANT MELANOMA**

White mass behind the pupil — in infancy → **RETINOBLASTOMA**

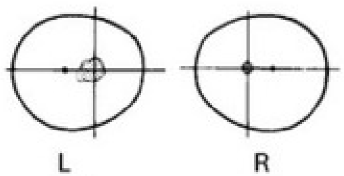
Examine the visual fields

If ophthalmoscopic examination is normal, or if optic atrophy is evident, then visual field examination is essential. Visual confrontation is useful for detecting large defects, but smaller defects require visual field charting with a Goldmann perimeter (page 10).

In interpreting the results of examination it is important to remember that the ocular system *reverses* the image. The nasal side of the fundus picks up the temporal image and vice versa. Damage, therefore, to the nasal side of the retina will produce a temporal visual field defect.



Central scotoma Characteristic of most optic nerve lesions.



OPTIC NEURITIS – associated papillitis may be evident on fundoscopy; may be first sign of *multiple sclerosis*.

OPTIC NERVE COMPRESSION

CT/MRI scan (orbital/intracranial)

Orbital lesion (usually with proptosis)
- *tumour*
- *granuloma*

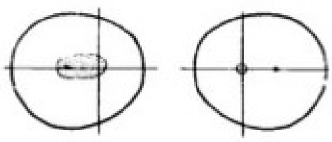
Lesion within optic canal
- *tumour,* e.g. meningioma
- *granuloma*
- *hyperostosis*, e.g. Paget's disease, fibrous dysplasia

Intracranial lesions
- *tumour*, e.g. meningioma (chordoma, dermoid)
- *granuloma*, e.g. tuberculoma, sarcoid (rare)
- *aneurysm*, e.g. ophthalmic → angiography confirms

Pupil response may be impaired (Marcus-Gunn pupil, see page 146)

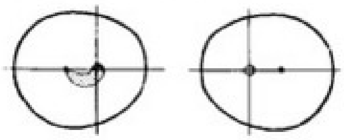
OPTIC NERVE GLIOMA → CT/MRI → exploration
LEBER'S OPTIC ATROPHY → large bilateral scotoma

Centro-caecal scotoma



The scotoma extends to involve the blind spot. Characteristic of *toxic amblyopia* – alcohol, tobacco.

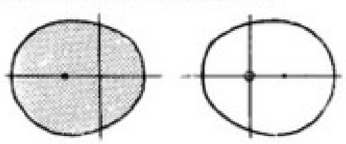
Arcuate scotoma



The scotoma extends from the blind spot following the course of nerve fibres.

Characteristic of *glaucoma*; seen also in small lesions close to the optic disc such as *choroiditis*.

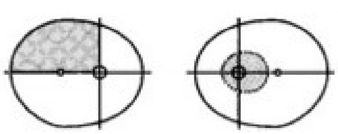
Monocular blindness



The end result of an inflammatory, vascular or compressive optic nerve lesion.

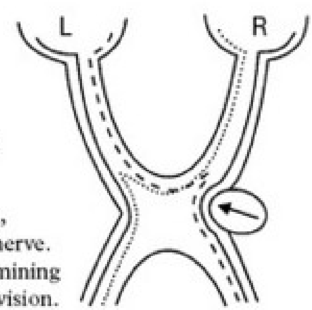
Direct pupillary response absent; consensual present.

Junctional scotoma



– indicates the presence of an *optic nerve lesion immediately anterior to the chiasma.*

Nasal fibres not only decussate in the chiasma, but also loop forward into the opposite optic nerve. This lesion emphasises the importance of examining the 'normal' eye in monocular impairment of vision.



Bitemporal hemianopia/quadrantanopia

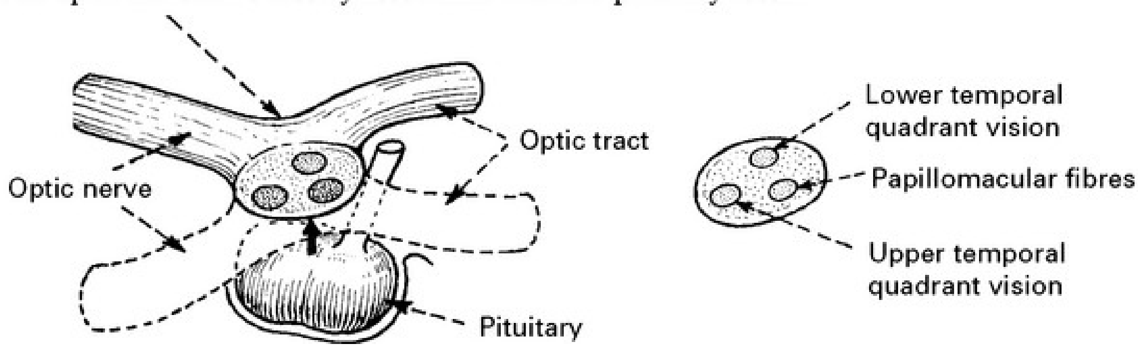
Involvement of the upper quadrants first indicates compression of the optic chiasma from below and suggests:
- PITUITARY ADENOMA
- NASOPHARYNGEAL CARCINOMA
- SPHENOID SINUS MUCOCELE

→ CT scan/MRI

Involvement of the lower quadrants first indicates compression of the optic chiasma from above and suggests:
- CRANIOPHARYNGIOMA
- THIRD VENTRICULAR TUMOUR

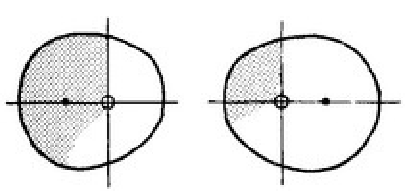
↓
CT scan/MRI

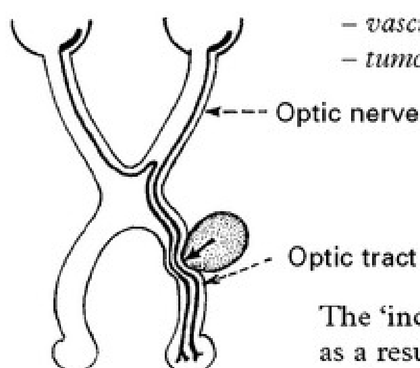
The optic chiasma is closely associated with the pituitary fossa.



Homonymous hemianopia

An incongruous homonymous hemianopia (i.e. one eye more affected than the other) suggests a compressive lesion of the **optic tract** near the chiasma.





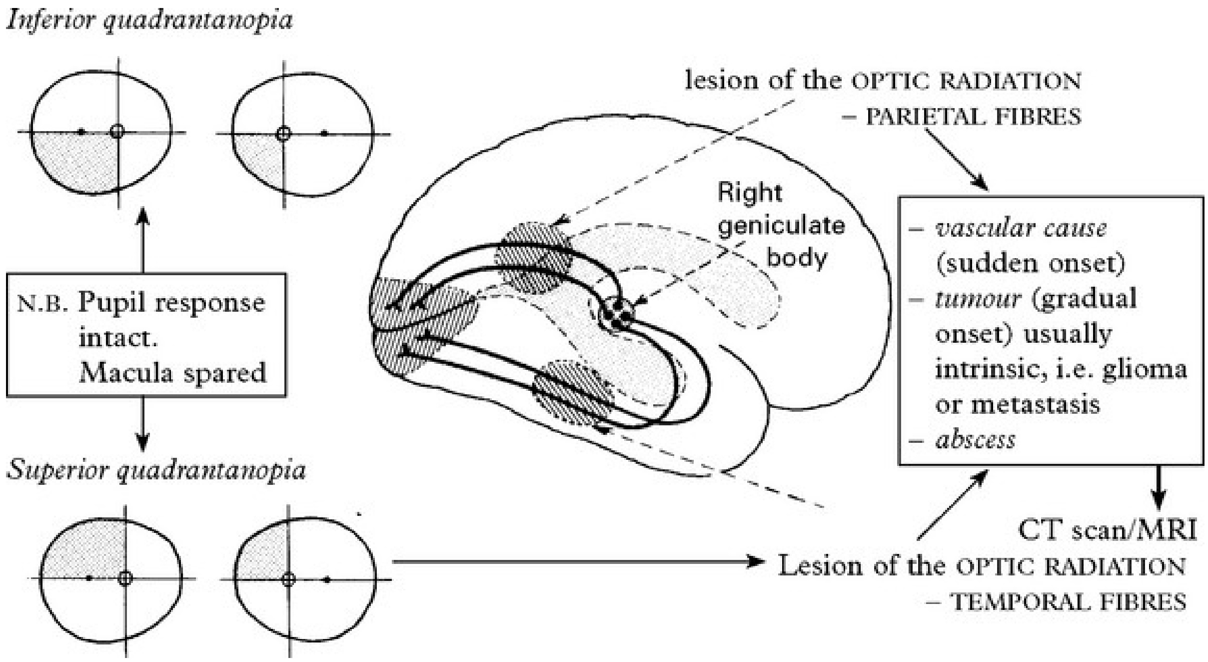
- *vascular cause* (sudden onset)
- *tumour* (gradual onset)

→ CT scan/MRI

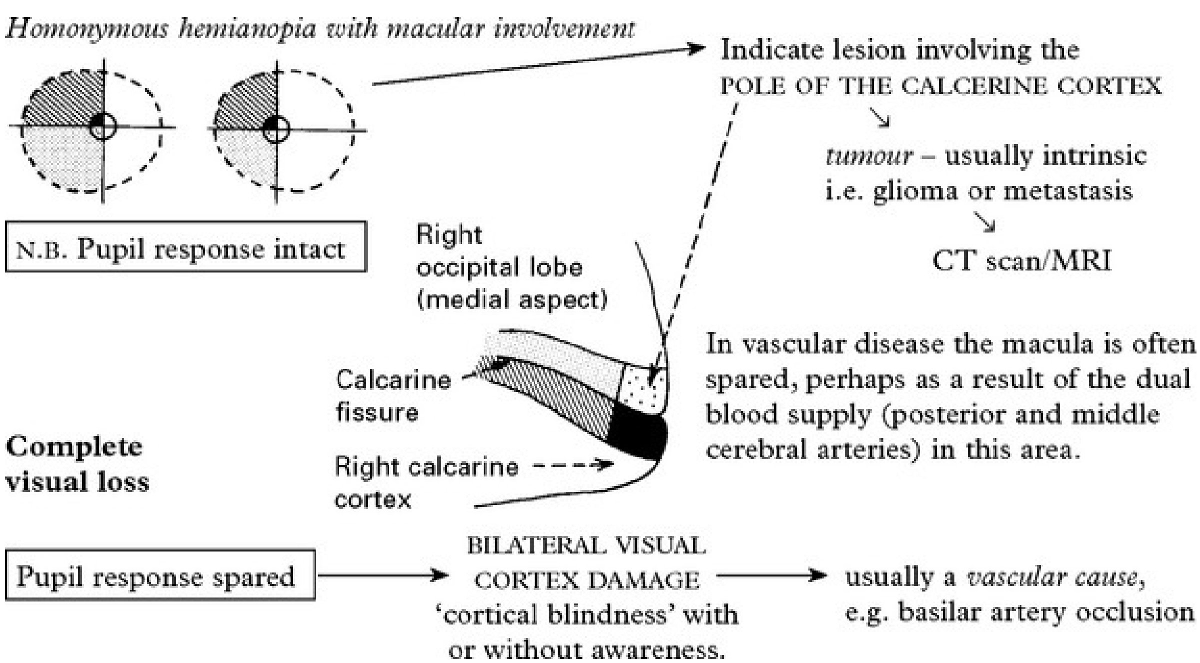
N.B. Pupil response may be impaired when light is shone from affected field

The 'incongruous' defect occurs as a result of rotation of nasal and temporal fibres.

Congruous homonymous hemianopia (fields can be exactly superimposed)



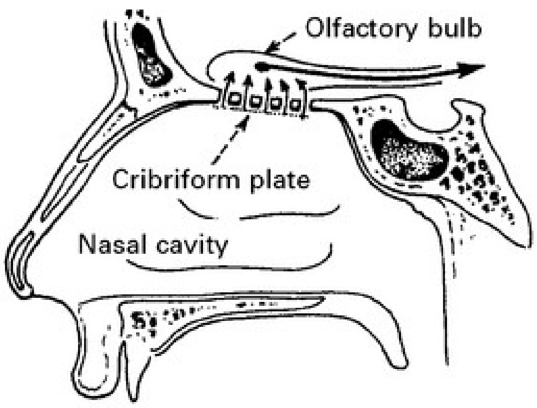
At the temporo-parietal junction where fibres meet, lesions produce a complete 'homonymous hemianopia'.



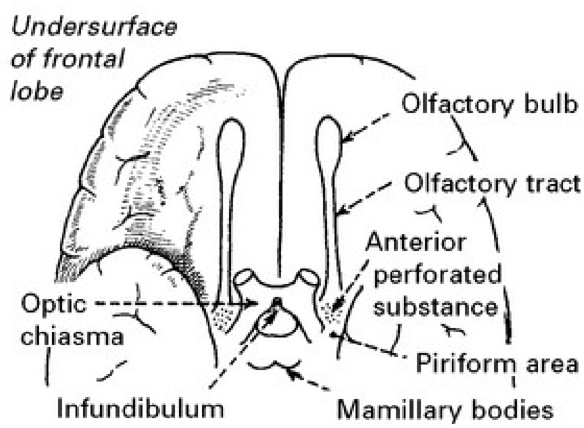
The interpretation of the visual image and its integration with other cortical functions is discussed under ‘Higher cortical function’.

DISORDERS OF SMELL

OLFACTORY (I) cranial nerve conveys the sensation of smell.

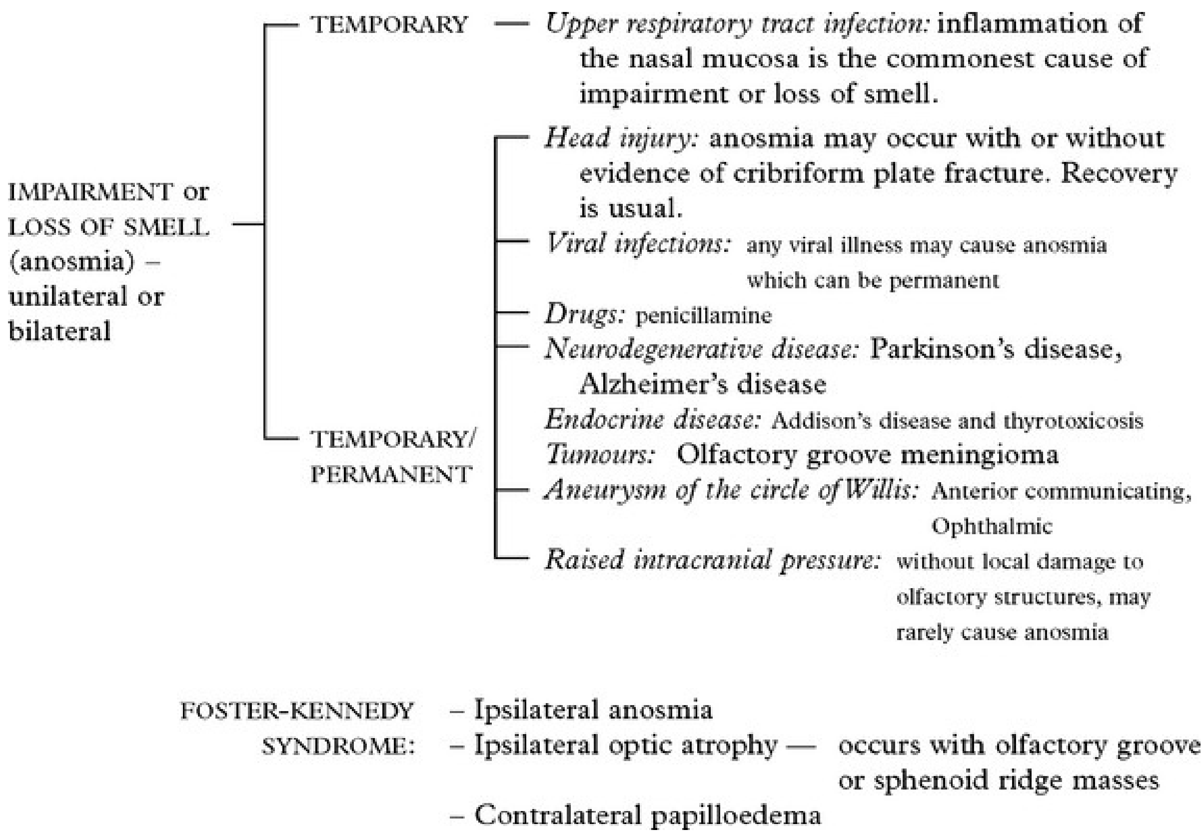


A number of fine nerves arising from receptor cells in the nasal mucosa pierce the *cribriform plate* of the ethmoid bone. These pass to the *olfactory bulb* where they synapse with neurons of the olfactory tract.



The axons partially decussate as they pass back in the olfactory tract to the *piriform area* of the temporal lobe and the *amygdaloid nucleus*.

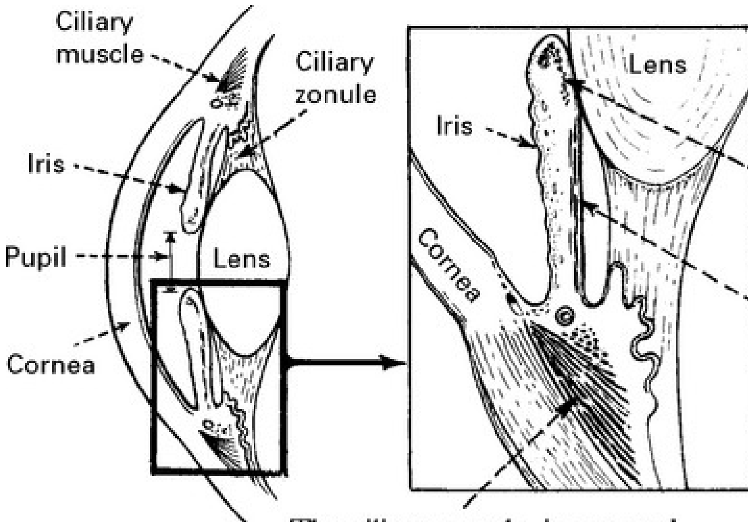
Differential diagnosis



OLFACTORY HALLUCINATIONS – occur in complex partial seizures and migraine.

PUPILLARY DISORDERS

ANATOMY/PHYSIOLOGY



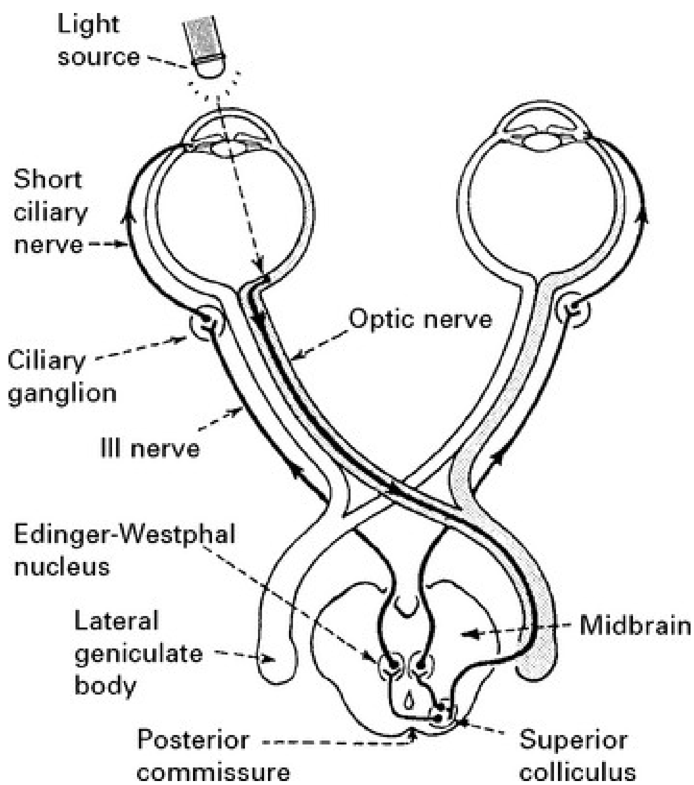
The ciliary muscle, innervated by the parasympathetic, controls the degree of convexity of the lens through the ciliary zonule.

The iris controls the size of the pupil. It contains two groups of smooth muscle fibre:

1. *Sphincter pupillae*: a circular constrictor, innervated by the parasympathetic nervous system.
2. *Dilator pupillae*: a radial dilator, innervated by the sympathetic nervous system.

Pupillary size (normal 2–6 mm) depends on the balance between sympathetic and parasympathetic tone.

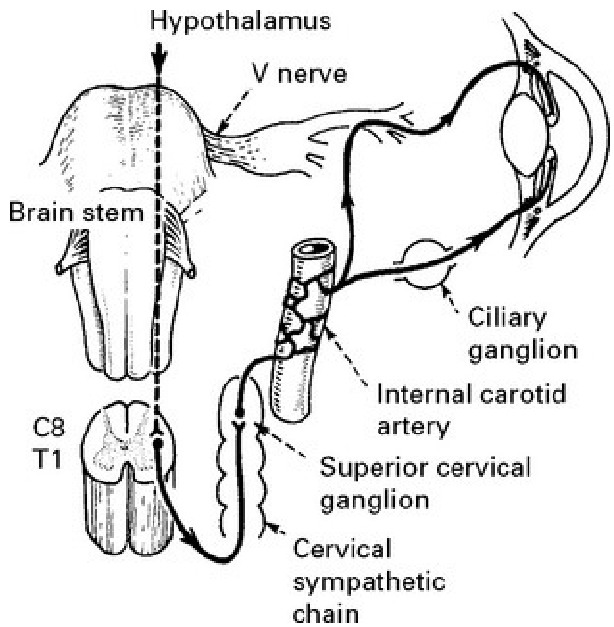
Pathway of pupillary constriction and the light reflex (parasympathetic)



A stimulus, such as a bright light shone in the left eye, will send an afferent impulse along the **optic nerve** to the midbrain (superior colliculus); here a second order fibre passes to the *Edinger-Westphal nucleus* (part of the III nerve nucleus) on the same and opposite side (through the posterior commissure). Efferent fibres leave in the **oculomotor nerve**, pass to the ciliary ganglion and thence, in the short ciliary nerve, to the constrictor fibres of the sphincter pupillae muscle.

If all pathways are intact, shining a light in one eye will constrict *both pupils* at an equal rate and to a similar degree.

Pathway of pupillary dilatation (sympathetic)



Sympathetic fibres descend from the ipsilateral hypothalamus through the lateral aspect of the brain stem into the spinal cord. The pupillary fibres pass out in the anterior roots of C8 and T1, enter the sympathetic chain and, in the superior cervical ganglion, give rise to postganglionic fibres which ascend on the wall of the internal carotid artery to enter the cranium. The fibres eventually leave the intracranial portion of the internal carotid artery and pass directly through the ciliary ganglion to the iris or join the cranial nerves III, IV, V and VI, running to the eye and iris. Sudomotor fibres (concerned with sweating) run up the external carotid artery to the dermis of the face.

Interruption of sympathetic supply affects:

1. Dilator pupillae causing a small pupil (miosis)

2. Levator palpebrae muscle (30% supplied by sympathetic) causing drooping of eyelid (ptosis)

3. Vasoconstrictor fibres to orbit, eyelid and face causing absence of sweating.

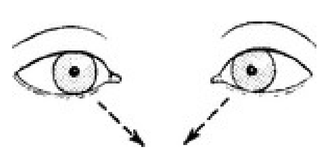
Interruption of parasympathetic supply affects:

Sphincter pupillae causing a large pupil (mydriasis)

Mechanism of accommodation

When gaze is focused on a near object the medial rectus muscles contract, producing convergence, the ciliary muscles contract enabling the lens to produce a more convex shape and the pupil constricts (accommodation for near vision).

The pathway is poorly understood but must involve the visual cortex, Edinger-Westphal nuclei and both medial rectus components of the III nerve nucleus in the midbrain.



Inability of the pupil to constrict during accommodation need not always be associated with impairment of convergence, though usually this is the case.

Pupillary inequality (anisocoria)

A difference in pupil size occurs in 20% of the normal population and is distinguished from pathological states by a normal response to bright light.

PUPIL DILATATION – CAUSES

III nerve lesion

Examination of the light reflex (page 11) distinguishes lesions of the optic (II) and oculomotor (III) nerves. Failure of the pupil to constrict when light is shone into either the affected or the contralateral eye indicates a lesion of the parasympathetic component of the III nerve.

Look for – **ptosis** – 70% of levator palpebrae muscle is supplied by the oculomotor nerve

– **impaired eye movements**.

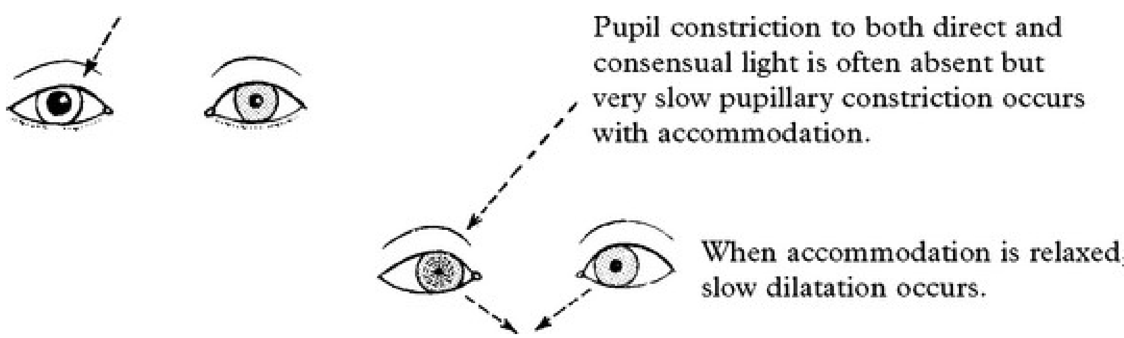
Causes of a III nerve lesion are described on page 153.

In comatose patients, pupil dilatation and failure to react to light is the simplest way of detecting a III nerve lesion; after head injury or in patients with raised intracranial pressure this is an important sign of transtentorial herniation.

The tonic pupil – Adie's pupil

This is a benign condition usually affecting young women. Onset is usually acute and unilateral in 80%.

The pupil dilates and the patient complains of mistiness in the affected eye.



Occasionally the pupil appears completely unreactive to both light and accommodation. When the pupil is associated with reduced or absent limb reflexes this is termed the *Holmes-Adie* syndrome. More widespread autonomic dysfunction – orthostatic hypotension, segmental disturbance of sweating and diarrhoea can co-exist.

Diagnosis: confirmed by pupillary response to pilocarpine (0.1% or 0.05%) – the tonic pupil will constrict (denervation hypersensitivity); the normal eye is not affected.

The cause is unknown; the lesion probably lies in the midbrain or ciliary ganglion.

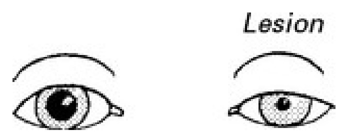
Migraine: Mydriasis persisting for some hours can accompany headache.

Drugs: Mydriasis occurs with anticholinergic drugs (atropine), tricyclic antidepressants, non-steroidal anti-inflammatories, antihistamines and oral contraceptives. Mydriasis can precipitate an attack of acute angle-closure glaucoma.

PUPIL CONSTRICTION – CAUSES

Horner's syndrome

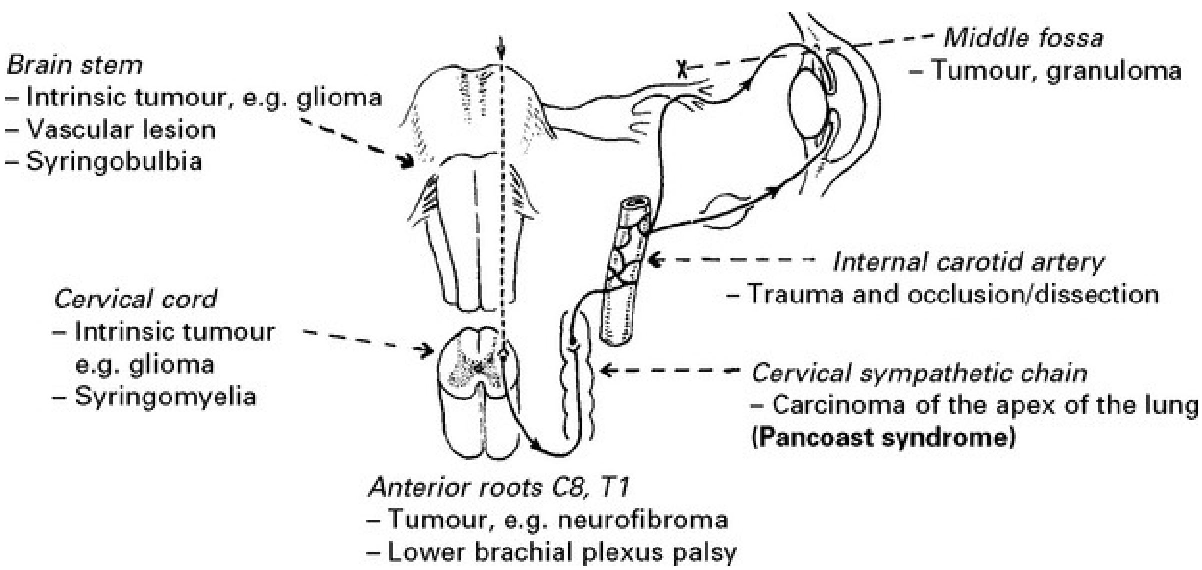
MIOSIS: the affected pupil is smaller than the opposite pupil. It does not dilate when the eye is shaded.



PTOSIS: the affected eyelid droops and may be slightly raised voluntarily. Ptosis is less marked than with a III nerve palsy.

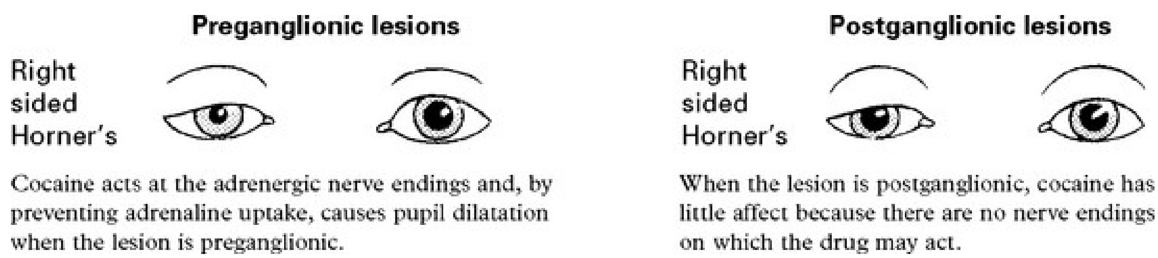
DISTURBANCE OF SWEATING: depends on the site of the lesion. Absence of sweating occurs when the lesion is proximal to fibre separation along the internal and external carotid arteries.

Horner's syndrome may result from sympathetic damage at the following sites:



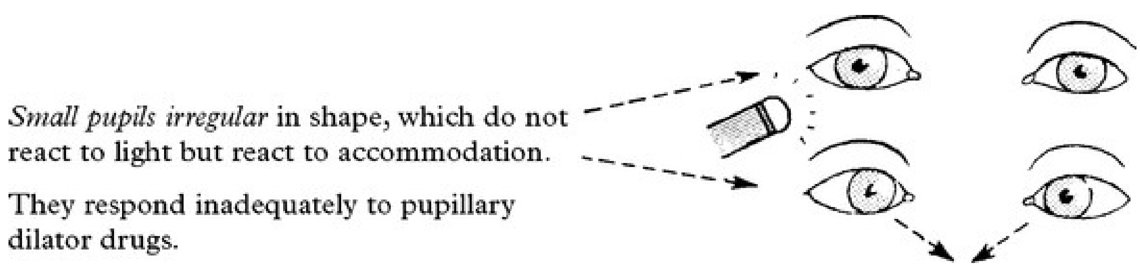
The congenital or familial form exists, often associated with lack of pigmentation of the iris. The lesion site is unknown.

Distinguish peripheral and central lesions by instilling drugs, *e.g.* 1% cocaine in eyes.



Investigative approach: depends on associated signs. Chest X-ray is mandatory to exclude an apical lung tumour.

The Argyll-Robertson pupil



The Argyll-Robertson pupil has also been described in *diabetes* and in *alcoholic neuropathy* as well as following infectious mononucleosis. The lesion could lie in the midbrain, involving fibres passing to the Edinger-Westphal nucleus, in the posterior commissure, or alternatively, in the ciliary ganglion. A central lesion seems most likely.

Investigative approach: – look for associated signs of neurosyphilis – blood serology – VDRL, Captia G.

Drugs

Parasympathomimetic drugs – Carbachol, phenothiazines and opiates produce miosis.

N.B. Do not confuse with small pupils, normally occurring in the elderly.

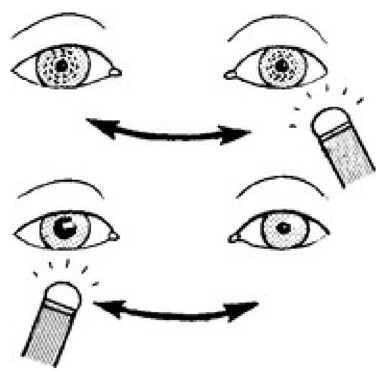
OTHER PUPILLARY DISORDERS

Failure of accommodation and convergence

Impaired accommodation and convergence are of limited diagnostic value since other clinical features are usually more prominent

Causes – extrapyramidal disease, *e.g.* Parkinson's

– tumours of the pineal region.



The Marcus Gunn pupil (pupillary escape)

Illumination of one eye normally produces pupillary constriction with a degree of waxing and waning (hippus).

When afferent transmission in the optic nerve is impaired, this 'escape' becomes more evident.

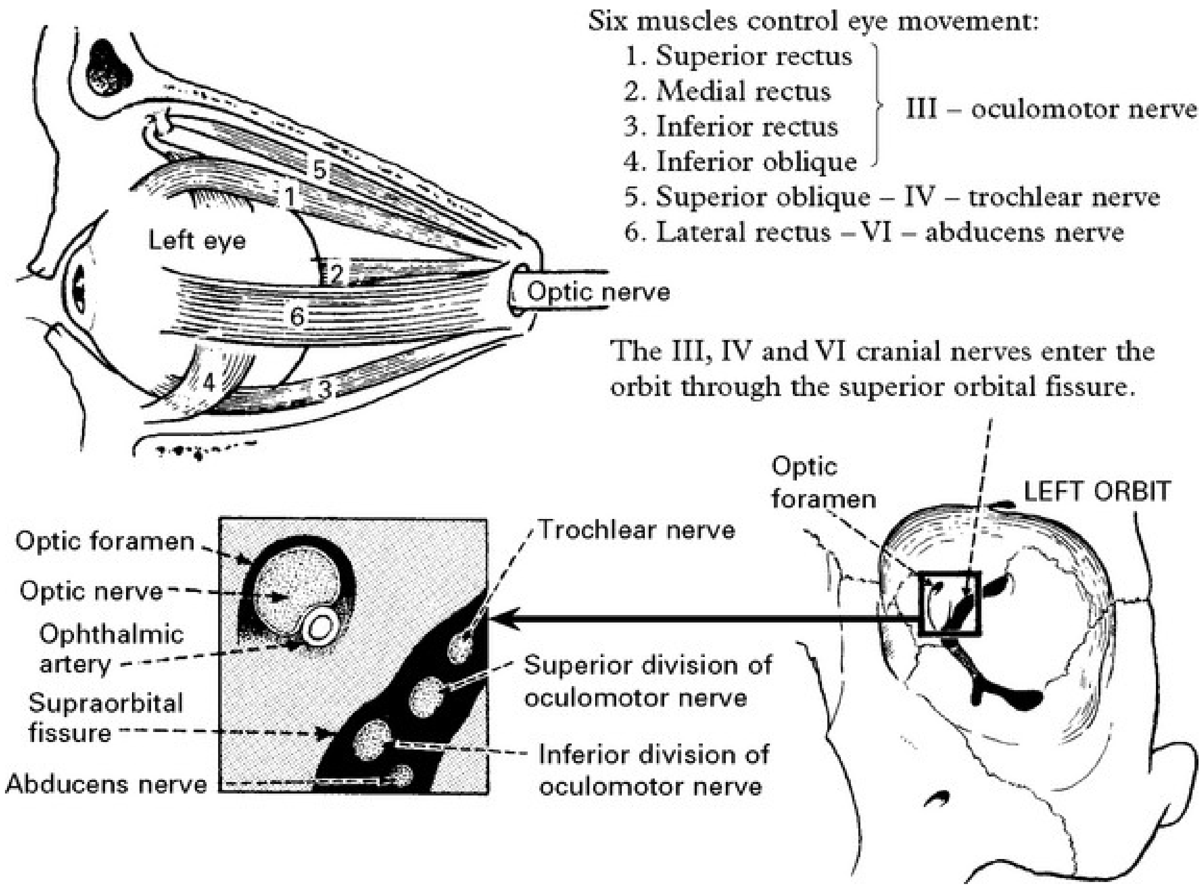
If the light source is 'swung' from eye to eye, dwelling 2–3 seconds on each, the affected pupil may eventually, paradoxically, dilate – a 'Marcus Gunn' pupil.

The swinging light test is a sensitive test of optic nerve damage but is also abnormal in retinal or macular disease.

DIPLOPIA – IMPAIRED OCULAR MOVEMENT

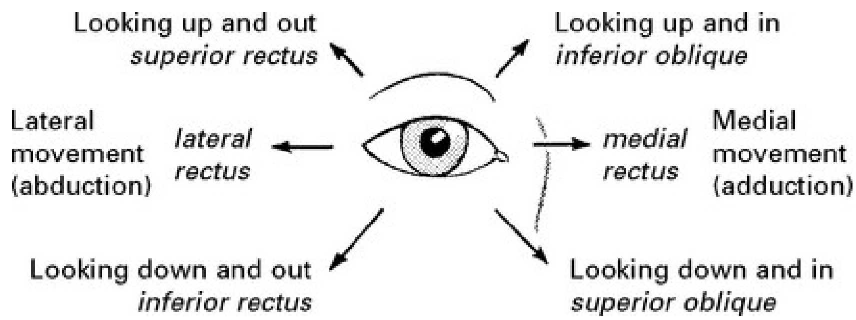
Diplopia or double vision results from impaired ocular movement.

RELATED ANATOMY AND PHYSIOLOGY



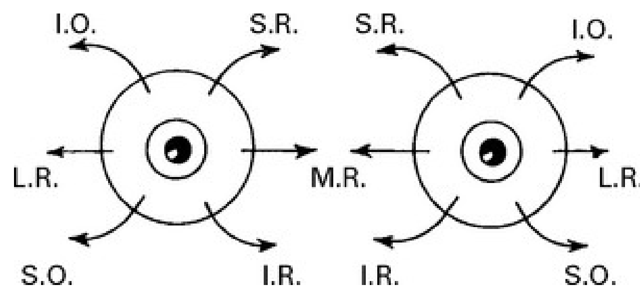
The line of action of individual ocular muscles

Eye movements result from a continuous interplay of all the ocular muscles, but each muscle has a direction of maximal efficiency. The oblique muscles move the eye up and down when it is turned in. The superior and inferior recti move the eye up and down when it is turned out.



Eye movements are examined in the six different directions of gaze representing individual muscle action.

As a result of the angle of insertion into the globe, the inferior and superior recti and the oblique muscles also have a rotatory or torsion effect.

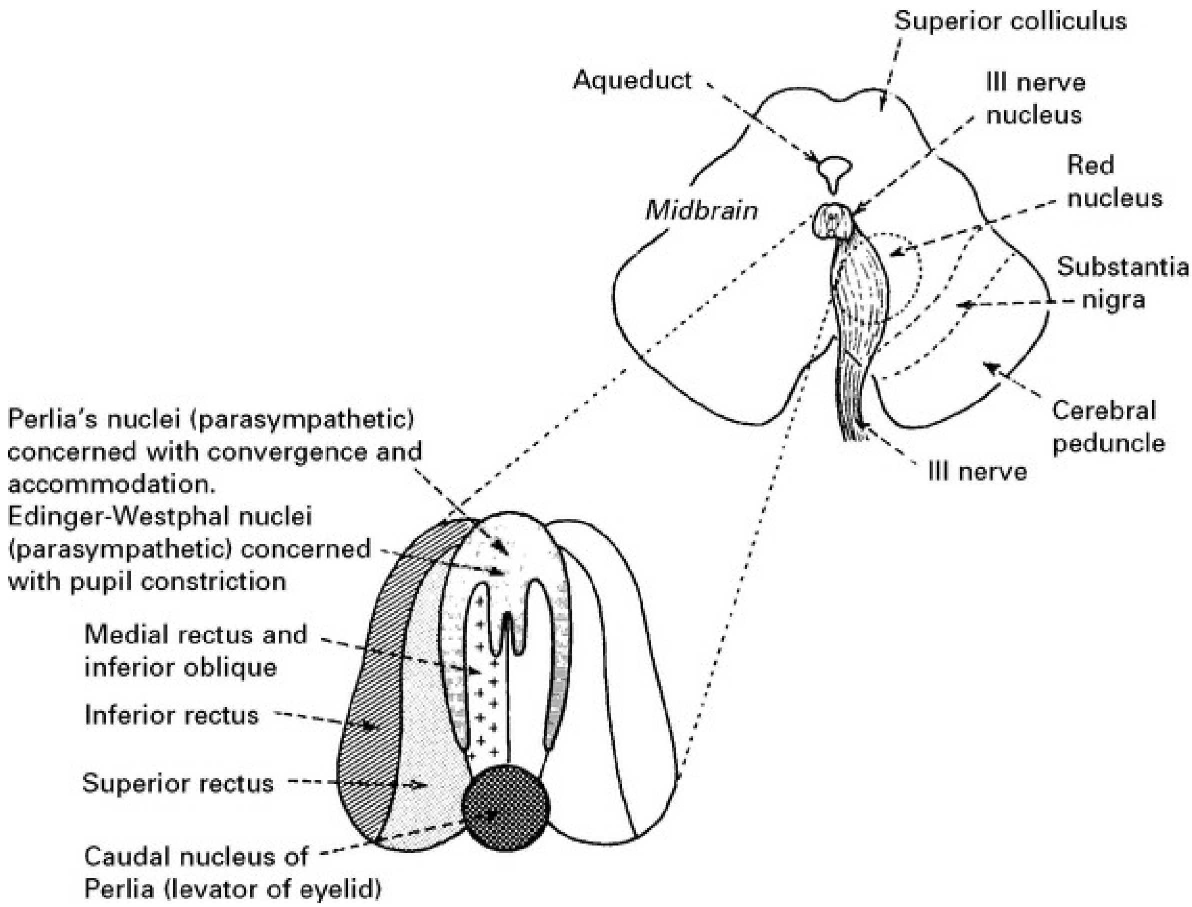


When the eye is turned out, the oblique muscles rotate the globe; when turned in, the inferior or superior recti rotate the globe.

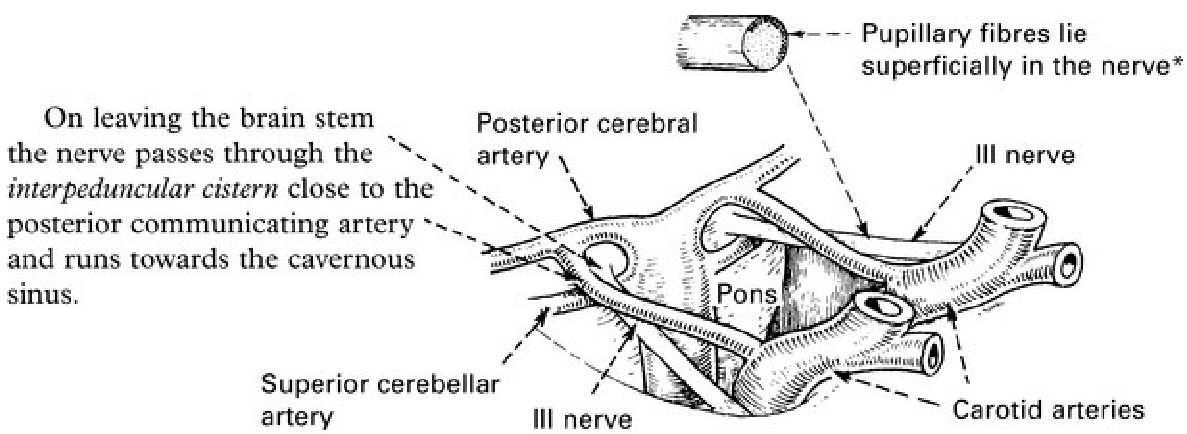
OCULOMOTOR (III) nerve

The oculomotor nucleus lies in the *ventral periaqueductal grey matter* at the level of the *superior colliculus*. Nerve fibres pass through the *red nucleus* and *substantia nigra* and emerge medial to the *cerebral peduncle*.

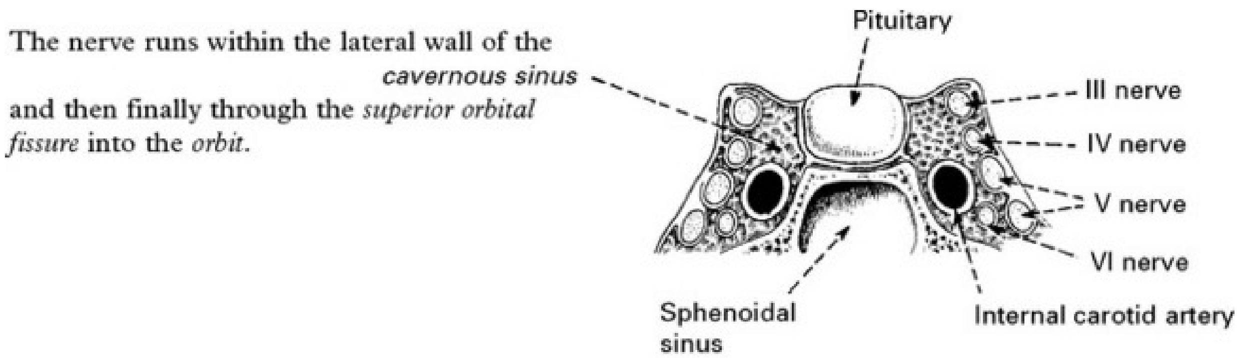
The nucleus has a complex structure:

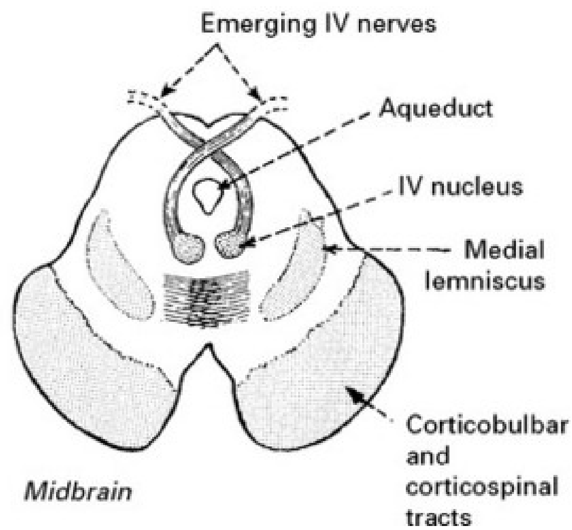


The nucleus is a paired structure which lies close to the midline, the portion representing the medial rectus abutting its neighbour.



*This in part explains early pupillary involvement with III nerve compression and pupillary sparing with nerve infarction in hypertension and diabetes.





Here it divides into:

1. Superior branch to the levator of the eyelid and the superior rectus.
2. Inferior branch to the inferior oblique, medial and inferior recti.

TROCHLEAR (IV) nerve

This nerve supplies the *superior oblique muscle* of the eye.

The nucleus lies in the midbrain at the level of the *inferior colliculus*, near the ventral *periaqueductal grey matter*. The nerve passes laterally and dorsally around the central grey matter and decussates in the dorsal aspect of the brain stem in close proximity to the *anterior medullary* *velum* of the cerebellum.

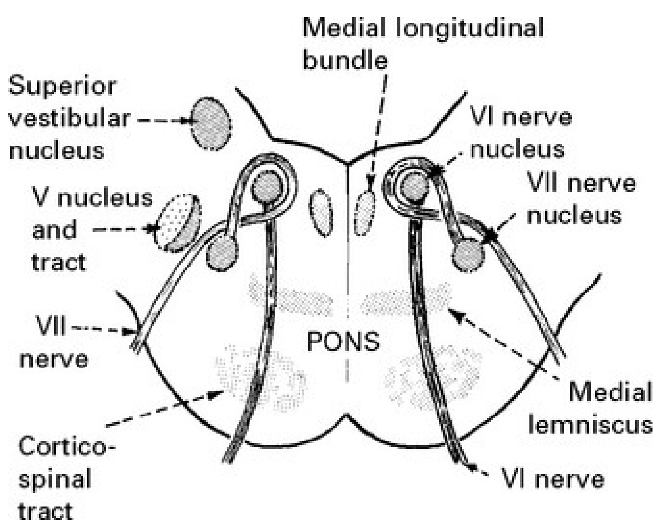
Emerging from the brain stem the nerve passes laterally around the *cerebral peduncle* and pierces the dura to lie in the lateral wall of the *cavernous sinus*. Finally, it passes through the *superior orbital fissure* into the orbit.

ABDUCENS (VI) nerve

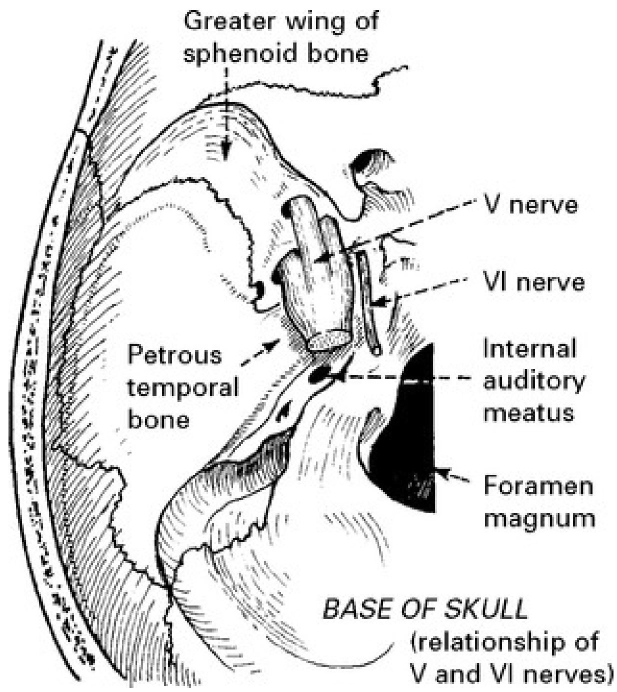
This nerve supplies the *lateral rectus muscle* of the eye.

The nucleus lies in the floor of the IV ventricle within the lower portion of the *pons*. The axons pass ventrally through the pons without decussating.

Note the close association of the VI and VII nuclei.



Emerging from the brain stem the nerve runs up anterior to the pons for approximately 15 mm before piercing the dura overlying the *basilar portion* of the *occipital bone*.

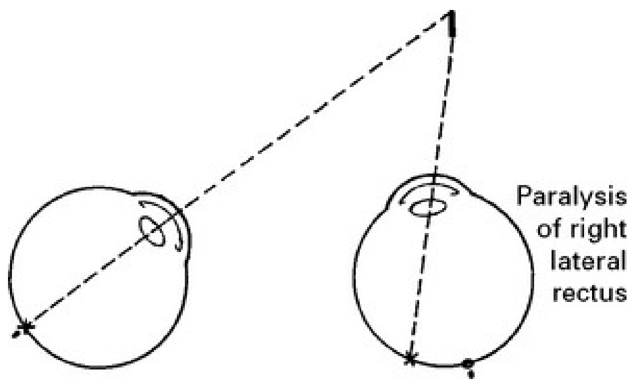


Under the dura the nerve runs up the *petrous portion* of the temporal bone and from its apex passes on to the lateral wall of the *cavernous sinus* and finally through the *superior orbital fissure*.

Note the long intracranial course and the proximity of the VI to the V cranial and greater superficial petrosal nerves at the apex of the petrous temporal bone.

DIPLOPIA

When the eyes fix on an image, impairment of movement of one eye results in projection of the image upon the macular area in the normal eye and to one side of the macula in the paretic eye; two images of the single object are thus perceived.



The image seen by the paretic eye is the false image; that seen by the normal eye is the *true image*. The false image is always outermost; this may lie in the vertical or the horizontal plane.

CLINICAL ASSESSMENT

1. **Examine the orbits**

- proptosis (forward displacement of the globe)
  - *orbital tumour or granuloma* → *Investigations*: CT/MRI scan
  - *carotid cavernous fistula* → CT/MR angiography
  - *cavernous sinus thrombosis* → CT/MR sinuses (contiguous infection)
  - *thyrotoxicosis* (unilateral exophthalmos) → CT/MRI (muscle enlargement) thyroid function
- globe fixation
  - *orbital fracture* with tethering of the globe → X-ray/CT

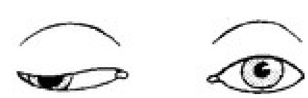
2. **Examine ocular movement** (page 12)
– note the presence of a squint or *strabismus* i.e. when the axes of the eyes are not parallel. →

Differentiate

- **Concomitant** squint (heterotropia) – an ocular disorder. The eyes adopt an abnormal position in relation to each other and the deviation is constant in all directions of gaze. Such squints develop in the first few years of life before binocular vision is established. Usually they are convergent (esotropia), occasionally divergent (exotropia). Suppression of vision from one eye (*amblyopia ex anopsia*) results in *absence of diplopia*.
  Occasionally patients subconsciously alternate vision from one eye to the other, retaining equal visual function in both – *strabismus alternans*. Correction of an underlying hypermetropia with convex lenses may offset the tendency for the eyes to converge.
- **Paralytic** squint:
  – Affected eye shows limited movement.
  – Angle of eye deviation and diplopia greatest when looking in the direction controlled by the weak muscle.
  – Diplopia is always present.
  – The patient may assume a head tilt posture to minimise the diplopia.
  Paralytic squint results from disturbance of function of nerves or muscles.

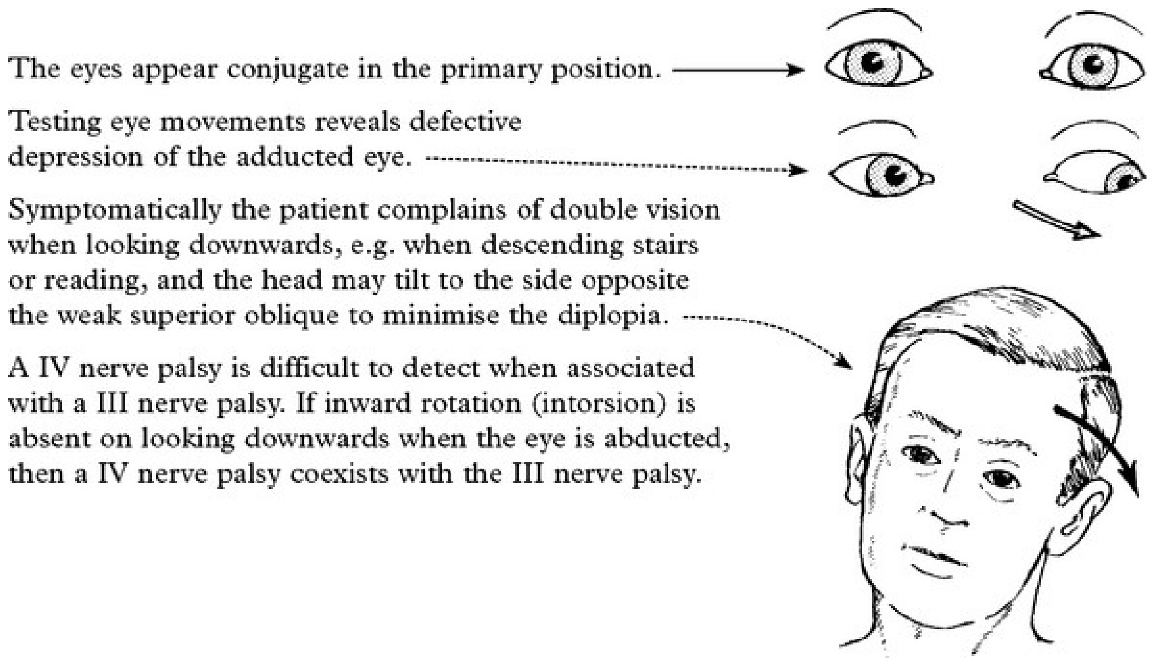
III NERVE LESION

In the primary position, the affected eye deviates laterally (due to unopposed action of the lateral rectus) and **ptosis** and **pupil dilatation** are evident. →

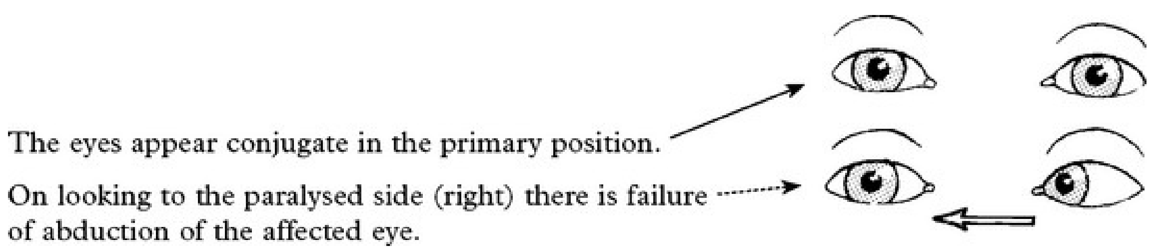


(Ptosis may be complete, unlike the partial ptosis of a Horner’s syndrome which disappears on looking up.)

IV NERVE LESION



VI NERVE LESION



Diplopia is horizontal (true and fake image side by side), is present only when looking to the paralysed side and is maximal at the extreme of binocular lateral vision.

NOTE: In partial oculomotor palsies, the patient may be aware of diplopia, although eye movements appear normal. When this occurs:

– check diplopia is ‘true’ by noting its disappearance on covering one eye.

– determine the direction of maximal image displacement and the eye responsible for the outermost image (see page 13).

This information is sufficient to differentiate a III, IV and VI nerve lesion.

OCULAR MUSCLES

If the limitation of eye movement is not restricted to one muscle, or group of muscles with a common innervation, and affects both eyes, look for:

- involvement of extraocular muscles (levator palpebrae superioris, orbicularis oculi)
- signs of fatigue on repeated testing

myasthenia gravis
ocular myopathy

CAUSES OF III NERVE LESION

Midbrain

When BILATERAL → oculomotor nucleus

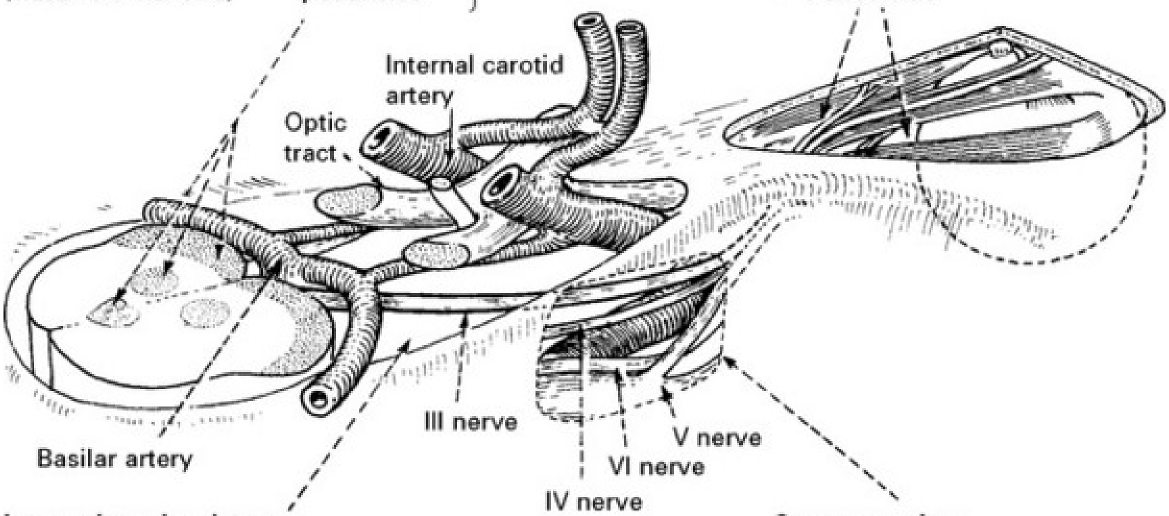
When III nerve lesion is associated with

TREMOR → red nucleus
OR
CONTRALATERAL HEMIPARESIS (WEBER'S SYNDROME) → cerebral peduncles

Infarction, demyelination, intrinsic tumour, e.g. glioma, basilar aneurysm compression

Orbital fissure/orbit

Look for PROPTOSIS and associated involvement of the IV, VI and FIRST DIVISION of the V NERVES
- *Orbital tumour, granuloma,*
- *Periosteitis*



Interpeduncular cistern

WHEN III NERVE LESION IS ASSOCIATED WITH:

DETERIORATION OF CONSCIOUS LEVEL → *Transtentorial herniation*

RETRO-ORBITAL PAIN ± SUBARACHNOID HAEMORRHAGE → *Aneurysm compression (posterior communicating or basilar aneurysm)*

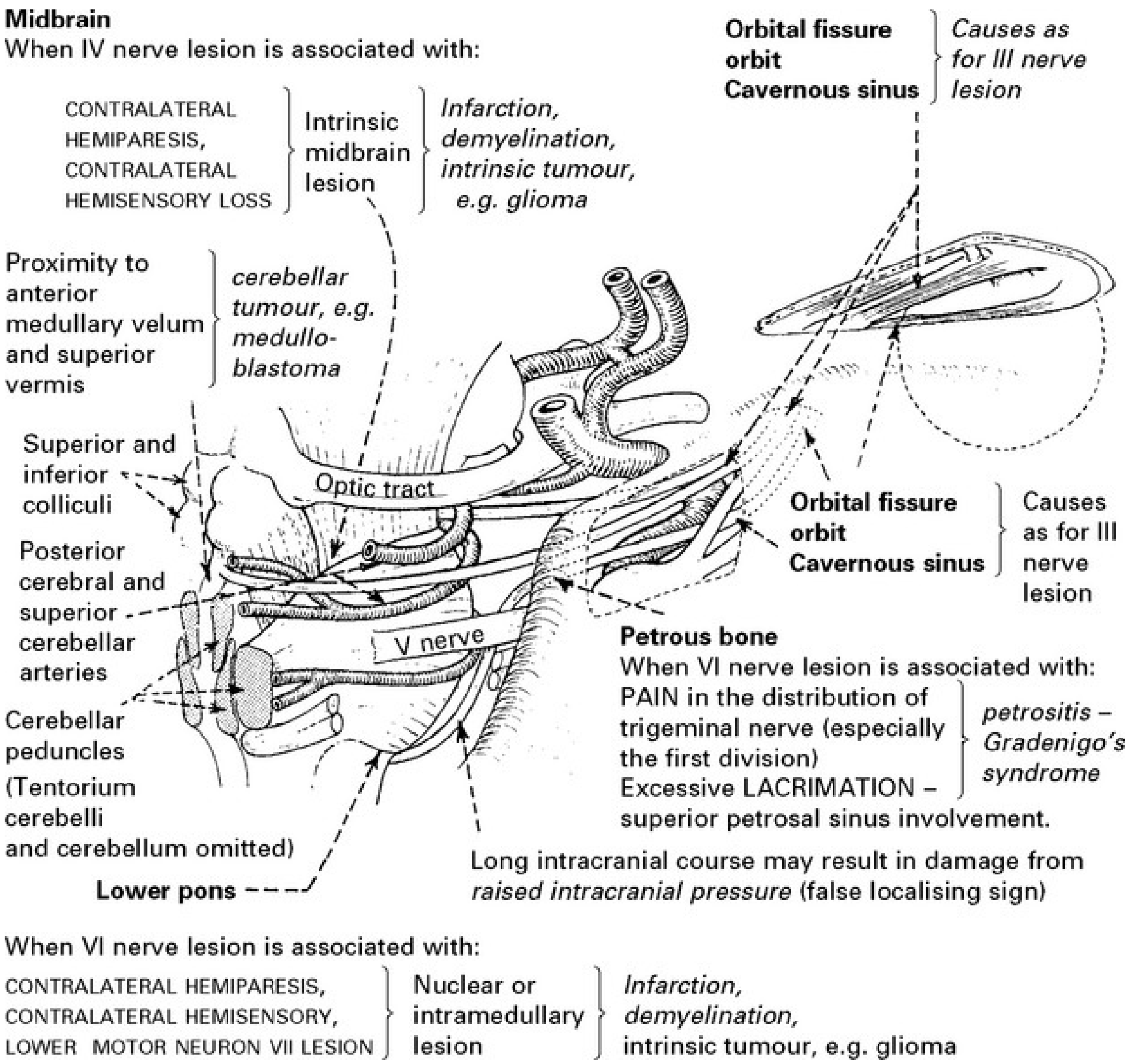
MENINGISM + OTHER CRANIAL NERVE PALSIES → *Basal meningitis*
- *TB, syphilitic, bacterial, fungal*
- *carcinomatous*

PUPIL REACTION SPARED SUDDEN ONSET → Nerve trunk infarction
- *hypertension,*
- *diabetes,*
- *polyarteritis nodosa,*
- *SLE*

Cavernous sinus

Look for associated involvement of IV, VI and 1st DIVISION OF V NERVE
- *Tumour e.g. pituitary adenoma, meningioma, metastasis, nasopharyngeal carcinoma*
- *Intracavernous aneurysm*
- *Cavernous sinus thrombosis*

CAUSES OF IV AND VI NERVE LESIONS



NOTE: Infective or carcinomatous meningitis and nerve trunk infarction may also involve the IV and VI nerves, although less often than the III nerve.

Investigative approach

III, IV or VI nerve lesions require investigation with *MRI (or CT)*; a III nerve lesion needs urgent investigation with *CT/MR angiography* to look for an enlarging aneurysm. Further investigation with *inflammatory markers* and *CSF cytology*, as directed by clinical circumstances. Elderly hypertensive or diabetic patients with complete pupillary sparing III nerve lesions will not need vascular imaging. Prognosis will depend on cause.

If myopathy or myasthenia gravis is suspected then acetyl choline receptor antibodies, EMG studies and perhaps muscle biopsy may be needed.

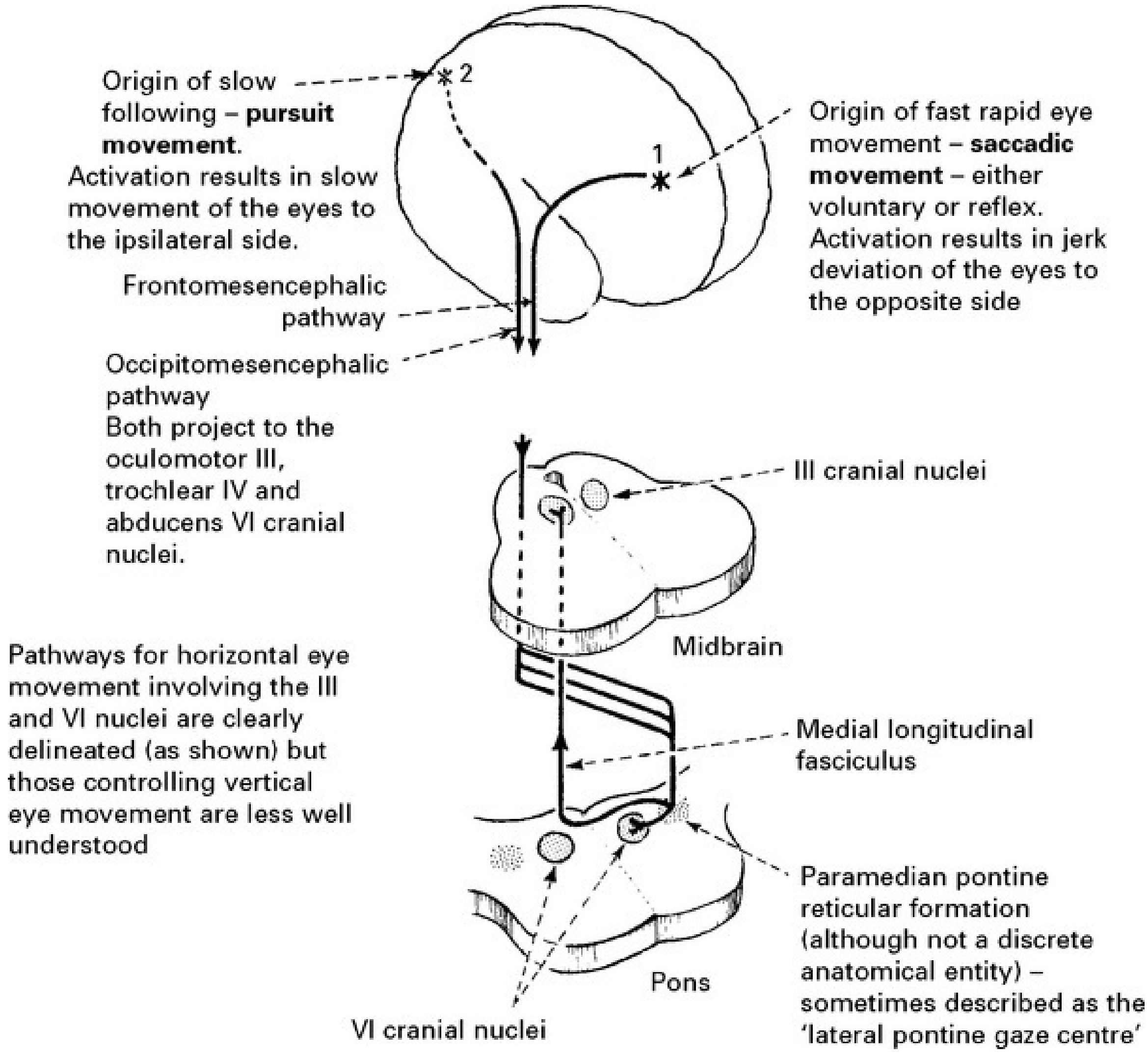
DISORDERS OF GAZE

ANATOMY AND PHYSIOLOGY

Two cortical centres of ocular control are recognised:

1. Middle gyrus of frontal lobe (frontal eye field).

2. Occipital cortex.



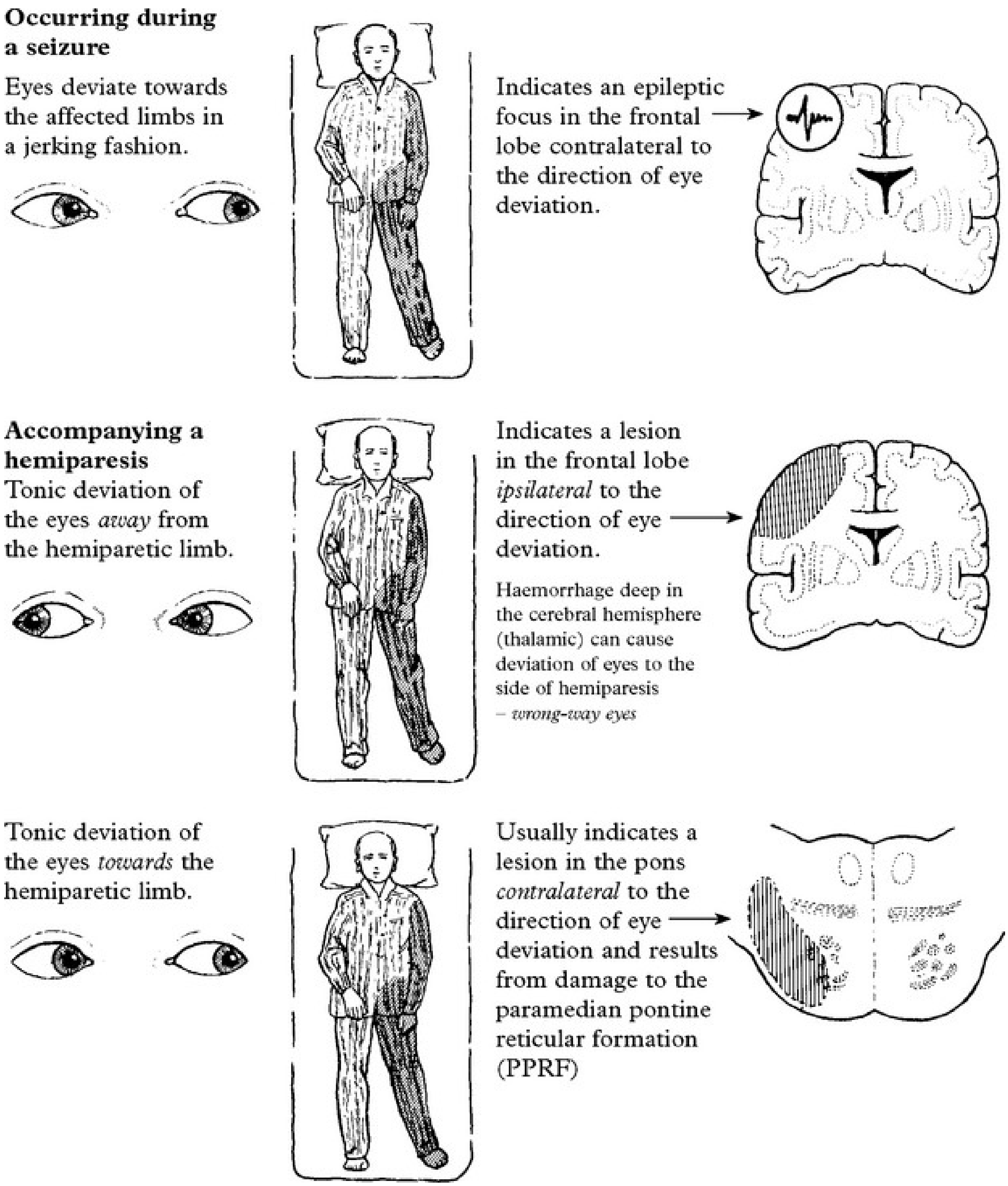
Note that the cortical descending pathways from one side activate the ipsilateral III nucleus and the contralateral VI nucleus thus swinging the direction of gaze to the opposite side.

It is important to distinguish between **saccadic** and **pursuit** movement. When following an object a slow pursuit movement maintains the image on the macular area of the retina. To fixate on a new object, rapid saccadic movement aligns the new target on the macular area. When locked into the new target, pursuit movement maintains fixation.

Eye movement occurs voluntarily in a conjugate (parallel) manner in any direction. Eye movements also occur reflexly to labyrinthine stimulation – **the vestibular ocular reflex**.

Gaze disorders usually follow vascular episodes (infarct or haemorrhage) but may also occur in traumatic, inflammatory or neoplastic disease. In gaze palsy eye movements are symmetrically limited in one direction.

CONJUGATE DEVIATION OF THE EYES



VERTICAL GAZE PALSY

Midbrain or pontine lesions may produce failure of upward or downward gaze. Disturbed downward gaze alone occurs with periaqueductal (Sylvian aqueduct) lesions. Impaired vertical eye movement is common in extrapyramidal disease (Progressive supranuclear palsy, page 366).

PARINAUD'S SYNDROME

This syndrome is characterised by impaired upward eye movements in association with a dorsal midbrain lesion **(+)**.

– upward gaze and convergence are lost

– the pupils may dilate and the response to light and accommodation is impaired

Causes:

Third ventricular tumours

Pineal region tumours

Hydrocephalus

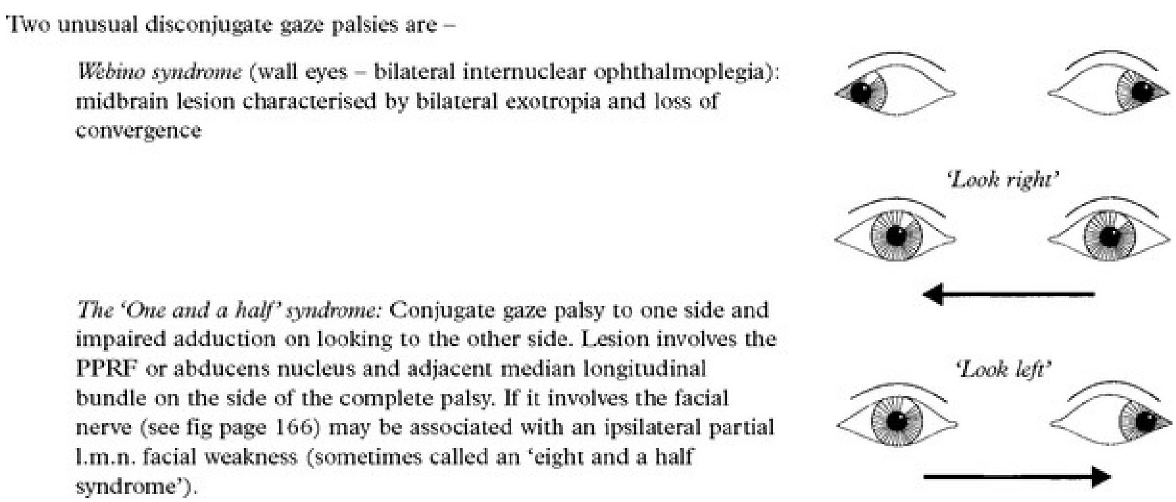
Multiple sclerosis

Wernicke's encephalopathy

Encephalitis

INTERNUCLEAR OPHTHALMOPLEGIA (ataxic nystagmus)

This disorder, caused by damage to the medial longitudinal bundle, is dealt with on page 186. It is an internuclear disorder of eye movement and produces a disconjugate gaze palsy.



OCULAR APRAXIA

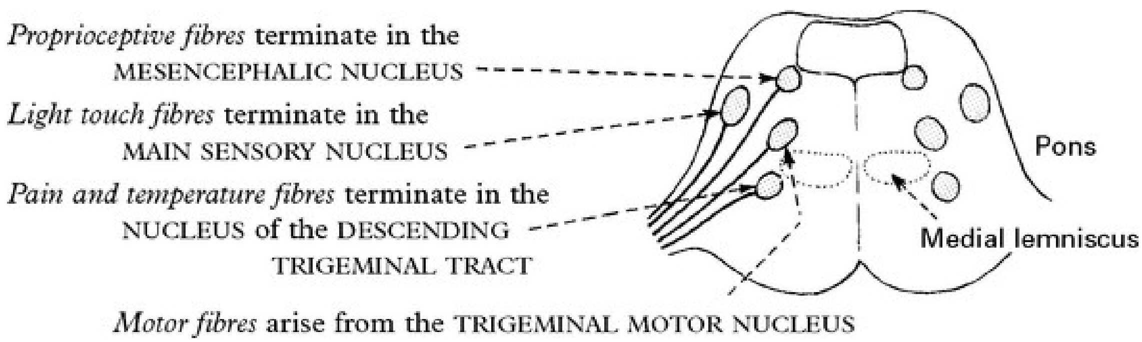
Bilateral prefrontal motor cortex damage will produce this unusual finding in which the patient does not move the eyes voluntarily to command, yet has a full range of random eye movement.

FACIAL PAIN AND SENSORY LOSS

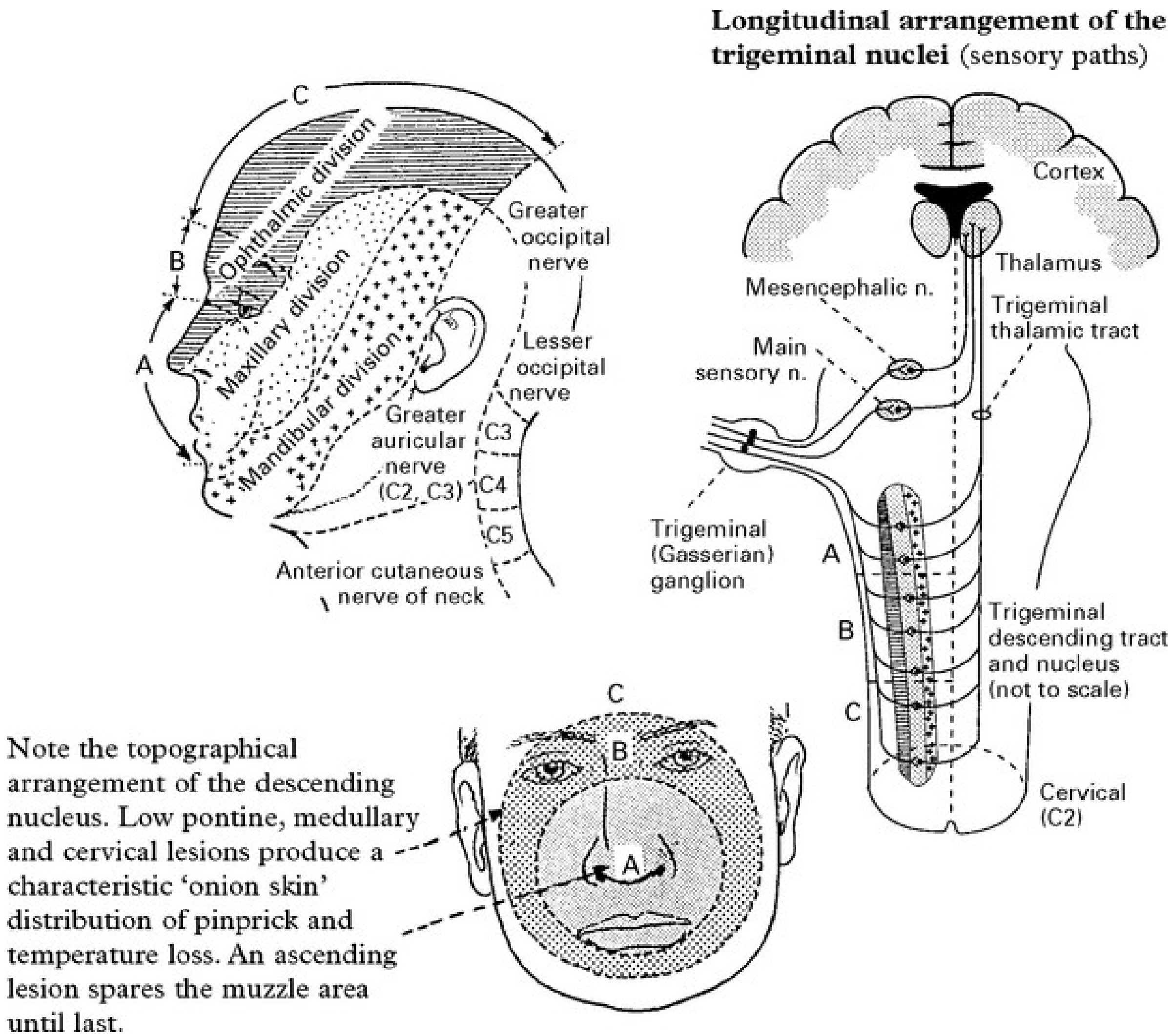
The fifth cranial nerve subserves facial sensation and innervates the muscles of mastication.

Anatomy

The anatomical arrangement of the trigeminal central connections are complex.

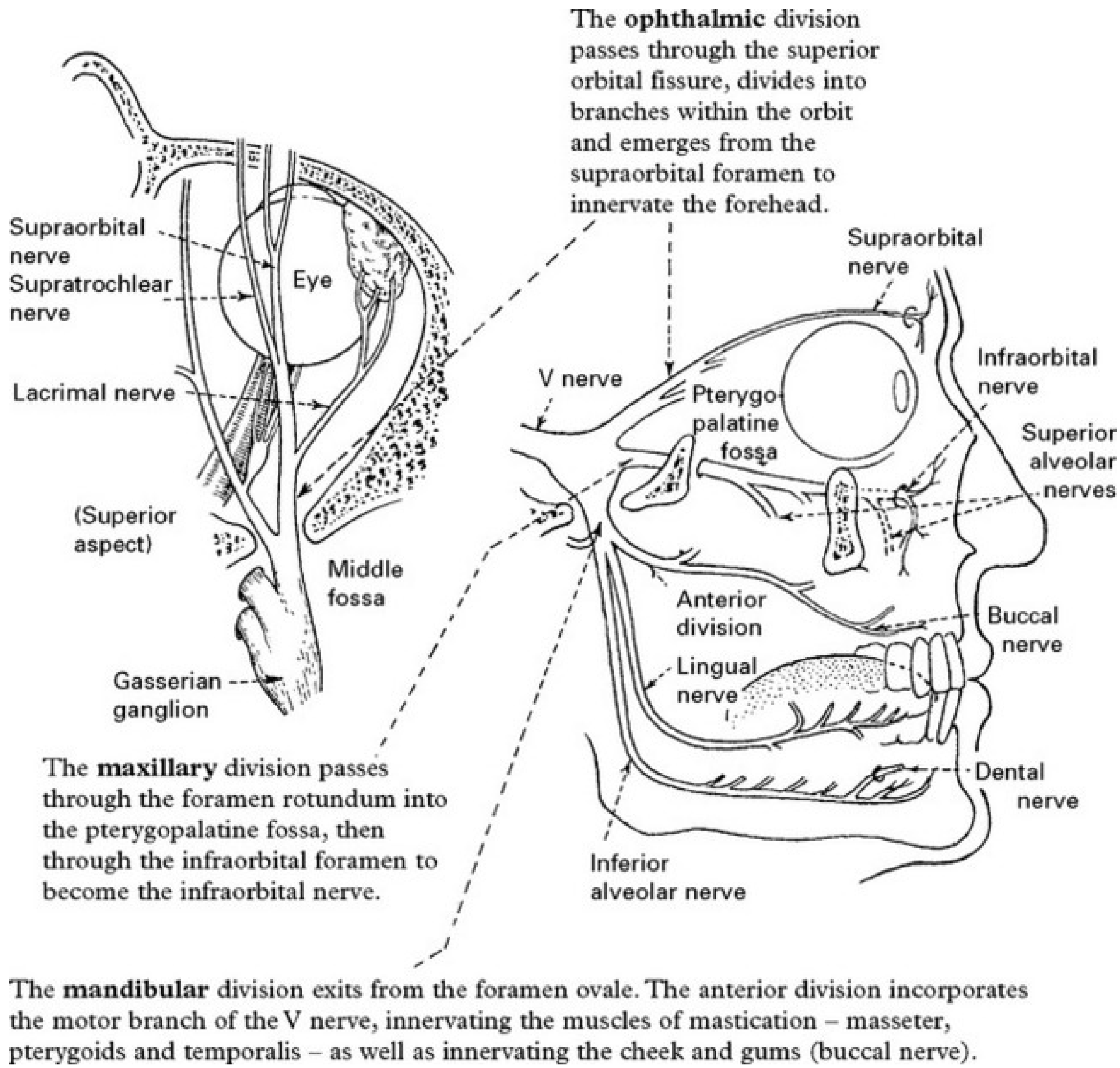


The separate location of the main sensory nucleus and nucleus of the descending trigeminal tract account for **dissociated sensory loss**, *i.e.* a low pontine or medullary lesion will result in loss of pain and temperature sensation with preservation of light touch.



The peripheral course of the V nerve

The motor and sensory nerve roots emerge separately from the lateral aspect of the brain stem at the midpontine level. The Gasserian ganglion of the sensory root contains bipolar sensory nuclei and lies on the apex of the petrous bone in the middle fossa. Here the three divisions of the trigeminal nerve merge. Each passes through its own foramen and carries sensation from a specific area of the face.



The lingual branch of the posterior trunk innervates the anterior two-thirds of the tongue (and is joined by the chordi tympani from the facial nerve carrying salivary secretomotor fibres and taste from the anterior two-thirds of the tongue).

EXAMINATION OF TRIGEMINAL NERVE FUNCTION

This should include examination of the corneal reflex and masticatory muscle function (page 14).

Note *pattern* of sensory loss → Divisional (i.e. V_1, V_2 or V_3) → *root or peripheral nerve lesion*
or
→ 'onion skin' → *brain stem lesion*

Note the *type* of sensory loss → Dissociated sensory loss (i.e. pain and temperature sensation lost, touch retained) → *brain stem lesion*

Note the presence of limb motor and/or sensory signs → With cranial nerve palsies → favours an intrinsic brain stem lesion, but does not exclude a cerebellopontine angle mass, causing brain stem distortion
→ Without cranial nerve palsies → supratentorial lesion

CAUSES OF V NERVE LESIONS

Pons

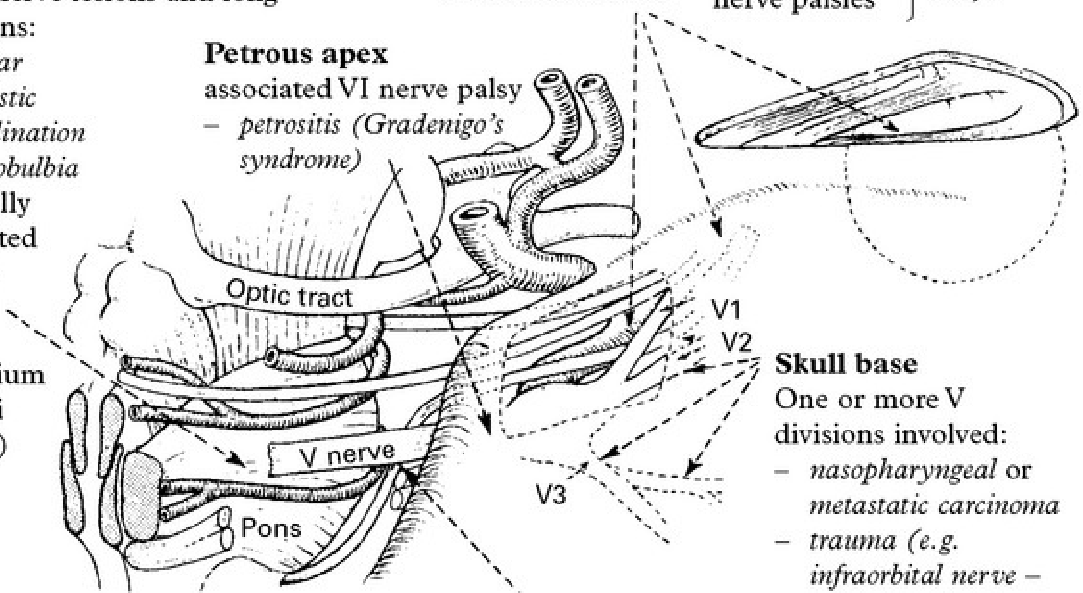
When associated with other cranial nerve lesions and long tract signs:
- *vascular*
- *neoplastic*
- *demyelination*
- *syringobulbia* (especially dissociated sensory loss)

(Tentorium cerebelli omitted)

Petrous apex
associated VI nerve palsy
- *petrositis (Gradenigo's syndrome)*

Orbital fissure
Orbit
Cavernous sinus

First division of V nerve ± III, IV and VI nerve palsies *(see III nerve lesions, page 153).*



Skull base
One or more V divisions involved:
- *nasopharyngeal* or *metastatic carcinoma*
- *trauma (e.g. infraorbital nerve* – malar fracture)

Other causes
- *diabetes*
- *SLE*

Cerebello-pontine angle
When associated with other cranial nerve lesions ± long tract signs:
- *acoustic neuroma*
- *trigeminal neuroma*
- *subacute (chronic) meningitis*

Sensory trigeminal neuropathy:

Progressive, painless loss of trigeminal sensation. Normally unilateral and without trigeminal motor weakness, the sensory loss may affect one or all trigeminal divisions. This condition is often associated with established connective tissue disease (scleroderma, Sjögren's syndrome and mixed connective tissue disease (MCTD)). Diagnosis requires exclusion of intracranial granuloma and tumour compressing the trigeminal nerve – meningioma, schwannoma, epidermoid – by contrast enhanced MRI.

Mental neuropathy (numb chin syndrome):

Caused by a lesion of the mandibular nerve or inferior alveolar or mental branches, usually the result of metastatic compression of the nerve within the mandible. Bone scans or an enhanced CT/MRI combined with image-guided aspiration is diagnostic.

Infraorbital neuropathy (numb cheek syndrome) has similar etiology.

Gradenigo's syndrome:

Lesions located at the petrous-temporal bone apex (osteitis or meningitis associated with otitis media) irritate the ophthalmic division of the trigeminal and abducens (VI) nerve. Forehead pain is accompanied by ipsilateral lateral rectus palsy and a Horner's syndrome if sympathetic fibres are also involved. Tumours and trauma can also produce this syndrome.

Neuropathic keratitis

Corneal anaesthesia from a central or peripheral V nerve lesion may lead to a neuropathic keratitis. The corneal surface becomes hazy, ulcerated and infected and blindness may follow.

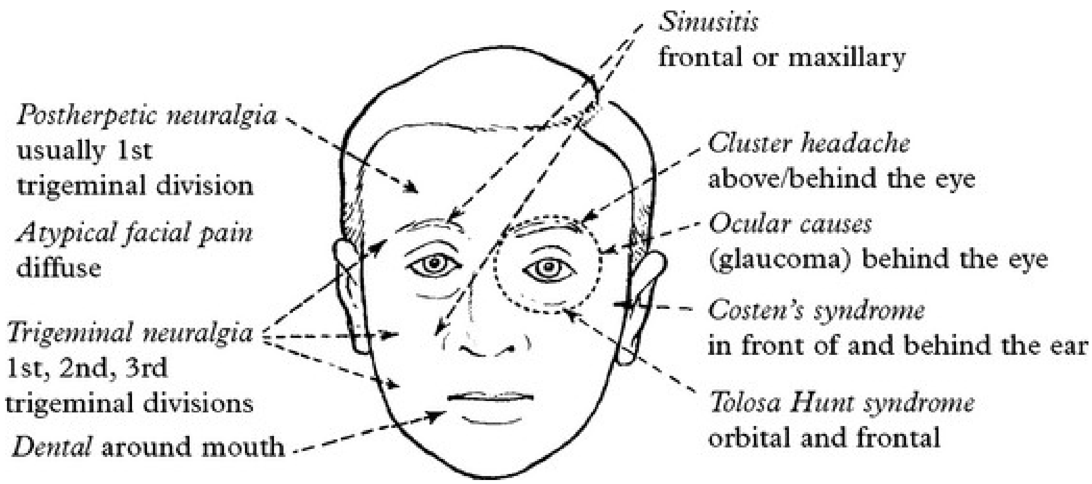
Patients with absent corneal sensation should wear a protective shield, attached to the side of spectacles, when out of doors.

FACIAL PAIN – DIAGNOSTIC APPROACH

Pain in the face may result from many different disorders and often presents as a diagnostic problem to the neurologist or neurosurgeon.

Consider:

1. **Site of pain**



2. **Quality of pain**

Trigeminal neuralgia	– sharp, stabbing, shooting, paroxysmal
Atypical facial pain	– dull, persisting
Postherpetic neuralgia	– dull, burning, persisting, occasional paroxysm
Dental	– dull
Sinusitis	– sharp, boring, worse in the morning
Ocular	– dull, throbbing
Costen's syndrome	– severe aching, aggravated by chewing
Cluster headache	– sharp, intermittent

3. **Associated symptoms/signs**

Trigeminal neuralgia Atypical facial pain Postherpetic neuralgia Dental Sinusitis Ocular Costen's syndrome Cluster headache	– often no neurological deficit, but occasional blunting of pinprick over involved region – accompanying features of depressive illness – evidence of scarring associated with sensory loss – swelling of lips/face – puffy appearance around eyes, tenderness to percussion over involved sinus – glaucoma: associated visual symptoms – blurring/haloes/loss – tenderness over temporomandibular joint – associated lacrimation/rhinorrhoea

Investigations

– guided by clinical suspicion

Blood tests:	ESR, FBC, biochemistry.
Imaging:	CT/MRI, dental X-rays, isotope bone scan.

FACIAL PAIN – TRIGEMINAL NEURALGIA

TRIGEMINAL NEURALGIA (tic douloureux)

Trigeminal neuralgia is characterised by paroxysmal attacks of severe, short, sharp, stabbing pain affecting one or more divisions of the trigeminal nerve. The pain involves the second or third divisions more often than the first; it rarely occurs bilaterally and never simultaneously on each side, occasionally more than one division is involved. Paroxysmal attacks last for several days or weeks; they are often superimposed on a more constant ache. When the attacks settle, the patient may remain pain free for many months.

Chewing, speaking, washing the face, tooth-brushing, cold winds, or touching a specific 'trigger spot', *e.g.* upper lip or gum, may all precipitate an attack of pain.

Trigeminal neuralgia more commonly affects females and patients over 50 years of age.

Aetiology

Trigeminal pain may be symptomatic of disorders which affect the nerve root or its entry zone.

Root or root entry zone compression

– *arterial vessels* often abut and sometimes clearly indent the trigeminal nerve root at the entry-zone into the pons, causing ephaptic transmission (short circuiting).

– *tumours* of the cerebellopontine angle lying against the V nerve roots, *e.g.* meningioma, epidermoid cyst, frequently present with trigeminal pain.

Demyelination – such a lesion in the pons should be considered in a 'young' person with trigeminal neuralgia. Trigger spots are rare. Remission occurs infrequently and the response to drug treatment is poor.

In some patients the cause remains unexplained, as do the long periods of remission.

Investigation

MR scan to exclude a cerebellopontine angle lesion or demyelination.

Management

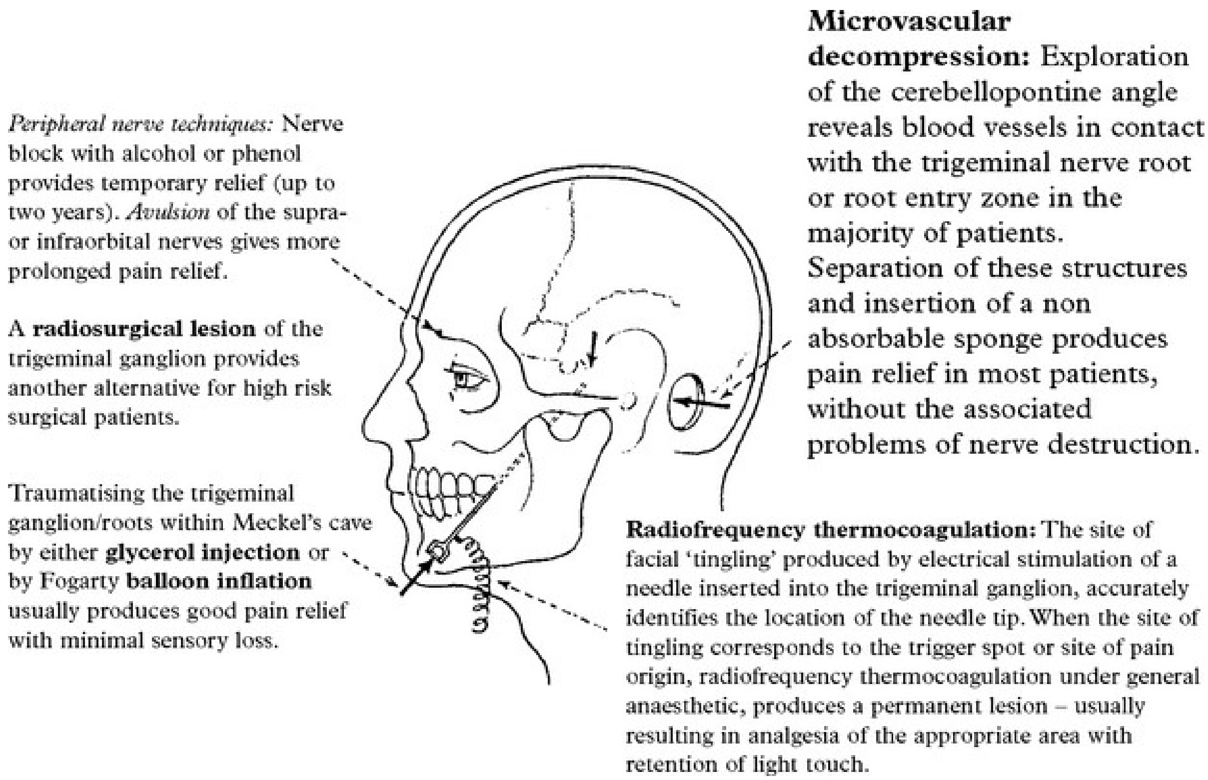
Drug therapy

CARBAMAZEPINE proves effective in most patients (and helps confirm the diagnosis). Provided toxicity does not become troublesome, *i.e.* drowsiness, ataxia, the dosage is increased until pain relief occurs (600–1600 mg/day). When remission is established, drug treatment can be discontinued.

If pain control is limited, other drugs – BACLOFEN, LAMOTRIGINE, GABAPENTIN, PHENYTOIN – may benefit.

Persistence of pain on full drug dosage or an intolerance of the drugs, indicates the need for more radical measures.

The choice lies between a range of lesional techniques, which all produce some damage to the trigeminal nerve with some consequent sensory loss, or microvascular decompression, which does not damage the nerve but has the risks associated with open neurosurgery.



Results and complications

Pain relief – no comparative trials have been done so accurate comparison of the wide variety of techniques used for trigeminal neuralgia is difficult. Microvascular decompression seems to be more likely to provide pain control with fewer relapses. Overall 80–85% of patients remain pain free for a 5-year period. Results of peripheral nerve avulsion are less satisfactory with pain recurring in 50% within 2 years.

Dysaesthesia/Anaesthesia dolorosa – this troublesome sensory disturbance follows any destructive technique to nerve or root in 5–30% of patients. Microvascular decompression avoids this.

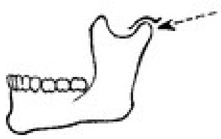
Corneal anaesthesia – this occurs when root section or thermocoagulation involves the first division and keratitis may result.

Mortality – microvascular decompression and open root section carry a very low mortality (< 1%), but this must not be ignored when comparing results with safer methods.

Treatment selection: This depends on discussion of the differing risks with the patient. In younger patients the absence of sensory complications make microvascular decompression the procedure of first choice. Frail and elderly patients may tolerate glycerol injection, balloon compression and thermocoagulation more easily than other procedures.

FACIAL PAIN – OTHER CAUSES

Temporomandibular joint dysfunction (Costen's syndrome)

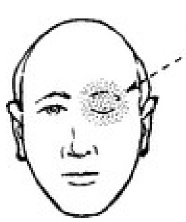


Aching pain occurring around the ear, aggravated by chewing; due to malalignment of one temporomandibular joint as a consequence of dental loss with altered 'bite' or involvement of the joint in rheumatoid arthritis. This condition requires dental treatment with realignment.

Raeder's syndrome (the paratrigeminal syndrome)

Pain and sensory loss in 1st and 2nd trigeminal divisions, maximal around the eye and associated with a sympathetic paresis (ptosis and small pupil). Sweating in the lower face is preserved. This may be associated with involvement of the other cranial nerves (IV & VI). This rare syndrome occurs with lesions of the middle fossa, *e.g.* nasopharyngeal carcinoma, granulomas and infection.

Tolosa Hunt syndrome



A condition in which an inflammatory process involving the cavernous sinus or superior orbital fissure presents with pain, loss of ocular movement and ophthalmic division sensory loss. The diagnosis is based on exclusion of tumour and response to steroids. Pathological examination confirms non-specific granulomatous change.

Atypical facial pain

The patient, often a young or middle-aged woman, experiences a dull, persistent pain, spreading diffusely over one or both sides of the face. These symptoms often result from an underlying depression and may respond well to antidepressant therapy.

Herpes zoster

Frequently affects the trigeminal territory, especially the ophthalmic division producing a painful 'herpetic rash' and often involving the cornea. The acute symptoms may resolve but lead to a chronic postherpetic neuralgia which slowly improves. Surgical procedures such as trigeminal root section do not help. The incidence of postherpetic neuralgia is not influenced by treatment with antiviral agents (acyclovir) in the acute phase.

Carotid artery dissection

This presents as acute retro-orbital pain with a Horner's syndrome (page 145) and may be associated with ipsilateral amaurosis fugax or contralateral hemisphere symptoms.

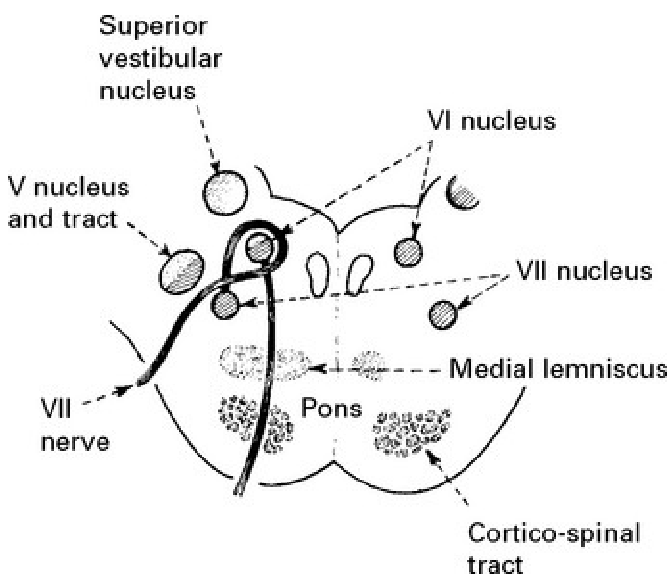
'Cluster' headaches – see page 73.

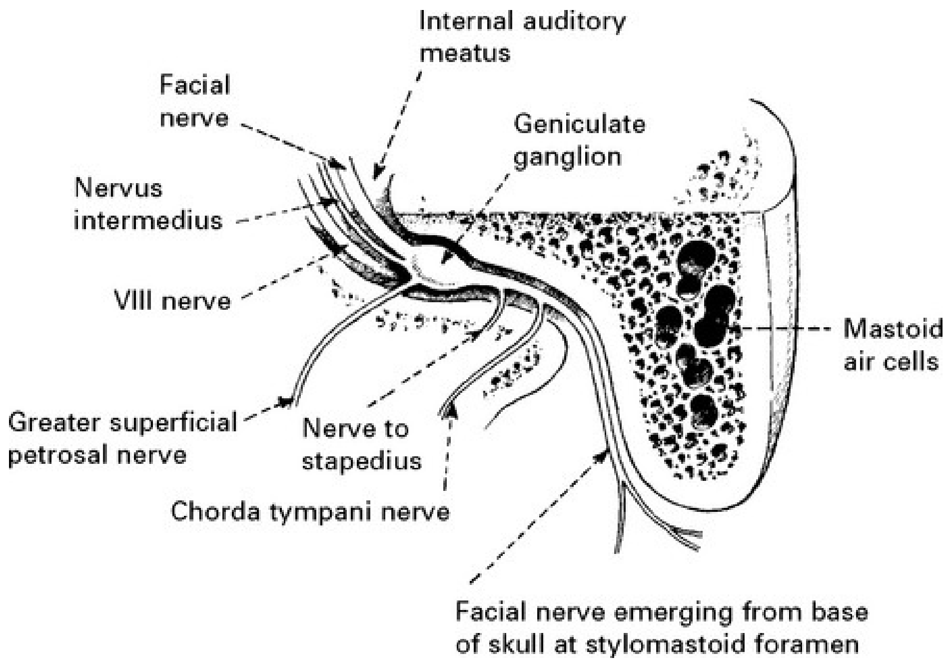
FACIAL WEAKNESS

Related anatomy

The facial (VII) nerve contains mainly motor fibres supplying the muscles of facial expression, but also visceral efferent (parasympathetic) and visceral afferent (taste) fibres.

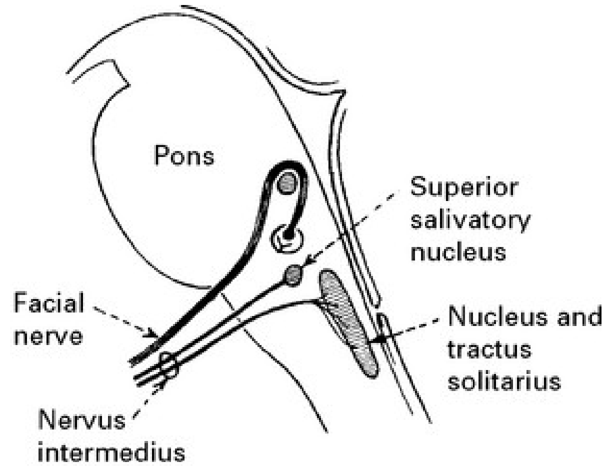
The motor nucleus lies in the lower pons medial to the descending nucleus and tract of the Vth cranial nerve. Axons from the motor nucleus wind around the nucleus of the VIth cranial nerve. The facial nerve and its visceral root (*nervus intermedius*) exit from the lateral aspect of the brain stem and cross the cerebellopontine angle immediately adjacent to the VIII cranial nerve. They enter the internal auditory meatus and, passing through the facial canal of the temporal bone, lie in close proximity to the inner ear and tympanic membrane. The facial nerve gives off several branches before exiting from the skull through the stylomastoid foramen.

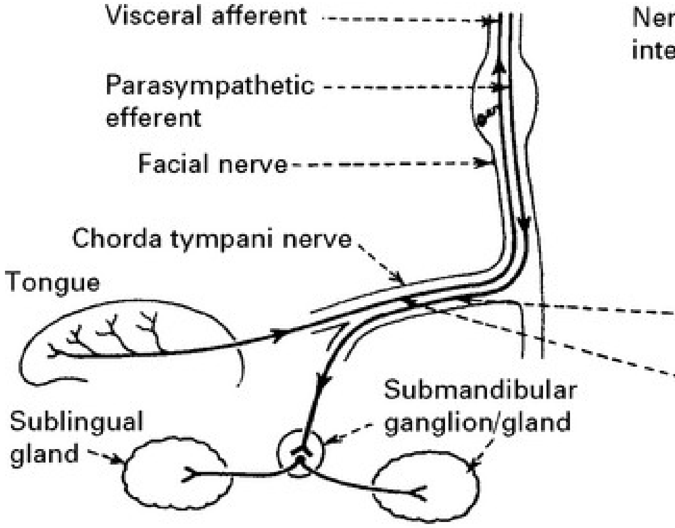




Visceral efferent and visceral afferent fibres arise and terminate in the superior salivary nucleus and nucleus/tractus solitarius respectively.

They run together as the nervus intermedius and accompany the facial nerve to the internal auditory meatus. The parasympathetic fibres (visceral efferent) pass in the greater petrosal nerve to the sphenopalatine ganglion and thence to the lacrimal gland to produce tears and in the chorda tympani nerve to the submandibular ganglion.





The chorda tympani nerve contains both parasympathetic efferent and visceral afferent fibres. Parasympathetic fibres are responsible for salivation. Visceral afferent fibres convey sensations of taste from the anterior two-thirds of the tongue. The geniculate ganglion contains the bipolar cell bodies of these afferent fibres.

Supranuclear control of facial muscles

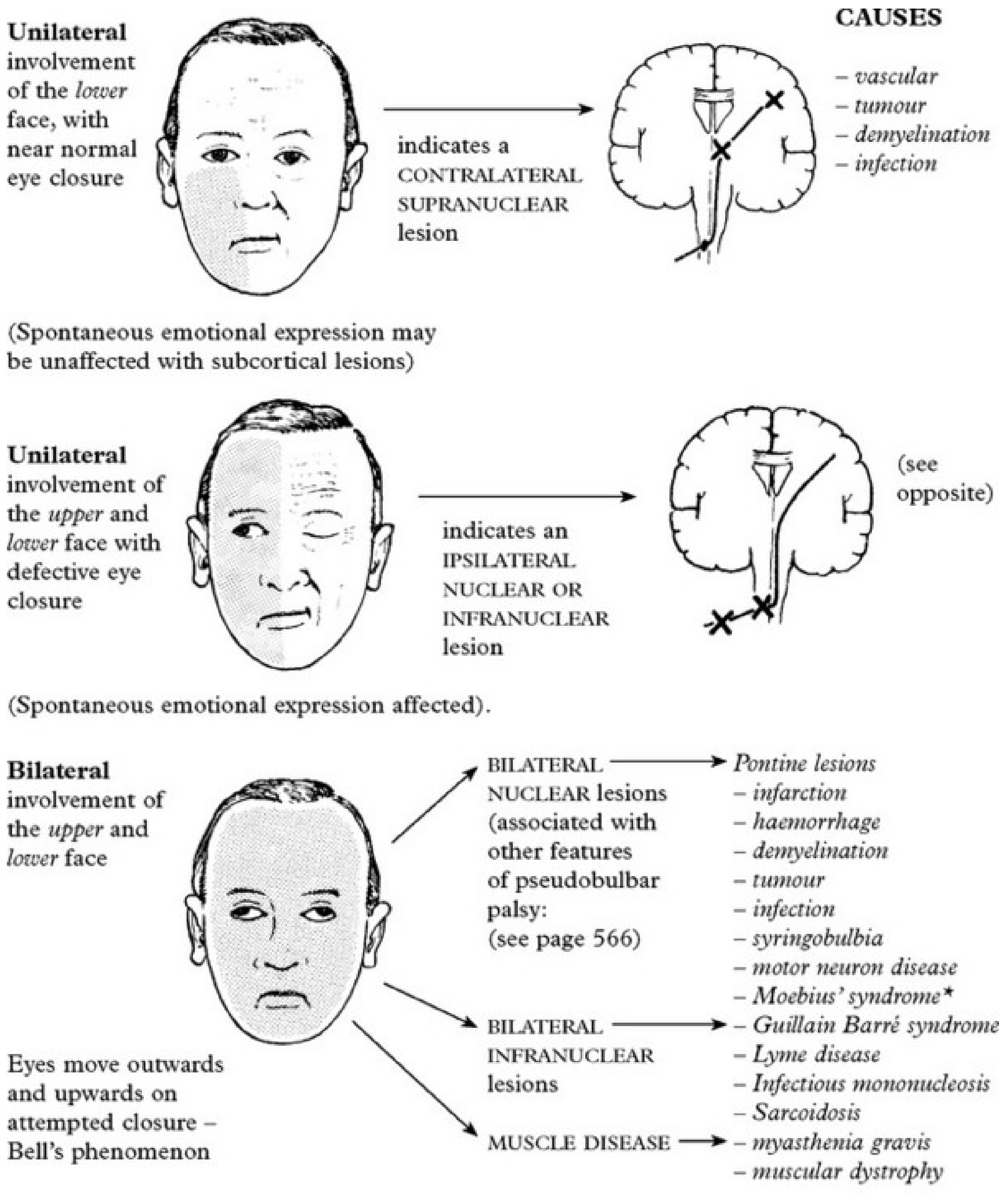
The muscles in the lower face are controlled by the contralateral hemisphere, whereas those in the upper face receive control from both hemispheres (bilateral representation). Hence a lower motor neuron lesion paralyses all facial muscles on that side, but an upper motor neuron (supranuclear) lesion paralyses only the muscles in the lower half of the face on the opposite side.

Clinical examination of the facial nerve (see page 15)

In addition to examining for facial weakness and taste impairment, also note whether the patient comments on reduced lacrimation or salivation on one side, or hyperacusis (exaggeration of sounds due to loss of the stapedius reflex).

LESION, LOCALISATION AND CAUSE

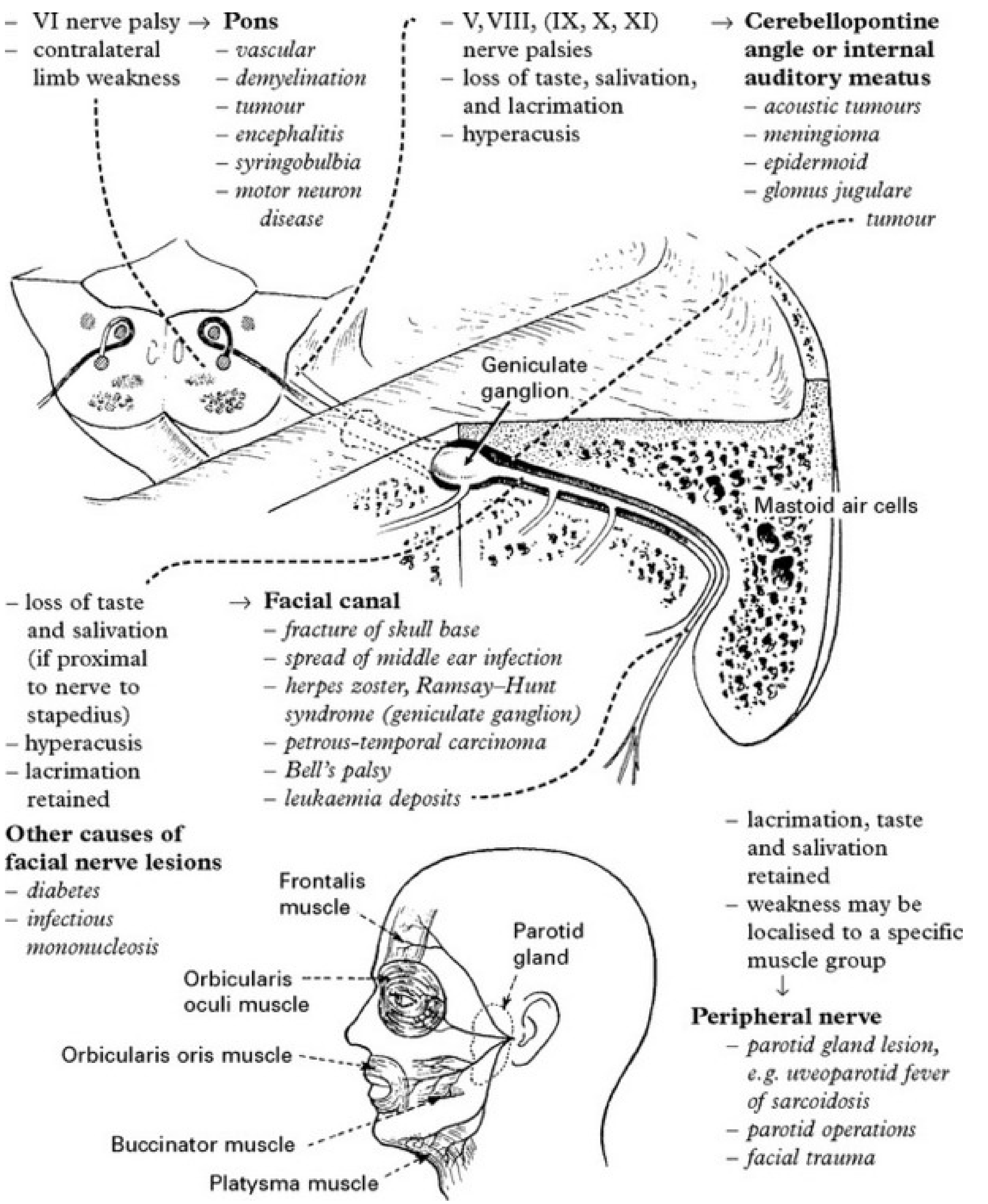
Note the *distribution:*



* Moebius' syndrome: a congenital failure of the development of the facial and abducens nuclei (bilateral).

NUCLEAR/INFRANUCLEAR LESIONS

The following features (if present) help in lesion location:



BELL'S PALSY

Bell's palsy is characterised by an acute paralysis of the face related to 'inflammation' and swelling of the facial nerve within the facial canal or at the stylomastoid foramen. It is usually unilateral, rarely bilateral, and may occur repetitively. In some, a family history of the condition is evident. Incidence 25/100,000/year.

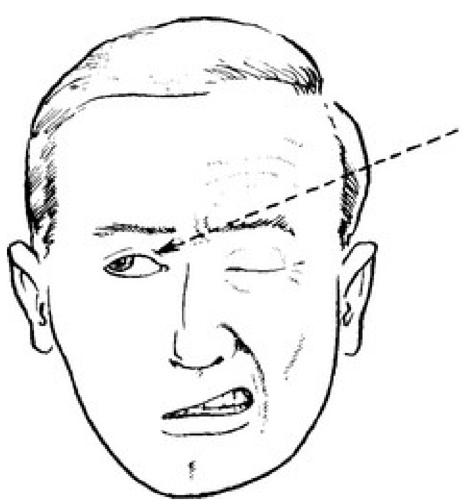
Aetiology

Uncertain, but may be associated with viral infections, *e.g.* herpes simplex and varicella-zoster; epidemics of Bell's palsy occur sporadically.

Symptoms

Pain of variable intensity over the ipsilateral mastoid precedes weakness, which develops over a 48-hour period.

Impairment of taste, hyperacusis and salivation depend on the extent of inflammation and will be lost in more severe cases. Lacrimation is seldom affected.



On attempting to close the eyes and show the teeth, the one eye does not close and the eyeball rotates upwards and outwards – Bell's phenomenon (normal eyeball movement on eye closure).

Diagnosis
Based on typical presentation and exclusion of middle ear disease, diabetes, sarcoidosis and Lyme disease.

Treatment

During the acute stage protect the exposed eye during sleep.

There is good evidence prednisolone given in high dosage in the acute stage (50 mg per day for 10 days) improves recovery. The role of antiviral therapy is less clear as conflicting results have been found in recent large trials. Eye care (shielding and artificial tears) is important in preventing corneal abrasion.

Prognosis

Most patients (70%) recover in 4–8 weeks without treatment. In the remainder, residual facial asymmetry may require corrective surgery. Incomplete paralysis indicates a good prognosis. In patients with complete paralysis, electrical absence of denervation on electromyography is an optimistic sign.

Occasionally aberrant reinnervation occurs – movement of the angle of the mouth on closing the eyes (jaw winking) or lacrimation when facial muscles contract (crocodile tears).

OTHER FACIAL NERVE DISORDERS

RAMSAY HUNT SYNDROME

Herpes zoster infection of the geniculate (facial) ganglion causes sudden severe facial weakness with a typical zoster vesicular eruption within the external auditory meatus. Pain is a major feature and may precede the facial weakness. Serosanguinous fluid may discharge from the ear.

Deafness may result from VIII involvement. Occasionally, other cranial nerves from V–XII are affected.

Treatment

Antiviral agents (acyclovir) may help.

HEMIFACIAL SPASM

This condition is characterised by unilateral clonic spasms beginning in the orbicularis oculi and spreading to involve other facial muscles. The stapedius muscle can be affected producing a subjective ipsilateral clicking sound.

Contractions are irregular, intermittent and worsened by emotional stress and fatigue.

Onset usually occurs in middle to old age and women are preferentially affected.

Most cases arise from vascular compression of the facial nerve at the root entry zone (in the same way as trigeminal neuralgia). In some, compression is caused by a tumour. Occasionally hemifacial spasm follows a Bell's palsy or traumatic facial injury.

The clinician must distinguish hemifacial spasm from milder habit spasms or tics which tend to be familial, and also from 'focal' seizures selectively affecting the face.

Investigations

MR scan of the posterior fossa excludes the presence of a cerebellar pontine angle lesion and may show an ectatic basilar artery.

Treatment

Drugs – Local infiltration with botulinum toxin of involved muscles is helpful. However, effect only lasts about 3 months and may produce temporary weakness.

Surgery – Posterior fossa exploration and microvascular decompression *i.e.* dissecting blood vessels off the facial nerve root entry zone, gives excellent results (cure rate 80%), but carries the risk of producing deafness and rarely brain stem damage.

TONIC FACIAL SPASM

Less common than hemifacial spasm. Occurs with cerebellar pontine angle lesions. It produces tonic elevation of the corner of the mouth with narrowing of the eye. The diagnosis is confirmed by CT/MR scanning and treatment is surgical.

FACIAL MYOKYMIA

A rare condition seen most often in multiple sclerosis. Flickering of facial muscles results from spontaneous discharge in the facial motor nucleus. Other brain stem signs are present. The facial movements respond to carbamazepine.

MYOCLONUS

Rhythmic facial movement associated with similar palatal movements and characteristic of dentate or olivary nucleus disease.

BLEPHAROSPASM

Spasmodic closing or screwing up of eyes (see page 371).

DEAFNESS, TINNITUS AND VERTIGO

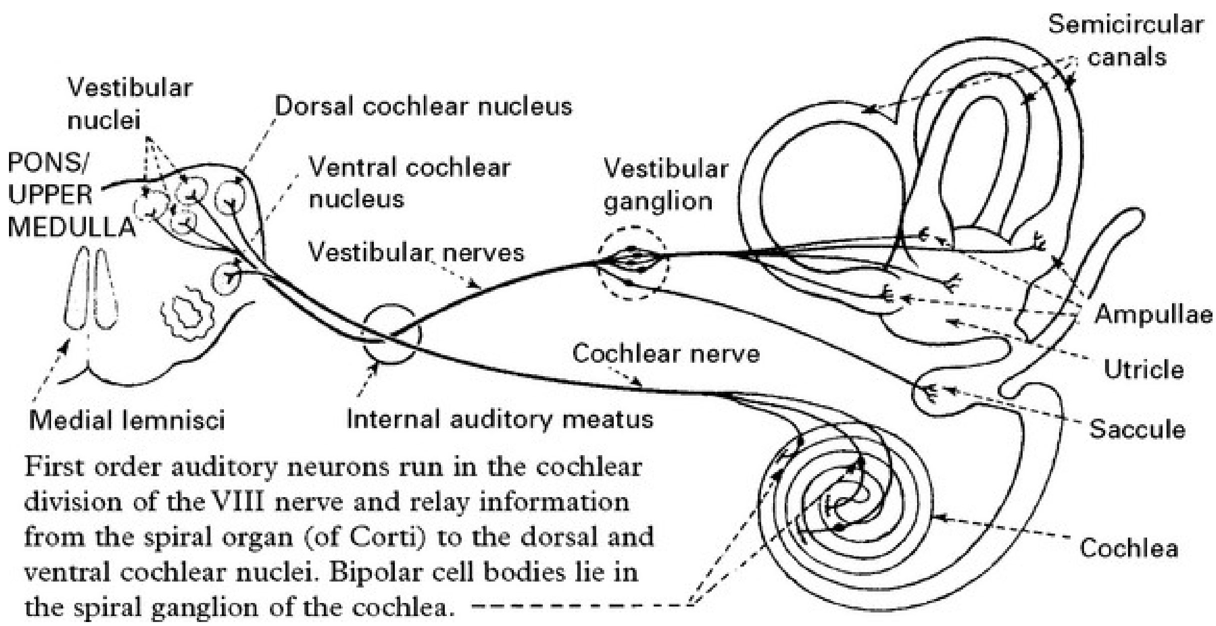
Deafness, tinnitus and vertigo result from disorders affecting the auditory and vestibular apparatus or their central connections transmitted through the VIII cranial nerve.

MECHANISMS OF AUDITORY AND VESTIBULAR FUNCTION

Auditory function: the cochlea converts sound waves into action potentials in	**Vestibular function:** the vestibular system responds to rotational and linear acceleration (including gravity) and along with a visual and proprioceptive input maintains

cochlear neurons. Sound waves are transmitted by the tympanic membrane and the ossicles to the oval window, setting up waves in the perilymph of the cochlea. The action of the waves on the spiral organ (of Corti) generates action potentials in the cochlear division of the VIII cranial nerve.	equilibrium and body orientation in space. Relative inertia of the endolymph within the semicircular canals during angular acceleration displaces hair cells imbedded in the cupula, activates the hair cells and transmits action potentials to the vestibular division of the VIII cranial nerve. Linear acceleration results in displacement of the otoliths within the utricle or saccule. This distorts the hair cells and increases or decreases the frequency of action potentials in the vestibular division of the VIII cranial nerve.

CENTRAL CONNECTIONS



First order vestibular neurons lie in the vestibular division of the VIII nerve and relay information from the utricle, saccule and semicircular canals to the vestibular nuclei (superior, inferior, medial and lateral). Bipolar cell bodies lie in the vestibular ganglion.

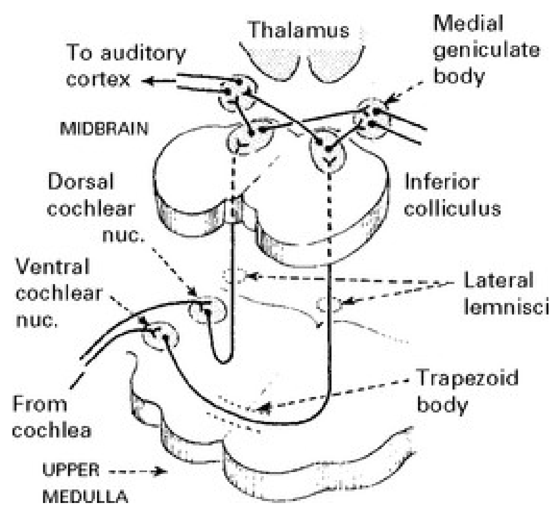
The cochlear (acoustic) and vestibular divisions travel together through the petrous bone to the internal auditory meatus where they emerge to pass through the subarachnoid space in the cerebellopontine angle, each entering the brain stem separately at the pontomedullary junction.

Auditory: From the cochlear nucleus, second order neurons either pass upwards in the lateral lemniscus to the ipsilateral inferior colliculus or decussate in the trapezoid body and pass up in the lateral lemniscus to the contralateral inferior colliculus.

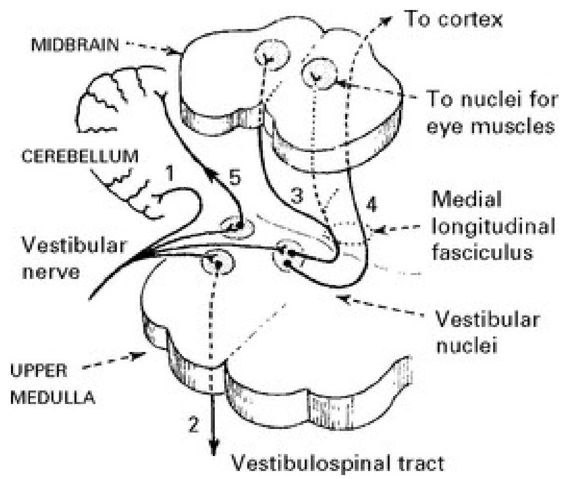
Third order neurons from the inferior colliculus on each side run to the medial geniculate body on both sides.

Fourth order neurons pass through the internal capsule and auditory radiation to the auditory cortex.

The bilateral nature of the connections ensures that a unilateral central lesion will not result in lateralised hearing loss.



Vestibular



1. Directly to cerebellum.

2. Second order neurons arise in the vestibular nucleus and descend in the ipsilateral vestibulospinal tract.

3. Second order neurons project to the oculomotor nuclei (III, IV, VI) through the medial longitudinal fasciculus.

4. Second order neurons project to the cortex (temporal lobe). The pathway is unclear.

5. Second order neurons project to the cerebellum.

(There is a bilateral feedback loop to the vestibular nuclei from the cerebellum though the fastigial nucleus.)

DEAFNESS: Three types of hearing loss are recognised:

1. *Conductive deafness:* failure of sound conduction to the cochlea.

2. *Sensorineural deafness:* failure of action potential production or transmission due to disease of the cochlea, cochlear nerve or cochlear central connections.

Further subdivision into cochlear and retrocochlear deafness helps establish the causative lesion.

3. *Pure word or cortical deafness:* a bilateral or dominant posterior temporal lobe (auditory cortex) lesion produces a failure to understand spoken language despite preserved hearing.

TINNITUS: a sensation of noise of ringing, buzzing, pulsing, hissing or singing quality.

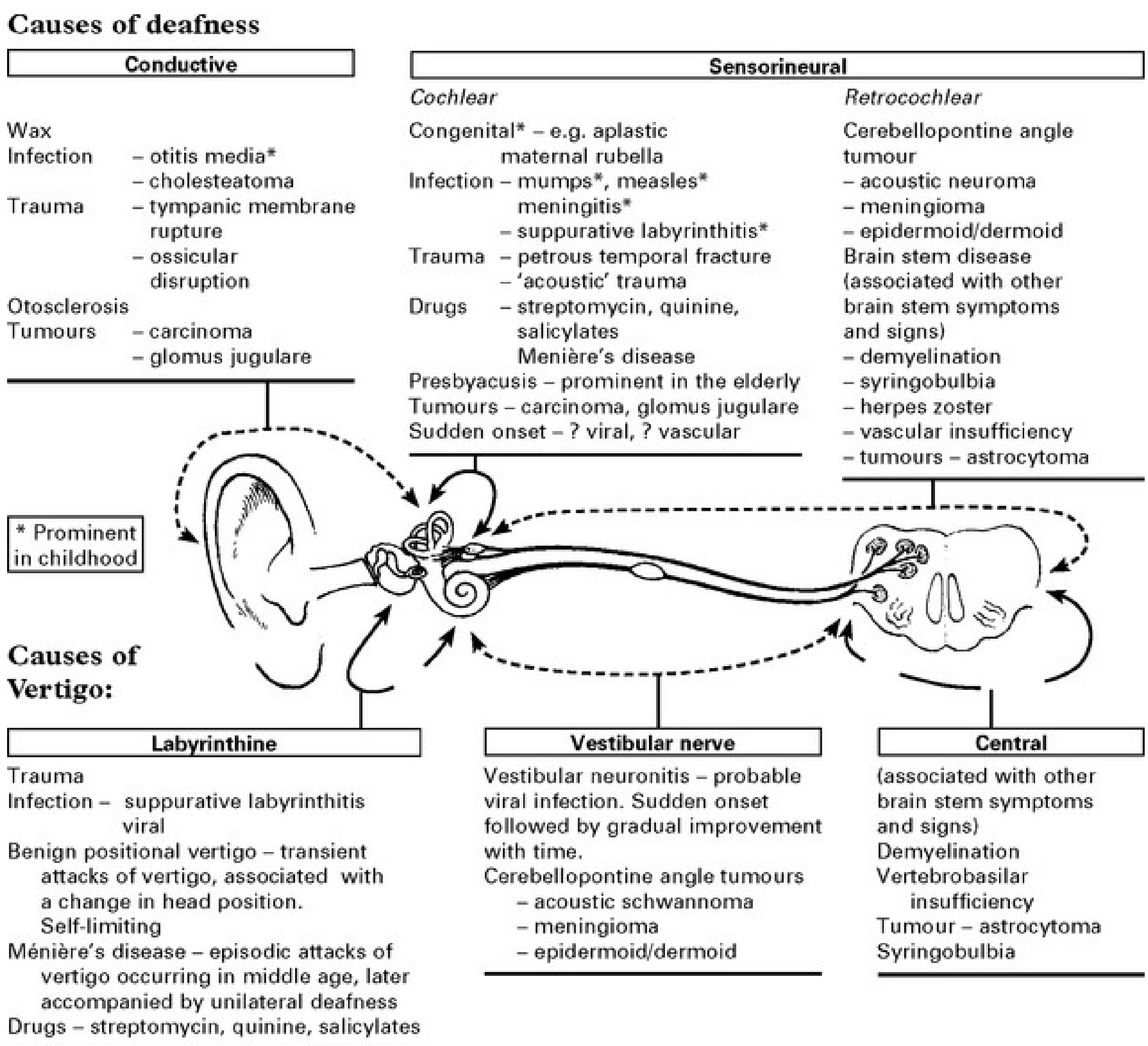
Tinnitus may be (i) continuous or intermittent, (ii) unilateral or bilateral, (iii) high or low pitch.

As a rule, when hearing loss is accompanied by tinnitus, conductive deafness is associated with low pitch tinnitus – sensorineural deafness is associated with high pitch tinnitus, except Menière's disease where tinnitus is low pitch. Pulsing tinnitus may have a vascular cause. In most patients no cause is found.

VERTIGO: an illusion of rotatory movement due to disturbed orientation of the body in space. The sufferer may sense that the environment is moving. Vertigo may result from disease of the labyrinth, vestibular nerve or their central connections.

Clinical examination

Examination of the external auditory meatus, tympanic membrane and eye movements (for nystagmus) and Weber's and Rinne's tests (page 16) provide valuable information, but more detailed neuro-otological tests (pages 62, 63) are usually required to determine the exact nature of the auditory or vestibular dysfunction and to locate the lesion site. The results of these tests may indicate the need for further investigation (e.g. CT/MR scan).



Causes of tinnitus

Any lesion causing deafness may also cause tinnitus. Occasionally patients perceive a vibratory noise inside the head, transmitted from an arteriovenous malformation or carotid stenosis. A lesion is more likely with unilateral tinnitus. No cause is found in most patients with bilateral tinnitus.

Patients with non-specific disease, *e.g.* anaemia, fever, hypertension, occasionally complain of tinnitus.

DISORDERS OF THE LOWER CRANIAL NERVES

NINTH (GLOSSOPHARYNGEAL) CRANIAL NERVE

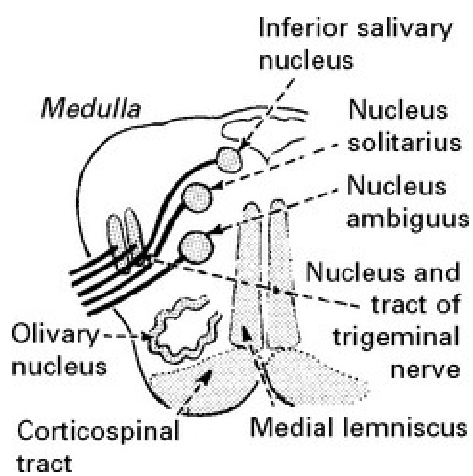
This is a mixed nerve with motor, sensory and parasympathetic functions.

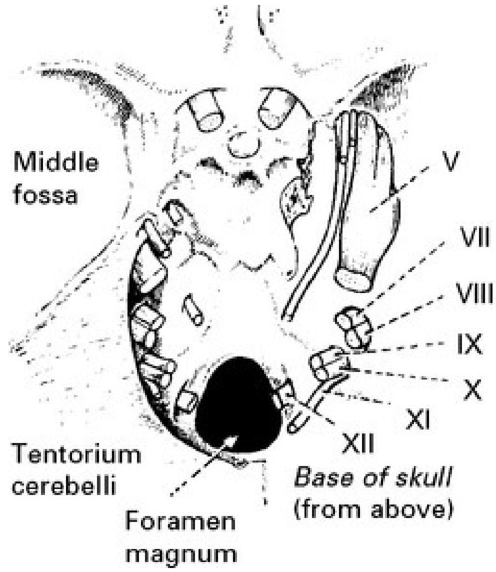
1. Motor fibres to stylopharyngeus muscle arise in the nucleus ambiguus.

2. Preganglionic parasympathetic fibres arise in the inferior salivatory nucleus and pass to the otic ganglion. From there postganglionic fibres innervate the parotid gland.

3. General somatic sensory fibres innervate the area of skin behind the ear, pass to the superior ganglion and end in the nucleus and tract of the trigeminal nerve.

4. Sensory fibres innervate the posterior third of the tongue (taste), pharynx, eustachian tube and carotid body/sinus and terminate centrally in the nucleus solitarius. The cell bodies lie in the inferior ganglion.





The IX nerve emerges as 5 or 6 rootlets from the medulla, dorsal to the olivary nucleus and passes with the vagus and accessory nerves through the jugular foramen in the neck.

Within the neck the nerve lies in close proximity to the internal carotid artery and internal jugular vein.

The superior and inferior ganglia lie in the jugular foramen, the otic ganglion in the neck below the foramen ovale.

Clinical examination (see page 17)

Disorders of the glossopharyngeal nerve

Glossopharyngeal palsy from either medullary or nerve root lesions does not occur in isolation. When associated with X and XI cranial nerve lesions, this constitutes the *jugular foramen syndrome*. Lesions producing this syndrome are listed on page 179.

GLOSSOPHARYNGEAL NEURALGIA

Short, sharp, lancinating attacks of pain, identical to trigeminal neuralgia in nature but affecting the posterior part of the pharynx or tonsillar area. The pain often radiates towards the ear and is triggered by swallowing. Reflex bradycardia and syncope occur due to stimulation of vagal nuclei by discharges from glossopharyngeal. As with trigeminal neuralgia, carbamazepine often provides effective relief – if not microvascular decompression or section of the IX nerve roots or nerve give good results.

TENTH (VAGUS) CRANIAL NERVE

This is a mixed nerve with motor, sensory and parasympathetic functions.

The central connections are complex though similar to those of the glossopharyngeal nerve.

1. Motor fibres supplying the pharynx, soft palate and larynx arise in the nucleus ambiguus.

2. Preganglionic parasympathetic fibres arise in the dorsal motor nucleus. Postganglionic fibres supply the thoracic and abdominal viscera.

3. Afferent fibres from the pharynx, larynx and external auditory meatus have cell bodies in the jugular ganglion and end in the nucleus and tract of the trigeminal nerve.

4. Afferent fibres from abdominal and thoracic viscera have cell bodies in the nodose ganglion and end in the nucleus solitarius. Taste perception in the pharynx ends similarly.